SONGNODES AND THE MUSIC LIBRARY

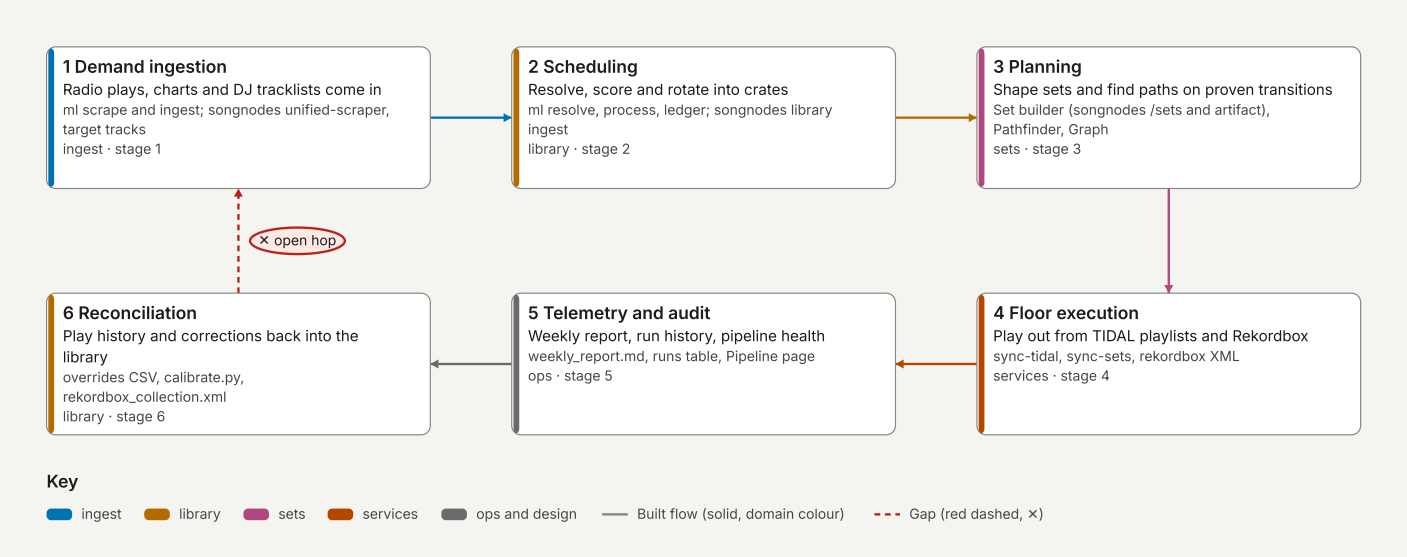
Data flow report: **7 Nov 2026**

The estate serves one loop: radio plays and DJ tracklists come in, become a curated club library, feed set building and the graph of proven transitions, play out in TIDAL and Rekordbox, and should come back as play history and corrections. Today the loop is open at the last step.

115	87	28	8
Pages and entry points	Numbered data flows	Critical data gaps	Personas

D1: **Closed loop**

The loop is open at one hop: reconciliation does not feed demand ingestion automatically.



D1 · n = 6 · Unit: stages · ● Built · Source: loop[] (name, mapping, systems, domain, gap) · Method: hand-placed ring of 6 cards, orthogonal hops; 0 crossings
Built 7 Nov 2026, 10:36 AEDT by Claude (Opus 5.5) for Markus, session 01NvWNfTGjhUeGYwDZx2H948 · songnodes 7429821 on docs/data-flow-investigation · musicLibrary snapshot 7 Nov 2026 · design system 1.10.0 · findings.json sha256 e469b41763f1

Contents

1 · Executive summary 3
2 · Architecture and numbered flows 6
3 · Page catalogue and standing rules 24
4 · Stores, signals and contracts 38
5 · Synchronisation protocols, timings and limits 107
6 · Personas and journeys 131
7 · Technical debt, risk and roadmap 148
8 · Gaps, proposed modules and storage backbone 158
9 · Built and Projected 177
10 · Sister-project tasks 180
11 · Verification register 182
12 · Methods, assumptions and open decisions 190
13 · Handover 195

Figures

D1 · Closed loop 5
D2a · System architecture – ingest, musiclibrary 10
D2b · System architecture – ingest, songnodes 11
D2c · System architecture – library, musiclibrary 12
D2d · System architecture – library, songnodes 13
D2e · System architecture – graph and sets 14
D2f · System architecture – services 15
D2g · System architecture – ops and design 16
D3a · Storage topology – musicLibrary stores 17
D3b · Storage topology – external services 18
D3c · Storage topology – songnodes medallion tables 19
D3d · Storage topology – songnodes operational tables 20
D3e · Storage topology – songnodes services and queues 21
D3f · Storage topology – songnodes browser stores 22
D4a · Page link graph – songnodes frontend 35
D4b · Page link graph – songnodes backend 36
D4c · Page link graph – musicLibrary 37
D5 · Persona overview 138
D5-P1 · Swimlane – P1 DJ performing 144
D5-P2 · Swimlane – P2 Set planner 139
D5-P3 · Swimlane – P3 Library curator 140
D5-P4 · Swimlane – P4 Pipeline operator 141
D5-P5 · Swimlane – P5 Developer 142
D5-P6 · Swimlane – P6 Scraper (system actor) 146
D5-P7 · Swimlane – P7 Weekly-run agent (system actor) 143
D5-P8 · Swimlane – P8 Beatport sync (assisted) 147
D6a · Sequence – Scrape to bronze, silver and projections 124
D6b · Sequence – Weekly musicLibrary run to TIDAL and songnodes 129
D6c · Sequence – TIDAL and Spotify OAuth connect 125
D7a · State machine – Scrape and ingestion job 128
D7b · State machine – Track rotation 126
D7c · State machine – OAuth token 127
D8a · Gap map – musicLibrary and cross-project 173
D8b · Gap map – songnodes backend 174
D8c · Gap map – songnodes frontend 175
D9 · Built to Projected change map 179
D10 · Storage backbone before and after 176

D11 · Roadmap timeline 157
D12 · Sister-project dependency graph 23
C1 · Pages by domain and status 24
C2 · Page and store access matrix 38
C3 · Signal primitives, built and projected 40
C4 · Store fan-in and fan-out 39
C5 · Timing constants on a log axis 130
C6 · Polling load by target store (Estimate) 115
C7 · Sizes against limits 34
C8 · Standing-rule conformance 33
C9 · Personas by page, with step numbers 145
C10 · Risk matrix, severity by effort 148
C11 · Verification outcome by area 182
C12 · Sister tasks by project, type and status 180

Section 1: **Executive summary**

Read as: one estate of two projects. songnodes (git, main @7429821) is the source of truth for SongNodes; E:\musicLibrary (no git) is read from a snapshot taken 7 Nov 2026 without its 58 GB data folder. Queued = open PRs #576–#580, musicLibrary plan/jules_payloads and unmerged briefs.

115
Pages and entry points

87
Numbered data flows

28
Critical data gaps

▲ Fix first

8
Personas

More figures 6

48
Stores and external systems

404
Signals extracted

17
Standing-rule violations

× To fix

42
Doc claims corrected or refuted

▲ Review

14
Queued items

16
Sister-project tasks

What holds

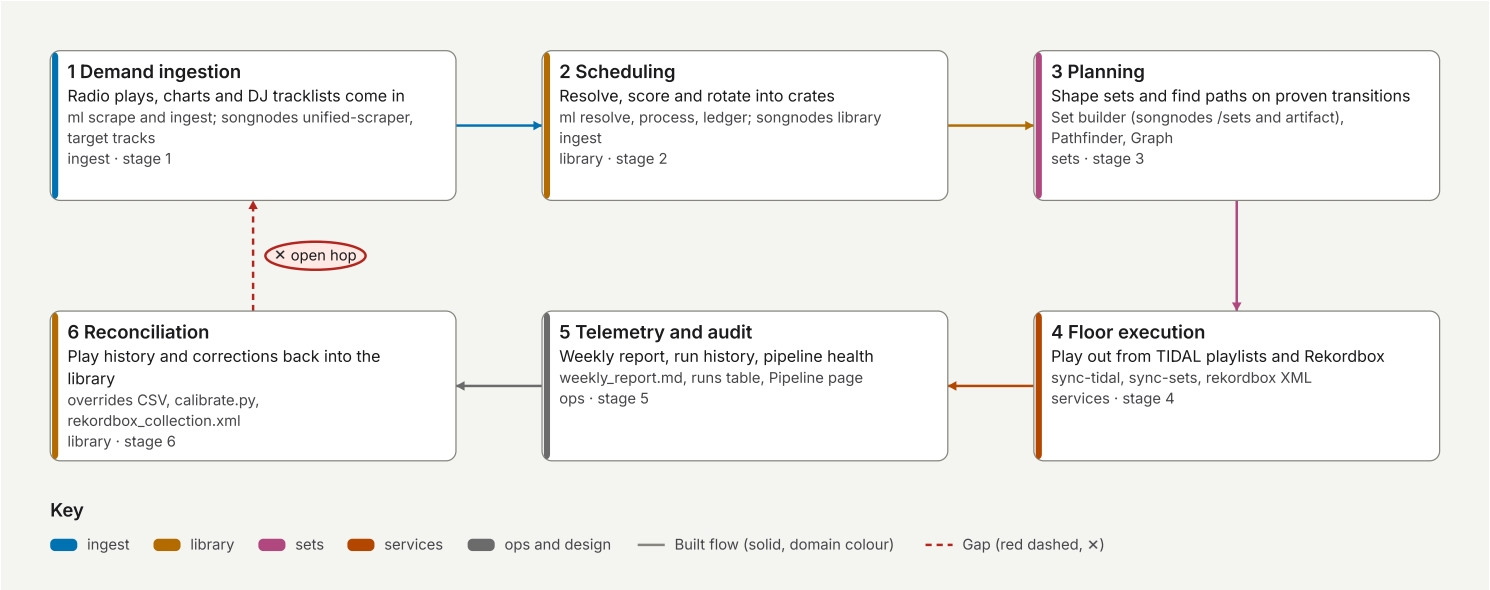
- songnodes is Git-first with one integration branch, a scope guard and design-system CI; the frontend copy of the design system is in sync and its checkers report zero findings.
- The scraping core is one unified-scraper container with complete-playlist rules, idempotency keys on ingest jobs and FOR UPDATE SKIP LOCKED job claiming.
- musicLibrary's weekly run is a clear eight-step file pipeline with an append-only ledger, a resumable resolve step and a full SQLite rebuild every run.
- Both projects already share one design system, one TIDAL developer app and one saved-set idea, so the join points are few and nameable.

What strains

- Contracts between the two projects have drifted: the documented sync script and file do not exist, and the two Set Builders no longer share a saved-set shape.
- The medallion model in CLAUDE.md is not what runs: the scraper writes silver directly and no gold or operational job runs in compose.
- Feedback loops are missing: play history, overrides and proven transitions never flow back into curation or set building, and no track has a BPM or key yet.
- Security posture is not ready for the queued tunnel: write and admin routes are open and the queued login falls back to a default account.

D1: **Closed loop**

The loop is open at one hop: reconciliation does not feed demand ingestion automatically.



D1 · n = 6 · Unit: stages · ● Built · Source: loop[] (name, mapping, systems, domain, gap) · Method: hand-placed ring of 6 cards, orthogonal hops; 0 crossings

Section 2: **Architecture and numbered flows** 87 flows

Every flow in the table is drawn in D2 with the same number.

Table: **Numbered flows** 87 flows

87 rows

FLOW	FROM	TO	DATA CONTRACT	CHANNEL	DOMAIN	STATE	EVIDENCE
1	ext-3rrr	ml-cli-scrape	3RRR episode tracks	–	ingest	Built	–
2	ext-abc-plays	ml-cli-scrape	triple j Doof plays	–	ingest	Built	–
3	ext-aria	ml-cli-scrape	ARIA Club rank and weeks	–	ingest	Built	–
4	ext-balance-wp	ml-cli-scrape	Balance tracklists	–	ingest	Built	–
5	ext-deezer	ml-cli-resolve	Deezer versions, BPM, ISRC, genre, label	–	ingest	Built	–
6	ext-metabrainz	ml-store-data	MusicBrainz dumps	–	ingest	Built	–
7	ext-pbs-airnet	ml-cli-scrape	PBS playlists	–	ingest	Built	–
8	ext-tidal	ml-cli-resolve	TIDAL versions	–	ingest	Built	–
9	ml-cli-ingest	ml-cli-resolve	to_resolve list	–	ingest	Built	–
10	ml-cli-scrape	ml-cli-ingest	raw rows	–	ingest	Built	–
11	ml-store-inbox	ml-cli-scrape	manual adds	–	ingest	Built	–
12	sn-be-svc-rest-api	sn-be-tbl-bronze	manual tracklist import	–	ingest	Built	–
13	sn-be-svc-unified-scraper	sn-be-store-redis	daily quota + setlist dedupe	–	ingest	Built	–
14	sn-be-svc-unified-scraper	sn-be-tbl-bronze	bronze playlists/tracks	–	ingest	Built	–
15	sn-be-svc-unified-scraper	sn-be-tbl-silver	layer-skipping silver writes	–	ingest	Built	–
16	sn-be-svc-worker	sn-be-svc-unified-scraper	scrape request	–	ingest	Built	–
17	sn-be-tbl-bronze	sn-be-tbl-silver	ingest_pending bronze→silver + transitions	–	ingest	Built	–
18	sn-be-tbl-silver	sn-be-tbl-outbox	track/artist/setlist events	–	ingest	Built	–
19	sn-be-tbl-target-tracks	sn-be-tbl-ingestion-jobs	scrape_targets hourly fan-out per source	–	ingest	Built	–
20	sn-fe-library	sn-be-svc-rest-api	mark library track as target (+ local sn-target-tracks-v1)	–	ingest	Built	–
21	sn-fe-library	sn-be-svc-rest-api	musicLibrary library.json export → ingest job	–	ingest	Built	–
22	sn-fe-pipeline	sn-be-svc-rest-api	scrape trigger (source, search_query, Idempotency-Key)	–	ingest	Built	–

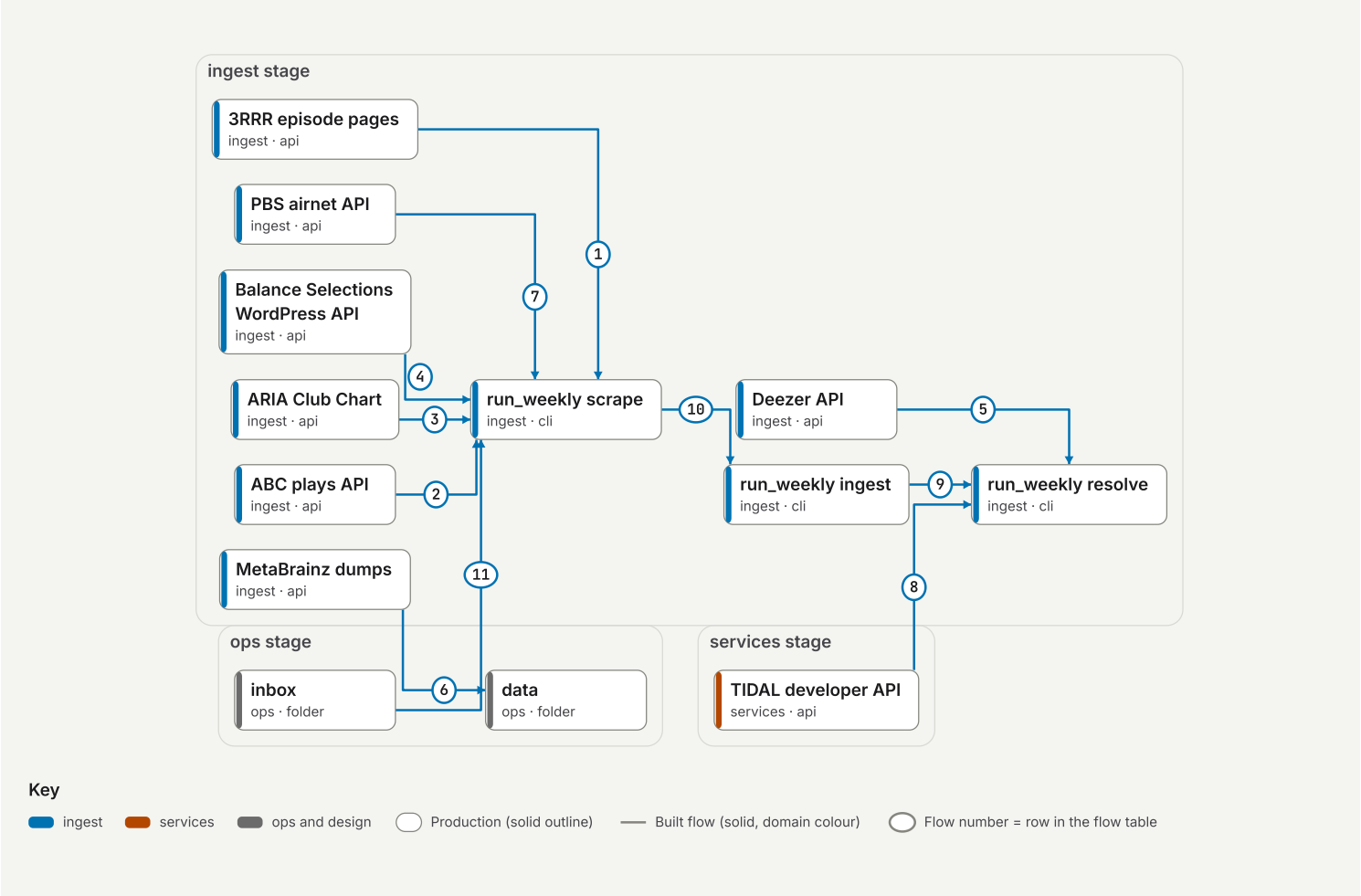
23	sn-fe-sources-import	sn-be-svc-rest-api	AI-found setlists → import-tracklist	–	ingest	Built	–
24	sn-fe-sources-targets	sn-be-svc-rest-api	target track CRUD	–	ingest	Built	–
25	sn-fe-sources-targets	sn-fe-sources-import	Import a tracklist with AI link	–	ingest	Built	–
26	ml-agent-beatport-sync	ml-cli-confirm	Beatport sync result	–	library	Built	–
27	ml-agent-weekly-run	ml-store-artifact-files	republish library.json	–	library	Built	–
28	ml-cli-build-db	ml-store-outputs	library.json	–	library	Built	–
29	ml-cli-build-db	ml-store-sqlite	SQLite rebuild	–	library	Built	–
30	ml-cli-db	sn-be-svc-rest-api	library.json ingest	–	library	Built	–
31	ml-cli-process	ext-rekordbox	Rekordbox XML	–	library	Built	–
32	ml-cli-process	ml-agent-beatport-sync	sync plan and buy list	–	library	Built	–
33	ml-cli-process	ml-store-ledger	tracks, pending, rejected, rotation	–	library	Built	–
34	ml-cli-resolve	ml-cli-process	resolved versions	–	library	Built	–
35	ml-cli-sync-songnodes	sn-be-svc-rest-api	library.json bulk ingest	–	library	Built	–
36	ml-cli-sync-tidal	ext-tidal	crate playlists	–	library	Built	–
37	ml-cli-sync-tidal	ml-cli-confirm	sync result	–	library	Built	–
38	ml-store-artifact-files	ml-page-set-builder	LibraryFile	–	library	Built	–
39	ml-store-ledger	ml-cli-build-db	full ledger	–	library	Built	–
40	ml-store-ledger	ml-cli-sync-tidal	crates	–	library	Built	–
41	ml-store-overrides	ml-cli-process	hand corrections	–	library	Built	–
42	sn-be-cli-import-music-library	sn-be-svc-rest-api	library.json via import-music-library.sh	–	library	Built	–
43	sn-be-svc-rest-api	sn-fe-library	library tracks (uuid ids)	–	library	Built	–
44	sn-be-svc-rest-api	sn-fe-library	library.tracks rows (compact keys)	–	library	Built	–
45	sn-be-svc-worker	sn-be-tbl-library	library_ingest	–	library	Built	–

46	sn-be-tbl-library	sn-be-tbl-target-tracks	auto-promote new library tracks	–	library	● Built	–
47	sn-be-tbl-outbox	sn-be-tbl-projection-tracks	projection_consume	–	library	● Built	–
48	sn-be-tbl-projection-tracks	sn-be-svc-rest-api	library search	–	library	● Built	–
49	sn-fe-home	sn-fe-library	AppHeader search / Ctrl K → /library?focus=search	–	library	● Built	–
50	ml-cli-sync-songnodes	sn-be-svc-rest-api	setlists.v1 (shape mismatch)	–	graph	● Built	–
51	sn-be-svc-graph-api	sn-fe-graph	graph nodes/edges for PIXI	–	graph	● Built	–
52	sn-be-svc-graph-api	sn-fe-graph	neighbourhood view loaded from API (view=neighbourhood&center&steps &max) instead of full graph	–	graph	▲ Queued	songnodes/STATUS.md:2505 (GV2 Next) +2
53	sn-be-tbl-outbox	sn-be-store-memgraph	graph_projection_consume (disabled)	–	graph	● Built	–
54	sn-be-tbl-silver	sn-be-svc-graph-api	graph read (silver direct)	–	graph	● Built	–
55	sn-fe-paths	sn-fe-graph	Open Pathfinder in Graph	–	graph	● Built	–
56	ml-agent-weekly-run	ml-cli-build-db	artifact_sets.json	–	sets	● Built	–
57	ml-cli-build-db	ml-cli-sync-sets	sets marked tidal	–	sets	● Built	–
58	ml-cli-sync-sets	ext-tidal	SET · <name> playlists	–	sets	● Built	–
59	ml-page-set-builder	ml-store-artifact-db	SavedSet documents	–	sets	● Built	–
60	ml-store-artifact-db	ml-agent-weekly-run	saved sets rows	–	sets	● Built	–
61	sn-be-svc-rest-api	sn-be-tbl-library	saved sets	–	sets	● Built	–
62	sn-fe-library	ml-store-inbox	artifact_sets.json download (manual copy)	–	sets	● Built	–
63	sn-fe-library	sn-be-tbl-library	saved sets	–	sets	● Built	–
64	sn-fe-library	sn-fe-sets	crate filters + pool=crate in URL; draft slots in sn-set-draft	–	sets	● Built	–
65	sn-fe-sets	ml-store-inbox	artifact_sets.json download (manual copy to musicLibrary inbox)	–	sets	● Built	–
66	sn-fe-sets	sn-be-svc-rest-api	SavedSet upsert {name,description,tracks[{track_id}]}	–	sets	● Built	–
67	ml-svc-ref-mb	sn-be-store-refmb	MusicBrainz reference lookups	–	services	● Built	–
68	sn-be-svc-rest-api	sn-be-legacy-metadata-enrichment	music search proxy (dead target)	–	services	● Built	–

69	sn-be-svc-rest-api	sn-be-store-redis	OAuth state	–	services	● Built	–
70	sn-be-svc-rest-api	sn-be-tbl-oauth	OAuth tokens and saved keys	–	services	● Built	–
71	sn-be-svc-worker	sn-be-svc-audio-analysis	analyse Deezer preview	–	services	● Built	–
72	sn-be-svc-worker	sn-be-tbl-silver	enrichment evidence/consensus	–	services	● Built	–
73	sn-be-tbl-oauth	sn-be-svc-worker	saved provider keys override env	–	services	● Built	–
74	sn-fe-oauth-callback	sn-be-svc-rest-api	tokens stored server-side for enrichment	–	services	● Built	–
75	sn-fe-oauth-callback	sn-fe-settings	oauth-success tokens (pop-up only)	–	services	● Built	–
76	sn-fe-sets	sn-be-svc-rest-api	send set to TIDAL	–	services	● Built	–
77	sn-fe-settings	sn-fe-oauth-callback	OAuth redirect with state in sessionStorage	–	services	● Built	–
78	ml-plan-jules	ext-jules	work-package sessions	–	ops	● Built	–
79	sn-be-q-store-cloudflare-tunnel	sn-be-svc-rest-api	public HTTPS via frontend nginx /api proxy	–	ops	▲ Queued	–
80	sn-be-svc-rest-api	sn-be-svc-unified-scraper	live scrape progress poll	–	ops	● Built	–
81	sn-be-svc-rest-api	sn-be-tbl-ingestion-jobs	scrape/library_ingest job enqueue (idempotency key)	–	ops	● Built	–
82	sn-be-svc-rest-api	sn-fe-pipeline	jobs, outbox status, stages (polled)	–	ops	● Built	–
83	sn-be-svc-worker	sn-be-store-rabbitmq	Celery beat ticks	–	ops	● Built	–
84	sn-fe-home	sn-fe-sets	LauncherCard links to every module via MODULE_ROUTES	–	ops	● Built	–
85	sn-fe-work-queue-tracker	ml-store-artifact-db	window.claude db collections meta, tasks (seeded from seed.json)	–	ops	● Built	–
86	sn-fe-library	ml-page-set-builder	compiled pages via build.mjs	–	design	● Built	–
87	sn-fe-settings	sn-fe-index-html	display prefs read before first paint	–	design	● Built	–

D2a: **System architecture – ingest, musiclibrary**

Every ingest, musiclibrary flow, numbered as in the flow table; all of them are built.

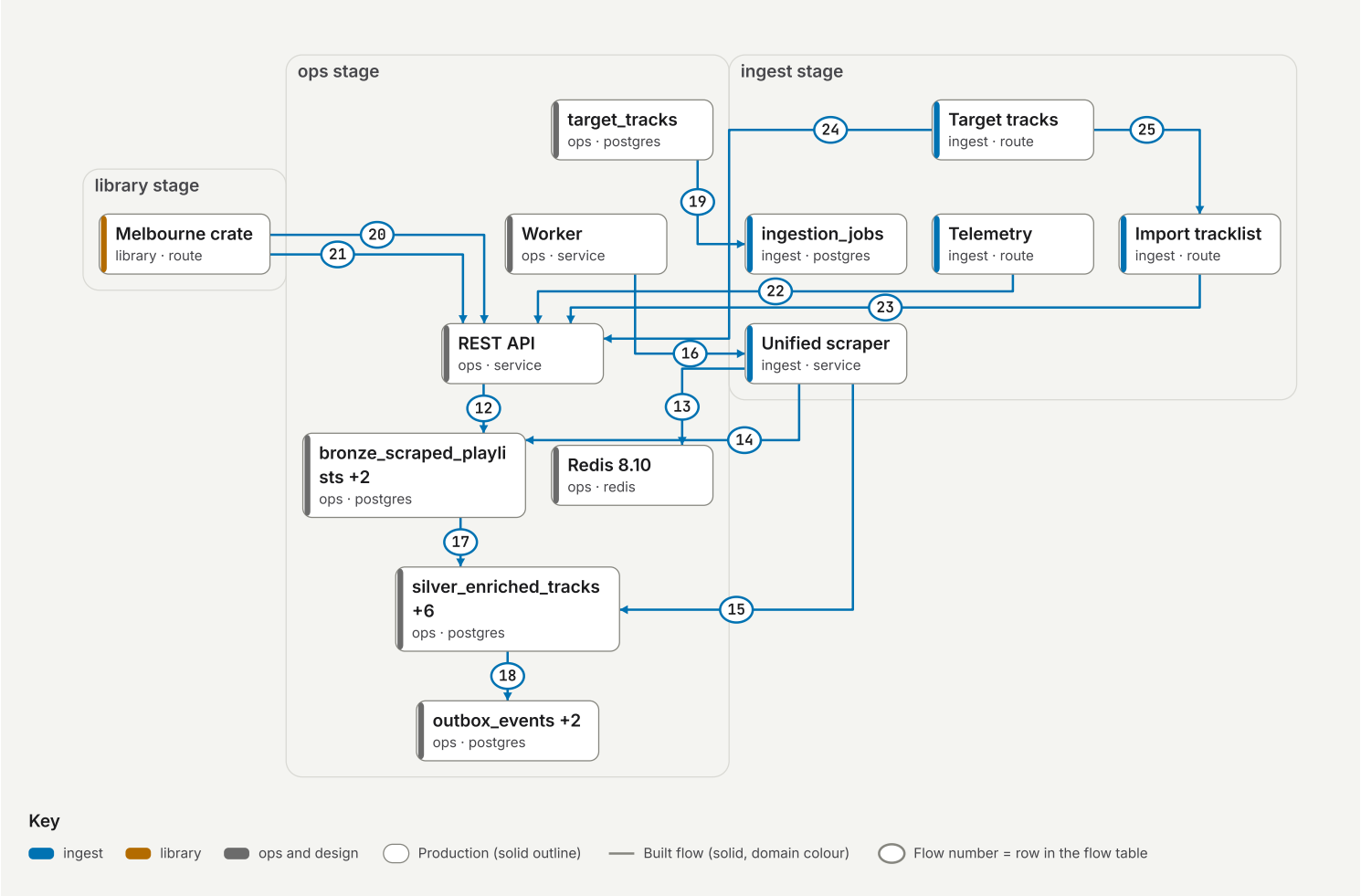


D2a · n = 11 · Unit: flows · ● Built

· Source: flows[] (n, id, from, to, label, domain, state); pages[]/stores[] (title, name, domain, kind, classification, state); domain_order; colour_map · Method: graphviz dot layered layout (LR, order reverse; 12 rank directions and input orders tried, scored by legibility then crossings), splines=ortho, own number badges, type floor 12 units; 1 crossings, 0 edges through nodes

D2b: **System architecture – ingest, songnodes**

Every ingest, songnodes flow, numbered as in the flow table; all of them are built.

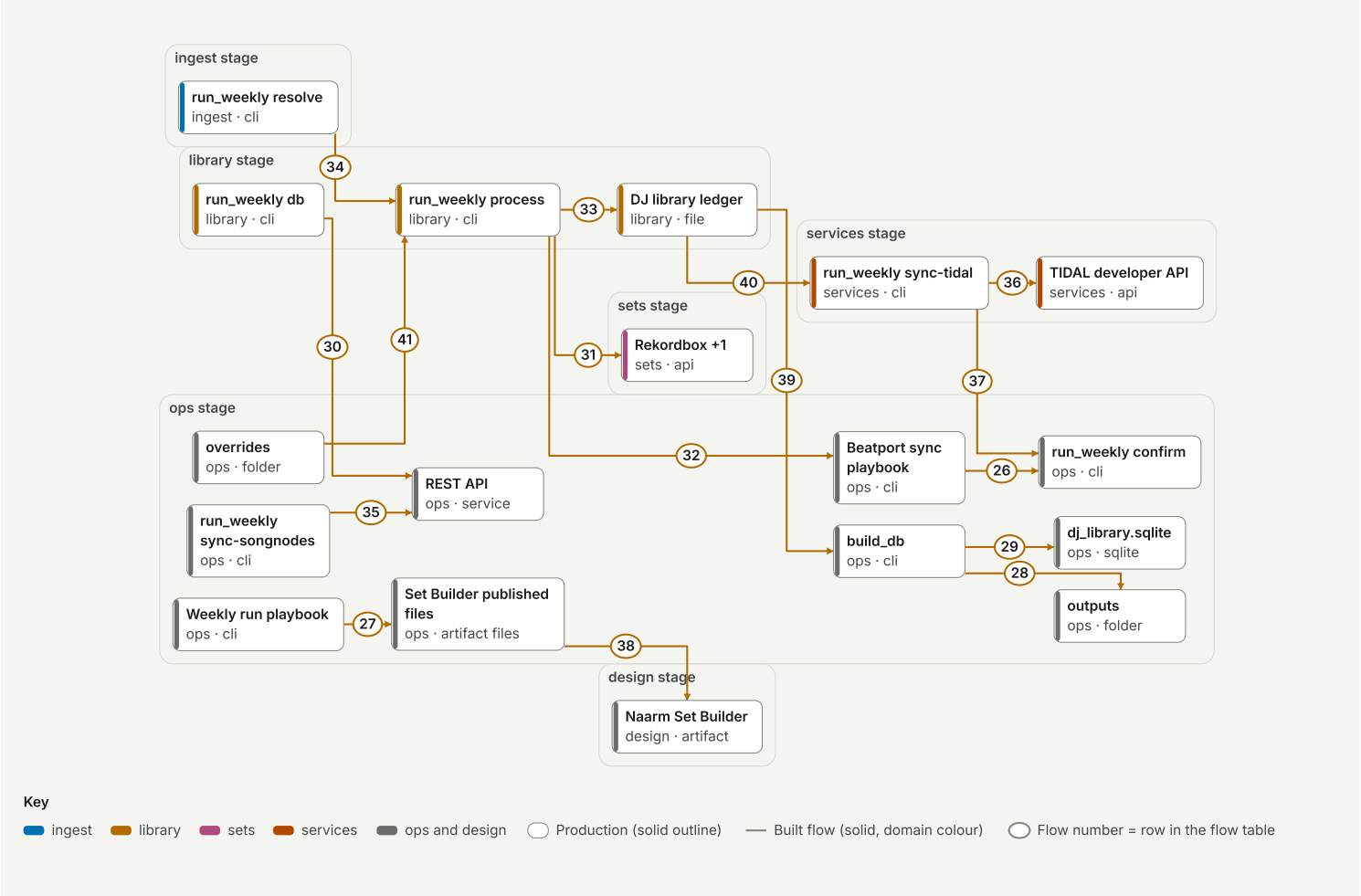


D2b · n = 14 · Unit: flows · ● Built

· Source: flows[] (n, id, from, to, label, domain, state); pages[]/stores[] (title, name, domain, kind, classification, state); domain_order; colour_map · Method: graphviz dot layered layout (TB, order given; 12 rank directions and input orders tried, scored by legibility then crossings), splines=ortho, own number badges, type floor 12 units; 7 crossings, 0 edges through nodes

D2c: **System architecture – library, musiclibrary**

Every library, musiclibrary flow, numbered as in the flow table; all of them are built.

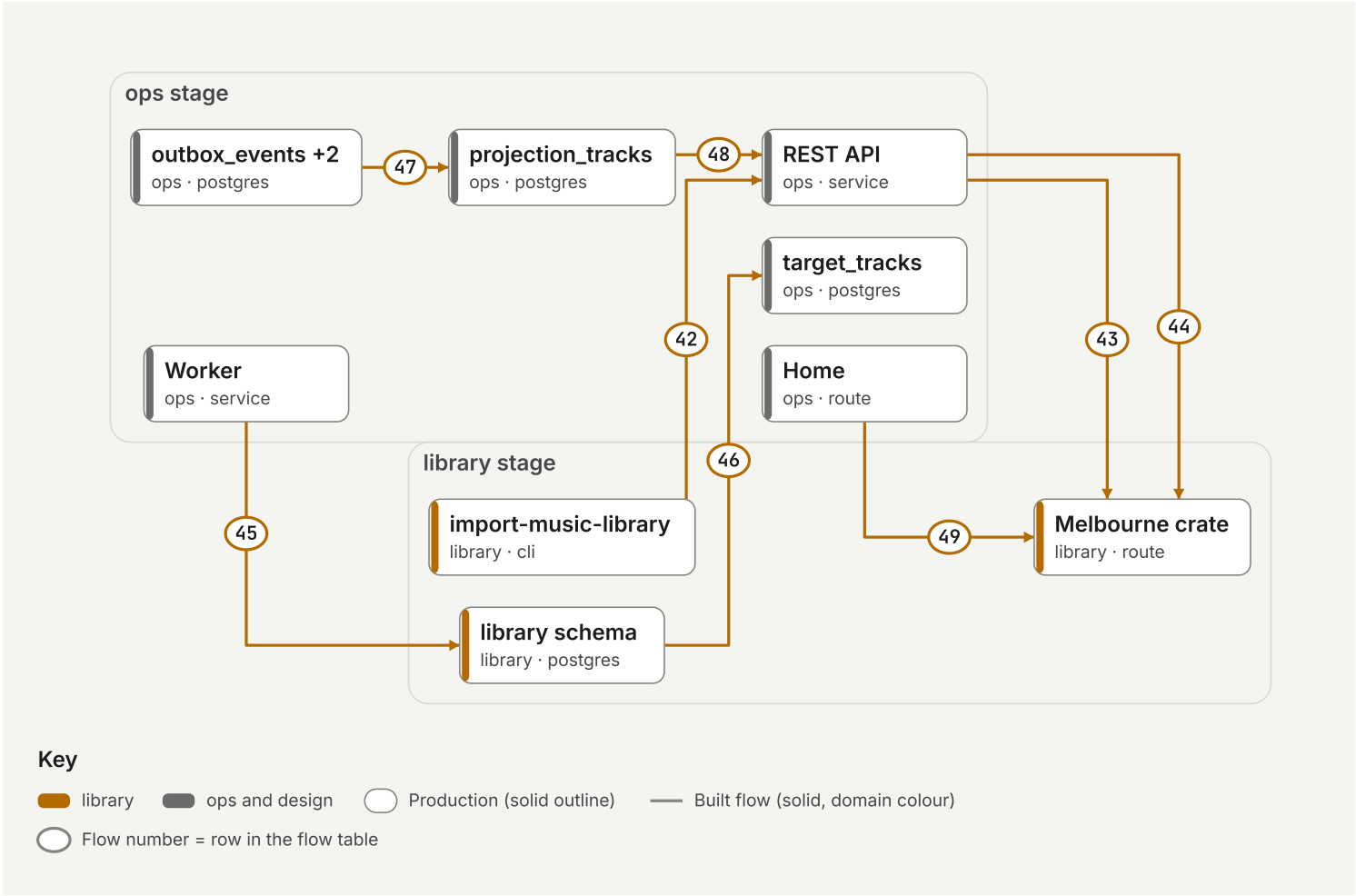


D2c · n = 16 · Unit: flows · ● Built

· Source: flows[] (n, id, from, to, label, domain, state); pages[]/stores[] (title, name, domain, kind, classification, state); domain_order; colour_map · Method: graphviz dot layered layout (LR, order given; 12 rank directions and input orders tried, scored by legibility then crossings), splines=ortho, own number badges, type floor 15 units; 3 crossings, 0 edges through nodes

D2d: **System architecture – library, songnodes**

Every library, songnodes flow, numbered as in the flow table; all of them are built.

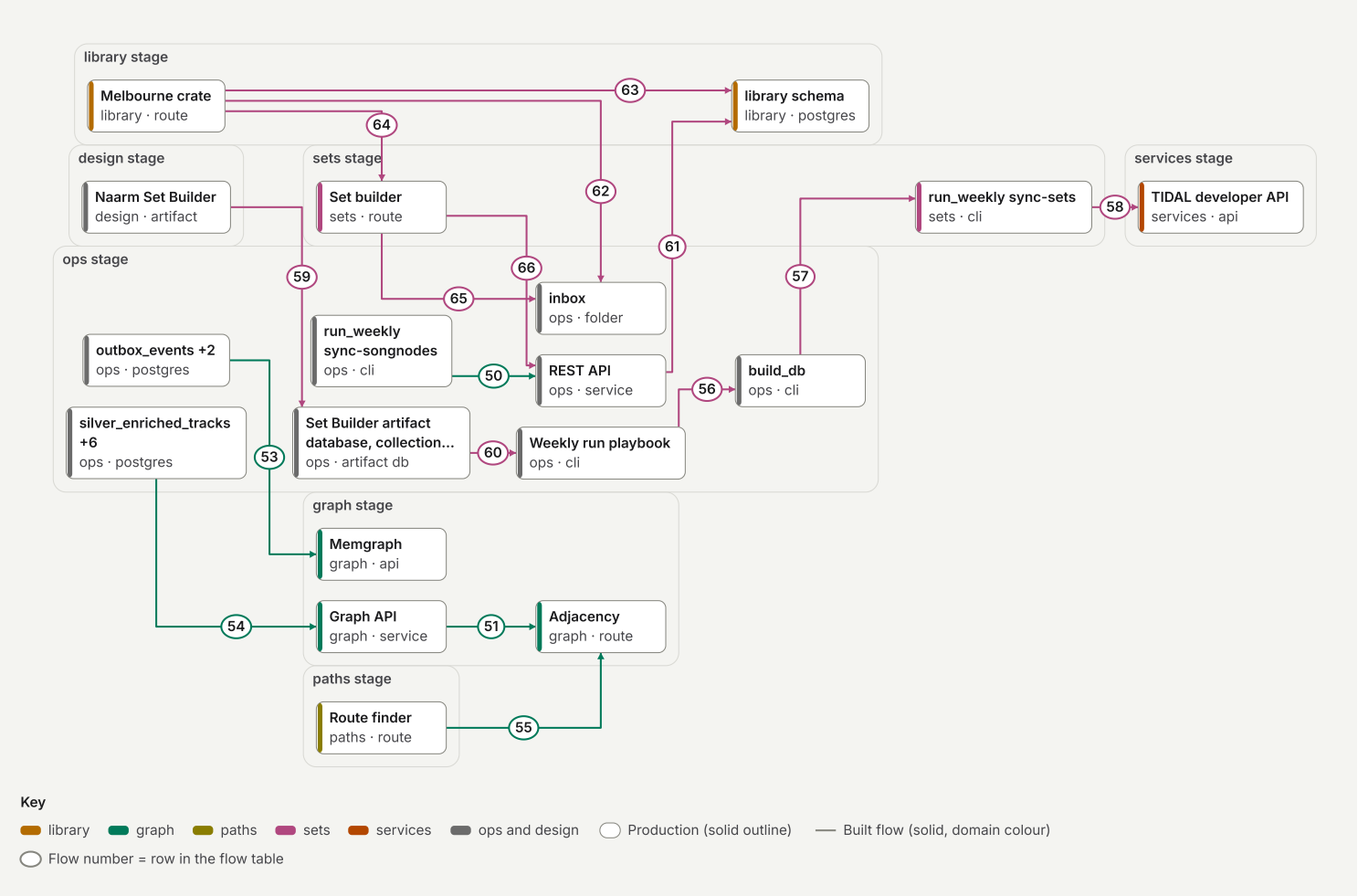


D2d · n = 8 · Unit: flows · ● Built

· Source: flows[] (n, id, from, to, label, domain, state); pages[]/stores[] (title, name, domain, kind, classification, state); domain_order; colour_map · Method: graphviz dot layered layout (LR, order given; 12 rank directions and input orders tried, scored by legibility then crossings), splines=ortho, own number badges, type floor 12 units; 0 crossings, 0 edges through nodes

D2e: **System architecture – graph and sets**

Every graph and sets flow, numbered as in the flow table; 1 queued flow appears only in the Projected view.

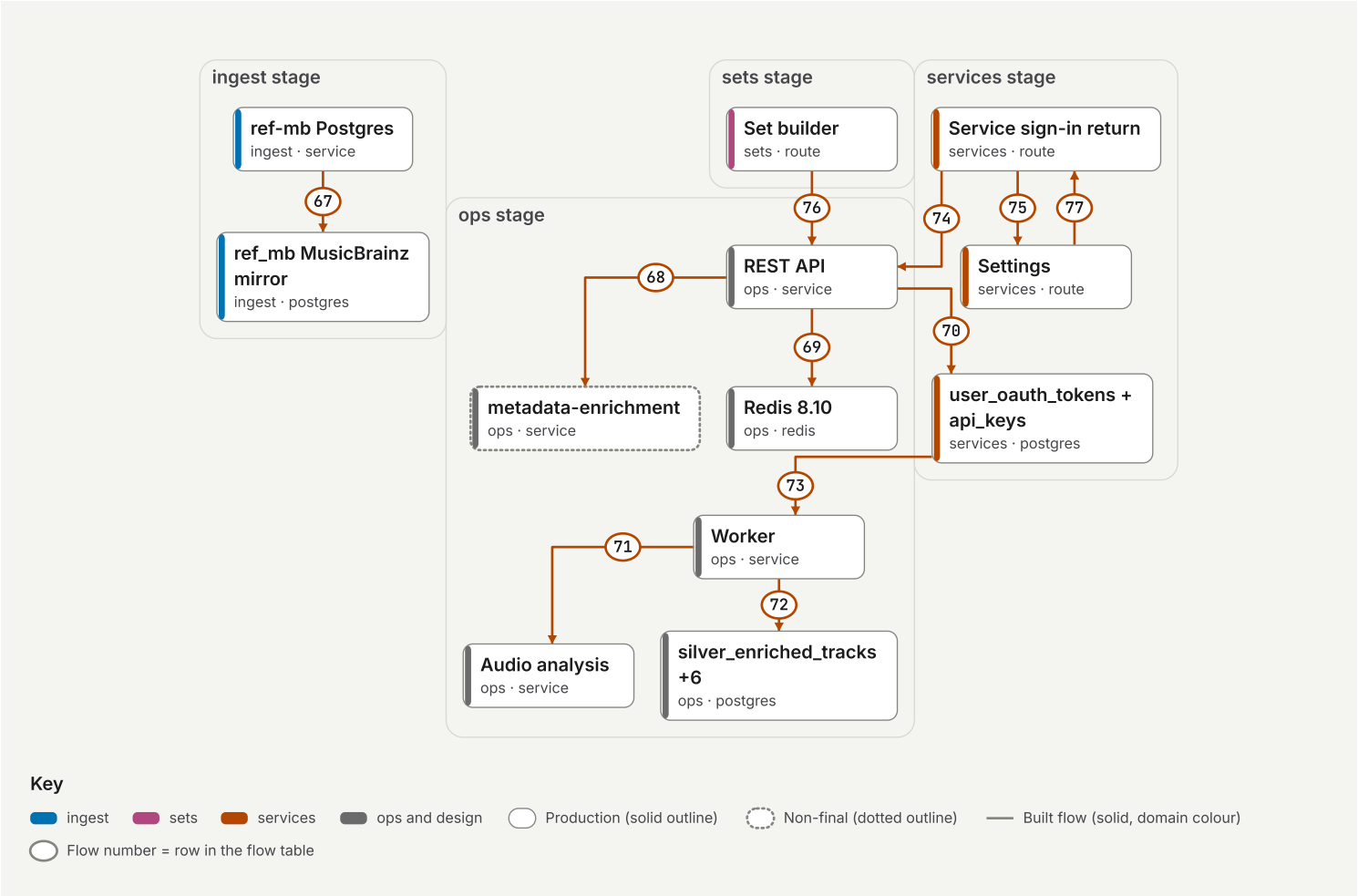


D2e · n = 17 · Unit: flows · `i built+queued`

· Source: flows[] (n, id, from, to, label, domain, state); pages[]/stores[] (title, name, domain, kind, classification, state); domain_order; colour_map · Method: graphviz dot layered layout (LR, order given; 12 rank directions and input orders tried, scored by legibility then crossings), splines=ortho, own number badges, type floor 15 units; 1 crossings, 0 edges through nodes

D2f: **System architecture – services**

Every services flow, numbered as in the flow table; all of them are built.

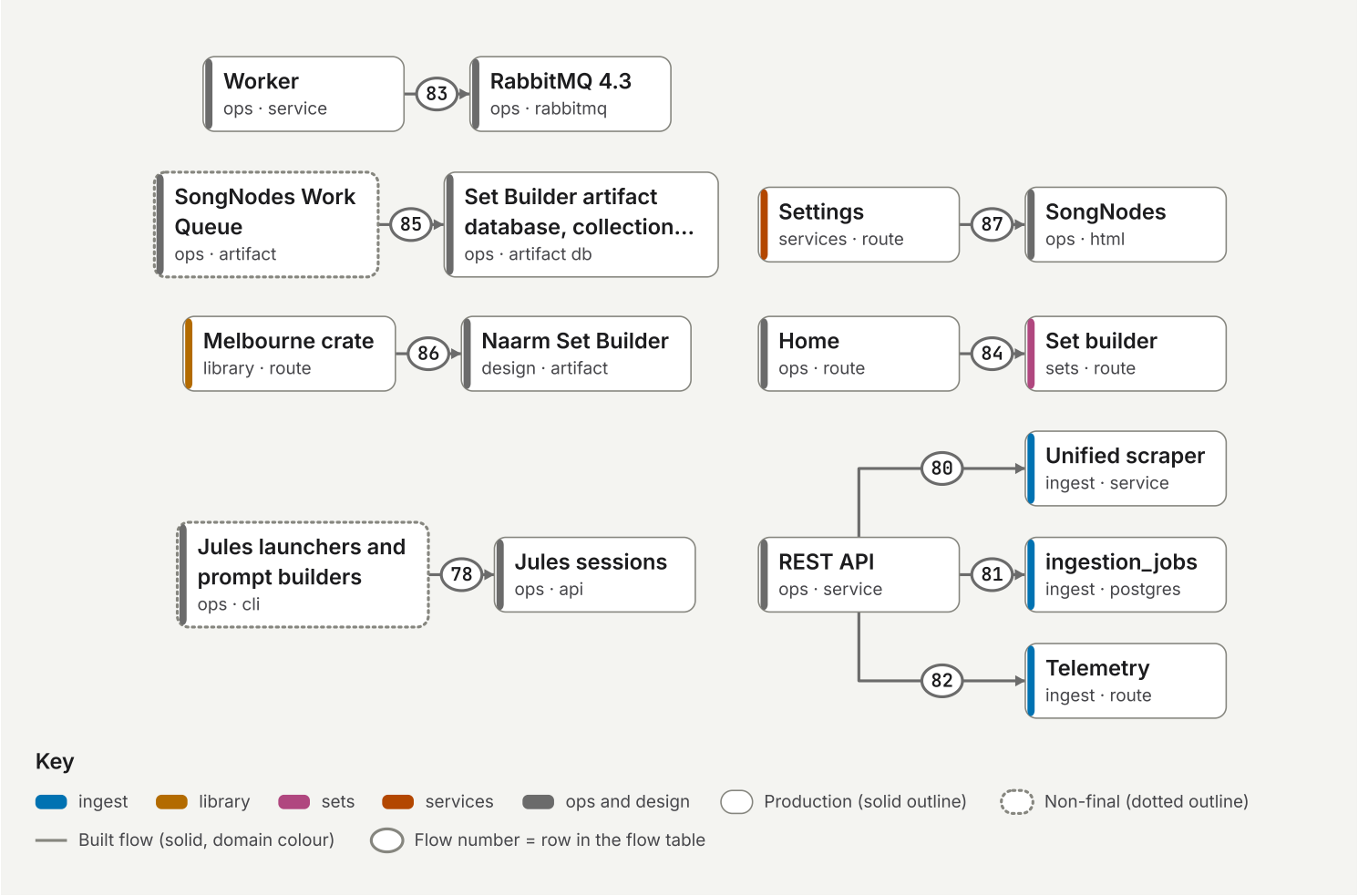


D2f · n = 11 · Unit: flows · ● Built

· Source: flows[] (n, id, from, to, label, domain, state); pages[]/stores[] (title, name, domain, kind, classification, state); domain_order; colour_map · Method: graphviz dot layered layout (TB, order given; 12 rank directions and input orders tried, scored by legibility then crossings), splines=ortho, own number badges, type floor 12 units; 0 crossings, 0 edges through nodes

D2g: **System architecture – ops and design**

Every ops and design flow, numbered as in the flow table; 1 queued flow appears only in the Projected view.

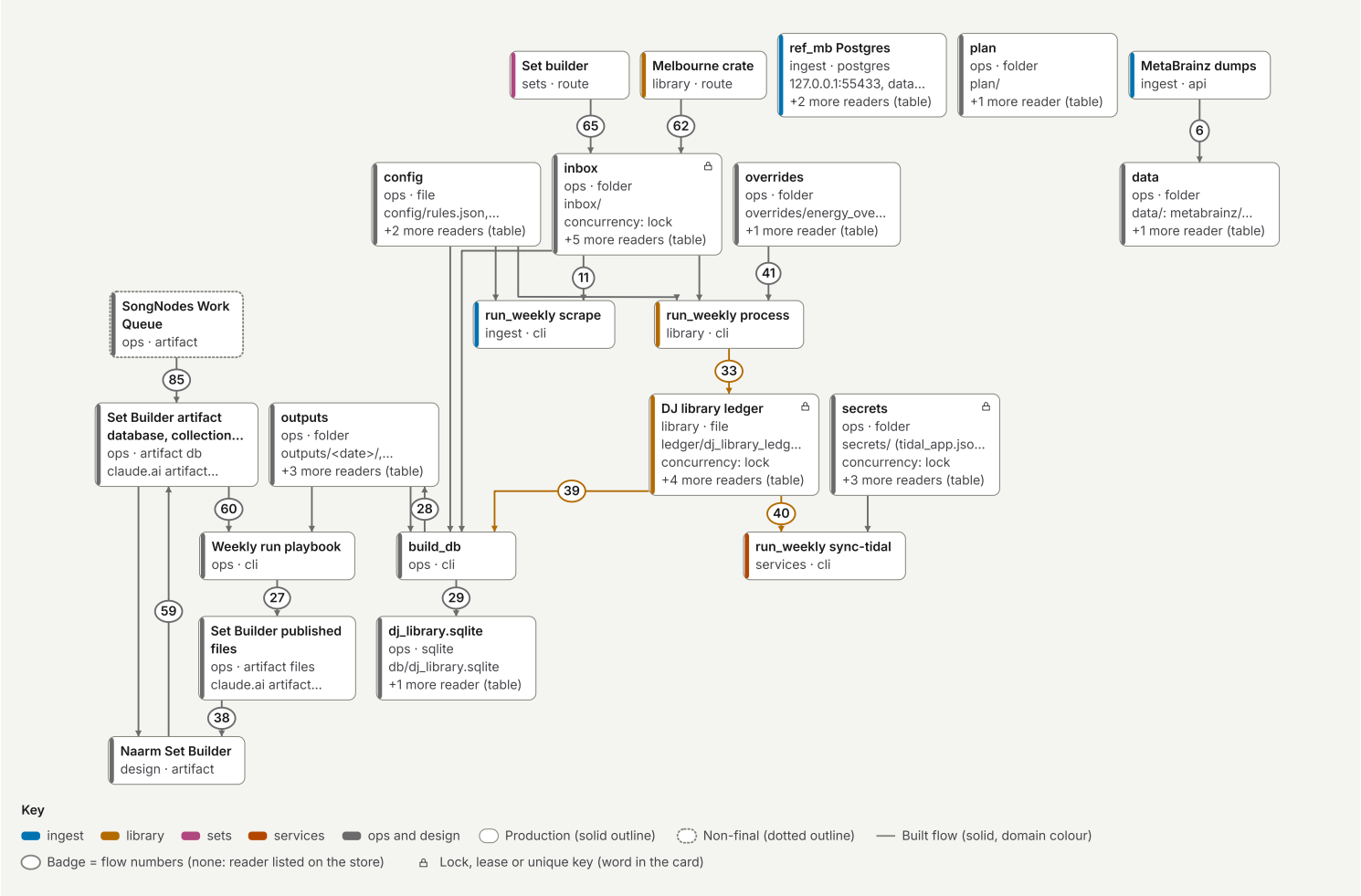


D2g · n = 10 · Unit: flows · `i built+queued`

· Source: flows[] (n, id, from, to, label, domain, state); pages[]/stores[] (title, name, domain, kind, classification, state); domain_order; colour_map · Method: graphviz dot layered layout (LR, order given, components packed 4 across, ungrouped; 12 rank directions and input orders tried, scored by legibility then crossings), splines=ortho, own number badges, type floor 12 units; 0 crossings, 0 edges through nodes

D3a: **Storage topology – musicLibrary stores**

3 of 12 musicLibrary stores carry a lock, lease or unique-key rule; the rest rely on whole-file overwrite or last writer wins.

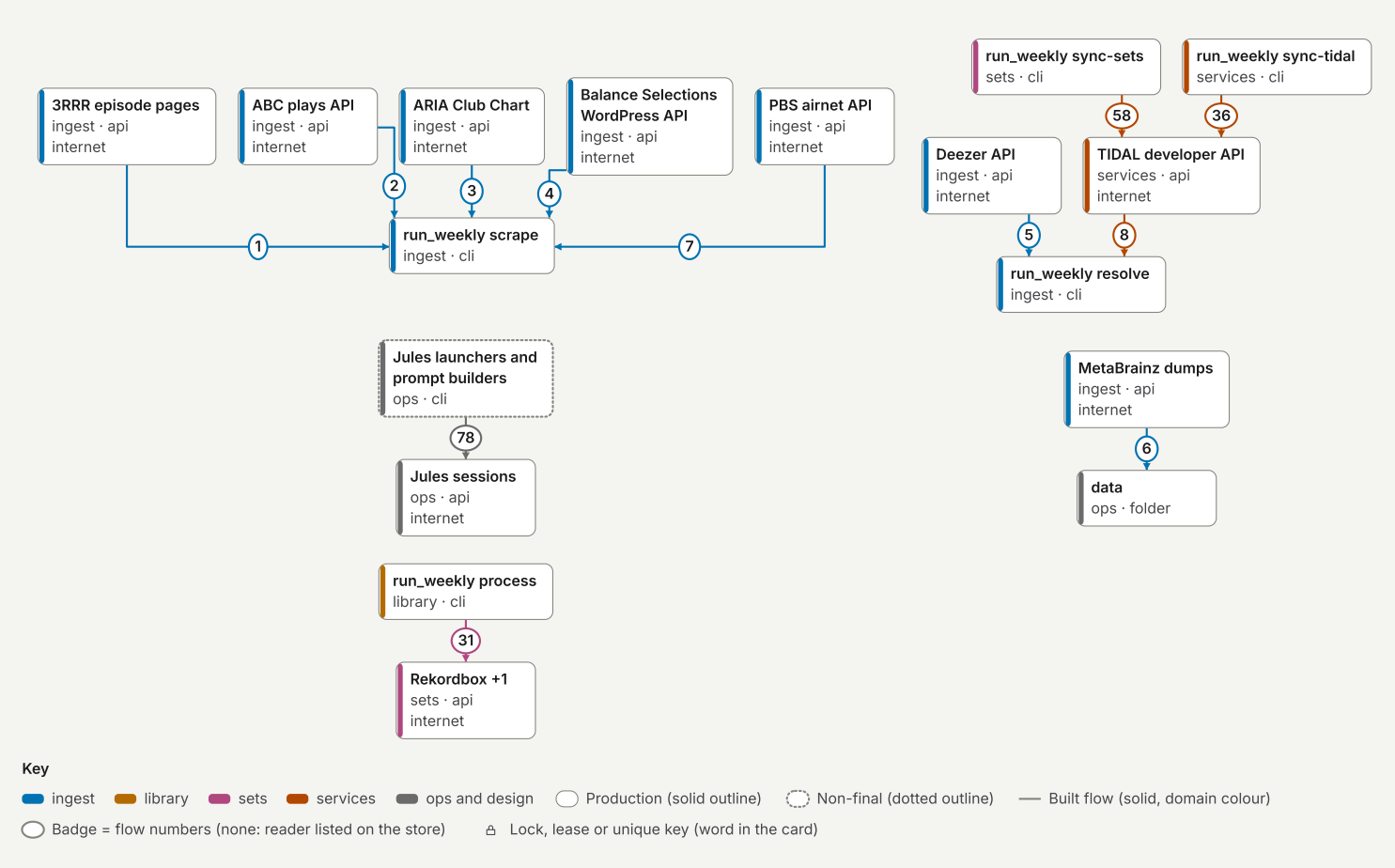


D3a · n = 12 · Unit: stores · ● Built

· Source: stores[] (kind, location, owner, readers, concurrency); flows[] touching stores · Method: graphviz dot layered layout (TB, order reverse, ungrouped; 12 rank directions and input orders tried, scored by legibility then crossings), splines=ortho, own number badges, type floor 15 units; 2 crossings, 0 edges through nodes

D3b: **Storage topology – external services**

0 of 10 external services carry a lock, lease or unique-key rule; the rest rely on whole-file overwrite or last writer wins.

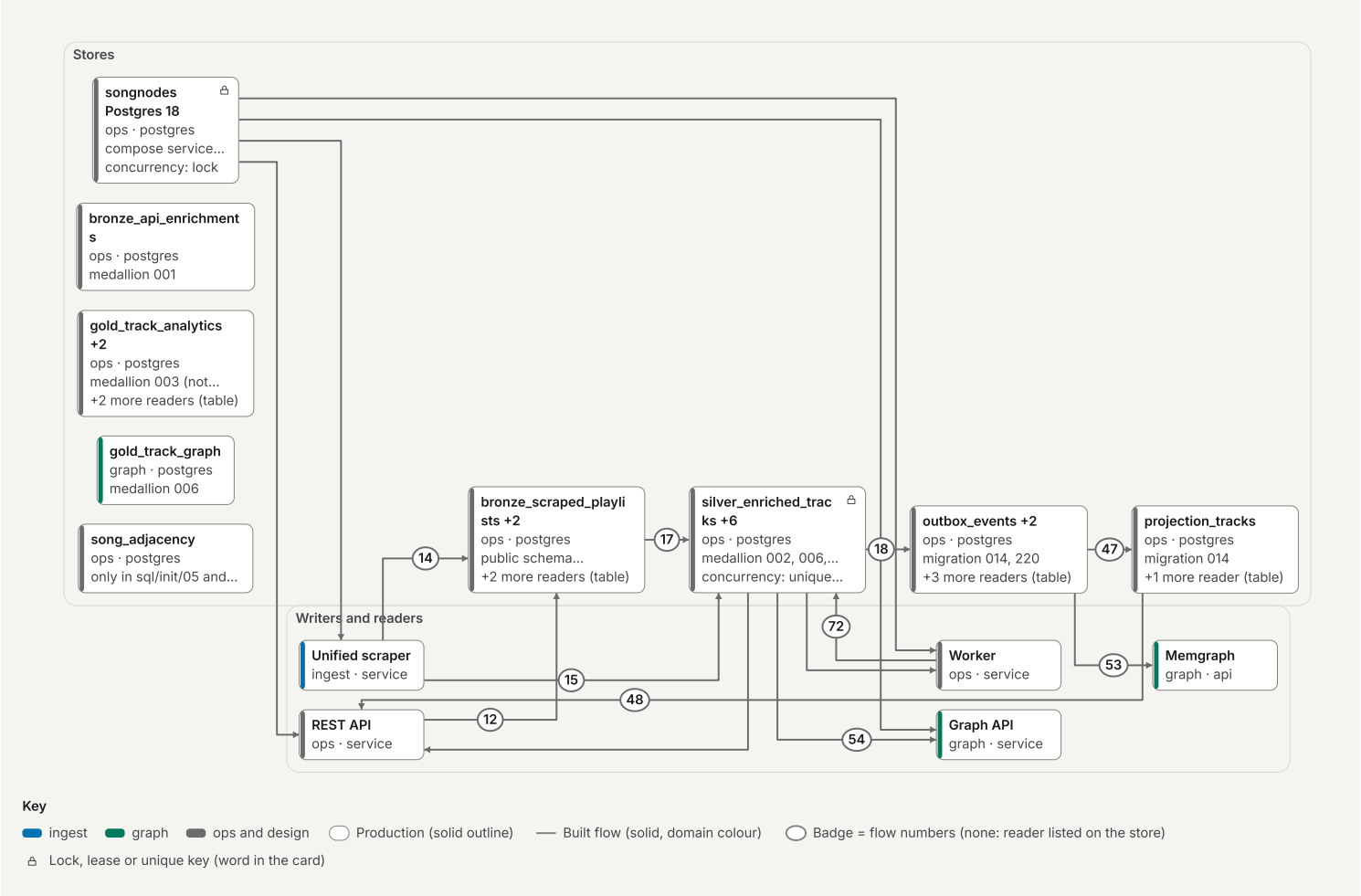


D3b · n = 10 · Unit: stores · ● Built

· Source: stores[] (kind, location, owner, readers, concurrency); flows[] touching stores · Method: graphviz dot layered layout (TB, order given, components packed 2 across, ungrouped; 12 rank directions and input orders tried, scored by legibility then crossings), splines=ortho, own number badges, type floor 15 units; 0 crossings, 0 edges through nodes

D3c: **Storage topology – songnodes medallion tables**

2 of 9 songnodes medallion tables carry a lock, lease or unique-key rule; the rest rely on whole-file overwrite or last writer wins.

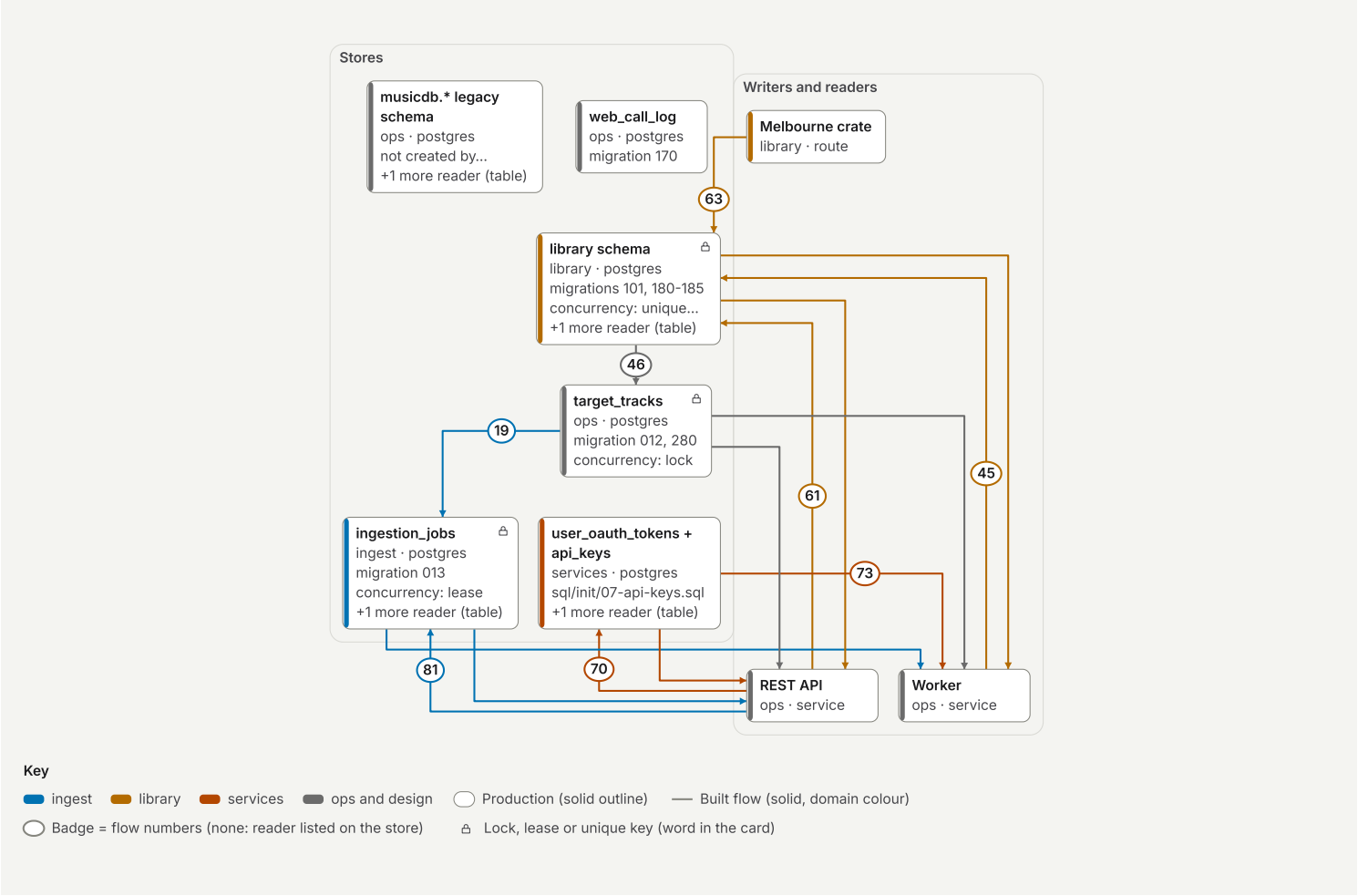


D3c · n = 9 · Unit: stores · ● Built

· Source: stores[] (kind, location, owner, readers, concurrency); flows[] touching stores · Method: graphviz dot layered layout (LR, order given; 12 rank directions and input orders tried, scored by legibility then crossings), splines=ortho, own number badges, type floor 15 units; 10 crossings, 0 edges through nodes

D3d: **Storage topology – songnodes operational tables**

3 of 6 songnodes operational tables carry a lock, lease or unique-key rule; the rest rely on whole-file overwrite or last writer wins.

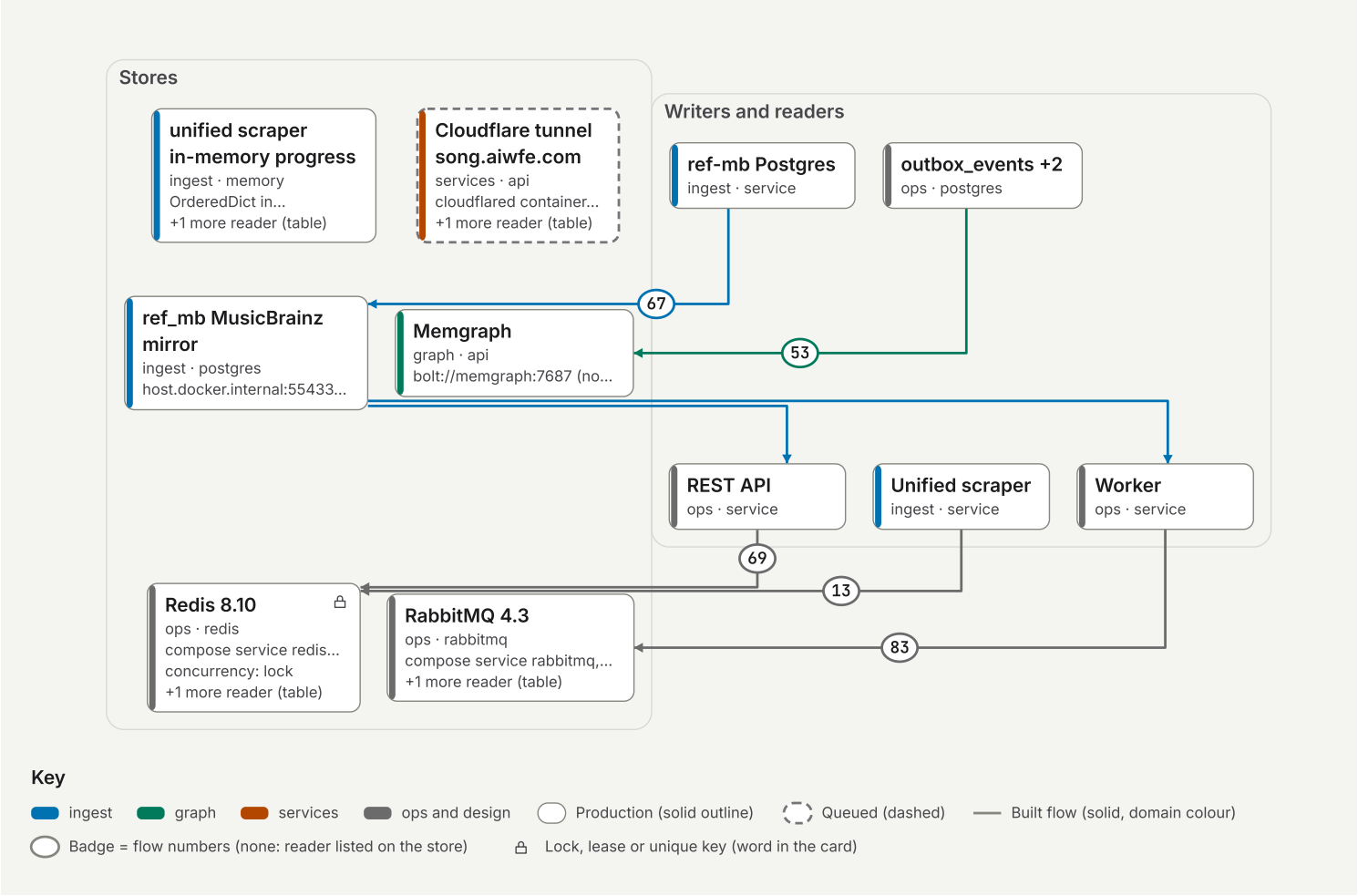


D3d · n = 6 · Unit: stores · ● Built

· Source: stores[] (kind, location, owner, readers, concurrency); flows[] touching stores · Method: graphviz dot layered layout (TB, order reverse; 12 rank directions and input orders tried, scored by legibility then crossings), splines=ortho, own number badges, type floor 15 units; 12 crossings, 0 edges through nodes

D3e: **Storage topology – songnodes services and queues**

1 of 6 songnodes services and queues carry a lock, lease or unique-key rule; the rest rely on whole-file overwrite or last writer wins.

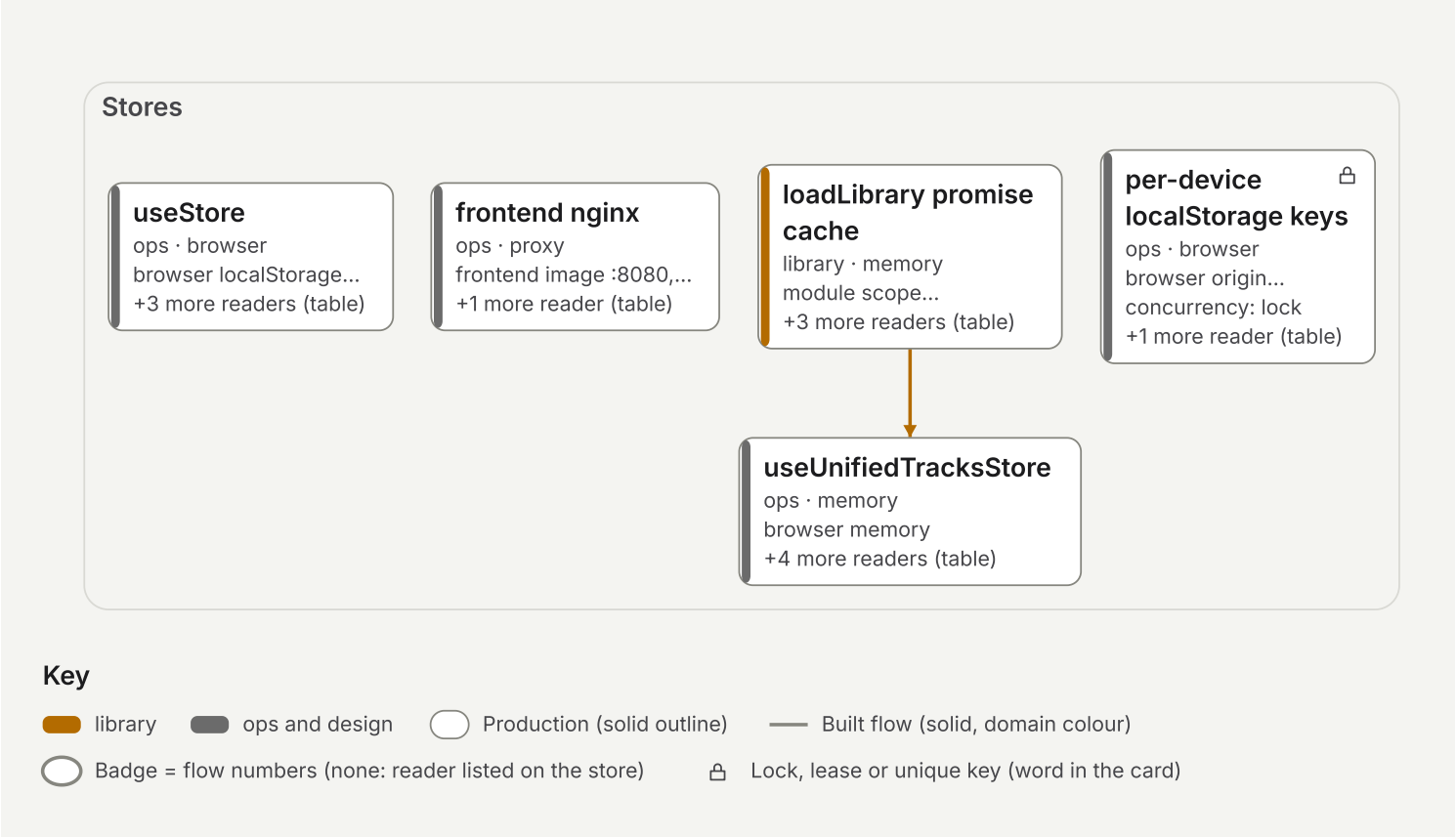


D3e · n = 6 · Unit: stores · ● Built

· Source: stores[] (kind, location, owner, readers, concurrency); flows[] touching stores · Method: graphviz dot layered layout (TB, order given; 12 rank directions and input orders tried, scored by legibility then crossings), splines=ortho, own number badges, type floor 12 units; 0 crossings, 0 edges through nodes

D3f: **Storage topology – songnodes browser stores**

1 of 5 songnodes browser stores carry a lock, lease or unique-key rule; the rest rely on whole-file overwrite or last writer wins.

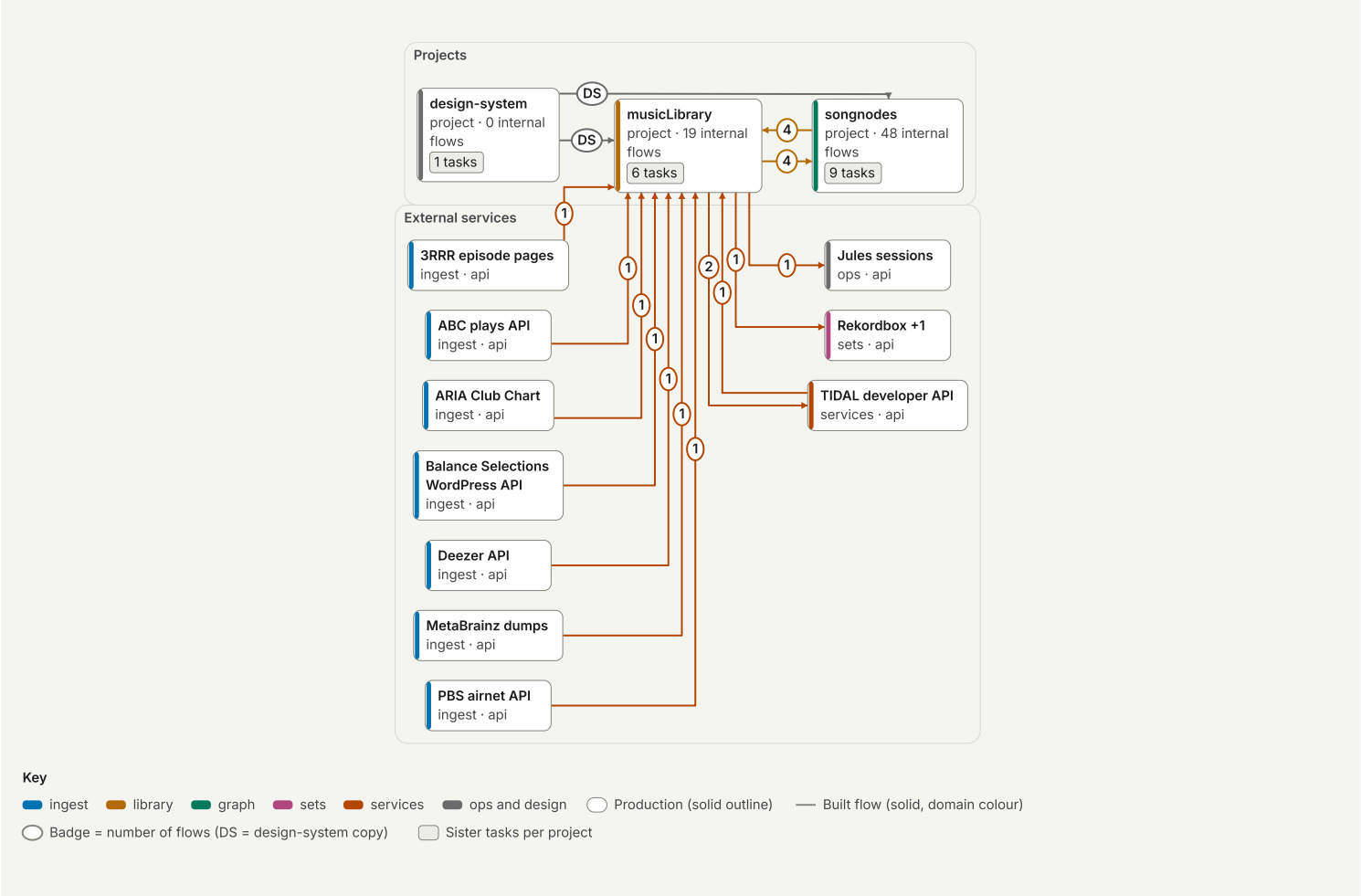


D3f · n = 5 · Unit: stores · ● Built

· Source: stores[] (kind, location, owner, readers, concurrency); flows[] touching stores · Method: graphviz dot layered layout (TB, order given; 12 rank directions and input orders tried, scored by legibility then crossings), splines=ortho, own number badges, type floor 12 units; 0 crossings, 0 edges through nodes

D12: **Sister-project dependency graph**

Only 8 of 87 flows cross between songnodes and musicLibrary; the rest stay inside one project or reach an external service.



D12 · n = 15 · Unit: project pairs · ● Built

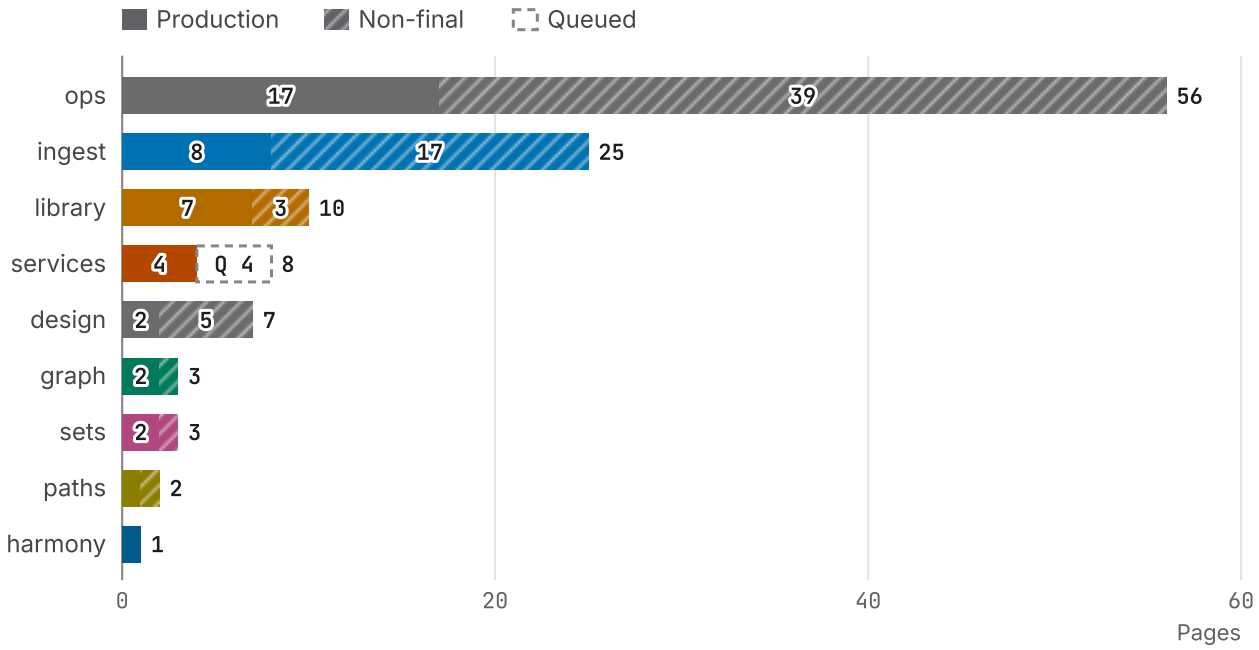
· Source: flows[] aggregated by repo pair (pages[]/stores[].repo); tasks[].project; design-system pages · Method: graphviz dot layered layout (LR, order given; 12 rank directions and input orders tried, scored by legibility then crossings), splines=ortho, own number badges, type floor 15 units; 0 crossings, 0 edges through nodes

Section 3: **Page catalogue and standing rules** 115 pages

44 production, 71 non-final, 17 rule violations.

C1: **Pages by domain and status**

ops holds the most pages (56 of 115); 67 pages are non-final and 4 queued.



C1 · n = 115 · Unit: pages · i Projected

· Source: pages[].domain, pages[].classification, pages[].state · Method: Count of pages per domain; built pages split by classification, queued pages counted separately; sorted by total.

Rules: **Standing rules audited** 8 rules

8 rows

RULE	RULE TEXT	SOURCE
R1	The NavDrawer is the only site navigation.	songnodes/CLAUDE.md §4 Rules: Interaction
R2	One primary button per view.	songnodes/CLAUDE.md §4 Rules: Interaction
R3	Never alert, confirm or prompt; use ConfirmDialog and Toast.	songnodes/CLAUDE.md §4 Rules: Interaction
R4	Tokens and sn-* classes only: no hex, rgb, named colours, px font sizes, ad-hoc radii, shadows or z-index numbers.	songnodes/CLAUDE.md §4 Rules: Colour and values
R5	No emoji in the interface; en-AU sentence case; missing values are an en dash.	songnodes/CLAUDE.md §4 Rules: Writing
R6	Pages register their route (G2) and REST routers register themselves (G3); do not wire App.tsx or main.py.	songnodes/AGENTS.md rule 12
R7	unified-scraper is the only scraper container.	songnodes/CLAUDE.md Critical requirement 3
R8	Valid artists on both endpoints of every transition; no NULL or Unknown Artist in the graph.	songnodes/CLAUDE.md Critical requirement 1

Table: **Page catalogue** 115 pages

115 rows

PAGE	REPO	DOMAIN	CATEGORY	STATE	BYTES	RULES	EVIDENCE
Naarm Set Builder (design-system build) `artifact/build/index.html`	musicLibrary	design	Production	Built	134	R5 accepted exception; R6 accepted exception	`musicLibrary/agent/weekly_run.md:8` +5
Naarm Set Builder (original hand-written page) `artifact/index.html`	musicLibrary	sets	i Non-final superseded-prototype	Built	42,185	R1 violation; R4 violation; R5 violation	`musicLibrary/artifact/index.html:231` +4
Naarm Set Builder (Vite entry) `artifact/app/index.html`	musicLibrary	ops	i Non-final dev-tool	Built	179	All checked rules pass	`musicLibrary/artifact/app/build.mjs:145-156` +1

PAGE	REPO	DOMAIN	CATEGORY	STATE	BYTES	RULES	EVIDENCE
run_weekly weekly `pipeline/run_weekly.py weekly`	musicLibrary	ops	Production	Built	32,519	–	`musicLibrary/pipeline/run_weekly.py:625-632` +1
run_weekly scrape `pipeline/run_weekly.py scrape`	musicLibrary	ingest	Production	Built	32,519	–	`musicLibrary/pipeline/run_weekly.py:370-399` +1
run_weekly ingest `pipeline/run_weekly.py ingest`	musicLibrary	ingest	Production	Built	32,519	–	`musicLibrary/pipeline/run_weekly.py:52-67` +1
run_weekly resolve `pipeline/run_weekly.py resolve`	musicLibrary	ingest	Production	Built	32,519	–	`musicLibrary/pipeline/run_weekly.py:414-463` +1
run_weekly process `pipeline/run_weekly.py process`	musicLibrary	library	Production	Built	32,519	–	`musicLibrary/pipeline/run_weekly.py:185-314` +1
run_weekly sync-tidal `pipeline/run_weekly.py sync-tidal`	musicLibrary	services	Production	Built	32,519	–	`musicLibrary/pipeline/run_weekly.py:467-521` +1
run_weekly confirm `pipeline/run_weekly.py confirm`	musicLibrary	ops	Production	Built	32,519	–	`musicLibrary/pipeline/run_weekly.py:339-366` +1
run_weekly db `pipeline/run_weekly.py db`	musicLibrary	library	Production	Built	32,519	–	`musicLibrary/pipeline/run_weekly.py:619-622` +1
run_weekly sync-sets `pipeline/run_weekly.py sync-sets`	musicLibrary	sets	Production	Built	32,519	–	`musicLibrary/pipeline/run_weekly.py:524-564` +1

run_weekly sync-songnodes `pipeline/run_weekly.py sync-songnodes`	musicLibrary	ops	● Production	● Built	32,519	–	`musicLibrary/pipeline/run_weekly.py:567-616` +1
run_weekly status `pipeline/run_weekly.py status`	musicLibrary	library	● Production	● Built	32,519	–	`musicLibrary/pipeline/run_weekly.py:635-639` +1
build_db `pipeline/build_db.py`	musicLibrary	ops	● Production	● Built	13,928	–	`musicLibrary/pipeline/build_db.py:109-187` +2
calibrate `pipeline/calibrate.py`	musicLibrary	library	● Production	● Built	4,409	–	`musicLibrary/pipeline/calibrate.py:22` +2
tidal_cli `pipeline/tidal_cli.py`	musicLibrary	services	● Production	● Built	2,166	–	`musicLibrary/pipeline/tidal_cli.py:18-41` +1
test_pipeline `pipeline/test_pipeline.py`	musicLibrary	ops	i Non-final test-harness	● Built	4,525	–	`musicLibrary/pipeline/test_pipeline.py` +1
fetch_metabrainz (py and sh) `pipeline/fetch_metabrainz.py`	musicLibrary	ingest	i Non-final dev-tool	● Built	5,097	–	`musicLibrary/pipeline/fetch_metabrainz.py:5-8` +3
ref-mb Postgres `ref_mb/compose.yaml`	musicLibrary	ingest	● Production	● Built	2,179	–	`musicLibrary/ref_mb/compose.yaml` +3
ref_mb load and maintenance scripts `ref_mb/scripts/load_ref_mb.ps1`	musicLibrary	ingest	i Non-final dev-tool	● Built	2,968	–	`musicLibrary/ref_mb/scripts` +1
ref_mb accuracy evaluation `ref_mb/scripts/run_eval.ps1`	musicLibrary	ingest	i Non-final test-harness	● Built	1,327	–	`musicLibrary/ref_mb/scripts/run_eval.ps1` +1
Weekly run playbook `agent/weekly_run.md`	musicLibrary	ops	● Production	● Built	5,567	–	`musicLibrary/agent/weekly_run.md`
Beatport sync playbook `agent/beatport_sync.md`	musicLibrary	ops	● Production	● Built	1,744	–	`musicLibrary/agent/beatport_sync.md`
Jules launchers and prompt builders `plan/launch_jules.py`	musicLibrary	ops	i Non-final dev-tool	● Built	14,261	–	`musicLibrary/plan/launch_jules.py:18` +2
songNodes weekend progress run prompt `plan/scheduled_run_prompt.md`	musicLibrary	ops	i Non-final doc-reference	● Built	3,548	–	`musicLibrary/plan/scheduled_run_prompt.md`

songnodes pytest report `tests/reports/junit.xml`	musicLibrary	ops	i Non-final doc-reference	● Built	1,734	–	`musicLibrary/tests/reports/junit.xml`
REST API `services/rest_api`	songnodes	ops	● Production	● Built	30,385	R6 accepted exception	`songnodes/compose.dev.yaml:52` +2
Graph API `services/graph-visualization-api`	songnodes	graph	● Production	● Built	40,500	All checked rules pass	`songnodes/compose.dev.yaml:36` +1
Worker `services/worker/celery_app.py`	songnodes	ops	● Production	● Built	5,008	All checked rules pass	`songnodes/compose.dev.yaml:97` +1
Unified scraper `scrapers/unified_scraper_api.py`	songnodes	ingest	● Production	● Built	24,173	R8 violation	`songnodes/compose.dev.yaml:145` +2
Audio analysis `services/audio-analysis/main.py`	songnodes	ops	● Production	● Built	25,523	All checked rules pass	`songnodes/compose.dev.yaml:186` +1
Postgres 18 `compose.dev.yaml#postgres`	songnodes	ops	● Production	● Built	8,505	–	`songnodes/compose.dev.yaml:4` +1
api-gateway `services/api-gateway`	songnodes	ops	i Non-final superseded-prototype	● Built	-	–	`songnodes/services/api-gateway` +1
api-gateway-internal `services/api-gateway-internal`	songnodes	ops	i Non-final superseded-prototype	● Built	17,030	–	`songnodes/services/api-gateway-internal/main.py` +1
browser-collector `services/browser-collector`	songnodes	ops	i Non-final superseded-prototype	● Built	26,662	–	`songnodes/services/browser-collector/main.py` +2
data-transformer `services/data-transformer`	songnodes	ops	i Non-final superseded-prototype	● Built	52,495	–	`songnodes/services/data-transformer/bronze_to_silver_etl.py:617` +2
data-validator `services/data-validator`	songnodes	ops	i Non-final superseded-prototype	● Built	57,259	–	`songnodes/services/data-validator/main.py` +1
db-connection-pool `services/db-connection-pool`	songnodes	ops	i Non-final superseded-prototype	● Built	16,452	–	`songnodes/services/db-connection-pool/main.py` +1
dlq-manager `services/dlq-manager`	songnodes	ops	i Non-final superseded-prototype	● Built	22,041	–	`songnodes/services/dlq-manager/main.py` +2
enhanced-visualization-service `services/enhanced-visualization-service`	songnodes	ops	i Non-final superseded-prototype	● Built	-	–	`songnodes/services/enhanced-visualization-service` +1
gold-to-operational-etl `services/gold-to-operational-etl`	songnodes	ops	i Non-final superseded-prototype	● Built	23,029	–	`songnodes/services/gold-to-operational-etl/gold_to_operational_etl.py:239` +1

gold-layer-processor `services/gold_layer_processor.py`	songnodes	library	i Non-final superseded-prototype	● Built	14,800	–	`songnodes/services/gold_layer_processor.py:237` +1
graph-projection `services/graph-projection`	songnodes	graph	i Non-final dev-tool	● Built	1,405	–	`songnodes/services/graph-projection/cli.py:5` +1
health-monitor `services/health-monitor`	songnodes	ops	i Non-final superseded-prototype	● Built	18,384	–	`songnodes/services/health-monitor/health-check.js` +1
metadata-enrichment `services/metadata-enrichment`	songnodes	ops	i Non-final superseded-prototype	● Built	69,816	–	`songnodes/services/metadata-enrichment/main.py` +3
nlp-processor `services/nlp-processor`	songnodes	library	i Non-final superseded-prototype	● Built	25,410	–	`songnodes/services/nlp-processor/main.py` +2
reference-data `services/reference-data`	songnodes	ops	i Non-final superseded-prototype	● Built	654	–	`songnodes/services/reference_data/resolver/resolver.py` +1
serato-integration `services/serato-integration`	songnodes	ops	i Non-final migration-repair	● Built	16,312	–	`songnodes/services/serato-integration/batch_import.py:187` +1
websocket-api `services/websocket-api`	songnodes	ops	i Non-final superseded-prototype	● Built	35,590	–	`songnodes/services/websocket-api/main.py` +1
database `services/database`	songnodes	ops	i Non-final doc-reference	● Built	-	–	`songnodes/services/database/migrations/008_cooldown_queue_up.sql` +1
scraper-orchestrator `scraper-orchestrator`	songnodes	ingest	i Non-final migration-repair	● Built	9,991	R7 accepted exception	`songnodes/scraper-orchestrator/enrich_existing_tracks.py:190` +2
scraper_service.py `scrapers/scraper_service.py`	songnodes	ingest	i Non-final superseded-prototype	● Built	7,957	R7 accepted exception	`songnodes/scrapers/scraper_service.py`
scraper_api_real.py `scrapers/scraper_api_real.py`	songnodes	ingest	i Non-final superseded-prototype	● Built	9,484	R7 accepted exception	`songnodes/scrapers/scraper_api_real.py`
real_data_scraper.py `scrapers/real_data_scraper.py`	songnodes	ingest	i Non-final superseded-prototype	● Built	19,724	–	`songnodes/scrapers/real_data_scraper.py`
raw_data_processor.py `scrapers/raw_data_processor.py`	songnodes	ingest	i Non-final migration-repair	● Built	23,683	–	`songnodes/scrapers/raw_data_processor.py`
recover_raw_data.py `scrapers/recover_raw_data.py`	songnodes	ingest	i Non-final migration-repair	● Built	35,415	–	`songnodes/scrapers/recover_raw_data.py`

enrich_existing_tracks.py `scrapers/enrich_existing_tracks.py`	songnodes	ingest	i Non-final migration-repair	● Built	10,550	–	songnodes/scrapers/enrich_existing_tracks.py
backfill_artist_names.py `scrapers/backfill_artist_names.py`	songnodes	ingest	i Non-final migration-repair	● Built	12,228	–	songnodes/scrapers/backfill_artist_names.py
batch_scrape_all_playlists.py `scrapers/batch_scrape_all_playlists.py`	songnodes	ingest	i Non-final dev-tool	● Built	3,754	–	songnodes/scrapers/batch_scrape_all_playlists.py
import_nodes_data.py `scrapers/import_nodes_data.py`	songnodes	ingest	i Non-final migration-repair	● Built	7,983	–	songnodes/scrapers/import_nodes_data.py
import_setlist_data.py `scrapers/import_setlist_data.py`	songnodes	ingest	i Non-final migration-repair	● Built	18,599	–	songnodes/scrapers/import_setlist_data.py
enrichment_queue_publisher.py `scrapers/pipelines/enrichment_queue_publisher.py`	songnodes	ingest	i Non-final superseded-prototype	● Built	10,538	–	songnodes/scrapers/pipelines/enrichment_queue_publisher.py
004_create_target_track_searches.sql `scrapers/migrations/004_create_target_track_searches.sql`	songnodes	ingest	i Non-final superseded-prototype	● Built	1,785	–	songnodes/scrapers/migrations/004_create_target_track_searches.sql
load_283_tracks.sql `load_283_tracks.sql`	songnodes	ops	i Non-final migration-repair	● Built	69,728	–	songnodes/load_283_tracks.sql
assign_proper_artists.py `assign_proper_artists.py`	songnodes	ops	i Non-final migration-repair	● Built	8,341	–	songnodes/assign_proper_artists.py
analyze-adjacency-problem.js `analyze-adjacency-problem.js`	songnodes	ops	i Non-final dev-tool	● Built	4,419	–	songnodes/analyze-adjacency-problem.js
backup_volumes.sh `backup_volumes.sh`	songnodes	ops	i Non-final superseded-prototype	● Built	7,525	–	songnodes/backup_volumes.sh
monitor_pipeline.sh `monitor_pipeline.sh`	songnodes	ops	i Non-final superseded-prototype	● Built	5,355	–	songnodes/monitor_pipeline.sh
setup-autostart.sh `setup-autostart.sh`	songnodes	ops	i Non-final release-installer	● Built	900	–	songnodes/setup-autostart.sh

songnodes.service songnodes.service	songnodes	ops	i Non-final release-installer	● Built	602	–	songnodes/songnodes.service
start-docker.sh start-docker.sh	songnodes	ops	i Non-final superseded-prototype	● Built	7,437	–	songnodes/start-docker.sh
skaffold.yaml skaffold.yaml	songnodes	ops	i Non-final superseded-prototype	● Built	8,011	–	songnodes/skaffold.yaml
QUICK_START_QUERIES.sql QUICK_START_QUERIES.sql	songnodes	ops	i Non-final doc-reference	● Built	10,392	–	songnodes/QUICK_START_QUERIES.sql
backup_database.sh scripts/backup_database.sh	songnodes	ops	i Non-final superseded-prototype	● Built	2,644	–	songnodes/scripts/backup_database.sh
restore_and_promote.sh scripts/restore_and_promote.sh	songnodes	ops	i Non-final migration-repair	● Built	8,297	–	songnodes/scripts/restore_and_promote.sh
upgrade-postgres-18.sh scripts/upgrade-postgres-18.sh	songnodes	ops	i Non-final migration-repair	● Built	3,548	–	songnodes/scripts/upgrade-postgres-18.sh
remove_same_artist_edges.sql scripts/remove_same_artist_edges.sql	songnodes	ops	i Non-final migration-repair	● Built	3,618	–	songnodes/scripts/remove_same_artist_edges.sql
clean_bracketed_artists.sql scripts/clean_bracketed_artists.sql	songnodes	ops	i Non-final migration-repair	● Built	4,987	–	songnodes/scripts/clean_bracketed_artists.sql
load-library-targets.sh scripts/load-library-targets.sh	songnodes	library	i Non-final dev-tool	● Built	1,474	–	songnodes/scripts/load-library-targets.sh
run_weekly.py pipeline/run_weekly.py	songnodes	ops	i Non-final trial-fork	● Built	7,022	–	songnodes/pipeline/run_weekly.py
helm deploy/helm	songnodes	ops	i Non-final superseded-prototype	● Built	36,560	–	songnodes/deploy/helm
k8s k8s	songnodes	ops	i Non-final superseded-prototype	● Built	9,886	–	songnodes/k8s
start-dev scripts/start-dev.sh	songnodes	ops	● Production	● Built	6,324	–	songnodes/scripts/start-dev.sh:31 +1
import-music-library scripts/import-music-library.sh	songnodes	library	● Production	● Built	3,555	–	songnodes/scripts/import-music-library.sh:1
sync-design-system scripts/sync-design-system.sh	songnodes	design	● Production	● Built	3,079	–	songnodes/.github/workflows/design-system.yml:36

scope_guard (G1) `scripts/scope_guard.py`	songnodes	ops	● Production	● Built	9,008	–	`songnodes/.github/workflows/scope-guard.yml:31`
e2e-docker `scripts/e2e-docker.sh`	songnodes	ops	● Production	● Built	1,365	–	`songnodes/CLAUDE.md:140`
migrate.py `services/worker/migrate.py`	songnodes	ops	● Production	● Built	4,367	–	`songnodes/services/worker/migrate.py:1`
library reconcile CLI `services/library/reconcile/cli.py`	songnodes	library	● Production	● Built	2,014	–	`songnodes/services/library/reconcile/cli.py:45`
Duo auth router (DUO1) `services/rest_api/routers/user_auth.py`	songnodes	services	i Non-final trial-fork	▲ Queued	–	All checked rules pass	PR #578 `services/rest_api/routers/user_auth.py`
cloudflared tunnel (DUO3) `compose.dev.yaml#cloudflared`	songnodes	services	i Non-final trial-fork	▲ Queued	–	–	PR #577 `compose.dev.yaml` +10 +1
Duo and tunnel guide (DUOAUTH) `docs/security/cisco-duo-auth-guide.md`	songnodes	services	i Non-final doc-reference	▲ Queued	–	–	PR #576 `docs/security/cisco-duo-auth-guide.md` +310
Home `songnodes/frontend/src/pages/HomePage.tsx`	songnodes	ops	● Production	● Built	3,492	R6 accepted exception	`songnodes/frontend/src/routes/registry.tsx:14` +1
Adjacency `songnodes/frontend/src/pages/GraphPage.tsx`	songnodes	graph	● Production	● Built	2,074	R4 accepted exception; R6 accepted exception; R8 violation	`songnodes/frontend/src/pages/GraphPage.tsx:38-58` +2
Melbourne crate `songnodes/frontend/src/features/musicLibrary/LibraryPage.tsx`	songnodes	library	● Production	● Built	27,441	All checked rules pass	`songnodes/frontend/src/features/musicLibrary/route.tsx:7` +2
Set builder `songnodes/frontend/src/features/musicLibrary/SetBuilderPage.tsx`	songnodes	sets	● Production	● Built	39,010	R2 violation	`songnodes/frontend/src/features/musicLibrary/route.tsx:8` +2
Route finder `songnodes/frontend/src/features/pathfinder/PathfinderPage.tsx`	songnodes	paths	● Production	● Built	14,782	R2 violation; R5 violation; R8 violation	`songnodes/frontend/src/features/pathfinder/route.tsx:4` +2
Route finder (planned) `songnodes/frontend/src/pages/ModulePages.tsx`	songnodes	paths	i Non-final superseded-prototype	● Built	3,338	–	`songnodes/frontend/src/pages/ModulePages.tsx:29-52` +1

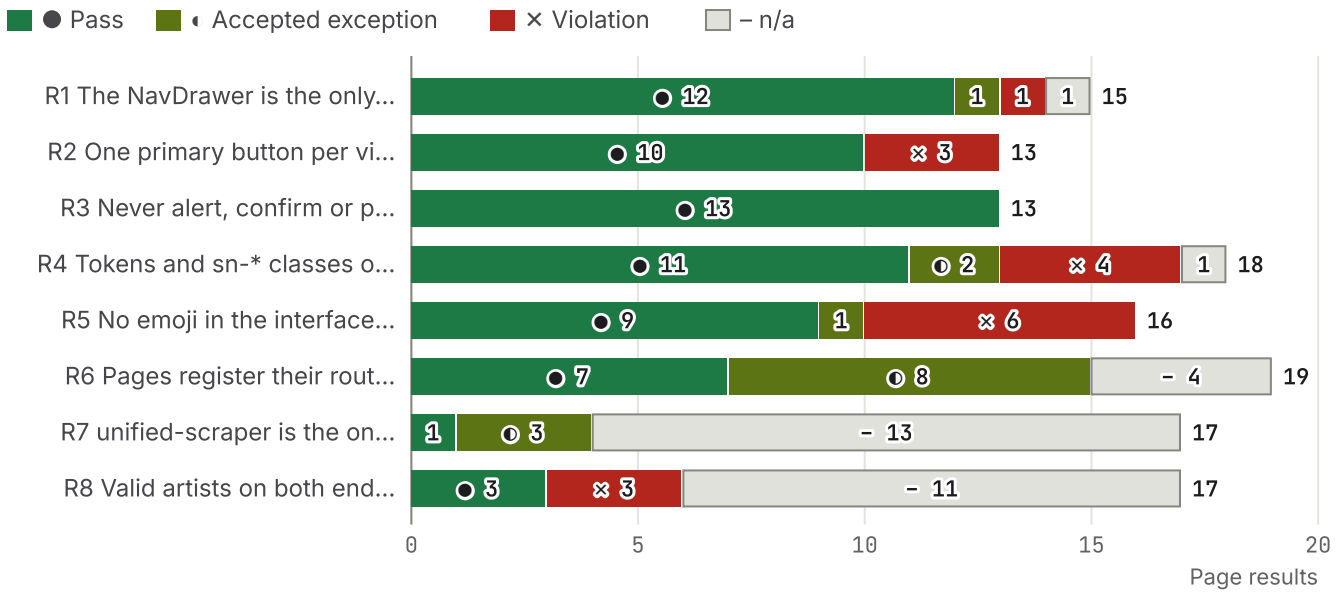
Camelot wheel songnodes/frontend/src/features/camelot/CamelotPage.tsx	songnodes	harmony	● Production	● Built	9,119	All checked rules pass	songnodes/frontend/src/features/camelot/route.tsx:6 +1
Target tracks songnodes/frontend/src/pages/ModulePages.tsx	songnodes	ingest	● Production	● Built	3,338	R6 accepted exception	songnodes/frontend/src/routes/registry.tsx:17-18 +2
Import tracklist songnodes/frontend/src/features/aiSetlists/AISetlistsPage.tsx	songnodes	ingest	● Production	● Built	18,853	R2 violation; R4 violation; R5 violation	songnodes/frontend/src/features/aiSetlists/route.tsx:7 +2
Telemetry songnodes/frontend/src/pages/ScraperProgressPage.tsx	songnodes	ingest	● Production	● Built	26,117	R4 violation; R6 accepted exception	songnodes/frontend/src/pages/ScraperProgressPage.tsx:18-120 +1
Settings songnodes/frontend/src/pages/SettingsPage.tsx	songnodes	services	● Production	● Built	44,111	R6 accepted exception	songnodes/frontend/src/routes/registry.tsx:20-21 +2
Service sign-in return songnodes/frontend/src/components/OAuthCallback.tsx	songnodes	services	● Production	● Built	13,171	R1 accepted exception; R5 violation	songnodes/frontend/src/App.tsx:30-33 +1
Page not found songnodes/frontend/src/pages/ModulePages.tsx	songnodes	ops	● Production	● Built	3,338	R6 accepted exception	songnodes/frontend/src/routes/registry.tsx:22 +1
SongNodes songnodes/frontend/index.html	songnodes	ops	● Production	● Built	1,304	All checked rules pass	songnodes/frontend/index.html:9-20 +1
Design system reference · SongNodes songnodes/docs/design-system/reference.html	songnodes	design	i Non-final doc-reference	● Built	93,789	R5 violation	songnodes/docs/design-system/reference.html +1
Design system reference · SongNodes (standalone) songnodes/docs/design-system/reference.standalone.html	songnodes	design	i Non-final doc-reference	● Built	633,370	–	songnodes/docs/design-system/reference.standalone.html +1
Component previews (98 files) songnodes/design-system/components/*/preview.html	songnodes	design	i Non-final doc-reference	● Built	1,068,167	All checked rules pass	songnodes/design-system/components/Toast/preview.html:1 +2

Design-system source previews (39 files) `songnodes/design-system/source/previews/{navsrc,charts-src,pagesA/src,pagesb/src}/*.html`	songnodes	design	i Non-final doc-reference	● Built	403,625	–	songnodes/design-system/source/previews/navsrc/PreferencesPanel.html +1
Violations fixture `songnodes/design-system/fixtures/violations.html`	songnodes	design	i Non-final test-harness	● Built	568	R4 accepted exception	songnodes/design-system/INDEX.md:50 +1
SongNodes Work Queue `songnodes/plan/work-queue-2026-10-30/tracker.html`	songnodes	ops	i Non-final dev-tool	● Built	11,468	R4 violation	songnodes/plan/work-queue-2026-10-30/tracker.html +1
Test DJ - Awesome Mix | MixesDB `songnodes/tests/unit/scrapers/fixtures/mixesdb_two_lists.html`	songnodes	ingest	i Non-final test-harness	● Built	849	–	songnodes/plan/agents/SC3_mixesdb_every_track.md:19
Sign in `songnodes/frontend/src/features/auth/LoginPage.tsx`	songnodes	services	i Non-final trial-fork	▲ Queued	–	R5 violation	gh api repos/zvirb/songnodes/pulls/579/files: frontend/src/features/auth/{AuthContext.tsx,LoginPage.tsx,route.tsx} +1

Category uses at least two independent signals per page (naming, live-data wiring, route registration or inbound links, docs status); hover a pill for the first two.

C8: **Standing-rule conformance**

17 of 128 page results are violations; R5 has the most (6).

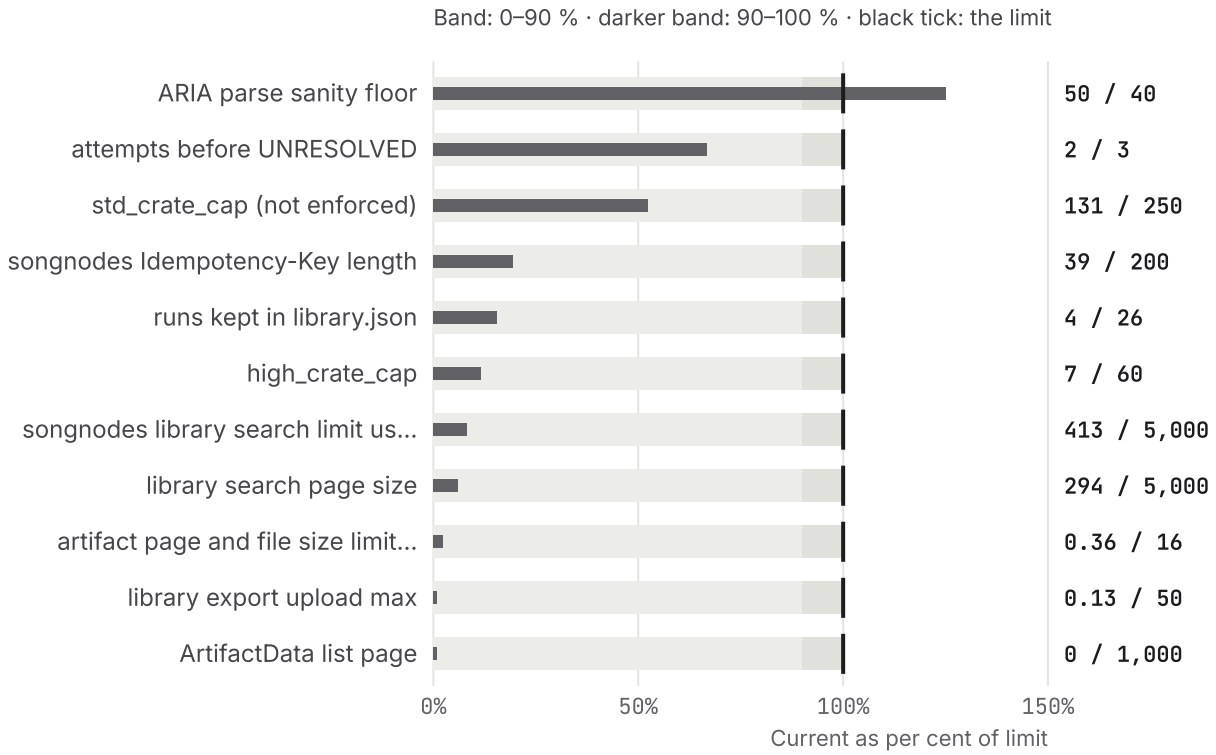


C8 · n = 128 · Unit: page results · ● Built

· Source: rule_results[] (rule, result), standing_rules[] · Method: Count of page results per rule R1–R8 (ordinal, not sorted) split by result.

C7: **Sizes against limits**

None of the 11 measured limits is within 10 % of its limit; the one with least headroom is ARIA parse sanity floor at 125 % of its floor.

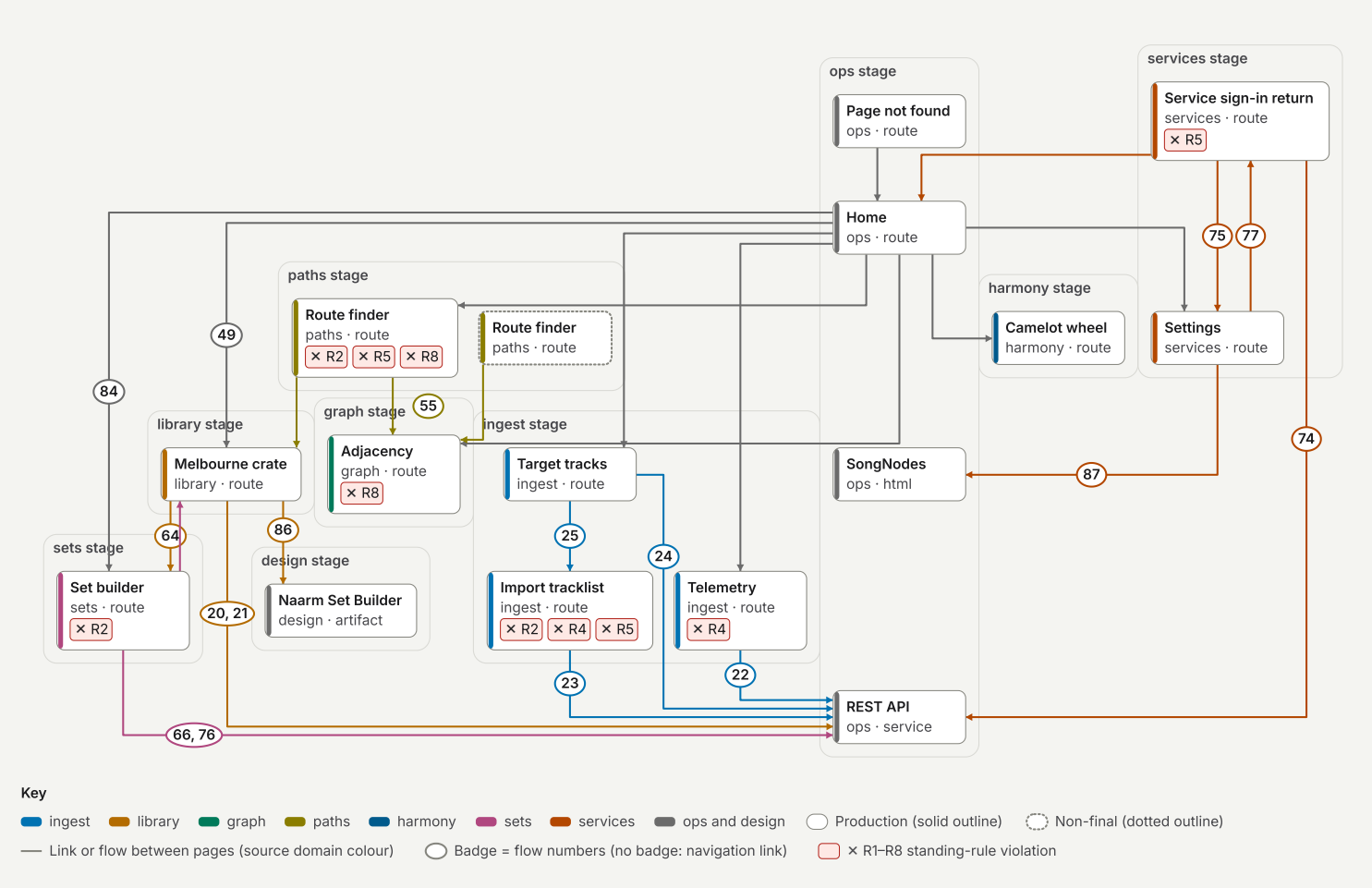


C7 · n = 11 · Unit: per cent of limit · ● Built

· Source: limits[] with current (value, unit, current) · Method: Bullet chart: current / limit. Text currents take their first number; a file map takes its largest file (the limit is per file), bytes to MB at 1,048,576. Floors (name says floor or minimum) are near when under 110 %, others when at or over 90 %.

D4a: **Page link graph – songnodes frontend**

7 of 17 songnodes frontend pages carry at least one standing-rule violation.

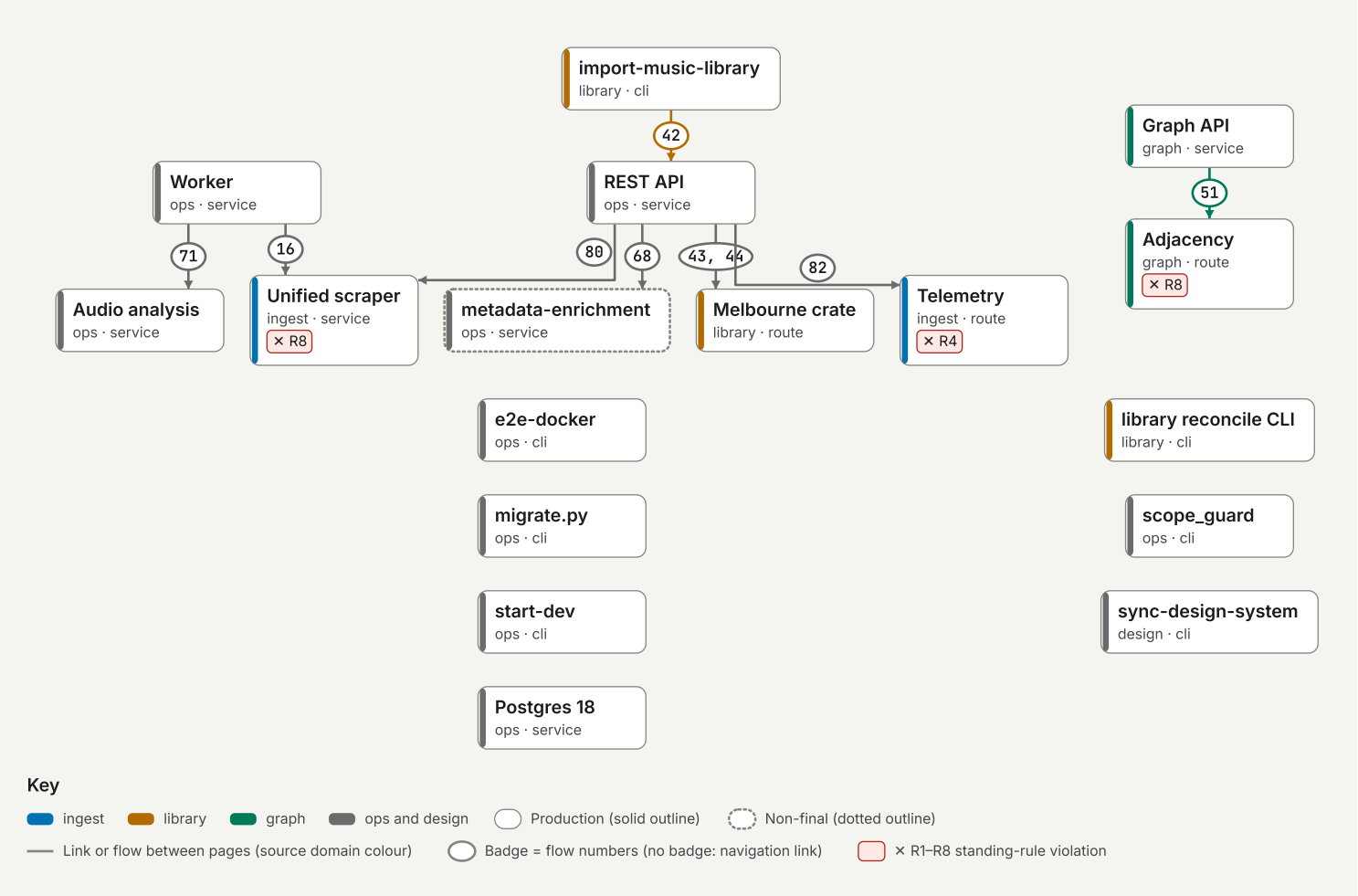


D4a · n = 17 · Unit: pages · **i built+queued**

· Source: pages[] (links_out, classification, state, domain); flows[] between pages; rule_results[] (result = violation) · Method: graphviz dot layered layout (TB, order given; 12 rank directions and input orders tried, scored by legibility then crossings), splines=ortho, own number badges, type floor 15 units; 4 crossings, 0 edges through nodes

D4b: **Page link graph – songnodes backend**

3 of 20 songnodes backend pages carry at least one standing-rule violation.

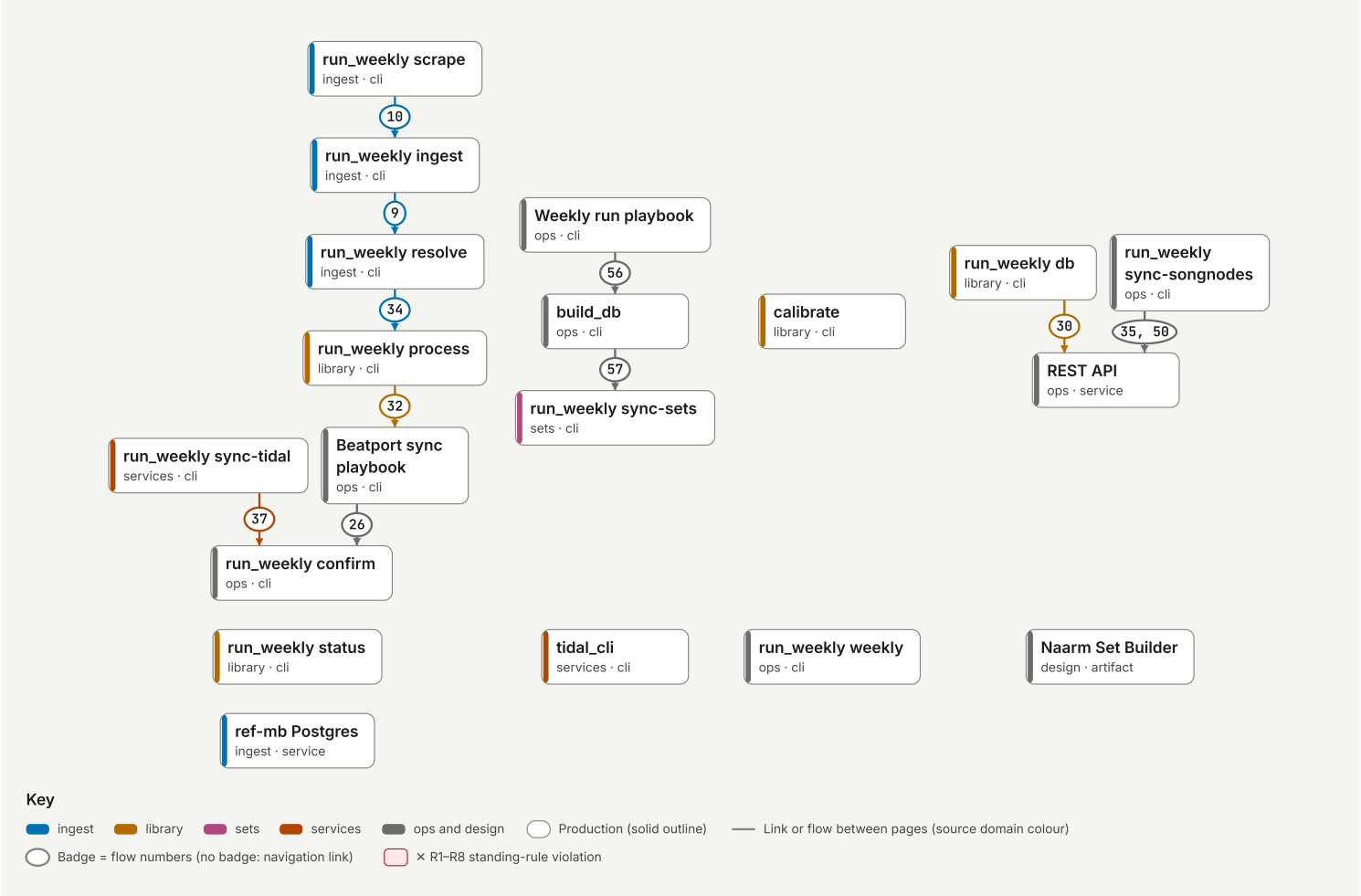


D4b · n = 20 · Unit: pages · `i built+queued`

· Source: pages[] (links_out, classification, state, domain); flows[] between pages; rule_results[] (result = violation) · Method: graphviz dot layered layout (TB, order degree, components packed 2 across, ungrouped; 12 rank directions and input orders tried, scored by legibility then crossings), splines=ortho, own number badges, type floor 12 units; 0 crossings, 0 edges through nodes

D4c: **Page link graph – musicLibrary**

0 of 19 musicLibrary pages carry at least one standing-rule violation.

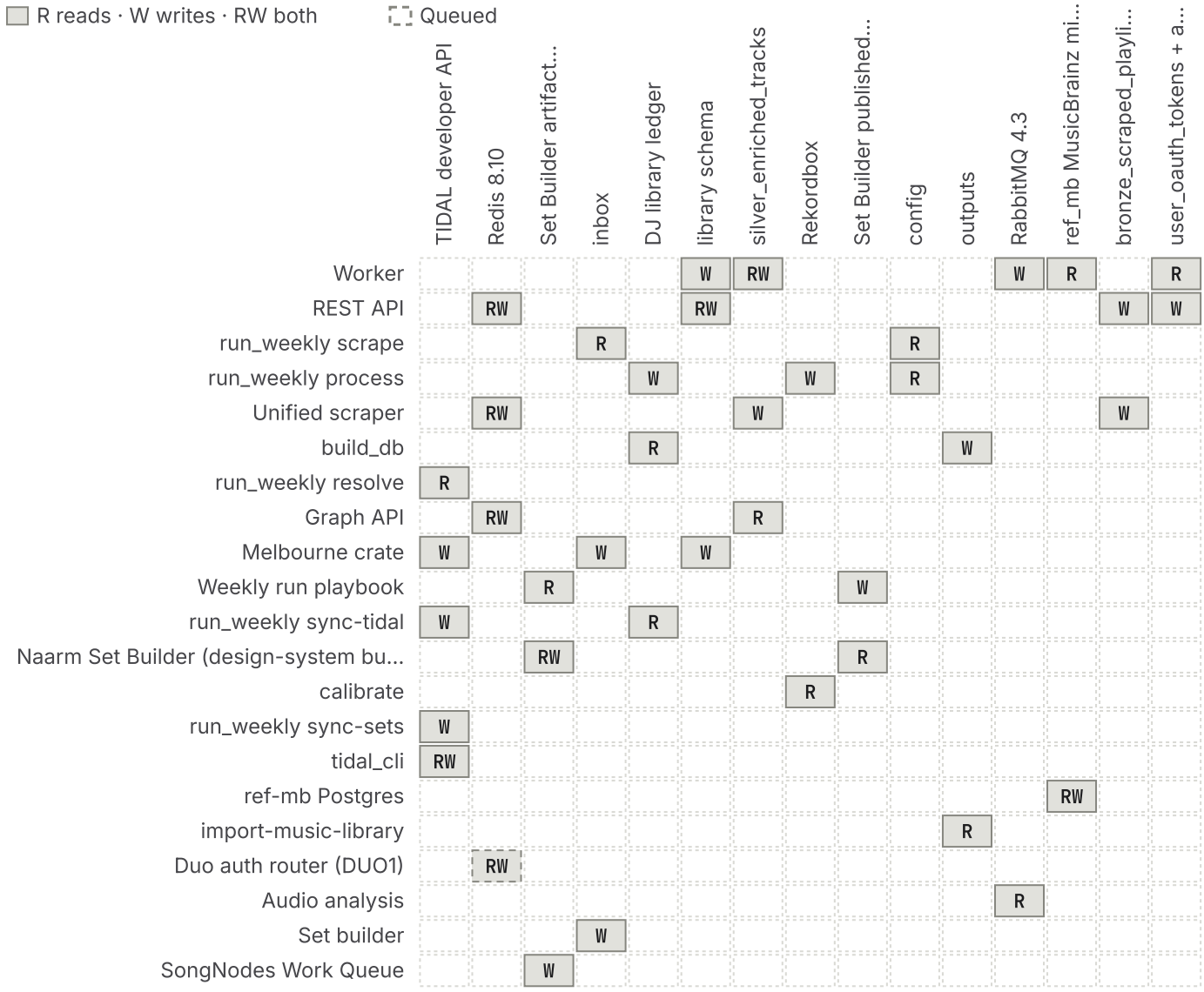


D4c · n = 19 · Unit: pages · ● Built

· Source: pages[] (links_out, classification, state, domain); flows[] between pages; rule_results[] (result = violation) · Method: graphviz dot layered layout (TB, order given, components packed 4 across, ungrouped; 12 rank directions and input orders tried, scored by legibility then crossings), splines=ortho, own number badges, type floor 12 units; 0 crossings, 0 edges through nodes

C2: **Page and store access matrix**

The 21 most connected pages against the 15 most used stores (of 23 pages and 35 stores with 61 access links); letters give the access, R read, W write, RW both.

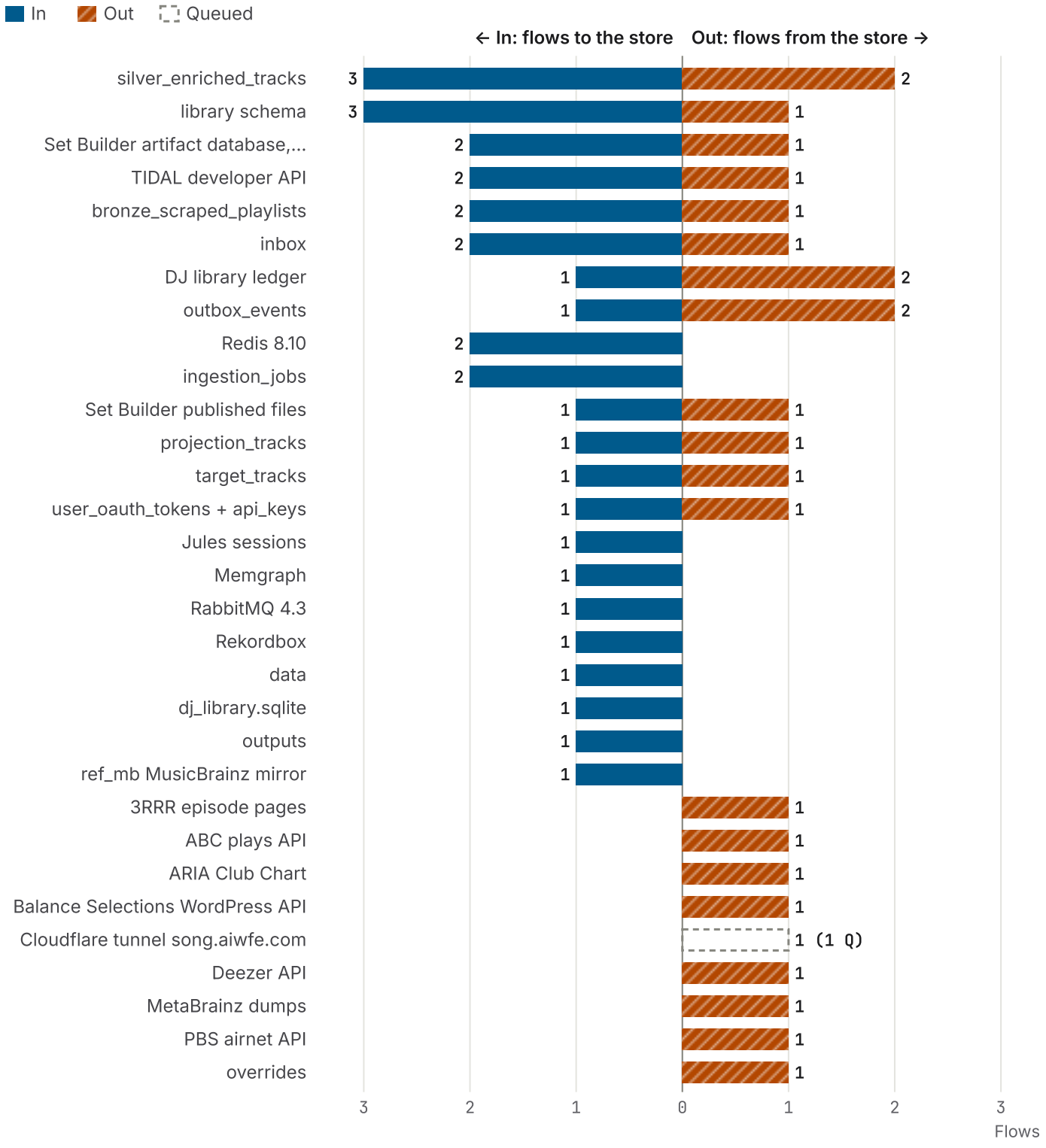


C2 · n = 61 · Unit: page–store links · **i Projected**

· Source: signals[] (from, to, direction, state), flows[] (from, to, state) · Method: A link is any signal or flow between a page and a store; signal direction gives R/W/RW, a flow into a store is W and out of a store is R. Rows and columns ranked by distinct connections (ties by id) and capped at 25 by 15.

C4: **Store fan-in and fan-out**

silver_enriched_tracks is the busiest store (3 flows in, 2 out); 31 of 48 stores sit on a numbered flow.

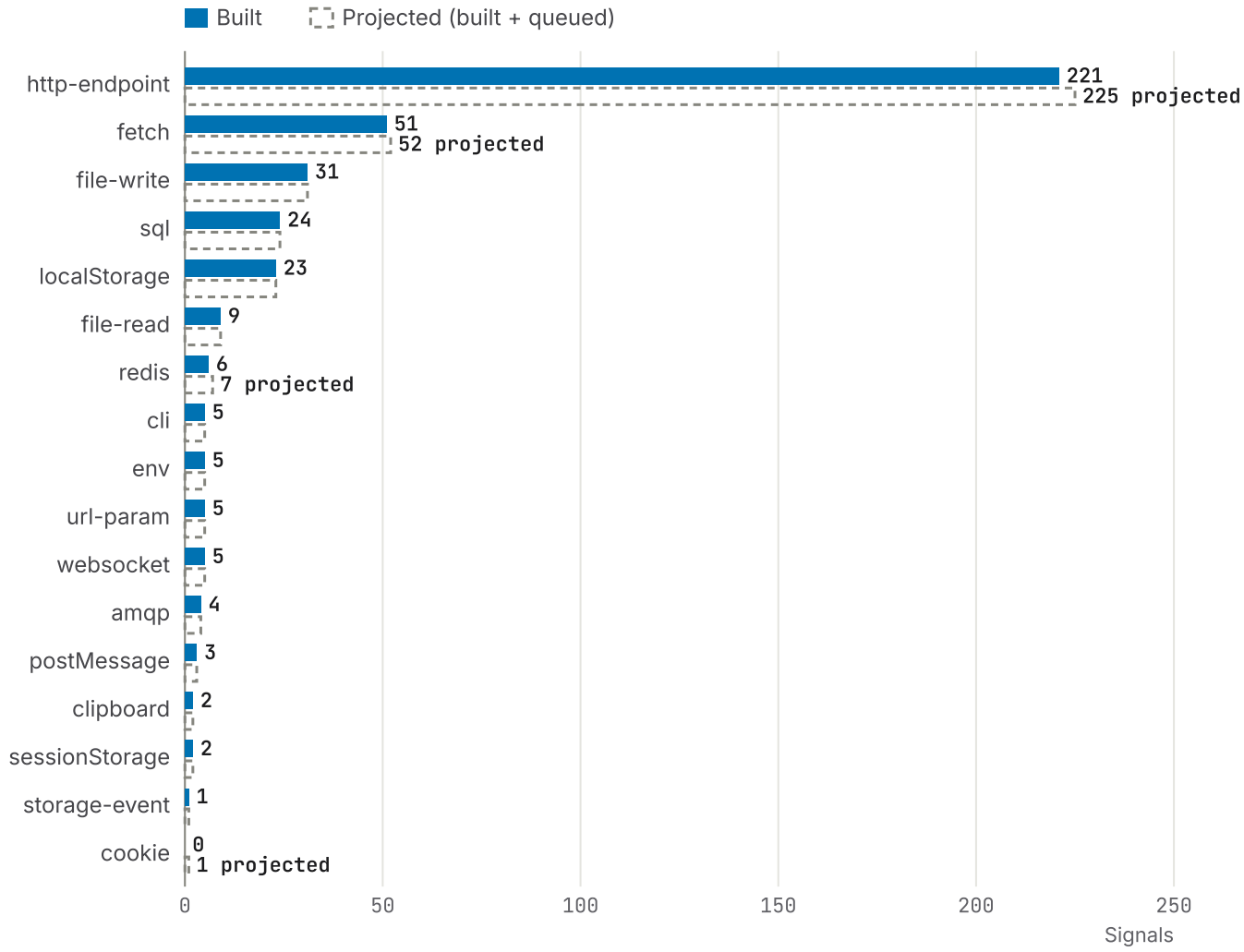


C4 · n = 49 · Unit: flows · i Projected

· Source: flows[] (from, to, channel, state), stores[], signals[] for the contract · Method: In = flows whose target is the store, out = flows whose source is the store; main contract = the signal named by the store's most frequent flow channel.

C3: **Signal primitives, built and projected**

http-endpoint carries 225 of 404 signals; queued work adds 7 signals (cookie, fetch, http-endpoint, redis).



C3 · n = 404 · Unit: signals · i Projected

· Source: signals[].type, signals[].state · Method: Count of signals per type; Built = state built, Projected = built + queued; sorted by projected count.

Table: **Stores and contracts** `48 stores`

48 rows

STORE	KIND	LOCATION	WRITER	READERS	CONCURRENCY	STATE	EVIDENCE
config `ml-store-config`	file	`config/rules.json`, `config/sources.json`	Markus (hand-edited; weekly agent must not edit thresholds)	ml-cli-scrape, ml-cli-ingest, ml-cli-process, ml-cli-build-db, ml-cli-test-pipeline	none; read-only at run time	● Built	`musicLibrary/pipeline/run_weekly.py:45-48` +1
inbox `ml-store-inbox`	folder	`inbox/`	scrape, ingest, resolve, sync-tidal, weekly agent (artifact_sets.json), beatport agent (resolved/sync_result), Markus (manual_adds.csv)	ml-cli-ingest, ml-cli-resolve, ml-cli-process, ml-cli-confirm, ml-cli-build-db, ml-cli-sync-sets, ml-cli-sync-songnodes	date-keyed files, whole-file overwrite (dump_json); resolved_<date> is incremental (load, skip done keys, flush every 20). No lock between the Beatport agent and the weekly run writing the same resolved_/sync_result_<date> file.	● Built	`musicLibrary/pipeline/run_weekly.py:39-42` +2
DJ library ledger `ml-store-ledger`	file	`ledger/dj_library_ledger.json` (+ `ledger/history/`)	run_weekly ingest/process/confirm	ml-cli-build-db, ml-cli-calibrate, ml-cli-sync-tidal, ml-cli-sync-sets, ml-cli-status, ml-agent-beatport-sync	single mutable JSON document (not append-only): write to .tmp then os.replace (atomic); process copies the old file to history/ledger_before_<date>.json first (overwritten if process re-runs the same date). No run lock; two runs would last-writer-win.	● Built	`musicLibrary/pipeline/djlib/ledger.py:62-70`
outputs `ml-store-outputs`	folder	`outputs/<date>/`, `outputs/crates/`, `outputs/artifact/library.json`, `outputs/rekordbox_musicLibrary.xml`	process, build_db, calibrate	ml-cli-build-db, ml-agent-weekly-run, ml-agent-beatport-sync, sn-scripts, ml-persona-curator	overwrite per run; deletes avoided on the synced drive (empty crates truncated)	● Built	`musicLibrary/pipeline/djlib/export.py:288-290`
overrides `ml-store-overrides`	folder	`overrides/energy_overrides.csv`, `overrides/rekordbox_collection.xml` (absent)	Markus	ml-cli-process, ml-cli-calibrate	re-applied every process run; header only today	● Built	`musicLibrary/pipeline/run_weekly.py:152-182`
dj_library.sqlite `ml-store-sqlite`	sqlite	`db/dj_library.sqlite`	build_db	ml-persona-curator (DB Browser)	full rebuild in a temp dir then shutil.copyfile over the target (copy, not atomic rename); row count and integrity_check afterwards; journal_mode DELETE. Current: 413 tracks, 0 sets, 4 runs, v_review_queue 413.	● Built	`musicLibrary/pipeline/build_db.py:113-186`

secrets ml-store-secrets	folder	secrets/ (tidal_app.json, tidal_user_token.json, tidal_pkce_pending.json, tidal_playlists.json); ref_mb/secrets/*.txt	tidal_cli, sync-tidal, sync-sets; Markus	ml-cli-resolve, ml-cli-sync-tidal, ml-cli-sync-sets, ml-svc-ref-mb	plain files, rewritten on refresh; pending cleared by writing {} (deletes blocked on synced drive). musicLibrary/secrets/ absent from this copy.	Built	musicLibrary/pipeline/djlib/tidal.py:112-113 +2
Set Builder artifact database, collection sets ml-store-artifact-db	artifact-db	claude.ai artifact GcqZTGjk6obdpjWaU7WFRq	Markus via the page (only-you-write rule)	ml-page-set-builder, ml-agent-weekly-run	page writes queued one document at a time; init merges shared-wins-by-id with local-only uploads; weekly agent read-only	Built	musicLibrary/artifact/app/artifact-data.ts:38-72 +2
Set Builder published files ml-store-artifact-files	artifact-files	claude.ai artifact GcqZTGjk6obdpjWaU7WFRq: index.html, app.js, assets/*, library.json	weekly agent (library.json), supervised rebuild (app.js)	ml-page-set-builder	publish refused unless read first; agent stops if live shell differs	Built	musicLibrary/agent/weekly_run.md:8
ref_mb Postgres ml-store-ref-mb	postgres	127.0.0.1:55433, data E:\musicLibrary\data\ref_mb_pgdata	ref_mb load scripts	sn-be-refmb, ml-cli-ref-mb-eval	read-only role default_transaction_read_only=on; stop with -t 600	Built	musicLibrary/ref_mb/compose.yaml +1
data (58 GB, not copied) ml-store-data	folder	data/: metabrainz/ (canonical, json, postgres dumps, logs), metabrainz_scratch/ (inventory, slice, sql benches, tools, a MusicBrainz server git checkout mbs/, PROGRESS.md), ref_mb_pgdata/pgdata (Postgres cluster, postmaster.pid 2026-11-06)	fetch_metabrainz.*, ref_mb scripts and compose, MB investigation agents	ml-svc-ref-mb	Postgres owns pgdata; downloads resumable by size	Built	musicLibrary/_data_manifest.txt
plan ml-store-plan	folder	plan/	prompt builders and launchers	ext-jules	none	Built	musicLibrary/plan/scheduled_run_prompt.md
songnodes Postgres 18 sn-be-store-postgres	postgres	compose service postgres, volume songnodes-dev_dev-postgres-18 (not published)	rest-api, worker, unified-scraper, graph-api	sn-be-svc-rest-api, sn-be-svc-graph-api, sn-be-svc-worker, sn-be-svc-unified-scraper	row locks: FOR UPDATE SKIP LOCKED on ingestion_jobs and target_tracks; FOR UPDATE on bronze/silver playlist during ingest; outbox ordered by (txid,id) below pg_snapshot_xmin	Built	songnodes/compose.dev.yaml:4 +3

bronze_scraped_playlists / bronze_scraped_tracks / bronze_scraped_artists `sn-be-tbl-bronze`	postgres	`public schema (medallion 001)`	unified-scraper persistence pipeline; rest-api import-tracklist	sn-be-svc-worker (ingest), sn-be-svc-rest-api (catalog_reads, jobs)	upsert on source keys; per-row savepoints; bronze_incomplete marker	● Built	`songnodes/sql/migrations/medallion/001_bronze_layer_up.sql:14` +1
bronze_api_enrichments `sn-be-tbl-bronze-api-enrichments`	postgres	`medallion 001`	none	–	none	● Built	`songnodes/sql/migrations/medallion/001_bronze_layer_up.sql:134`
silver_enriched_tracks / _artists / _playlists / silver_playlist_tracks / silver_track_transitions / silver_track_metadata_evidence / silver_camelot_corrections `sn-be-tbl-silver`	postgres	`medallion 002, 006, migration 014, 270`	worker datalayer (ingest, evidence, identity); also scraper persistence pipeline and rest-api catalog writes	sn-be-svc-graph-api, sn-be-svc-rest-api, sn-be-svc-worker	transitions ON CONFLICT (from_track_id,to_track_id); silver_enriched_tracks has no unique key (lookup-then-insert in upsert_track)	● Built	`songnodes/services/common/datalayer/ingest.py:104` +1
gold_track_graph (materialized view) `sn-be-tbl-gold-track-graph`	postgres	`medallion 006`	created at init; refreshed only by restore_and_promote.sh and remove_same_artist_edges.sql	–	REFRESH CONCURRENTLY needs migration 008 unique index, which compose does not apply	● Built	`songnodes/sql/migrations/medallion/006_track_adjacency_edges_up.sql:83` +1
gold_track_analytics / gold_playlist_analytics / gold_artist_analytics `sn-be-tbl-gold-analytics`	postgres	`medallion 003 (not applied by compose)`	data-transformer and gold_layer_processor (not deployed)	legacy gold-to-operational-etl, metadata-enrichment medallion_metrics	n/a	● Built	`songnodes/sql/migrations/medallion/003_gold_layer_up.sql` +1
projection_tracks `sn-be-tbl-projection-tracks`	postgres	`migration 014`	worker projection_consume / rebuild	sn-be-svc-rest-api (library_search, preview)	outbox consumer projection_tracks; keyed upserts; reconcile job	● Built	`songnodes/services/common/datalayer/projection.py:48` +1

ingestion_jobs sn-be-tbl-ingestion-jobs	postgres	migration 013	rest-api (enqueue, retry), worker (schedule, claim, settle), start-dev.sh (transition_scoring seed)	sn-be-svc-rest-api, sn-be-svc-worker, import-music-library.sh via API	unique (kind, idempotency_key); SKIP LOCKED claim; lease 300s (scrape 1800s); attempts/max_attempts; exponential backoff to 3600s; dead state; prune succeeded after 14 days	Built	songnodes/services/common/datalayer/jobs.py:20 +2
outbox_events / outbox_consumers / outbox_dead_letters sn-be-tbl-outbox	postgres	migration 014, 220	every catalog write in the same transaction	projection_tracks consumer, graph_projection consumer, GET /api/v1/outbox/status	at-least-once; handler+offset in one transaction; 3 in-process attempts then dead letter; rebase_after_restore for txid reset	Built	songnodes/services/common/datalayer/outbox.py:1 +1
library schema (tracks, track_xref, appearances, chart_positions, rotation_history, pending, rejected, beatport_check, sources, runs, sets, set_tracks, v_crates) sn-be-tbl-library	postgres	migrations 101, 180-185	worker library_ingest; rest-api sets routes	sn-be-svc-graph-api, sn-be-svc-rest-api, sn-be-svc-worker	one transaction per ingest payload; job idempotency key	Built	songnodes/services/library/importer/import_library.py:95 +1
target_tracks sn-be-tbl-target-tracks	postgres	migration 012, 280	rest-api target_tracks router; library ingest auto-promotion; scrape_targets	sn-be-svc-worker, sn-be-svc-rest-api	scheduler UPDATE … FOR UPDATE SKIP LOCKED; playlists_found under row lock; unique live (artist,title) once 280 applies	Built	songnodes/services/common/datalayer/target_scrape.py:72 +1
web_call_log sn-be-tbl-web-call-log	postgres	migration 170	common/providers/web_call_log.py	–	append-only	Built	songnodes/services/common/providers/web_call_log.py:24
user_oauth_tokens + api_keys sn-be-tbl-oauth	postgres	sql/init/07-api-keys.sql	rest-api music_auth and api_keys routers	sn-be-svc-rest-api, sn-be-svc-worker (saved keys)	upsert on (service,user_id); single user default_user	Built	songnodes/sql/init/07-api-keys.sql:316 +1

song_adjacency sn-be-tbl-song-adjacency	postgres	only in sql/init/05 and scrapers/music_schema.sql (not applied)	scraper persistence pipeline	–	n/a	Built	songnodes/scrapers/pipelines/persistence_pipeline.py:1731
musicdb.* legacy schema (tracks, user_setlists, tracks_audio_analysis, target_track_searches) sn-be-tbl-musicdb-legacy	postgres	not created by compose	graph-api user-setlists, audio-analysis queue path, root SQL scripts	graph-api suggestions/test routes	none	Built	songnodes/services/graph-visualization-api/main.py:392 +1
Redis 8.10 sn-be-store-redis	redis	compose service redis (anonymous volume /data)	rest-api (OAuth state), graph-api (batch status), unified-scraper spiders (quota, dedupe)	same writers	TTL keys only; no locks	Built	songnodes/compose.dev.yaml:26
RabbitMQ 4.3 sn-be-store-rabbitmq	rabbitmq	compose service rabbitmq, volume dev-rabbitmq-4, guest/guest	Celery beat in worker	Celery worker	acks_late, reject_on_worker_lost, prefetch 1; task expiry; remote control/gossip/mingle off	Built	songnodes/services/worker/celery_app.py:44
Memgraph sn-be-store-memgraph	external-api	bolt://memgraph:7687 (no service in compose)	graph_projection consumer when enabled	–	n/a	Built	songnodes/services/common/datalayer/graph_projection.py:22
ref_mb MusicBrainz mirror sn-be-store-refmb	postgres	host.docker.internal:55433 db ref_mb (optional, MB_LOCAL_ENABLED=false)	external (musicLibrary fetch_metabrainz)	sn-be-svc-worker, sn-be-svc-rest-api	read-only role, 2000ms statement timeout	Built	songnodes/compose.dev.yaml:61
unified scraper in-memory progress sn-be-store-scraper-progress	memory	OrderedDict in unified_scraper_api process	unified-scraper	rest-api job stages	ContextVar per request; lost on restart	Built	songnodes/scrapers/unified_scraper_api.py:109

Cloudflare tunnel song.aiwfe.com `sn-be-q-store-cloudflare-tunnel`	external-api	`cloudflared container (queued)`	operator	public internet	n/a	▲ Queued	`PR #577 compose.dev.yaml`
useStore (Zustand, persist songnodes-store, devtools "SongNodes Store") `sn-fe-store-zustand`	localStorage	`browser localStorage songnodes-store`	frontend store slices (graph, credentials, pathfinding, search)	sn-fe-graph, sn-fe-settings, sn-fe-sets	none: last writer per tab wins, no storage-event rehydrate across tabs	● Built	`songnodes/frontend/src/store/useStore.ts:340,1478-1553`
useUnifiedTracksStore (in-memory zustand) `sn-fe-store-unified-tracks`	memory	`browser memory`	hooks/useUnifiedTracks.ts refresh()	sn-fe-paths, sn-fe-harmony, sn-fe-sources-import, sn-fe-library	none; local sn-target-tracks-v1 overrides backend target state	● Built	`songnodes/frontend/src/hooks/useUnifiedTracks.ts:62-110`
loadLibrary() promise cache `sn-fe-store-library-cache`	memory	`module scope features/musicLibrary/data.ts`	loadLibrary	sn-fe-home, sn-fe-library, sn-fe-sets, sn-fe-store-unified-tracks	single shared promise, reset on error or forceRefresh	● Built	`songnodes/frontend/src/features/musicLibrary/data.ts:71,107-135`
per-device localStorage keys (sn-*, songnodes_*) `sn-fe-store-browser-ls`	localStorage	`browser origin http://127.0.0.1:3006`	see signals sn-fe-ls-*	all routes	append/overwrite; only sn-favs-library-crate listens to storage events; token refresh uses navigator.locks with localStorage mutex fallback	● Built	`songnodes/frontend/src/features/musicLibrary/storage.ts:8-19` +1
frontend nginx (inline default.conf) `sn-fe-store-nginx`	proxy	`frontend image :8080, published 127.0.0.1:3006`	frontend/Dockerfile	browser	n/a; resolver 127.0.0.11 valid=10s	● Built	`songnodes/frontend/Dockerfile:28-102` +1
PBS airnet API `ext-pbs-airnet`	external-api	`internet`	third party	–	rate limits and retries on the caller side	● Built	`see signals that reference ext-pbs-airnet`
3RRR episode pages `ext-3rrr`	external-api	`internet`	third party	–	rate limits and retries on the caller side	● Built	`see signals that reference ext-3rrr`
ABC plays API (triple j, Double J) `ext-abc-plays`	external-api	`internet`	third party	–	rate limits and retries on the caller side	● Built	`see signals that reference ext-abc-plays`

ARIA Club Chart `ext-aria`	external-api	internet	third party	–	rate limits and retries on the caller side	Built	see signals that reference ext-aria
Balance Selections WordPress API `ext-balance-wp`	external-api	internet	third party	–	rate limits and retries on the caller side	Built	see signals that reference ext-balance-wp
TIDAL developer API `ext-tidal`	external-api	internet	third party	–	rate limits and retries on the caller side	Built	see signals that reference ext-tidal
Deezer API `ext-deezer`	external-api	internet	third party	–	rate limits and retries on the caller side	Built	see signals that reference ext-deezer
MetaBrainz dumps `ext-metabrainz`	external-api	internet	third party	–	rate limits and retries on the caller side	Built	see signals that reference ext-metabrainz
Rekordbox / Lexicon (XML) `ext-rekordbox`	external-api	internet	third party	–	rate limits and retries on the caller side	Built	see signals that reference ext-rekordbox
Jules sessions (Google) `ext-jules`	external-api	internet	third party	–	rate limits and retries on the caller side	Built	see signals that reference ext-jules

Table: **Signals** `404 signals`

404 rows

PRIMITIVE	KEY OR ENDPOINT	DIRECTION	FROM	TO	FORMAT	SCHEMA	STATE	METHODS	EVIDENCE
file-read	config/sources.json	read	ml-store-config	ml-cli-scrape	json	{_comment, sources:[{id, kind airnet|rrr_html|abc_plays|aria_html|wp_tracklist|manual, enabled, energy_prior, pillar_hint, club_only, programs, episodes_per_program, station, weekdays, start_hour_local, hours, max_play_seconds, url, api, search}]}	Built	M1, M2, M3	musicLibrary/pipeline/run_weekly.py:45-48 +1
file-read	config/rules.json	read	ml-store-config	ml-cli-process	json	gates, genre_map, pillar_short, energy{bpm_bands,weights,review_spread, cue_adjust}, tiers, stars_by_score, rotation{promote_*, high_*, std_archive_*, high_crate_cap, std_crate_cap(unused)}, heritage_artists, generic_genres, exclude_genres, bpm_range	Built	M1, M2	musicLibrary/pipeline/run_weekly.py:45-48 +2
fetch	https://airnet.org.au/rest/stations/3pbs/programs/<slug>/episodes and <episodeRestUrl>/playlists	read	ext-pbs-airnet	ml-cli-scrape	json	episode{start, episodeRestUrl}; playlist item{type:'track', artist, title|track}	Built	M1, M2	musicLibrary/pipeline/djlib/scrapers.py:45-63

PRIMITIVE	KEY OR ENDPOINT	DIRECTION	FROM	TO	FORMAT	SCHEMA	STATE	METHODS	EVIDENCE
fetch	https://www.rrr.org.au/explore/programs/<slug>[/episodes/<id>-<slug>]	read	ext-3rrr	ml-cli-scrape	html	li.audio-summary__track with __track-title and __track-artist	Built	M1, M2	musicLibrary/pipeline/djlib/scrapers.py:128-155
fetch	https://music.ab	read	ext-	ml-	json	items[].rec	Built	M1, M2	musicLibr

	cradio.net.au/api/v1/plays/search.json?station&from&to&limit=100		abc-plays	cli-scrape		ording{title, artists[{name,type}], duration}			ary/pipeline/djlib/scrapers.py:82-104
fetch	ARIA Club Chart page (sources.json url)	read	ext-aria	ml-cli-scrape	html	c-chart-item__rank/__title/__artist and 'weeks in'	Built	M1, M2	musicLibrary/pipeline/djlib/scrapers.py:108-124
fetch	Balance Selections WordPress REST posts (sources.json api)	read	ext-balance-wp	ml-cli-scrape	json	posts[] {content.rendered numbered 'Artist – Title', title.rendered, date, link}	Built	M1, M2	musicLibrary/pipeline/djlib/scrapers.py:159-171
file-write	inbox/manual_adds.csv	both	ml-persona-curator	ml-cli-scrape	csv	artist,title,mix,source_id,show,date,url,processed (processed stamped with run date by scrape)	Built	M1, M3	musicLibrary/pipeline/run_weekly.py:378-394 +1
file-write	inbox/raw_<date>.json	both	ml-cli-scrape	ml-cli-ingest	json	[{artist,title,remix_info,source_id,show,broadcast_date,url,[chart_rank,chart_weeks,saves]}]	Built	M1, M2, M3	musicLibrary/pipeline/run_weekly.py:395 +2
file-write	outputs/<date>/scrape_log.txt	write	ml-cli-scrape	ml-agent-weekly-run	text	'<source_id>: N rows' and FAILED lines	Built	M1, M2	musicLibrary/pipeline/run_weekly.py:396-398 +1
file-write	inbox/to_resolve_<date>.json	both	ml-cli-ingest	ml-cli-resolve	json	[{norm_key,artist,title,remix_info}]	Built	M1, M2, M3	musicLibrary/pipeline/run_weekly.py:62-64 +1

file-write	inbox/resolved_<date>.json	both	ml-cli-resolve	ml-cli-process	json	{norm_key:{tidal_versions[{platform,id,url,title,mix_name,duration_seconds,bpm,key,isrc,artist,popularity}],deezer_versions[{platform,id,url,title,mix_name,duration_seconds,isrc,artist,(+bpm,isrc,release_date,label on longest)}],genre,deezer_genres?,beatport_versions? (written by beatport_sync agent)}}	Built	M1, M2, M3	musicLibrary/pipeline/run_weekly.py:417-462 +2
fetch	https://openapi.tidal.com/v2/searchResults, /tracks, /tracks/{id}/relationships/genres (client credentials)	read	ext-tidal	ml-cli-resolve	json:api	Tracks_Attributes{title,version,duration ISO8601,bpm,key,keyScale,isrc,popularity}; M3 data: 0 of 1007 returned versions carry bpm or key	Built	M1, M3	musicLibrary/pipeline/djlib/tidal.py:207-234 +1
fetch	https://api.deezer.com /search, /track/{id}, /album/{id}	read	ext-deezer	ml-cli-resolve	json	search data[] {id,title,title_short,title_version, duration,isrc,artist.name,link}; track{bpm, isrc,release_date,album.id}; album{genres.data[].name,label}	Built	M1, M2	musicLibrary/pipeline/djlib/deezer.py:7-50

fetch	https://auth.tidal.com/v1/oauth2/token and https://login.tidal.com/authorize (PKCE, redirect http://127.0.0.1:3006/oauth/callback)	both	ml-cli-tidal	ext-tidal	form	grant_type client_credentials|authorization_code|refresh_token, client_secret in body	Built	M1, M2	musicLibrary/pipeline/djlib/tidal.py:62-114 +1
file-read	secrets/tidal_app.json	read	ml-store-secrets	ml-cli-resolve	json	{client_id, client_secret, redirect_uri?, scopes?} copied from songnodes .env TIDAL_CLIENT_ID/SECRET	Built	M1	musicLibrary/pipeline/djlib/tidal.py:52-56 +1
file-write	secrets/tidal_user_token.json and tidal_pkce_pending.json	both	ml-cli-tidal	ml-cli-sync-tidal	json	token{access_token, refresh_token, expires_in, scope, obtained_at}; pending{verifier,state,created,redirect}	Built	M1	musicLibrary/pipeline/djlib/tidal.py:57-58 +3
file-write	secrets/tidal_playlists.json	both	ml-cli-sync-tidal	ml-cli-sync-sets	json	{<crate name>: <playlist id>, 'set:<set id>': <playlist id>}	Built	M1	musicLibrary/pipeline/run_weekly.py:476-496 +1
fetch	openapi.tidal.com/v2 POST /playlists, GET/POST/DELETE /playlists/{id}/relationships/items (user token, Idempotency-Key uuid5)	write	ml-cli-sync-tidal	ext-tidal	json:api	playlists UNLISTED; items {id,type:'tracks'}; onDuplicates SKIP; DELETE needs meta.itemId	Built	M1, M2	musicLibrary/pipeline/djlib/tidal.py:237-272

file-write	ledger/dj_library_ledger.json	both	ml-cli-process	ml-cli-build-db	json	{schema_version:1, updated, tracks{track_id: {track_id,norm_key,artist,title,mix_version,remixer,duration_seconds,bpm,camelot_key,camelot_boost_plus2/minus5, genre,label,release_year,isrc,genre_pillar,pillar_method,wave_energy_tier,energy_score_1_to_10,energy_methods,energy_auto?,energy_override?,star_rating_1_to_5,color_code,functional_tags, sources_detected,appearances[],chart{},saves,rotation_status,rotation_since,rotation_history[],rotation_locked?,repromoted?,first_seen,added,platform_sync_state,synced_platforms, beatport,tidal,deezer, beatport_search_url,local_path} }, pending{}, rejected{}, beatport_check{}}	Built	M1, M2, M3	musicLibrary/pipeline/djlib/ledger.py:45-70 +4

file-write	ledger/history/ledger_before_<date>.json	write	ml-cli-process	ml-persona-developer	json	copy of the ledger file as it was before process saved	Built	M1, M2	musicLibrary/pipeline/djlib/ledger.py:67-69 +1
file-read	overrides/energy_overrides.csv	read	ml-persona-curator	ml-cli-process	csv	track_id,artist,title,energy_score,genre_pillar,rotation_status,tags(;),local_path,note — header only today	Built	M1, M3	musicLibrary/pipeline/run_weekly.py:152-182 +1
file-write	outputs/<date>/sync_plan.json	write	ml-cli-process	ml-agent-beatport-sync	json	{playlists[], actions[{op add|remove, playlist, track_id, artist, title, mix_version, duration_seconds, beatport_id, beatport_url, tidal_id, platforms[]}], run_date}	Built	M1, M2	musicLibrary/pipeline/djlib/export.py:308-332 +2
file-write	outputs/crates/<crate>.csv	write	ml-cli-process	ml-persona-curator	csv	artist,title,mix_version,bpm,camelot_key,camelot_boost_plus2,camelot_boost_minus5,energy_score_1_to_10,star_rating_1_to_5,color_code,functional_tags,sources_detected,rotation_since,platform_sync_state,beatport_url,tidal_url; emptied crates truncated to 0 bytes	Built	M1, M2	musicLibrary/pipeline/djlib/export.py:285-305

file-write	outputs/<date>/buy_list.csv	write	ml-cli-process	ml-agent-beatport-sync	csv	artist,title,mix_version,label,sources,energy,platform_sync_state,beatport_url (High Rotation without local_path)	Built	M1, M2	musicLibrary/pipeline/djlib/export.py:335-346 +1
file-write	outputs/<date>/review_queue.csv	write	ml-cli-process	ml-persona-curator	csv	track_id,artist,title,mix_version,bpm,genre,energy_score,bpm_method,source_method,crate,energy_score_override (blank, copied by hand to overrides)	Built	M1, M2	musicLibrary/pipeline/run_weekly.py:298-304
file-write	outputs/<date>/rejected.csv	write	ml-cli-process	ml-persona-curator	csv	artist,title,reason,versions_seen	Built	M1, M2	musicLibrary/pipeline/run_weekly.py:305-308
file-write	outputs/<date>/weekly_report.md	both	ml-cli-process	ml-cli-build-db	markdown	headline lines parsed by regex in build_db.parse_report into runs	Built	M1, M2	musicLibrary/pipeline/run_weekly.py:310-312 +2
file-write	outputs/rekordbox_musicLibrary.xml	write	ml-cli-process	ext-rekordbox	xml	DJ_PLAYLISTS 1.0.0; COLLECTION TRACK(TrackID,Name,Artist,Genre,Label,TotalTime,AverageBpm,Tonality,Rating,Colour,Comments,Grouping,Location file://localhost); PLAYLISTS musicLibrary/<crate> nodes; only tracks with local_path (0 today)	Built	M1, M2	musicLibrary/pipeline/djlib/export.py:356-385

file-read	overrides/rekordbox_collection.xml	read	ext-rekordbox	ml-cli-calibrate	xml	TRACK DateAdded, LastPlayed|DateModified, PlayCount	Built	M1	musicLibrary/pipeline/calibrate.py:55-81
file-write	outputs/calibration_suggestions.json	write	ml-cli-calibrate	ml-persona-curator	json	{generated, energy{overrides_available, band_shift_by_pillar, suggested_energy_prior_by_source, mean_abs_error_auto}, duration_floor_seconds{pillar:{p10,median,n}}, rotation{...}}	Built	M1	musicLibrary/pipeline/calibrate.py:24 +1
file-write	inbox/sync_result_<date>.json	both	ml-cli-sync-tidal	ml-cli-confirm	json	[{track_id, platforms{tidal|beatport:true}, tidal_id?, tidal_url?, tidal_duration_seconds?, tidal_short_only?}]	Built	M1, M3	musicLibrary/pipeline/run_weekly.py:518 +2
file-write	outputs/<date>/ALERT_tidal.txt, ALERT_tidal_sets.txt	write	ml-cli-sync-tidal	ml-agent-weekly-run	text	TidalError message	Built	M1, M2	musicLibrary/pipeline/run_weekly.py:514-515 +2
file-write	inbox/artifact_sets.json	both	ml-agent-weekly-run	ml-cli-build-db	json	ArtifactData rows [{id, data:{name, shape, minutes, mix, pool, tidal, seed, tracks[{id,tgt,a,t,m,e,k,bpm,tidal(url),move}], updated, created}}] or a plain list; today '[]'	Built	M1, M2, M3	musicLibrary/agent/weekly_run.md:5 +3

sql	db/dj_library.sqlite	write	ml-cli-build-db	ml-persona-curator	sqlite	tables meta, sources, tracks, track_tags, appearances, chart_positions, rotation_history, pending, rejected, beatport_check, sets, set_tracks, runs; views v_crates, v_source_yield, v_review_queue, v_buy_list	Built	M1, M3	musicLibrary/pipeline/build_db.py:22-73 +1
file-write	outputs/artifact/library.json	write	ml-cli-build-db	ml-page-set-builder	json	{generated UTC, ledger_updated, counts{tracks,pending,rejected,beatport_check}, sources[{id,prior,club,on}], runs[{date,added,rejected,promoted,demoted,archived,review_queue}] (last 26), tracks[{id,a,t,m,d,bpm,k,e,p,rot 1|2|3,tags,src,first,last,g,lbl,sync,tidal,dz,bp,chart{rank,weeks,peak}|null,rev,ovr,local}]} excluding 04_Pruned_Archive	Built	M1, M2, M3	musicLibrary/pipeline/build_db.py:190-229

file-write	artifact/build/library.json	both	ml-agent-weekly-run	ml-page-set-builder	json	byte-identical copy of outputs/artifact/library.json (M3 compare: equal)	Built	M1, M3	musicLibrary/agent/weekly_run.md:8
fetch	library.json (relative, cache no-cache)	read	ml-store-artifact-files	ml-page-set-builder	json	LibraryFile	Built	M1, M3	musicLibrary/artifact/app/artifact-data.ts:14-23 +1
postMessage	window.claude.use('db') collection 'sets' (get, doc.set, doc.delete; legacy page also onSnapshot and use('user').can('data.write'))	both	ml-page-set-builder	ml-store-artifact-db	json	doc id = set id; body = SavedSet without id	Built	M1, M3	musicLibrary/artifact/app/artifact-data.ts:25-72 +1
cli	ArtifactData list collection 'sets' limit 1000 (weekly agent)	read	ml-store-artifact-db	ml-agent-weekly-run	json	[{id, data}]	Built	M1	musicLibrary/agent/weekly_run.md:5
cli	Artifact publish url=GcqZTGjk6obdpjWaU7WFRq file_path=weekly_publish/index.html files{library.json}	write	ml-agent-weekly-run	ml-store-artifact-files	html+json	carry-forward app.js, assets, capabilities	Built	M1	musicLibrary/agent/weekly_run.md:8
localStorage	sn-music-library-sets	both	ml-page-set-builder	ml-page-set-builder	json	SavedSet[] cache of the shared store	Built	M1, M3	musicLibrary/artifact/app/artifact-data.ts:11 +2
localStorage	sn-set-draft	both	ml-page-set-builder	ml-page-set-builder	json	Draft (sanitiseDraft)	Built	M1, M3	songnodes/frontend/src/features/musicLibrary/storage.ts:24
localStorage	sn-filters-library	both	ml-page-set-builder	ml-page-set-builder	json	library filters and sort	Built	M1, M3	songnodes/frontend/src/features/musicLibrary/storage.ts:22

localStorage	sn-density	both	ml-page-set-builder	ml-page-set-builder	string	SN.prefs density (design system)	Built	M3	musicLibrary/artifact/app/main.tsx:90
localStorage	sn-banner-keyed—<page>	both	ml-page-set-builder	ml-page-set-builder	json	dismissed banner count	Built	M1	songnodes/frontend/src/features/musicLibrary/storage.ts:30
localStorage	nsb:view, nsb:filters, nsb:draft	both	ml-page-set-builder-legacy	ml-page-set-builder-legacy	json	view name, filters, SET draft	Built	M1	musicLibrary/artifact/index.html:231
clipboard	navigator.clipboard.writeText tracklist	write	ml-page-set-builder-legacy	ext-clipboard	text	'NN. Artist – Title (Mix) [E bpm key]'	Built	M1	musicLibrary/artifact/index.html:525-530
http-endpoint	POST http://127.0.0.1:8082/api/v1/library/ingest (Idempotency-Key import—<md5>)	write	ml-cli-sync-songnodes	sn-be-svc-rest-api	json	library.json body; job library_ingest maps short keys a/t/m/... via _from_export into library.tracks plus library.track_xref(source_system, source_id=h:/bp: id → canonical uuid)	Built	M1, M2	musicLibrary/pipeline/run_weekly.py:574-597 +3

http-endpoint	POST /api/v1/tracks/import-tracklist with songnodes.setlists.v1 documents	write	ml-cli-sync-songnodes	sn-be-svc-rest-api	json	sent {schema:'songnodes.setlists.v1', setlists[{dj,event,date,venue,genre,source_url,tracks[{position,artist,title,mix,label}]}]}; endpoint expects TracklistImportRequest{tracks[], dj_artist?, name?}	Built	M1, M2	musicLibrary/pipeline/run_weekly.py:599-616 +2
file-read	JohnCourse.json, kiki.json (root), JohnCourse.json.bak	read	ml-persona-developer	ml-cli-sync-songnodes	json	songnodes.setlists.v1: 29 and 5 DJ setlists from 1001tracklists/MixesDB; .bak is invalid JSON	Built	M1, M3	musicLibrary/JohnCourse.json +2
cli	songnodes scripts/import-music-library.sh (MUSIC_LIBRARY_DIR default ../musicLibrary)	read	sn-be-cli-import-music-library	ml-store-outputs	json	reads outputs/artifact/library.json; requires a tracks array	Built	M1, M2	songnodes/scripts/import-music-library.sh:13-46
http-endpoint	GET /api/v1/library/search?limit=5000 (songnodes /library, /sets)	read	sn-be-svc-rest-api	sn-fe-library	json	LibraryTrack with id = canonical uuid; bp NULL; generated null; runs/sources empty	Built	M1, M2	songnodes/frontend/src/features/musicLibrary/data.ts:108-136 +1
http-endpoint	GET/PUT/DELETE /api/v1/library/sets[/id], GET /api/v1/library/sets/export	both	sn-fe-library	sn-store-library-sets	json	PUT body {name, description, tracks[{track_id uuid}]}; export {sets[{name,description,tracks[{track_id}]}]}	Built	M1, M2	songnodes/frontend/src/features/musicLibrary/data.ts:193-283 +2

file-write	artifact_sets.json download from songnodes /sets (Export sets for the weekly run)	write	sn-fe-library	ml-sig-artifact-sets-export	json	SavedSet[] with numeric string ids, track ids are songnodes uuids, no tidal flag	Built	M1	songnodes/frontend/src/features/musicLibrary/SetBuilderPage.tsx:223-225 +1
http-endpoint	POST /api/v1/tidal/sets/send (songnodes Send to TIDAL dialog)	write	sn-fe-library	ext-tidal	json	creates a TIDAL playlist from a set in play order, songnodes user token	Built	M1	songnodes/frontend/src/features/setBuilder/sendToTidal.ts:44 +1
sql	postgres://ref_mb_reader@host.docker.internal:55433/ref_mb (MB_LOCAL_*)	read	ml-svc-ref-mb	sn-be-store-refmb	sql	ref_mb schema (MusicBrainz core subset + canonical_* + load_manifest), read-only role	Built	M1, M2	songnodes/services/common/refmb/settings.py:21-34 +2
fetch	https://data.metabrainz.org/pub/musicbrainz (dumps 20261024-102842, 20261024-001001, 20261017-080003)	read	ext-metabrainz	ml-cli-fetch-metabrainz	tar.zst/tar.bz2	dump archives with SHA256SUMS	Built	M1, M2	musicLibrary/pipeline/fetch_metabrainz.py:6-8
fetch	https://www.beatport.com/search/tracks?q=… via Claude in Chrome (supervised)	both	ml-agent-beatport-sync	ext-beatport	html	playlist named exactly as crate	Built	M1	musicLibrary/agent/beatport_sync.md:3-4
fetch	POST https://jules.googleapis.com/v1alpha/sessions (X-Goog-Api-Key from JULES_API_KEY env)	write	ml-plan-jules	ext-jules	json	session payloads plan/jules_payloads/*.json; results plan/jules_sessions.json	Built	M1, M2	musicLibrary/plan/launch_jules.py:10 +1
env	SONGNODES_API_URL (run_weekly) vs SONGNODES_API (import script)	read	ml-store-env	ml-cli-sync-songnodes	string	default http://127.0.0.1:8082	Built	M1, M2	musicLibrary/pipeline/run_weekly.py:574 +1
http-endpoint	GET /health	both	client	sn-be-legacy-dlq-manager	json	handler health_check; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/dlq-manager/main.py:549

http-endpoint	GET /metrics	both	client	sn-be-legacy-dlq-manager	json	handler metrics; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/dlq-manager/main.py:560
http-endpoint	GET /dlq/messages	both	client	sn-be-legacy-dlq-manager	json	handler get_dlq_messages; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/dlq-manager/main.py:566
http-endpoint	GET /dlq/stats	both	client	sn-be-legacy-dlq-manager	json	handler get_dlq_stats; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/dlq-manager/main.py:587
http-endpoint	POST /dlq/replay/{message_id}	both	client	sn-be-legacy-dlq-manager	json	handler replay_message; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/dlq-manager/main.py:601
http-endpoint	POST /dlq/replay/batch	both	client	sn-be-legacy-dlq-manager	json	handler replay_batch; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/dlq-manager/main.py:623
http-endpoint	DELETE /dlq/message/{message_id}	both	client	sn-be-legacy-dlq-manager	json	handler delete_message; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/dlq-manager/main.py:640
http-endpoint	GET /dlq/errors/grouped	both	client	sn-be-legacy-dlq-manager	json	handler get_error_groups; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/dlq-manager/main.py:664
http-endpoint	GET /health	both	client	sn-be-legacy-nlp-processor	json	handler health_check; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/nlp-processor/main.py:349

http-endpoint	GET /metrics	both	client	sn-be-legacy-nlp-processor	json	handler metrics; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/nlp-processor/main.py:363
http-endpoint	POST /analyze	both	client	sn-be-legacy-nlp-processor	json	handler analyze_text; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/nlp-processor/main.py:395
http-endpoint	POST /extract-artists	both	client	sn-be-legacy-nlp-processor	json	handler extract_artists; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/nlp-processor/main.py:424
http-endpoint	POST /extract_tracklist	both	client	sn-be-legacy-nlp-processor	json	handler extract_tracklist; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/nlp-processor/main.py:450
http-endpoint	POST /analyze_text	both	client	sn-be-legacy-nlp-processor	json	handler analyze_text_structure; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/nlp-processor/main.py:478
http-endpoint	POST /extract_timestamps	both	client	sn-be-legacy-nlp-processor	json	handler extract_timestamps; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/nlp-processor/main.py:510
http-endpoint	POST /llm/extract_tracklist	both	client	sn-be-legacy-nlp-processor	json	handler llm_extract_tracklist; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/nlp-processor/main.py:544

http-endpoint	POST /llm/classify_genre	both	client	sn-be-legacy-nlp-processor	json	handler llm_classify_genre; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/nlp-processor/main.py:622
http-endpoint	POST /llm/generate	both	client	sn-be-legacy-nlp-processor	json	handler llm_generate; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/nlp-processor/main.py:695
http-endpoint	GET /health	both	client	sn-be-legacy-data-transformer	json	handler health_check; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/data-transformer/main.py:1043
http-endpoint	POST /transform	both	client	sn-be-legacy-data-transformer	json	handler submit_transformation_task; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/data-transformer/main.py:1185
http-endpoint	GET /tasks/{task_id}	both	client	sn-be-legacy-data-transformer	json	handler get_task_status; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/data-transformer/main.py:1205
http-endpoint	GET /tasks	both	client	sn-be-legacy-data-transformer	json	handler get_tasks; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/data-transformer/main.py:1216
http-endpoint	GET /stats	both	client	sn-be-legacy-data-transformer	json	handler get_transformation_stats; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/data-transformer/main.py:1231

http-endpoint	GET /metrics	both	client	sn-be-legacy-data-transformer	json	handler get_metrics; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/data-transformer/main.py:1273
http-endpoint	POST /normalize	both	client	sn-be-legacy-data-transformer	json	handler normalize_tracks; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/data-transformer/main.py:1278
http-endpoint	GET /fingerprints/duplicates	both	client	sn-be-legacy-data-transformer	json	handler find_duplicates; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/data-transformer/main.py:1298
http-endpoint	GET /api/graph/nodes	both	client	sn-be-svc-graph-api	json	handler get_nodes; production	Built	M1, M3	songnodes/services/graph-visualization-api/main.py:555
http-endpoint	GET /api/graph/edges	both	client	sn-be-svc-graph-api	json	handler get_edges; production	Built	M1, M3	songnodes/services/graph-visualization-api/main.py:563
http-endpoint	GET /api/graph/data	both	client	sn-be-svc-graph-api	json	handler get_graph_data; production	Built	M1, M3	songnodes/services/graph-visualization-api/main.py:574
http-endpoint	GET /api/graph/stats	both	client	sn-be-svc-graph-api	json	handler get_graph_stats; production	Built	M1, M3	songnodes/services/graph-visualization-api/main.py:593
http-endpoint	GET /api/graph/search	both	client	sn-be-svc-graph-api	json	handler search_graph; production	Built	M1, M3	songnodes/services/graph-visualization-api/main.py:598

http-endpoint	GET /api/v1/visualization/search	both	client	sn-be-svc-graph-api	json	handler search_graph; production	Built	M1, M3	songnodes/services/graph-visualization-api/main.py:599
http-endpoint	GET /test/adjacency	both	client	sn-be-svc-graph-api	json	handler test_adjacency; production service, test/debug route reading legacy tables (tracks, graph_nodes)	Built	M1, M3	songnodes/services/graph-visualization-api/main.py:608
http-endpoint	GET /test/nodes	both	client	sn-be-svc-graph-api	json	handler test_nodes; production service, test/debug route reading legacy tables (tracks, graph_nodes)	Built	M1, M3	songnodes/services/graph-visualization-api/main.py:631
http-endpoint	GET /health	both	client	sn-be-svc-graph-api	json	handler health_check; production	Built	M1, M3	songnodes/services/graph-visualization-api/main.py:704
http-endpoint	GET /metrics	both	client	sn-be-svc-graph-api	json	handler metrics; production	Built	M1, M3	songnodes/services/graph-visualization-api/main.py:797
http-endpoint	POST /api/user-setlists	both	client	sn-be-svc-graph-api	json	handler create_user_setlist; production	Built	M1, M3	songnodes/services/graph-visualization-api/main.py:808
http-endpoint	POST /api/v1/visualization/graph	both	client	sn-be-svc-graph-api	json	handler get_visualization_graph; production	Built	M1, M3	songnodes/services/graph-visualization-api/main.py:845

http-endpoint	POST /api/v1/visualization/nodes/batch	both	client	sn-be-svc-graph-api	json	handler create_nodes_batch; production	Built	M1, M3	songnodes/services/graph-visualization-api/main.py:892
http-endpoint	GET /api/v1/visualization/batch/{batch_id}/status	both	client	sn-be-svc-graph-api	json	handler get_batch_status; production	Built	M1, M3	songnodes/services/graph-visualization-api/main.py:902
websocket	WEBSOCKET /api/graph/ws/{room_id}	both	client	sn-be-svc-graph-api	json	handler websocket_endpoint; production	Built	M1, M3	songnodes/services/graph-visualization-api/main.py:915
http-endpoint	GET /api/graph/neighborhood/{node_id}	both	client	sn-be-svc-graph-api	json	handler get_node_neighborhood; production	Built	M1, M3	songnodes/services/graph-visualization-api/main.py:962
http-endpoint	GET /api/graph	both	client	sn-be-svc-graph-api	json	handler get_combined_graph_data; production	Built	M1, M3	songnodes/services/graph-visualization-api/main.py:978
http-endpoint	GET /api/v1/graph	both	client	sn-be-svc-graph-api	json	handler get_combined_graph_data; production	Built	M1, M3	songnodes/services/graph-visualization-api/main.py:979
http-endpoint	GET /health	both	client	sn-be-svc-rest-api	json	handler health_check; production	Built	M1, M3	songnodes/services/rest_api/main.py:282
http-endpoint	PATCH /api/tracks/{track_id}	both	client	sn-be-svc-rest-api	json	handler update_track_metadata; production	Built	M1, M3	songnodes/services/rest_api/main.py:419
http-endpoint	GET /api/v1/tracks/{track_id}/preview	both	client	sn-be-svc-rest-api	json	handler get_track_preview; production	Built	M1, M3	songnodes/services/rest_api/main.py:461
http-endpoint	GET /api/graph/nodes	both	client	sn-be-svc-rest-api	json	handler get_graph_nodes; production	Built	M1, M3	songnodes/services/rest_api/main.py:501

http-endpoint	GET /api/graph/edges	both	client	sn-be-svc-rest-api	json	handler get_graph_edges; production	Built	M1, M3	songnodes/services/rest_api/main.py:570
http-endpoint	GET /api/v1/music-search/{service}	both	client	sn-be-svc-rest-api	json	handler music_search; production	Built	M1, M3	songnodes/services/rest_api/main.py:672
http-endpoint	GET /api/v1/library/search	both	client	sn-be-svc-rest-api	json	handler library_search; production	Built	M1, M3	songnodes/services/rest_api/routers/library_search.py:54
http-endpoint	GET /api/v1/library/tracks/{track_id}/preview	both	client	sn-be-svc-rest-api	json	handler track_preview; production	Built	M1, M3	songnodes/services/rest_api/routers/preview.py:80
http-endpoint	GET /api/v1/tracks	both	client	sn-be-svc-rest-api	json	handler list_tracks; production	Built	M1, M3	songnodes/services/rest_api/routers/catalog_reads.py:66
http-endpoint	GET /api/search/tracks	both	client	sn-be-svc-rest-api	json	handler search_tracks; production	Built	M1, M3	songnodes/services/rest_api/routers/catalog_reads.py:92
http-endpoint	GET /api/tracks/{track_id}	both	client	sn-be-svc-rest-api	json	handler get_track; production	Built	M1, M3	songnodes/services/rest_api/routers/catalog_reads.py:105
http-endpoint	GET /api/v1/tracks/{track_id}	both	client	sn-be-svc-rest-api	json	handler get_track; production	Built	M1, M3	songnodes/services/rest_api/routers/catalog_reads.py:106
http-endpoint	GET /api/v1/artists	both	client	sn-be-svc-rest-api	json	handler list_artists; production	Built	M1, M3	songnodes/services/rest_api/routers/catalog_reads.py:139

http-endpoint	GET /api/v1/artists/search	both	client	sn-be-svc-rest-api	json	handler search_artists; production	Built	M1, M3	songnodes/services/rest_api/routers/catalog_reads.py:146
http-endpoint	GET /api/v1/artists/{artist_id}	both	client	sn-be-svc-rest-api	json	handler get_artist; production	Built	M1, M3	songnodes/services/rest_api/routers/catalog_reads.py:162
http-endpoint	GET /api/v1/setlists	both	client	sn-be-svc-rest-api	json	handler list_setlists; production	Built	M1, M3	songnodes/services/rest_api/routers/catalog_reads.py:190
http-endpoint	GET /api/v1/playlists	both	client	sn-be-svc-rest-api	json	handler list_setlists; production	Built	M1, M3	songnodes/services/rest_api/routers/catalog_reads.py:191
http-endpoint	GET /api/v1/setlists/{playlist_id}	both	client	sn-be-svc-rest-api	json	handler get_setlist; production	Built	M1, M3	songnodes/services/rest_api/routers/catalog_reads.py:221
http-endpoint	GET /api/v1/playlists/{playlist_id}	both	client	sn-be-svc-rest-api	json	handler get_setlist; production	Built	M1, M3	songnodes/services/rest_api/routers/catalog_reads.py:222
http-endpoint	GET /api/v1/setlists/{playlist_id}/tracks	both	client	sn-be-svc-rest-api	json	handler get_setlist_tracks; production	Built	M1, M3	songnodes/services/rest_api/routers/catalog_reads.py:229
http-endpoint	POST /api/v1/scrape/trigger	both	client	sn-be-svc-rest-api	json	handler trigger_scrape; production	Built	M1, M3	songnodes/services/rest_api/routers/jobs.py:66
http-endpoint	GET /api/v1/jobs	both	client	sn-be-svc-rest-api	json	handler list_jobs; production	Built	M1, M3	songnodes/services/rest_api/routers/jobs.py:109

http-endpoint	GET /api/v1/jobs/{job_id}	both	client	sn-be-svc-rest-api	json	handler get_job; production	Built	M1, M3	songnodes/services/rest_api/routers/jobs.py:133
http-endpoint	POST /api/v1/jobs/{job_id}/retry	both	client	sn-be-svc-rest-api	json	handler retry_job; production	Built	M1, M3	songnodes/services/rest_api/routers/jobs.py:140
http-endpoint	GET /api/v1/jobs/{job_id}/stages	both	client	sn-be-svc-rest-api	json	handler job_stages; production	Built	M1, M3	songnodes/services/rest_api/routers/jobs.py:266
http-endpoint	POST /api/v1/spotify/playlists/create	both	client	sn-be-svc-rest-api	json	handler create_playlist; production	Built	M1, M3	songnodes/services/rest_api/routers/spotify_playlists.py:104
http-endpoint	GET /api/v1/spotify/playlists/list	both	client	sn-be-svc-rest-api	json	handler list_playlists; production	Built	M1, M3	songnodes/services/rest_api/routers/spotify_playlists.py:175
http-endpoint	POST /api/v1/spotify/playlists/{playlist_id}/tracks	both	client	sn-be-svc-rest-api	json	handler add_tracks_to_playlist; production	Built	M1, M3	songnodes/services/rest_api/routers/spotify_playlists.py:237
http-endpoint	POST /api/v1/spotify/tracks/search	both	client	sn-be-svc-rest-api	json	handler search_track; production	Built	M1, M3	songnodes/services/rest_api/routers/spotify_playlists.py:300
http-endpoint	POST /api/v1/spotify/tracks/check-availability	both	client	sn-be-svc-rest-api	json	handler check_track_availability; production	Built	M1, M3	songnodes/services/rest_api/routers/spotify_playlists.py:377
http-endpoint	GET /api/v1/artists/dirty	both	client	sn-be-svc-rest-api	json	handler get_dirty_artists; production	Built	M1, M3	songnodes/services/rest_api/routers/catalog_writes.py:42

http-endpoint	POST /api/v1/artists/{artist_id}/rename	both	client	sn-be-svc-rest-api	json	handler rename_artist; production	Built	M1, M3	songnodes/services/rest_api/routers/catalog_writes.py:58
http-endpoint	POST /api/v1/artists/{duplicate_id}/merge	both	client	sn-be-svc-rest-api	json	handler merge_artist; production	Built	M1, M3	songnodes/services/rest_api/routers/catalog_writes.py:66
http-endpoint	DELETE /api/v1/artists/{artist_id}	both	client	sn-be-svc-rest-api	json	handler delete_artist; production	Built	M1, M3	songnodes/services/rest_api/routers/catalog_writes.py:73
http-endpoint	GET /api/v1/tracks/missing-artist	both	client	sn-be-svc-rest-api	json	handler get_tracks_without_artist; production	Built	M1, M3	songnodes/services/rest_api/routers/catalog_writes.py:80
http-endpoint	POST /api/v1/tracks/{track_id}/assign-artist	both	client	sn-be-svc-rest-api	json	handler assign_artist_to_track; production	Built	M1, M3	songnodes/services/rest_api/routers/catalog_writes.py:90
http-endpoint	POST /api/v1/artists	both	client	sn-be-svc-rest-api	json	handler create_artist; production	Built	M1, M3	songnodes/services/rest_api/routers/catalog_writes.py:99
http-endpoint	POST /api/v1/tracks	both	client	sn-be-svc-rest-api	json	handler create_track; production	Built	M1, M3	songnodes/services/rest_api/routers/catalog_writes.py:114
http-endpoint	POST /api/v1/tracks/import-tracklist	both	client	sn-be-svc-rest-api	json	handler import_tracklist; production	Built	M1, M3	songnodes/services/rest_api/routers/catalog_writes.py:123
http-endpoint	GET /api/v1/api-keys/requirements	both	client	sn-be-svc-rest-api	json	handler get_api_key_requirements; production	Built	M1, M3	songnodes/services/rest_api/routers/api_keys.py:125

http-endpoint	GET /api/v1/api-keys	both	client	sn-be-svc-rest-api	json	handler get_api_keys; production	Built	M1, M3	songnodes/services/rest_api/routers/api_keys.py:151
http-endpoint	POST /api/v1/api-keys	both	client	sn-be-svc-rest-api	json	handler create_or_update_api_key; production	Built	M1, M3	songnodes/services/rest_api/routers/api_keys.py:204
http-endpoint	DELETE /api/v1/api-keys/{service_name}/{key_name}	both	client	sn-be-svc-rest-api	json	handler delete_api_key; production	Built	M1, M3	songnodes/services/rest_api/routers/api_keys.py:258
http-endpoint	POST /api/v1/api-keys/test	both	client	sn-be-svc-rest-api	json	handler test_api_key; production	Built	M1, M3	songnodes/services/rest_api/routers/api_keys.py:290
http-endpoint	POST /api/v1/library/ingest	both	client	sn-be-svc-rest-api	json	handler ingest_library; production	Built	M1, M3	songnodes/services/rest_api/routers/library_ingest.py:10
http-endpoint	GET /api/v1/library/sets	both	client	sn-be-svc-rest-api	json	handler list_sets; production	Built	M1, M3	songnodes/services/rest_api/routers/library_ingest.py:21
http-endpoint	PUT /api/v1/library/sets/{set_id}	both	client	sn-be-svc-rest-api	json	handler update_set; production	Built	M1, M3	songnodes/services/rest_api/routers/library_ingest.py:28
http-endpoint	DELETE /api/v1/library/sets/{set_id}	both	client	sn-be-svc-rest-api	json	handler remove_set; production	Built	M1, M3	songnodes/services/rest_api/routers/library_ingest.py:36
http-endpoint	GET /api/v1/library/sets/export	both	client	sn-be-svc-rest-api	json	handler export_sets; production	Built	M1, M3	songnodes/services/rest_api/routers/library_ingest.py:44

http-endpoint	POST /api/v1/pathfinder/find-path	both	client	sn-be-svc-rest-api	json	handler find_dj_path; production	Built	M1, M3	songnodes/services/rest_api/routers/pathfinder.py:480
http-endpoint	POST /api/v1/jobs	both	client	sn-be-svc-rest-api	json	handler enqueue_job; production	Built	M1, M3	songnodes/services/rest_api/routers/datalayer.py:39
http-endpoint	GET /api/v1/metadata/review	both	client	sn-be-svc-rest-api	json	handler metadata_review; production	Built	M1, M3	songnodes/services/rest_api/routers/datalayer.py:53
http-endpoint	GET /api/v1/tracks/{track_id}/canonical	both	client	sn-be-svc-rest-api	json	handler canonical_track; production	Built	M1, M3	songnodes/services/rest_api/routers/datalayer.py:72
http-endpoint	POST /api/v1/tracks/{track_id}/merge	both	client	sn-be-svc-rest-api	json	handler merge_track; production	Built	M1, M3	songnodes/services/rest_api/routers/datalayer.py:85
http-endpoint	GET /api/v1/merges	both	client	sn-be-svc-rest-api	json	handler list_merges; production	Built	M1, M3	songnodes/services/rest_api/routers/datalayer.py:94
http-endpoint	POST /api/v1/merges/{history_id}/revert	both	client	sn-be-svc-rest-api	json	handler revert_merge; production	Built	M1, M3	songnodes/services/rest_api/routers/datalayer.py:104
http-endpoint	GET /api/v1/outbox/status	both	client	sn-be-svc-rest-api	json	handler outbox_status; production	Built	M1, M3	songnodes/services/rest_api/routers/datalayer.py:114
http-endpoint	POST /api/v1/projections/tracks/reconcile	both	client	sn-be-svc-rest-api	json	handler reconcile_projection; production	Built	M1, M3	songnodes/services/rest_api/routers/datalayer.py:128
http-endpoint	GET /api/v1/attributions	both	client	sn-be-svc-rest-api	json	handler attributions; production	Built	M1, M3	songnodes/services/rest_api/routers/datalayer.py:135

http-endpoint	GET /api/v1/resolution/review	both	client	sn-be-svc-rest-api	json	handler resolution_review; production	Built	M1, M3	songnodes/services/rest_api/routers/datalayer.py:147
http-endpoint	POST /api/v1/resolution/{decision_id}/review	both	client	sn-be-svc-rest-api	json	handler resolution_verdict; production	Built	M1, M3	songnodes/services/rest_api/routers/datalayer.py:159
http-endpoint	GET /api/v1/scrape/window	both	client	sn-be-svc-rest-api	json	handler get_scrape_window; production	Built	M1, M3	songnodes/services/rest_api/routers/scrape_window.py:71
http-endpoint	PUT /api/v1/scrape/window	both	client	sn-be-svc-rest-api	json	handler put_scrape_window; production	Built	M1, M3	songnodes/services/rest_api/routers/scrape_window.py:82
http-endpoint	GET /api/v1/target-tracks/suggestions	both	client	sn-be-svc-rest-api	json	handler target_suggestions; production	Built	M1, M3	songnodes/services/rest_api/routers/target_tracks.py:52
http-endpoint	GET /api/v1/target-tracks	both	client	sn-be-svc-rest-api	json	handler list_targets; production	Built	M1, M3	songnodes/services/rest_api/routers/target_tracks.py:69
http-endpoint	POST /api/v1/target-tracks	both	client	sn-be-svc-rest-api	json	handler create_target; production	Built	M1, M3	songnodes/services/rest_api/routers/target_tracks.py:111
http-endpoint	PUT /api/v1/target-tracks/{track_id}	both	client	sn-be-svc-rest-api	json	handler update_target; production	Built	M1, M3	songnodes/services/rest_api/routers/target_tracks.py:129
http-endpoint	DELETE /api/v1/target-tracks/{track_id}	both	client	sn-be-svc-rest-api	json	handler delete_target; production	Built	M1, M3	songnodes/services/rest_api/routers/target_tracks.py:153

http-endpoint	POST /api/v1/target-tracks/{track_id}/search	both	client	sn-be-svc-rest-api	json	handler search_target; production	Built	M1, M3	songnodes/services/rest_api/routers/target_tracks.py:161
http-endpoint	POST /api/v1/music-auth/test/tidal	both	client	sn-be-svc-rest-api	json	handler test_tidal_auth; production	Built	M1, M3	songnodes/services/rest_api/routers/music_auth.py:150
http-endpoint	POST /api/v1/music-auth/test/spotify	both	client	sn-be-svc-rest-api	json	handler test_spotify_auth; production	Built	M1, M3	songnodes/services/rest_api/routers/music_auth.py:270
http-endpoint	GET /api/v1/music-auth/spotify/authorize	both	client	sn-be-svc-rest-api	json	handler spotify_authorize; production	Built	M1, M3	songnodes/services/rest_api/routers/music_auth.py:360
http-endpoint	GET /api/v1/music-auth/spotify/callback	both	client	sn-be-svc-rest-api	json	handler spotify_callback; production	Built	M1, M3	songnodes/services/rest_api/routers/music_auth.py:463
http-endpoint	POST /api/v1/music-auth/store-token	both	client	sn-be-svc-rest-api	json	handler store_oauth_token; production	Built	M1, M3	songnodes/services/rest_api/routers/music_auth.py:613
http-endpoint	POST /api/v1/music-auth/spotify/token	both	client	sn-be-svc-rest-api	json	handler spotify_exchange_token; production	Built	M1, M3	songnodes/services/rest_api/routers/music_auth.py:662
http-endpoint	POST /api/v1/music-auth/spotify/refresh	both	client	sn-be-svc-rest-api	json	handler spotify_refresh_token; production	Built	M1, M3	songnodes/services/rest_api/routers/music_auth.py:749
http-endpoint	GET /api/v1/music-auth/spotify/search	both	client	sn-be-svc-rest-api	json	handler search_spotify_tracks; production	Built	M1, M3	songnodes/services/rest_api/routers/music_auth.py:848
http-endpoint	POST /api/v1/music-auth/tidal/device/init	both	client	sn-be-svc-rest-api	json	handler init_tidal_device_flow; production	Built	M1, M3	songnodes/services/rest_api/routers/music_auth.py:915

http-endpoint	POST /api/v1/music-auth/tidal/device/poll	both	client	sn-be-svc-rest-api	json	handler poll_tidal_device_token; production	Built	M1, M3	songnodes/services/rest_api/routers/music_auth.py:996
http-endpoint	POST /api/v1/music-auth/tidal/oauth/init	both	client	sn-be-svc-rest-api	json	handler init_tidal_oauth; production	Built	M1, M3	songnodes/services/rest_api/routers/music_auth.py:1075
http-endpoint	GET /api/v1/music-auth/tidal/oauth/callback	both	client	sn-be-svc-rest-api	json	handler tidal_oauth_callback; production	Built	M1, M3	songnodes/services/rest_api/routers/music_auth.py:1158
http-endpoint	GET /api/v1/music-auth/reddit/callback	both	client	sn-be-svc-rest-api	json	handler reddit_callback; production	Built	M1, M3	songnodes/services/rest_api/routers/music_auth.py:1230
http-endpoint	GET /api/v1/music-auth/lastfm/callback	both	client	sn-be-svc-rest-api	json	handler lastfm_callback; production	Built	M1, M3	songnodes/services/rest_api/routers/music_auth.py:1235
http-endpoint	POST /api/v1/music-auth/tidal/oauth/refresh	both	client	sn-be-svc-rest-api	json	handler refresh_tidal_token; production	Built	M1, M3	songnodes/services/rest_api/routers/music_auth.py:1240
http-endpoint	POST /api/v1/music-auth/tidal/refresh	both	client	sn-be-svc-rest-api	json	handler refresh_tidal_token; production	Built	M1, M3	songnodes/services/rest_api/routers/music_auth.py:1241
http-endpoint	GET /api/v1/music-auth/health	both	client	sn-be-svc-rest-api	json	handler health_check; production	Built	M1, M3	songnodes/services/rest_api/routers/music_auth.py:1305
http-endpoint	GET /api/v1/music-auth/spotify/token/status	both	client	sn-be-svc-rest-api	json	handler get_spotify_token_status; production	Built	M1, M3	songnodes/services/rest_api/routers/music_auth.py:1310
http-endpoint	POST /api/v1/music-auth/spotify/token/refresh-manual	both	client	sn-be-svc-rest-api	json	handler manual_refresh_spotify_token; production	Built	M1, M3	songnodes/services/rest_api/routers/music_auth.py:1361

http-endpoint	GET /api/v1/youtube/quota	both	client	sn-be-svc-rest-api	json	handler get_quota_status; production	Built	M1, M3	songnodes/services/rest_api/routers/youtube_api.py:116
http-endpoint	GET /api/v1/youtube/search	both	client	sn-be-svc-rest-api	json	handler search_youtube; production	Built	M1, M3	songnodes/services/rest_api/routers/youtube_api.py:155
http-endpoint	POST /api/v1/youtube/quota/reset	both	client	sn-be-svc-rest-api	json	handler reset_quota; production	Built	M1, M3	songnodes/services/rest_api/routers/youtube_api.py:248
http-endpoint	POST /api/v1/ai/complete	both	client	sn-be-svc-rest-api	json	handler ai_complete; production	Built	M1, M3	songnodes/services/rest_api/routers/ai_proxy.py:56
http-endpoint	POST /api/v1/tidal/playlists/create	both	client	sn-be-svc-rest-api	json	handler create_playlist; production	Built	M1, M3	songnodes/services/rest_api/routers/tidal_playlists.py:66
http-endpoint	GET /api/v1/tidal/playlists/list	both	client	sn-be-svc-rest-api	json	handler list_playlists; production	Built	M1, M3	songnodes/services/rest_api/routers/tidal_playlists.py:114
http-endpoint	POST /api/v1/tidal/playlists/{playlist_id}/tracks	both	client	sn-be-svc-rest-api	json	handler add_tracks_to_playlist; production	Built	M1, M3	songnodes/services/rest_api/routers/tidal_playlists.py:162
http-endpoint	POST /api/v1/tidal/tracks/search	both	client	sn-be-svc-rest-api	json	handler search_track; production	Built	M1, M3	songnodes/services/rest_api/routers/tidal_playlists.py:196
http-endpoint	POST /api/v1/tidal/tracks/check-availability	both	client	sn-be-svc-rest-api	json	handler check_track_availability; production	Built	M1, M3	songnodes/services/rest_api/routers/tidal_playlists.py:244

http-endpoint	POST /api/v1/tidal/sets/send	both	client	sn-be-svc-rest-api	json	handler send_set; production	Built	M1, M3	songnodes/services/rest_api/routers/tidal_playlists.py:318
http-endpoint	GET /health	both	client	sn-be-legacy-metadata-enrichment	json	handler health_check; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/metadata-enrichment/main.py:616
http-endpoint	GET /metrics	both	client	sn-be-legacy-metadata-enrichment	json	handler prometheus_metrics; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/metadata-enrichment/main.py:697
http-endpoint	POST /enrich	both	client	sn-be-legacy-metadata-enrichment	json	handler enrich_track; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/metadata-enrichment/main.py:702
http-endpoint	POST /enrich/batch	both	client	sn-be-legacy-metadata-enrichment	json	handler enrich_batch; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/metadata-enrichment/main.py:731
http-endpoint	POST /enrich/trigger	both	client	sn-be-legacy-metadata-enrichment	json	handler trigger_enrichment; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/metadata-enrichment/main.py:778

http-endpoint	POST /enrich/reset-circuit-breaker/{service}	both	client	sn-be-legacy-metadata-enrichment	json	handler reset_circuit_breaker_failures; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/metadata-enrichment/main.py:813
http-endpoint	GET /stats	both	client	sn-be-legacy-metadata-enrichment	json	handler get_enrichment_stats; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/metadata-enrichment/main.py:956
http-endpoint	GET /admin/config	both	client	sn-be-legacy-metadata-enrichment	json	handler get_enrichment_config; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/metadata-enrichment/main.py:1146
http-endpoint	POST /admin/config/reload	both	client	sn-be-legacy-metadata-enrichment	json	handler reload_config; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/metadata-enrichment/main.py:1175
http-endpoint	GET /admin/config/field/{field_name}	both	client	sn-be-legacy-metadata-enrichment	json	handler get_field_config; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/metadata-enrichment/main.py:1209
http-endpoint	PUT /admin/config/field/{field_name}	both	client	sn-be-legacy-metadata-enrichment	json	handler update_field_config; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/metadata-enrichment/main.py:1259

http-endpoint	GET /admin/config/providers	both	client	sn-be-legacy-metadata-enrichment	json	handler get_available_providers; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/metadata-enrichment/main.py:1379
http-endpoint	POST /enrich/multi-tier-resolve	both	client	sn-be-legacy-metadata-enrichment	json	handler trigger_multi_tier_resolution; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/metadata-enrichment/main.py:1446
http-endpoint	POST /enrich/hunt-labels	both	client	sn-be-legacy-metadata-enrichment	json	handler trigger_label_hunter; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/metadata-enrichment/main.py:1523
http-endpoint	POST /enrich/cooldown/migrate	both	client	sn-be-legacy-metadata-enrichment	json	handler migrate_to_cooldown_queue; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/metadata-enrichment/main.py:1586
http-endpoint	POST /enrich/cooldown/process	both	client	sn-be-legacy-metadata-enrichment	json	handler process_cooldown_queue_endpoint; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/metadata-enrichment/main.py:1636
http-endpoint	GET /enrich/cooldown/stats	both	client	sn-be-legacy-metadata-enrichment	json	handler get_cooldown_stats; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/metadata-enrichment/main.py:1687

http-endpoint	POST /enrich/probabilistic-match	both	client	sn-be-legacy-metadata-enrichment	json	handler trigger_probabilistic_matching; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/metadata-enrichment/main.py:1725
http-endpoint	GET /api/v1/search/{service}	both	client	sn-be-legacy-metadata-enrichment	json	handler search; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/metadata-enrichment/search_routes.py:11
http-endpoint	GET /health	both	client	sn-be-legacy-api-gateway-internal	json	handler health_check; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/api-gateway-internal/main.py:308
http-endpoint	POST /api/{provider}/search	both	client	sn-be-legacy-api-gateway-internal	json	handler unified_search; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/api-gateway-internal/main.py:333
http-endpoint	GET /api/{provider}/track/{track_id}	both	client	sn-be-legacy-api-gateway-internal	json	handler get_track; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/api-gateway-internal/main.py:416
http-endpoint	GET /admin/circuit-breakers	both	client	sn-be-legacy-api-gateway-internal	json	handler get_circuit_breakers; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/api-gateway-internal/main.py:451

http-endpoint	POST /admin/circuit-breakers/{provider}/reset	both	client	sn-be-legacy-api-gateway-internal	json	handler reset_circuit_breaker; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/api-gateway-internal/main.py:469
http-endpoint	GET /admin/cache/stats	both	client	sn-be-legacy-api-gateway-internal	json	handler get_cache_stats; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/api-gateway-internal/main.py:487
http-endpoint	GET /health	both	client	sn-be-svc-audio-analysis	json	handler health_check; production	Built	M1, M3	songnodes/services/audio-analysis/main.py:202
http-endpoint	GET /metrics	both	client	sn-be-svc-audio-analysis	json	handler metrics; production	Built	M1, M3	songnodes/services/audio-analysis/main.py:292
http-endpoint	POST /analyze-url	both	client	sn-be-svc-audio-analysis	json	handler analyze_url; production	Built	M1, M3	songnodes/services/audio-analysis/main.py:360
http-endpoint	POST /analyze	both	client	sn-be-svc-audio-analysis	json	handler analyze_track; production	Built	M1, M3	songnodes/services/audio-analysis/main.py:393
http-endpoint	GET /health	both	client	sn-be-legacy-websocket-api	json	handler health_check; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/websocket-api/main.py:402
http-endpoint	GET /stats	both	client	sn-be-legacy-websocket-api	json	handler get_stats; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/websocket-api/main.py:481

http-endpoint	GET /metrics	both	client	sn-be-legacy-websocket-api	json	handler metrics; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/websocket-api/main.py:496
websocket	WEBSOCKET /ws/public	both	client	sn-be-legacy-websocket-api	json	handler public_websocket_endpoint; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/websocket-api/main.py:505
websocket	WEBSOCKET /ws/{user_id}	both	client	sn-be-legacy-websocket-api	json	handler websocket_endpoint; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/websocket-api/main.py:554
http-endpoint	POST /api/v1/graph/broadcast	both	client	sn-be-legacy-websocket-api	json	handler broadcast_graph_update; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/websocket-api/main.py:785
http-endpoint	POST /api/v1/graph/nodes/added	both	client	sn-be-legacy-websocket-api	json	handler broadcast_nodes_added; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/websocket-api/main.py:802
http-endpoint	POST /api/v1/graph/nodes/updated	both	client	sn-be-legacy-websocket-api	json	handler broadcast_nodes_updated; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/websocket-api/main.py:815
http-endpoint	POST /api/v1/graph/nodes/removed	both	client	sn-be-legacy-websocket-api	json	handler broadcast_nodes_removed; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/websocket-api/main.py:828

http-endpoint	POST /api/v1/broadcast/{room_id}	both	client	sn-be-legacy-websocket-api	json	handler broadcast_message; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/websocket-api/main.py:841
http-endpoint	GET /api/v1/rooms/{room_id}/history	both	client	sn-be-legacy-websocket-api	json	handler get_room_history; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/websocket-api/main.py:864
http-endpoint	GET /health	both	client	sn-be-legacy-data-validator	json	handler health_check; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/data-validator/main.py:1242
http-endpoint	POST /validate	both	client	sn-be-legacy-data-validator	json	handler submit_validation_task; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/data-validator/main.py:1351
http-endpoint	POST /validate/sync	both	client	sn-be-legacy-data-validator	json	handler validate_sync; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/data-validator/main.py:1371
http-endpoint	GET /tasks/{task_id}	both	client	sn-be-legacy-data-validator	json	handler get_task_status; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/data-validator/main.py:1382
http-endpoint	GET /tasks	both	client	sn-be-legacy-data-validator	json	handler get_tasks; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/data-validator/main.py:1393

http-endpoint	GET /rules	both	client	sn-be-legacy-data-validator	json	handler get_validation_rules; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/data-validator/main.py:1408
http-endpoint	GET /stats	both	client	sn-be-legacy-data-validator	json	handler get_validation_stats; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/data-validator/main.py:1419
http-endpoint	GET /metrics	both	client	sn-be-legacy-data-validator	json	handler get_metrics; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/data-validator/main.py:1466
http-endpoint	GET /health	both	client	sn-be-legacy-browser-collector	json	handler health_check; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/browser-collector/main.py:357
http-endpoint	POST /collect	both	client	sn-be-legacy-browser-collector	json	handler collect_data; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/browser-collector/main.py:422
http-endpoint	POST /extract/{raw_data_id}	both	client	sn-be-legacy-browser-collector	json	handler extract_from_raw_data; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/browser-collector/main.py:636
http-endpoint	GET /metrics	both	client	sn-be-legacy-browser-collector	json	handler metrics; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/browser-collector/main.py:685

http-endpoint	GET /health	both	client	sn-be-legacy-db-connection-pool	json	handler health_check; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/db-connection-pool/main.py:294
http-endpoint	GET /stats	both	client	sn-be-legacy-db-connection-pool	json	handler get_pool_stats; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/db-connection-pool/main.py:360
http-endpoint	GET /metrics	both	client	sn-be-legacy-db-connection-pool	json	handler get_metrics; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/db-connection-pool/main.py:368
http-endpoint	POST /reload	both	client	sn-be-legacy-db-connection-pool	json	handler reload_configuration; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/db-connection-pool/main.py:376
http-endpoint	POST /query	both	client	sn-be-legacy-db-connection-pool	json	handler execute_query; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/db-connection-pool/main.py:381
http-endpoint	GET /pool-info	both	client	sn-be-legacy-db-connection-pool	json	handler get_detailed_pool_info; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/services/db-connection-pool/main.py:386
http-endpoint	GET /health	both	client	sn-be-legacy-scraper-api-real	json	handler health_check; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/scrapers/scraper_api_real.py:85

http-endpoint	GET /metrics	both	client	sn-be-legacy-scraper-api-real	json	handler metrics; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/scrapers/scraper_api_real.py:89
http-endpoint	POST /scrape	both	client	sn-be-legacy-scraper-api-real	json	handler scrape_url; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/scrapers/scraper_api_real.py:97
http-endpoint	GET /	both	client	sn-be-legacy-scraper-service	json	handler root; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/scrapers/scraper_service.py:59
http-endpoint	GET /health	both	client	sn-be-legacy-scraper-service	json	handler health_check; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/scrapers/scraper_service.py:69
http-endpoint	GET /spiders	both	client	sn-be-legacy-scraper-service	json	handler list_spiders; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/scrapers/scraper_service.py:79
http-endpoint	POST /run/{spider_name}	both	client	sn-be-legacy-scraper-service	json	handler run_spider; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/scrapers/scraper_service.py:113
http-endpoint	GET /status/{spider_name}	both	client	sn-be-legacy-scraper-service	json	handler get_spider_status; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/scrapers/scraper_service.py:155

http-endpoint	POST /migrate	both	client	sn-be-legacy-scraper-service	json	handler run_migration; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/scrapers/scraper_service.py:217
http-endpoint	GET /config	both	client	sn-be-legacy-scraper-service	json	handler get_configuration; non-final service (not in compose.dev.yaml)	Built	M1, M3	songnodes/scrapers/scraper_service.py:229
http-endpoint	GET /health	both	client	sn-be-svc-unified-scraper	json	handler health_check; production	Built	M1, M3	songnodes/scrapers/unified_scraper_api.py:163
http-endpoint	GET /metrics	both	client	sn-be-svc-unified-scraper	json	handler metrics; production	Built	M1, M3	songnodes/scrapers/unified_scraper_api.py:173
http-endpoint	GET /scrape/progress/{job_id}	both	client	sn-be-svc-unified-scraper	json	handler scrape_progress; production	Built	M1, M3	songnodes/scrapers/unified_scraper_api.py:179
http-endpoint	GET /sources	both	client	sn-be-svc-unified-scraper	json	handler list_sources; production	Built	M1, M3	songnodes/scrapers/unified_scraper_api.py:188
http-endpoint	POST /scrape	both	client	sn-be-svc-unified-scraper	json	handler scrape; production	Built	M1, M3	songnodes/scrapers/unified_scraper_api.py:578
amqp	celery task songnodes.drain (beat every JOB_POLL_SECONDS=5s, expires 15s)	write	sn-be-svc-worker	sn-be-store-rabbitmq	celery	no payload; claims ingestion_jobs	Built	M1, M2	songnodes/services/worker/celery_app.py:47 +1

amqp	celery task songnodes.schedule (beat every SCHEDULE_SECONDS=30s)	write	sn-be-svc-worker	sn-be-store-rabbitmq	celery	enqueues tick:{kind}:{bucket} jobs	Built	M1, M2	songnodes/services/worker/celery_app.py:48 +2
amqp	audio_analysis_queue (vhost musicdb)	read	sn-be-store-rabbitmq	sn-be-svc-audio-analysis	json	analysis request; consumer only starts when RabbitMQ and DATABASE_URL are configured, which compose does not do	Built	M1, M2	songnodes/services/audio-analysis/main.py:151 +2
amqp	metadata-enrichment / dlq-manager / websocket-api queues	both	sn-be-legacy-metadata-enrichment	sn-be-legacy-dlq-manager	json	legacy queues, no deployed producer or consumer	Built	M1, M2	songnodes/services/metadata-enrichment/queue_init.py +3
redis	oauth:spotify:{state} / oauth:tidal:{state} / oauth:tidal:device:{device_code} (setex, deleted on callback)	both	sn-be-svc-rest-api	sn-be-store-redis	json	PKCE verifier, redirect, created_at	Built	M1, M2	songnodes/services/rest_api/routers/music_auth.py:413 +4
redis	batch:{batch_id} (setex 3600)	both	sn-be-svc-graph-api	sn-be-store-redis	json	batch status for POST /api/v1/visualization/nodes/batch (bulk insert itself is disabled)	Built	M1, M2	songnodes/services/graph-visualization-api/main.py:475 +2
redis	cache_result keys (CACHE_TTL 300s)	both	sn-be-svc-graph-api	sn-be-store-redis	json	decorator defined but applied to no route (grep finds no @cache_result)	Built	M1	songnodes/services/graph-visualization-api/main.py:332 +1
redis	scraped:setlists:{mixesdb|1001tracklists}:last_run (TTL SCRAPER_RUN_TTL_HOURS=24h)	both	sn-be-svc-unified-scraper	sn-be-store-redis	iso timestamp	daily run quota; CloseSpider(daily_quota_reached) unless force_run	Built	M1, M2	songnodes/scrapers/spiders/1001tracklists_spider.py:309 +3

redis	scraped:setlists:{source}:{sha digest} (TTL SCRAPER_SOURCE_TTL_DAYS=30d)	both	sn-be-svc-unified-scraper	sn-be-store-redis	iso timestamp	per-setlist cross-run dedupe	Built	M1, M2	songnodes/scrapers/spiders/mixesdb_spider.py:655 +2
redis	spotify spider token/expiry keys	both	sn-be-svc-unified-scraper	sn-be-store-redis	text	store spider (not in SPIDER_MAP)	Built	M1, M2	songnodes/scrapers/spiders/stores/spotify_spider.py:222
websocket	/api/graph/ws/{room_id} (graph_update, cursor_position, ping)	both	client	sn-be-svc-graph-api	json	in-memory room broadcast, no persistence	Built	M1, M2	songnodes/services/graph-visualization-api/main.py:915
http-endpoint	POST {UNIFIED_SCRAPER_URL}/scrape	write	sn-be-svc-worker	sn-be-svc-unified-scraper	json	{source, search_query|url, artist_name, max_pages, limit, job_id} → {bronze_playlists_persisted, bronze_tracks_persisted,...}; timeout 960s	Built	M1, M2	songnodes/services/common/datalayer/handlers.py:60 +2
http-endpoint	GET {UNIFIED_SCRAPER_URL}/scrape/progress/{job_id}	read	sn-be-svc-rest-api	sn-be-svc-unified-scraper	json	{pages, playlists, tracks, updated_at}; in-memory, lost on scraper restart	Built	M1, M2	songnodes/services/rest_api/routers/jobs.py:213 +2
http-endpoint	POST {AUDIO_ANALYSIS_URL}/analyze-url	write	sn-be-svc-worker	sn-be-svc-audio-analysis	json	Deezer preview URL → key, bpm	Built	M1, M2	songnodes/services/common/datalayer/audio.py +2

http-endpoint	GET {METADATA_ENRICHMENT_URL=http://metadata-enrichment:8086}/search...	read	sn-be-svc-rest-api	sn-be-legacy-metadata-enrichment	json	music search proxy; target service is not in compose so every call returns 503 search_service_unavailable	Built	M1, M2	songnodes/services/rest_api/main.py:692 +1
cli	scripts/import-music-library.sh → POST /api/v1/library/ingest (Idempotency-Key import-<md5>)	write	ml-cli-db	sn-be-svc-rest-api	json	musicLibrary outputs/artifact/library.json (export shape, contracts/library_ingest.md)	Built	M1, M2	songnodes/scripts/import-music-library.sh:1 +2
file-read	$MUSIC_LIBRARY_DIR/outputs/artifact/library.json	read	ml-cli-db	sn-be-cli-import-music-library	json	library export	Built	M1, M2	songnodes/scripts/import-music-library.sh:15
cli	python -m library.importer --sqlite <library.db> --api-url (default http://localhost:8080/...)	write	ml-library-db	sn-be-svc-rest-api	sqlite->json	reads musicLibrary SQLite and posts to ingest; default port 8080 is not the REST API (8082)	Built	M1, M2	songnodes/services/library/importer/__main__.py:15 +1
file-write	library.reconcile.cli --out <csv>	write	sn-be-store-postgres	operator	csv	library vs silver match candidates	Built	M1, M2	songnodes/services/library/reconcile/cli.py:27 +1
file-write	pipeline/build_db.py → db/library.db and outputs/artifact/library.json	write	sn-be-cli-pipeline	sn-be-file-read-library-json	sqlite/json	songnodes-side copy of the weekly pipeline	Built	M1, M2	songnodes/pipeline/build_db.py:10 +2

file-write	backup-YYYYMMDD.dump via compose exec pg_dump -U songnodes	write	sn-be-store-postgres	operator	pg custom dump	documented one-liner (CLAUDE.md section 7)	Built	M1	songnodes/CLAUDE.md:300
file-write	scripts/backup_database.sh → /backups/postgres/musicdb_backup_*.sql.gz; backup_volumes.sh → ./backups/volumes/*.tar.gz	write	sn-be-store-postgres	operator	gzip sql / tar.gz	target user musicdb_user, db musicdb and volumes songnodes_postgres_data etc., none of which exist in compose project songnodes-dev	Built	M1, M2	songnodes/scripts/backup_database.sh:52 +1
file-write	/tmp/worker-heartbeat	both	sn-be-svc-worker	docker-healthcheck	mtime	touched by drain/schedule; Dockerfile HEALTHCHECK fails if older than 90s	Built	M1, M2	songnodes/services/worker/celery_app.py:67 +1
env	TIDAL_CLIENT_ID/SECRET, SPOTIFY_CLIENT_ID/SECRET, LASTFM_API_KEY, DISCOGS_TOKEN, GETSONGBPM_API_KEY, MB_LOCAL_*	read	host-env/.env	sn-be-svc-worker	env	passed through compose with empty defaults	Built	M1, M2	songnodes/compose.dev.yaml:60 +1
env	ENRICHMENT_PROVIDERS (default none)	read	host-env/.env	sn-be-svc-worker	csv	gate for provider calls; audio-analysis Deezer calls governed separately by AUDIO_ANALYSIS_ENABLED (default true)	Built	M1, M2	songnodes/compose.dev.yaml:125 +1
sql	api_keys table via get_api_key(service,key,secret) → overrides env	read	sn-be-store-postgres	sn-be-svc-worker	pgp_sym encrypted base64	API_KEY_ENCRYPTION_SECRET not set in compose, so the SQL falls back to a hard-coded default key	Built	M1, M2	songnodes/services/common/providers/saved_keys.py:19 +2

env	SONGNODES_API_TOKEN	read	host-env/.env	sn-be-svc-rest-api	text	not set in compose -> all write/admin routes open	Built	M1, M2	songnodes/services/rest_api/security/auth.py:28 +1
sql	INSERT bronze_scraped_playlists / bronze_scraped_tracks / bronze_scraped_artists	write	sn-be-svc-unified-scraper	sn-be-tbl-bronze	postgres	upsert on (source, source_url, source_*_id); per-row savepoints; bronze_incomplete marker	Built	M1, M2	songnodes/scrapers/pipelines/persistence_pipeline.py:1117 +3
sql	INSERT silver_enriched_tracks / silver_enriched_playlists / silver_playlist_tracks directly from the scraper	write	sn-be-svc-unified-scraper	sn-be-tbl-silver	postgres	plain INSERT, no ON CONFLICT on tracks, artist defaults to Unknown Artist, no outbox event	Built	M1, M2	songnodes/scrapers/pipelines/persistence_pipeline.py:910 +4
sql	INSERT song_adjacency (scraper)	write	sn-be-svc-unified-scraper	sn-be-tbl-song-adjacency	postgres	table is not created by any migration compose applies	Built	M1, M2	songnodes/scrapers/pipelines/persistence_pipeline.py:1731 +1
sql	INSERT bronze_* from POST /api/v1/tracks/import-tracklist, then ingest_playlist synchronously	write	sn-be-svc-rest-api	sn-be-tbl-bronze	postgres	manual tracklist	Built	M1, M2	songnodes/services/rest_api/catalog_writes.py:300 +2
sql	ingest_pending: bronze playlists without silver marker datalayer-ingest-v1 → silver_enriched_* + silver_track_transitions (N→N+1) + outbox	both	sn-be-svc-worker	sn-be-tbl-silver	postgres	completeness check (positions 1..n, MAX_ENTRIES 5000); TRANSITION_UPSERT ON CONFLICT (from_track_id,to_track_id) occurrence_count+1	Built	M1, M2	songnodes/services/common/datalayer/ingest.py:28 +3

sql	projection_consume: outbox_events (track/artist) → projection_tracks	both	sn-be-svc-worker	sn-be-tbl-projection-tracks	postgres	keyed upsert/delete; reconcile compares source_updated_at	Built	M1, M2	songnodes/services/common/datalayer/projection.py:14 +3
sql	graph-api reads silver_enriched_tracks + silver_track_transitions + library.* (REPEATABLE READ snapshot)	read	sn-be-tbl-silver	sn-be-svc-graph-api	postgres	VISIBLE_TRACK filter excludes blank/unknown/various artists	Built	M1, M2	songnodes/services/graph-visualization-api/catalog.py:17 +3
sql	library_search / preview read projection_tracks + library.* + evidence	read	sn-be-tbl-projection-tracks	sn-be-svc-rest-api	postgres	–	Built	M1, M2	songnodes/services/rest_api/routers/library_search.py:54 +1
sql	library_ingest job → library.tracks/track_xref/appearances/rotation_history/runs/sets/set_tracks + target_tracks auto-promotion + outbox library.track.*	write	sn-be-svc-worker	sn-be-tbl-library	postgres	one transaction per payload	Built	M1, M2	songnodes/services/library/ingest/handler.py:6 +5
sql	library.sets / library.set_tracks via PUT/DELETE /api/v1/library/sets/{id}; save appends library.set.saved.v1	both	sn-be-svc-rest-api	sn-be-tbl-library	postgres	delete-then-insert set_tracks; delete emits no event	Built	M1, M2	songnodes/services/library/model/library_db.py:10 +2
sql	target_tracks CRUD (REST), scheduler stamps last_searched (SKIP LOCKED), scrape result writes playlists_found	both	sn-be-svc-rest-api	sn-be-tbl-target-tracks	postgres	–	Built	M1, M2	songnodes/services/rest_api/routers/target_tracks.py:111 +2
sql	scrape_window single row (PUT /api/v1/scrape/window) read by scrape_targets	both	sn-be-svc-rest-api	sn-be-svc-worker	postgres	starts_on, ends_on (Melbourne days, inclusive)	Built	M1, M2	songnodes/services/rest_api/routers/scrape_window.py:22 +1

sql	ingestion_jobs enqueue/claim/complete/fail/retry/prune	both	sn-be-svc-rest-api	sn-be-svc-worker	postgres	(kind, idempotency_key) unique; status queued|running|succeeded|dead	Built	M1, M2	songnodes/services/common/datalayer/jobs.py:26 +4
sql	outbox_events append / outbox_consumers offsets / outbox_dead_letters	both	sn-be-svc-rest-api	sn-be-svc-worker	postgres	(txid,id) ordering below pg_snapshot_xmin	Built	M1, M2	songnodes/services/common/datalayer/outbox.py:21 +2
sql	enrich_track / enrichment_backfill / audio_analysis → silver_track_metadata_evidence + silver_enriched_tracks consensus + backfill_checkpoints	write	sn-be-svc-worker	sn-be-tbl-silver	postgres	evidence then consensus; outbox track updated	Built	M1, M2	songnodes/services/common/datalayer/evidence.py:37 +3
sql	er_shadow → er_decisions; merges → entity_merge_history / entity_redirects	write	sn-be-svc-worker	sn-be-tbl-silver	postgres	shadow mode, merges nothing automatically	Built	M1, M2	songnodes/services/common/datalayer/er.py:159 +1
sql	INSERT web_call_log (provider calls)	write	sn-be-svc-worker	sn-be-tbl-web-call-log	postgres	written, no reader in any service	Built	M1, M2	songnodes/services/common/providers/web_call_log.py:24
sql	user_oauth_tokens upsert/update (plain text tokens, user_id default_user)	write	sn-be-svc-rest-api	sn-be-tbl-oauth	postgres	readers are metadata-enrichment (not in compose) and music_auth itself	Built	M1, M2	songnodes/services/rest_api/routers/music_auth.py:558 +3
sql	POST /api/user-setlists → CREATE TABLE IF NOT EXISTS musicdb.user_setlists (+tracks)	write	sn-be-svc-graph-api	sn-be-tbl-musicdb-legacy	postgres	musicdb schema is not created in the compose database; written, never read	Built	M1, M2	songnodes/services/graph-visualization-api/main.py:808 +2

sql	graph_projection_consume → Memgraph bolt://memgraph:7687 (only if MEMGRAPH_ENABLED/URI)	write	sn-be-svc-worker	sn-be-store-memgraph	postgres	Cypher MERGE; no-op in compose (no memgraph service, no env), offsets still advance	Built	M1, M2	songnodes/services/common/datalayer/graph_projection.py:22 +2
sql	MusicBrainz local mirror ref_mb (MB_LOCAL_* on host.docker.internal:55433, statement timeout 2000ms)	read	sn-be-store-refmb	sn-be-svc-worker	postgres	read-only role ref_mb_reader	Built	M1, M2	songnodes/compose.dev.yaml:61 +2
fetch	https://openapi.tidal.com, auth.tidal.com	both	sn-be-svc-worker	sn-be-ext-store-tidal	json	external API	Built	M1	songnodes/services/common/providers/tidal.py
fetch	api.deezer.com + cdnt-preview.dzcdn.net previews	read	sn-be-svc-worker	sn-be-ext-store-deezer	json	external API	Built	M1	songnodes/services/common/providers/deezer.py
fetch	api.discogs.com	read	sn-be-svc-worker	sn-be-ext-store-discogs	json	external API	Built	M1	songnodes/services/common/providers/discogs.py
fetch	ws.audioscrobbler.com	read	sn-be-svc-worker	sn-be-ext-store-lastfm	json	external API	Built	M1	songnodes/services/common/providers/lastfm.py
fetch	musicbrainz.org (MUSICBRAINZ_CONTACT UA)	read	sn-be-svc-worker	sn-be-ext-store-musicbrainz	json	external API	Built	M1	songnodes/services/common/providers/musicbrainz.py
fetch	getsongbpm.com (GETSONGBPM_ENABLED false)	read	sn-be-svc-worker	sn-be-ext-store-getsongbpm	json	external API	Built	M1	songnodes/services/common/providers/getsongbpm.py

fetch	api.spotify.com / accounts.spotify.com (rest-api OAuth and playlists)	both	sn-be-svc-rest-api	sn-be-ext-store-spotify	json	external API	Built	M1	songnodes/services/rest_api/routers/spotify_playlists.py
fetch	TIDAL user playlists (Bearer from client)	write	sn-be-svc-rest-api	sn-be-ext-store-tidal-user	json	external API	Built	M1	songnodes/services/rest_api/routers/tidal_playlists.py
fetch	www.googleapis.com YouTube Data API (quota counter)	read	sn-be-svc-rest-api	sn-be-ext-store-youtube	json	external API	Built	M1	songnodes/services/rest_api/routers/youtube_api.py
fetch	api.anthropic.com / api.openai.com / generativelanguage.googleapis.com	both	sn-be-svc-rest-api	sn-be-ext-store-ai	json	external API	Built	M1	songnodes/services/rest_api/routers/ai_proxy.py
fetch	www.mixesdb.com (Scrapy, robots crawl-delay)	read	sn-be-svc-unified-scraper	sn-be-ext-store-mixesdb	json	external API	Built	M1	songnodes/scrapers/spiders/mixesdb_spider.py
fetch	www.1001tracklists.com (Scrapy + Playwright)	read	sn-be-svc-unified-scraper	sn-be-ext-store-1001tracklists	json	external API	Built	M1	songnodes/scrapers/spiders/1001tracklists_spider.py
fetch	Spotify/MusicBrainz/Last.fm/Discogs/Reddit from scraper item pipelines	read	sn-be-svc-unified-scraper	sn-be-ext-store-scraper-enrichment	json	external API	Built	M1	songnodes/scrapers/settings.py

fetch	Ollama CAPTCHA/HTML analysis (CLAUDE.md 5.1)	read	sn-be-svc-unified-scraper	sn-be-ext-store-ollama	json	no ollama service in compose; ollama_html_analyzer.py / llm_scraper_engine.py not in ITEM_PIPELINES	● Built	M1, M2	songnodes/scrapers/ollama_html_analyzer.py +1
http-endpoint	POST /api/v1/auth/login	both	client	sn-be-q-user-auth-router	json	queued PR #578	▲ Queued	M1	PR #578 services/rest_api/routers/user_auth.py
http-endpoint	GET /api/v1/auth/duo/callback	both	client	sn-be-q-user-auth-router	json	queued PR #578	▲ Queued	M1	PR #578 services/rest_api/routers/user_auth.py
http-endpoint	GET /api/v1/auth/me	both	client	sn-be-q-user-auth-router	json	queued PR #578	▲ Queued	M1	PR #578 services/rest_api/routers/user_auth.py
http-endpoint	POST /api/v1/auth/logout	both	client	sn-be-q-user-auth-router	json	queued PR #578	▲ Queued	M1	PR #578 services/rest_api/routers/user_auth.py
redis	duo:auth_state:{state} (setex 600s; in-memory fallback dict)	both	sn-be-q-user-auth-router	sn-be-store-redis	json	queued PR #578	▲ Queued	M1	PR #578 services/rest_api/routers/user_auth.py
cookie	songnodes_session (HS256 JWT, 7 days)	write	sn-be-q-user-auth-router	browser	jwt	queued PR #578; signing key JWT_SECRET_KEY or JWT_SECRET	▲ Queued	M1	PR #578 services/rest_api/routers/user_auth.py

localStorage	sn-theme, sn-text-size, sn-density, sn-motion	both	sn-fe-settings	sn-fe-index-html	string	theme dark|light|booth|system; text size; density comfortable|compact; motion system|reduce|full	Built	M1, M2, M3	songnodes/design-system/components/bundle.js:611-624 +2
localStorage	songnodes-store	both	sn-fe-store-zustand	sn-fe-store-zustand	json	zustand persist v6: panelState, viewState (sets as arrays), searchFilters, savedSetlists, pathfindingState, musicCredentials (access/refresh tokens, client id/secret)	Built	M1, M2	songnodes/frontend/src/store/useStore.ts:1478-1545
localStorage	sn-music-library-sets	both	sn-fe-sets	sn-fe-home	json	SavedSet[] {id,name,shape,minutes,mix,pool,seed,tracks[{id,tgt,a,t,m,e,k,bpm,tidal,move,track_id}],updated,created,description}	Built	M1, M2	songnodes/frontend/src/features/musicLibrary/data.ts:138,343-357 +1
localStorage	sn-set-draft	both	sn-fe-library	sn-fe-sets	json	Draft {id,name,shape,minutes,mix,pool,slots,seed,crate} via sanitiseDraft	Built	M1, M2	songnodes/frontend/src/features/musicLibrary/storage.ts:24-27 +1
localStorage	sn-filters-library	both	sn-fe-library	sn-fe-library	json	{filters: CrateFilters, sort: {key,dir}}; URL params win (design-system/12-inputs.md:64)	Built	M1, M2	songnodes/frontend/src/features/musicLibrary/storage.ts:22 +1

localStorage	sn-banner-keyed-library, sn-banner-keyed-sets	both	sn-fe-library	sn-fe-sets	json number	count of keyed tracks at dismissal	Built	M1, M2	songnodes/frontend/src/features/musicLibrary/storage.ts:30 +2
localStorage	sn-favs-library-crate, sn-favs-pathfinder-waypoints	both	sn-fe-library	sn-fe-paths	json	FavoriteItem<T>[]	Built	M1, M2	songnodes/frontend/src/utils/favoriteQueries.ts:8-101 +2
localStorage	sn-recent-quick-search, sn-recent-music-service (and per-namespace AutocompleteSearchField)	both	sn-fe-graph	sn-fe-graph	json	recent search strings	Built	M1, M2	songnodes/frontend/src/utils/recentSearches.ts:5-103 +2
localStorage	sn-graph-view	both	sn-fe-graph	sn-fe-graph	string	library|neighbourhood	Built	M1, M2	songnodes/frontend/src/lib/graphView.ts:18-30
localStorage	sn-target-tracks-v1	both	sn-fe-library	sn-fe-harmony	json	Record<libraryTrackId, {isTarget,priority,pendingSync}>	Built	M1, M2	songnodes/frontend/src/services/trackSyncAdapter.ts:3-100 +1
localStorage	sn-dismissed-suggestions	both	sn-fe-sources-targets	sn-fe-sources-targets	json	string[] of suggestion ids	Built	M1, M2	songnodes/frontend/src/hooks/useTargetSuggestions.ts:23-40
localStorage	songnodes_ai_setlists_ledger	both	sn-fe-sources-import	sn-fe-sources-import	json	array of imported AI setlist ledger entries	Built	M1, M2	songnodes/frontend/src/features/aiSetlists/AISetlistsPage.tsx:57,147
localStorage	songnodes_ai_key_config, songnodes_ai_keys_server_v1	both	sn-fe-settings	sn-fe-sources-import	json	AIKeyConfig (preferred provider, no raw keys); migration flag removes legacy raw keys songnodes_key_*, ai_key_*, apiKey_*	Built	M1, M2	songnodes/frontend/src/services/apiKeyService.ts:10,170-223

localStorage	songnodes_ai_prompt_history	both	sn-fe-graph	sn-fe-graph	json	AIPromptHistoryItem[]; only used by unreachable AIAssistantModal	Built	M1, M2	songnodes/frontend/src/services/aiService.ts:62,373-395
localStorage	tidal_oauth_tokens, spotify_oauth_tokens	read	sn-fe-oauth-callback	sn-fe-graph	json	{access_token,refresh_token,expires_at}; legacy, migrated into songnodes-store then removed	Built	M1, M2	songnodes/frontend/src/pages/GraphPage.tsx:15-33 +2
localStorage	mutex_lock_token_refresh_<service>	both	sn-fe-graph	sn-fe-graph	json	{id,expiresAt}; fallback only when navigator.locks is missing	Built	M1, M2	songnodes/frontend/src/hooks/useTokenRefresh.ts:43-95
localStorage	sn-disclosure:<name>, sn-skip-moved-targets	both	sn-fe-ds-component-previews	sn-fe-ds-component-previews	string	preview-only state; no app code reads them	Built	M1, M3	songnodes/design-system/components/Disclosure/preview.html:46-49 +1
sessionStorage	spotify_auth_state (also checks <service>_auth_state, tidal_auth_state)	both	sn-fe-settings	sn-fe-oauth-callback	string	64 hex chars CSRF state	Built	M1, M2	songnodes/frontend/src/pages/SettingsPage.tsx:381-386 +2
sessionStorage	<service>_oauth_tokens	write	sn-fe-oauth-callback	–	json	{access_token,refresh_token,expires_in,expires_at,token_type,scope}; no reader	Built	M1, M2	songnodes/frontend/src/components/OAuthCallback.tsx:227-229
postMessage	oauth-success | oauth-error	write	sn-fe-oauth-callback	sn-fe-settings	json	{type, service, tokens|error}; targetOrigin window.location.origin; receiver checks event.origin	Built	M1, M2	songnodes/frontend/src/components/OAuthCallback.tsx:56-61,258-266 +2

storage-event	sn-favs-library-crate	read	sn-fe-library	sn-fe-library	json	cross-tab refresh of filter presets	Built	M1, M2	songnodes/frontend/src/features/musicLibrary/components/FilterPresets.tsx:38-45
url-param	/library?q,rot,pillar,tier,tags,src,checks,rev,sort,focus=search	both	sn-fe-sets	sn-fe-library	query string	filtersToParams/filtersFromParams; focus=search from AppHeader	Built	M1, M2	songnodes/frontend/src/features/musicLibrary/LibraryPage.tsx:71-140 +1
url-param	/sets?open=<setId>&pool=crate&<crate filters>	read	sn-fe-library	sn-fe-sets	query string	open set id or crate pool built from library filters	Built	M1, M2	songnodes/frontend/src/features/musicLibrary/SetBuilderPage.tsx:51,120,633 +1
url-param	/graph?mode=play|plan&tab=analysis|keymood|pathfinder|tidal|spotify|targets	read	sn-fe-paths	sn-fe-graph	query string	tab implies plan mode	Built	M1, M2	songnodes/frontend/src/components/DJInterface.tsx:95-122 +1
url-param	/settings/:settingId and /settings#display	read	sn-fe-settings	sn-fe-settings	path	settingId in SETTINGS ids	Built	M1, M2	songnodes/frontend/src/pages/SettingsPage.tsx:265 +1
url-param	?code&state (callback query)	read	–	sn-fe-oauth-callback	query string	OAuth authorisation code	Built	M1, M2	songnodes/frontend/src/components/OAuthCallback.tsx:68-76
file-read	library export .json (musicLibrary outputs/artifact/library.json)	read	–	sn-fe-library	json	LibraryFile {generated,ledger_updated,counts,sources,runs,tracks[]} max 50 MB; hashed to Idempotency-Key import-sha256-<hex>	Built	M1, M2	songnodes/frontend/src/features/musicLibrary/components/LibraryImportDialog.tsx:6-24,114-130 +1

file-read	tracklist .txt/.csv	read	–	sn-fe-graph	text	pasted/uploaded tracklist read with FileReader then POST /api/v1/tracks/import-tracklist	Built	M1, M2	songnodes/frontend/src/components/TracklistImporter.tsx:172,220,305
file-write	<set>.m3u8 | .csv | .json | .xml (Rekordbox)	write	sn-fe-sets	–	m3u8/csv/json/xml	exporters.ts exportM3U8, exportRekordbox; ExportDialog exportCsv/exportJson	Built	M1, M2	songnodes/frontend/src/features/setBuilder/ExportDialog.tsx:70-98 +1
file-write	artifact_sets.json	write	sn-fe-sets	ml-store-inbox	json	SavedSet[] (plain list, no tidal flag)	Built	M1, M2	songnodes/frontend/src/features/musicLibrary/SetBuilderPage.tsx:223-226 +2
file-write	pathfinder-result.json | pathfinder-playlist.m3u | pathfinder-result.csv	write	sn-fe-graph	–	json/m3u/csv	path segments with track name, artist, bpm, camelot_key, durations	Built	M1, M2	songnodes/frontend/src/components/PathfinderPanel/pathfinder.machine.ts:69-111
clipboard	navigator.clipboard.writeText	write	sn-fe-library	–	text	copy track/tracklist text (musicLibrary ui.tsx), AI setlist values, lib/utils copyToClipboard; tracker.html copies prompts	Built	M1, M2	songnodes/frontend/src/features/musicLibrary/components/ui.tsx:164 +3
env	VITE_API_BASE_URL, VITE_ALLOWED_ORIGINS, VITE_OAUTH_REDIRECT_ORIGIN	read	–	sn-fe-home	string	default same-origin /api	Built	M1, M2	songnodes/frontend/src/lib/apiBase.ts:1-100

fetch	GET /api/v1/library/search?limit=5000	read	sn-fe-library	sn-be-svc-rest-api	json	row → LibraryTrack compact keys a,t,m,d,bpm,k,e,p,rot,tags,src,first,last,g,lbl,sync,tidal,dz,bp,chart,rev,ovr,local,mbid	Built	M1, M2	songnodes/frontend/src/features/musicLibrary/data.ts:114
fetch	GET/PUT/DELETE /api/v1/library/sets[/{id}] (PUT /0 creates)	both	sn-fe-sets	sn-be-svc-rest-api	json	PUT body {name,description,tracks[{track_id}]} → {id}	Built	M1, M2	songnodes/frontend/src/features/musicLibrary/data.ts:196,264,290
fetch	POST /api/v1/library/ingest (Idempotency-Key) + GET /api/v1/jobs/{id} + POST /api/v1/jobs/{id}/retry	both	sn-fe-library	sn-be-svc-rest-api	json	{job_id,status,created}; poll every 2 s	Built	M1, M2	songnodes/frontend/src/features/musicLibrary/components/LibraryImportDialog.tsx:114,136,150
fetch	GET /api/v1/library/tracks/{id}/preview	read	sn-fe-library	sn-be-svc-rest-api	json	–	Built	M1, M2	songnodes/frontend/src/features/musicLibrary/components/LibraryRowPreview.tsx:27
fetch	GET/POST/PUT/DELETE /api/v1/target-tracks[/{id}], POST /{id}/search, GET /suggestions	both	sn-fe-sources-targets	sn-be-svc-rest-api	json	{track_id,artist,title,priority,is_active,search_terms,genres,archived}	Built	M1, M2	songnodes/frontend/src/components/TargetTracksManager.tsx:97-317 +5
fetch	GET /api/v1/music-search/{service}	read	sn-fe-sources-targets	sn-be-svc-rest-api	json	–	Built	M1, M2	songnodes/frontend/src/components/MusicServiceSearch.tsx:73
fetch	GET /api/graph/data?limit=10000 (fallback GET /api/graph)	read	sn-fe-graph	sn-be-svc-graph-api	json	{nodes[],edges[{id,source,target,weight,distance,type|edge_type}]}	Built	M1, M2	songnodes/frontend/src/hooks/useDataLoader.ts:28,82 +1

fetch	GET /api/graph/edges?node_ids=…&limit=2	read	sn-fe-graph	sn-be-svc-graph-api	json	–	Built	M1, M2	songnodes/frontend/src/components/EdgeDetailsModal.tsx:29
fetch	POST /api/v1/pathfinder/find-path	both	sn-fe-paths	sn-be-svc-rest-api	json	–	Built	M1, M2	songnodes/frontend/src/components/PathfinderPanel/pathfinder.machine.ts:47 +1
fetch	POST /api/v1/scrape/trigger?source&search_query (Idempotency-Key)	both	sn-fe-pipeline	sn-be-svc-rest-api	json	202 {created}	Built	M1, M2	songnodes/frontend/src/pages/ScraperProgressPage.tsx:105
fetch	GET/PUT /api/v1/scrape/window	both	sn-fe-pipeline	sn-be-svc-rest-api	json	{starts_on, ends_on,updated_at}	Built	M1, M2	songnodes/frontend/src/features/pipeline/ScrapeWindowCard.tsx:5,22
fetch	GET /api/v1/jobs?limit=200, ?kind=scrape&limit=50, GET /api/v1/outbox/status, GET /api/v1/jobs/{id}/stages, POST /jobs/{id}/retry	both	sn-fe-pipeline	sn-be-svc-rest-api	json	polled 10 s, visibility refresh	Built	M1, M2	songnodes/frontend/src/features/pipeline/usePipelineTelemetry.ts:275-277 +2
fetch	GET/POST /api/v1/api-keys, GET /requirements, POST /test, DELETE /{service}/{key}	both	sn-fe-settings	sn-be-svc-rest-api	json	–	Built	M1, M2	songnodes/frontend/src/pages/SettingsPage.tsx:306-442 +2

fetch	POST /api/v1/music-auth/tidal/oauth/init, GET spotify/authorize (navigation), GET spotify/callback | tidal/oauth/callback, POST store-token, POST {service}/refresh, POST test/{tidal|spotify}	both	sn-fe-settings	sn-be-svc-rest-api	json	–	Built	M1, M2	songnodes/frontend/src/pages/SettingsPage.tsx:362-394 +3
fetch	/api/v1/{tidal|spotify}/playlists/list|create|{id}/tracks, /tracks/search, /tracks/check-availability	both	sn-fe-graph	sn-be-svc-rest-api	json	–	Built	M1, M2	songnodes/frontend/src/components/TidalPlaylistManager.tsx:80-240 +1
fetch	POST /api/v1/tidal/sets/send (Bearer browser-held TIDAL token)	both	sn-fe-sets	sn-be-svc-rest-api	json	{name,description,tracks[{...,tidal}]}	Built	M1, M2	songnodes/frontend/src/features/setBuilder/sendToTidal.ts:38-44
fetch	POST /api/v1/ai/complete	both	sn-fe-sources-import	sn-be-svc-rest-api	json	–	Built	M1, M2	songnodes/frontend/src/services/aiProxyClient.ts:32
fetch	GET /api/v1/setlists	read	sn-fe-sources-import	sn-be-svc-rest-api	json	–	Built	M1, M2	songnodes/frontend/src/features/aiSetlists/AISetlistsPage.tsx:61
fetch	POST /api/v1/tracks/import-tracklist	both	sn-fe-sources-import	sn-be-svc-rest-api	json	–	Built	M1, M2	songnodes/frontend/src/features/aiSetlists/api/importRun.ts:29 +1

fetch	GET /api/v1/tracks/missing-artist, /artists/search, /artists/dirty; POST /tracks/{id}/assign-artist, /artists, /artists/{id}/merge|rename; DELETE /artists/{id}	both	sn-fe-graph	sn-be-svc-rest-api	json	–	Built	M1, M2	songnodes/frontend/src/components/ArtistAttributionManager.tsx:92-350
fetch	GET /api/v1/attributions	read	sn-fe-home	sn-be-svc-rest-api	json	–	Built	M1, M2	songnodes/frontend/src/utils/attributions.ts:28 +1
fetch	POST /api/v1/scrapers/target-tracks/search, POST /api/v1/scrapers/queue/clear	both	sn-fe-graph	sn-be-svc-rest-api	json	no matching backend route	Built	M1, M2	songnodes/frontend/src/components/DJInterface.tsx:360, 395, 432
fetch	/api/v1/search/*, /api/v1/pathfinding/*, /api/v1/tracks/batch, /api/v1/setlists POST/DELETE/export, /api/graph/tracks, /api/graph/positions, /api/v1/tracks/{id}/preview	both	sn-fe-graph	sn-be-svc-rest-api	json	graphApi/searchApi/pathfindingApi/setlistApi exported but not called (only trackApi via unreachable NowPlayingDeck)	Built	M1, M2	songnodes/frontend/src/services/api.ts:242-556

websocket	/api/ws	read	sn-fe-graph	sn-be-svc-rest-api	json	LiveDataLoader graph updates; component not imported anywhere	● Built	M1, M2	songnodes/frontend/src/components/LiveDataLoader.tsx:138-140
fetch	PATCH /api/tracks/{id}	write	sn-fe-graph	sn-be-svc-rest-api	json	EditTrackModal (unreachable)	● Built	M1, M2	songnodes/frontend/src/components/EditTrackModal.tsx:57
fetch	GET /api/v1/auth/me, POST /api/v1/auth/login, POST /api/v1/auth/logout (cookie, credentials: include)	both	sn-fe-login	sn-be-svc-rest-api	json	session cookie; user {email}	▲ Queued	M1	PR #579 frontend/src/features/auth/AuthContext.tsx:40-115 +1
postMessage	window CustomEvent 'sn-toast'	both	any page (showToast)	sn-fe-home	js object	{id,text,action}	● Built	M1	songnodes/frontend/src/components/ds/index.tsx:1423-1454

Section 5: **Synchronisation protocols, timings and limits** 21 protocols

Protocol: **Resumable resolve**

Participants	ml-cli-resolve, ml-agent-weekly-run
How it syncs	Load inbox/resolved_<date>.json if present; todo = to_resolve items whose norm_key is not yet a key; 4 worker threads; results merged in order and the whole file re-dumped every 20 songs and at the end. The agent loops 'timeout 160 ... resolve' until the final line appears.
Conflicts	first result per norm_key wins; an empty result (both APIs failed or no match) is stored and never retried for that date
Stale data	no TIDAL app secret: Deezer only
State	● Built

Evidence 2

```
musicLibrary/pipeline/run_weekly.py:414-463
musicLibrary/agent/weekly_run.md:3
```

Protocol: **Ledger save and snapshot**

Participants	ml-cli-ingest, ml-cli-process, ml-cli-confirm
How it syncs	json.dump to <ledger>.tmp; process first copy2's the existing ledger to ledger/history/ledger_before_<date>.json; os.replace tmp over the ledger.
Conflicts	last writer wins; no lock or version check
Stale data	history snapshots for manual restore
State	● Built

Evidence 1

```
musicLibrary/pipeline/djlib/ledger.py:62-70
```

Protocol: **TIDAL crate reconcile**

Participants	ml-cli-sync-tidal, ext-tidal, ml-cli-confirm
How it syncs	Desired = every active crate's tidal ids sorted by energy then BPM; playlists in tidal_playlists.json without 'set:' prefix and no longer a crate are emptied. For each: create if missing (Idempotency-Key uuid5('create:'+name)), list items, remove extras, add missing in chunks of 50 (onDuplicates SKIP, key uuid5(add:pid:date:name:n:i)). Then writes sync_result and runs confirm.
Conflicts	ledger is the source of truth; TIDAL order is not enforced for crates
Stale data	no user token: step skipped; TidalError: ALERT_tidal.txt and partial results
State	● Built

Evidence 2

```
musicLibrary/pipeline/run_weekly.py:467-521
musicLibrary/pipeline/djlib/tidal.py:237-272
```

Protocol: **Saved set to TIDAL**

Participants	ml-store-artifact-db, ml-agent-weekly-run, ml-cli-sync-sets, ext-tidal
How it syncs	Weekly agent copies ArtifactData rows into inbox/artifact_sets.json; load_sets flattens {id,data}; sets with tidal truthy map track ids through the ledger (fallback tr.tidal_id); playlist key 'set:<id>'; if current item list != wanted ids, remove all then add all.
Conflicts	page copy wins (full replace)
Stale data	no token: skipped with message
State	● Built

Evidence 3

```
musicLibrary/pipeline/run_weekly.py:524-564
musicLibrary/pipeline/build_db.py:93-106
musicLibrary/agent/weekly_run.md:5
```

Protocol: **Artifact saved-set mirror**

Participants	ml-page-set-builder, ml-store-artifact-db
How it syncs	initSavedSets: mem = localStorage; race use('db') and collection.get against 4 s; mem = remote + local-only; upload local-only. writeSavedSets: localStorage at once; changed docs set() and missing ids delete() on a serial promise queue with errors swallowed.
Conflicts	shared copy wins by id at load; local deletions made while the store was unreachable are never replayed
Stale data	no db: localStorage only
State	Built

Evidence 1

```
musicLibrary/artifact/app/artifact-data.ts:38-72
```

Protocol: **library.json to songnodes**

Participants	ml-cli-sync-songnodes, sn-scripts, sn-be-library-ingest
How it syncs	POST body = file bytes, Idempotency-Key 'import-'+md5(file); enqueue library_ingest job; worker maps short keys, upserts tracks, writes library.track_xref(source_id → canonical uuid), appearances, rotation_history, then queues projections and enrichment backfill. The songnodes script polls /api/v1/jobs/<id> and retries a dead job; run_weekly does not poll.
Conflicts	same file never imports twice; a changed file is a new job
Stale data	run_weekly prints a note on failure and carries on
State	Built

Evidence 3

```
musicLibrary/pipeline/run_weekly.py:567-597
songnodes/scripts/import-music-library.sh:29-70
songnodes/services/library/ingest/handler.py:6-17
```

Protocol: **TIDAL PKCE login through the songnodes app**

Participants	ml-cli-tidal, ext-tidal, songnodes frontend :3006
How it syncs	login-start stores verifier/state in tidal_pkce_pending.json and prints the authorize URL with redirect 127.0.0.1:3006/oauth/callback; the user copies the landing address; login-finish checks state and age (15 min) and exchanges the code with client_secret in the body.
Conflicts	if songnodes is running, its /oauth/callback route (App.tsx) consumes the single-use code with its own verifier, so musicLibrary's exchange fails
Stale data	refresh_token on expiry; else re-login
State	Built

Evidence 3

```
musicLibrary/pipeline/djlib/tidal.py:87-114
songnodes/frontend/src/App.tsx:33
songnodes/frontend/src/components/OAuthCallback.tsx:168
```

Protocol: **Cloud to laptop file transfer**

Participants	ml-agent-weekly-run
How it syncs	device_stage_files to bring files into the cloud workspace; device_commit_files with a unique staged name per run and md5 check both sides.
Conflicts	unique filenames avoid stale cached copies
Stale data	stop and report
State	● Built

Evidence 1

```
musicLibrary/agent/weekly_run.md:File transfer
```

Protocol: **Manual adds processed stamp**

Participants	ml-persona-curator, ml-cli-scrape
How it syncs	Rows without 'processed' are appended to the scrape rows, then the CSV is rewritten with processed=<run date> before raw_<date>.json is written.
Conflicts	stamped rows are never re-read
Stale data	none
State	● Built

Evidence 1

```
musicLibrary/pipeline/run_weekly.py:378-395
```

Protocol: **Durable job queue (ingestion_jobs)**

Participants	sn-be-svc-rest-api, sn-be-svc-worker, sn-be-tbl-ingestion-jobs
How it syncs	enqueue INSERT ON CONFLICT (kind,idempotency_key) DO NOTHING returning the original row; claim_next first dead-letters expired leases with no attempts left, then selects queued (run_after<=now) or lease-expired running rows FOR UPDATE SKIP LOCKED and increments attempts; handler and follow-up enqueues and complete run in one transaction; failure → fail() requeues with backoff 30s*2^(attempts-1) capped 3600s (x4 for transient), dead when attempts>=max_attempts or permanent 4xx (except 408/429).
Conflicts	idempotency key returns existing job; SKIP LOCKED prevents double claim; lease reclaim for crashed workers
Stale data	dead jobs revived by POST /api/v1/jobs/{id}/retry; succeeded pruned after 14 days
State	● Built

Evidence 3

```
songnodes/services/common/datalayer/jobs.py:26
songnodes/services/common/datalayer/jobs.py:39
songnodes/services/common/datalayer/runner.py:25
```

Protocol: **Recurring jobs via time-bucketed keys**

Participants	sn-be-svc-worker
How it syncs	songnodes.schedule every 30s enqueues projection_consume, graph_projection_consume (15s buckets), ingest_pending (60s), scrape_targets (3600s), audio_analysis (600s) with key tick:{kind}:{now//interval}.
Conflicts	overlapping beats collide on the same key
Stale data	missed buckets are simply skipped
State	● Built

Evidence 1

```
songnodes/services/worker/celery_app.py:83
```

Protocol: **Transactional outbox to projections**

Participants	all catalog writers, projection_tracks consumer, graph_projection consumer
How it syncs	append in the writer transaction; consumers read (txid,id) > offset and txid < pg_snapshot_xmin; handler + advance in one transaction; 3 attempts then outbox_dead_letters and advance; rebase_after_restore resets txids after dump/restore.
Conflicts	keyed upserts, at-least-once
Stale data	projection_reconcile / rebuild jobs; POST /api/v1/projections/tracks/reconcile
State	● Built

Evidence 2

```
songnodes/services/common/datalayer/outbox.py:1
songnodes/services/common/datalayer/projection.py:84
```

Protocol: **Library ingest idempotency**

Participants	import-music-library.sh, sn-be-svc-rest-api, sn-be-svc-worker
How it syncs	script computes md5 of library.json → Idempotency-Key import-<md5>; POST /api/v1/library/ingest enqueues library_ingest; script polls GET /api/v1/jobs/{id} and retries dead once; worker processes payload in one transaction and enqueues projection/backfill follow-ups.
Conflicts	same key returns the original job regardless of body (contract promises 409 for a different body: not implemented)
Stale data	none
State	● Built

Evidence 3

```
songnodes/scripts/import-music-library.sh:45
songnodes/services/rest_api/routers/library_ingest.py:10
songnodes/contracts/library_ingest.md:16
```

Protocol: **Scrape -> ingest -> projection ordering**

Participants	sn-be-svc-worker, sn-be-svc-unified-scraper
How it syncs	scrape handler (sync HTTP) → follow-up ingest_pending keyed by job → if changed, projection_consume, graph_projection_consume, enrichment_backfill; ingest_pending chains itself while examined>=limit; independent 60s tick also runs ingest_pending.
Conflicts	ingest_playlist locks bronze and silver playlist rows FOR UPDATE; INGEST_MARKER prevents re-ingest
Stale data	60s tick catches anything a follow-up missed
State	● Built

Evidence 2

```
songnodes/services/common/datalayer/handlers.py:96
songnodes/services/common/datalayer/handlers.py:111
```

Protocol: **OAuth state in Redis**

Participants	sn-be-svc-rest-api, sn-be-store-redis, Spotify, TIDAL
How it syncs	authorize stores PKCE verifier under oauth:{svc}:{state} with TTL; callback reads and deletes; tokens returned to client and optionally persisted to user_oauth_tokens (default_user); access log redacts code/state.
Conflicts	single-use state key
Stale data	TTL expiry → user restarts flow
State	● Built

Evidence 2

```
songnodes/services/rest_api/routers/music_auth.py:413
songnodes/services/rest_api/routers/access_log.py:1
```

Protocol: **Scheduled-scrape window**

Participants	sn-be-svc-rest-api, sn-be-svc-worker
How it syncs	scrape_targets reads one scrape_window row; outside Melbourne-day window it returns outside_window; manual scrapes ignore it.
Conflicts	single row
Stale data	missing table/row = no window
State	● Built

Evidence 1

```
songnodes/services/common/datalayer/target_scrape.py:12
```

Protocol: **Saved sets server/device merge**

Participants	sn-fe-sets, sn-fe-home, sn-be-rest-api, localStorage sn-music-library-sets, sn-set-draft
How it syncs	readSavedSetsAsync: GET /api/v1/library/sets; pair each server set with a local copy by id, else by content (name + ordered track ids) for local ids (non-numeric); server fields win; unpaired local-id sets kept; numeric ids missing on server dropped (deleted elsewhere); write merged list to localStorage and remap draft id. writeSavedSetsAsync diffs against serverSetsCache: DELETE removed, PUT changed (updated or name differs), PUT /0 creates and expects {id}; pendingCreates map dedupes concurrent creates in-tab.
Conflicts	server wins on name, description, tracks; local-only fields (shape, minutes, mix, pool, seed) kept from local copy
Stale data	GET failure → device copy, source "device"
State	● Built

Evidence 1

```
songnodes/frontend/src/features/musicLibrary/data.ts:166-337
```

Protocol: **Library target flag sync**

Participants	sn-fe-library, sn-fe-store-unified-tracks, sn-be-rest-api
How it syncs	syncTargetTrack POSTs /api/v1/target-tracks (or DELETEs by UUID) and records {isTarget,priority,pendingSync} in sn-target-tracks-v1; useUnifiedTracks hydrates library rows from local state first, then backend targets matched by lower-case artist|title.
Conflicts	local entry always beats backend state
Stale data	failure leaves pendingSync=true; no retry loop
State	● Built

Evidence 2

```
songnodes/frontend/src/services/trackSyncAdapter.ts:31-100
songnodes/frontend/src/hooks/useUnifiedTracks.ts:88-110
```

Protocol: **OAuth connect and token handoff**

Participants	sn-fe-settings, sn-fe-oauth-callback, sn-be-rest-api, sn-fe-store-zustand
How it syncs	Spotify: random 32-byte state in sessionStorage, navigate to /api/v1/music-auth/spotify/authorize; TIDAL: POST tidal/oauth/init then navigate. Callback page validates state, GETs backend callback, stores tokens in sessionStorage <service>_oauth_tokens, POSTs store-token, then postMessage to opener (same origin) or redirects to /.
Conflicts	n/a
Stale data	useTokenRefresh refreshes under navigator.locks (localStorage mutex fallback)
State	● Built

Evidence 3

```
songnodes/frontend/src/pages/SettingsPage.tsx:329-394
songnodes/frontend/src/components/OAuthCallback.tsx:60-290
songnodes/frontend/src/hooks/useTokenRefresh.ts:43-210
```

Protocol: **Pipeline telemetry polling**

Participants	sn-fe-pipeline, sn-be-rest-api
How it syncs	fetch jobs (200), scrape jobs (50), outbox status in parallel; setTimeout chain 10 s (30 s intended on error); refetch on visibilitychange.
Conflicts	n/a
Stale data	source 'unavailable' with explicit error
State	● Built

Evidence 1

```
songnodes/frontend/src/features/pipeline/usePipelineTelemetry.ts:270-
335
```

Protocol: **Idempotent library ingest and scrape trigger**

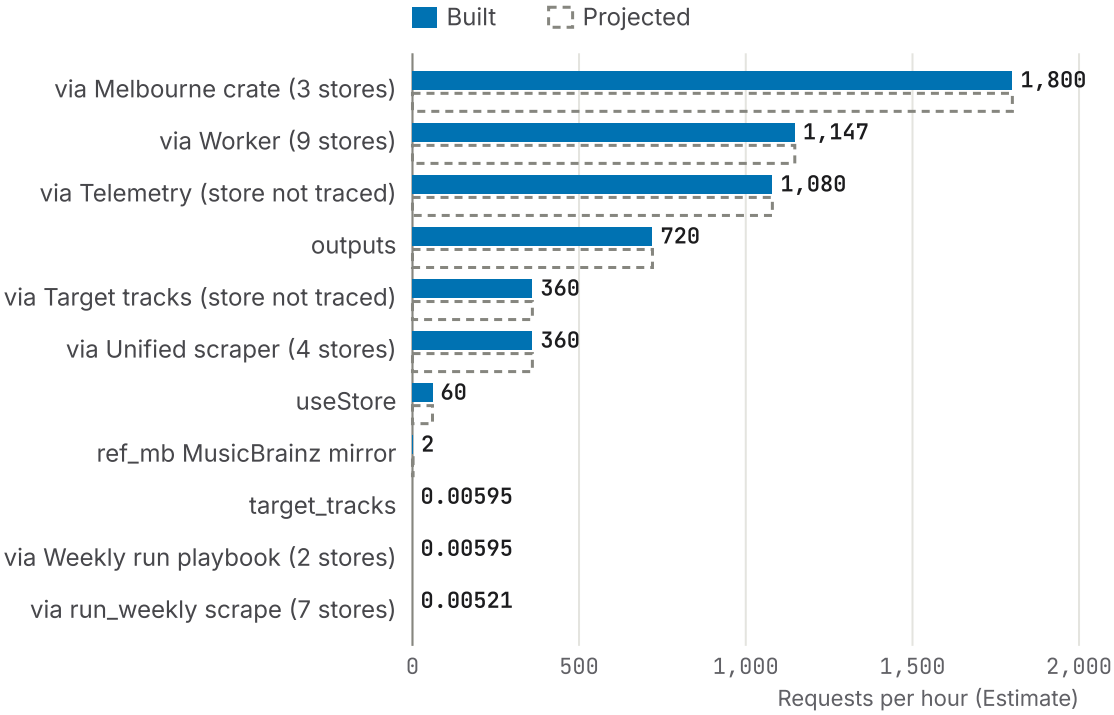
Participants	sn-fe-library, sn-fe-pipeline, sn-be-rest-api
How it syncs	library: Idempotency-Key import-sha256-<file hash> so the same file is never imported twice (matches scripts/import-music-library.sh scheme intent); scrape: one crypto.randomUUID per form submission reused on retry; 202 created=false shows "Already queued".
Conflicts	server-side idempotency (backend scope)
Stale data	job retry endpoint
State	● Built

Evidence 3

```
songnodes/frontend/src/features/musicLibrary/components/LibraryImportDialog.tsx:15-
21,114
songnodes/frontend/src/pages/ScraperProgressPage.tsx:94-128
songnodes/plan/agents/L8_library_import_dialog.md:9
```

C6: **Polling load by target store (Estimate)**

Estimate: the heaviest polling target is via Melbourne crate (3 stores), 1,800 requests per hour; queued work adds no polling.



C6 · n = 17 · Unit: requests per hour (Estimate) · Built

· Source: timings[] (kind poll|schedule, value_ms, where), signals[] and flows[] to resolve the store · Method: Estimate: requests per hour = 3,600,000 / value_ms per poll or schedule timing, summed by target store; the target is the timing's own store, else the single store linked to its component by signals or flows, else the component is named with its store count. Assumes every tick makes one request.

Table: **Timing constants** 105 constants

105 rows

CONSTANT	KIND	VALUE	WHERE	STATE	EVIDENCE
weekly run schedule	schedule	Sunday 7:37am (weekly)	ml-agent-weekly-run	Built	musicLibrary/agent/weekly_run.md:1 +1
scrape look-back window	schedule	--days 8	ml-cli-scrape	Built	musicLibrary/pipeline/run_weekly.py:646 +1
scraper polite delay	debounce	delay=1.0	ml-cli-scrape	Built	musicLibrary/pipeline/djlib/scrapers.py:19-20
3RRR episode delay	debounce	delay=2	ml-cli-scrape	Built	musicLibrary/pipeline/djlib/scrapers.py:146
scraper HTTP timeout	timeout	timeout=30	ml-cli-scrape	Built	musicLibrary/pipeline/djlib/scrapers.py:22
Deezer per-request sleep	debounce	time.sleep(0.15)	ml-cli-resolve	Built	musicLibrary/pipeline/djlib/deezer.py:10
Deezer quota backoff	retry	2 ** attempt s, 4 attempts (1,2,4,8 s)	ml-cli-resolve	Built	musicLibrary/pipeline/djlib/deezer.py:9-15
Deezer HTTP timeout	timeout	timeout=20	ml-cli-resolve	Built	musicLibrary/pipeline/djlib/deezer.py:11
TIDAL API and token HTTP timeout	timeout	timeout=30	ml-cli-resolve	Built	musicLibrary/pipeline/djlib/tidal.py:70 +1
TIDAL 429/5xx retry	retry	Retry-After or 2 ** attempt s, 5 attempts	ml-cli-sync-tidal	Built	musicLibrary/pipeline/djlib/tidal.py:177-188

TIDAL user token refresh margin	expiry	expires_in - 120	ml-store-secrets	Built	musicLibrary/pipeline/djlib/tidal.py:156
TIDAL client token refresh margin	expiry	expires_in - 60	ml-cli-resolve	Built	musicLibrary/pipeline/djlib/tidal.py:78
TIDAL token default lifetime	expiry	tok.get('expires_in', 3600)	ml-store-secrets	Built	musicLibrary/pipeline/djlib/tidal.py:156
PKCE login link max age	expiry	> 900	ml-cli-tidal	Built	musicLibrary/pipeline/djlib/tidal.py:101-102
one-step login callback wait	timeout	600 x 0.5 s	ml-cli-tidal	Built	musicLibrary/pipeline/djlib/tidal.py:139-142
resolve foreground chunk	timeout	timeout 160	ml-agent-weekly-run	Built	musicLibrary/agent/weekly_run.md:3
songnodes ingest POST timeout	timeout	timeout=30	ml-cli-sync-songnodes	Built	musicLibrary/pipeline/run_weekly.py:594 +1
import-music-library job poll	poll	sleep 5, 120 polls (10 min)	sn-be-cli-import-music-library	Built	songnodes/scripts/import-music-library.sh:55-70
artifact shared-store init timeout	timeout	setTimeout(..., 4000)	ml-page-set-builder	Built	musicLibrary/artifact/app/artifact-data.ts:59
legacy toast	timeout	2200	ml-page-set-builder-legacy	Built	musicLibrary/artifact/index.html
legacy delete confirm arm	timeout	3000	ml-page-set-builder-legacy	Built	musicLibrary/artifact/index.html:563
promotion window	ttl	promote_window_days 30	ml-cli-process	Built	musicLibrary/config/rules.json
early chart entry window	ttl	chart_entry_window_days 21	ml-cli-process	Built	musicLibrary/config/rules.json
chart last-seen freshness	ttl	timedelta(days=14)	ml-cli-process	Built	musicLibrary/pipeline/djlib/ledger.py:133
High Rotation tenure cap	ttl	high_tenure_max_days 42	ml-cli-process	Built	musicLibrary/config/rules.json
High Rotation silence demotion	ttl	high_silence_days 45	ml-cli-process	Built	musicLibrary/config/rules.json
Standard archive minimum age	ttl	std_archive_min_age_days 180	ml-cli-process	Built	musicLibrary/config/rules.json
Standard archive silence	ttl	std_archive_silence_days 60	ml-cli-process	Built	musicLibrary/config/rules.json
ref-mb stop grace period	timeout	stop_grace_period: 10m	ml-svc-ref-mb	Built	musicLibrary/ref_mb/compose.yaml
ref-mb healthcheck	heartbeat	interval 30s, timeout 5s, retries 5	ml-svc-ref-mb	Built	musicLibrary/ref_mb/compose.yaml
ref-mb checkpoint_timeout	schedule	checkpoint_timeout=30min	ml-svc-ref-mb	Built	musicLibrary/ref_mb/compose.yaml
songnodes ref_mb statement timeout	timeout	MB_LOCAL_STATEMENT_TIMEOUT_MS 2000	sn-be-store-refmb	Built	songnodes/services/common/refmb/settings.py:29
MetaBrainz HEAD retry	retry	5 tries, sleep 5, timeout 60	ml-cli-fetch-metabrainz	Built	musicLibrary/pipeline/fetch_metabrainz.py:11-17
MetaBrainz curl retry	retry	--retry 8 --retry-delay 5	ml-cli-fetch-metabrainz	Built	musicLibrary/pipeline/fetch_metabrainz.sh
Celery beat drain-jobs	poll	JOB_POLL_SECONDS default '5'	sn-be-svc-worker	Built	songnodes/services/worker/celery_app.py:25
drain task expires	expiry	max(POLL_SECONDS * 3, 15)	sn-be-svc-worker	Built	songnodes/services/worker/celery_app.py:47

Celery beat schedule-recurring	schedule	SCHEDULE_SECONDS default '30'	sn-be-svc-worker	Built	songnodes/services/worker/celery_app.py:26
schedule task expires	expiry	SCHEDULE_SECONDS * 3	sn-be-svc-worker	Built	songnodes/services/worker/celery_app.py:48
ingest_pending bucket	schedule	INGEST_INTERVAL_SECONDS '60'	sn-be-svc-worker	Built	songnodes/services/worker/celery_app.py:27
projection_consume and graph_projection_consume bucket	schedule	PROJECTION_INTERVAL_SECONDS '15'	sn-be-svc-worker	Built	songnodes/services/worker/celery_app.py:28
scrape_targets bucket	schedule	SCRAPE_TARGETS_INTERVAL_SECONDS 3600	sn-be-svc-worker	Built	songnodes/compose.dev.yaml:134 +1
audio_analysis bucket	schedule	AUDIO_ANALYSIS_INTERVAL_SECONDS 600	sn-be-svc-worker	Built	songnodes/compose.dev.yaml:128 +1
job lease	lease	DEFAULT_LEASE_SECONDS = 300	sn-be-tbl-ingestion-jobs	Built	songnodes/services/common/datalayer/jobs.py:21
scrape job lease	lease	KIND_LEASES = {'scrape': 1800.0}	sn-be-tbl-ingestion-jobs	Built	songnodes/services/common/datalayer/jobs.py:24
job retry backoff base (x4 for transient), doubled per attempt	retry	BACKOFF_SECONDS = 30; 30*2^(attempts-1)	sn-be-tbl-ingestion-jobs	Built	songnodes/services/common/datalayer/runner.py:16 +1
job retry backoff cap	retry	MAX_BACKOFF_SECONDS = 3600	sn-be-tbl-ingestion-jobs	Built	songnodes/services/common/datalayer/jobs.py:20
succeeded job retention	ttl	JOB_RETENTION_DAYS '14'	sn-be-tbl-ingestion-jobs	Built	songnodes/services/worker/celery_app.py:93
target re-search interval	schedule	TARGET_RESEARCH_DAYS 7	sn-be-tbl-target-tracks	Built	songnodes/compose.dev.yaml:136 +1
worker → /scrape HTTP timeout	timeout	SCRAPE_TIMEOUT_SECONDS = 960	sn-be-svc-worker	Built	songnodes/services/common/datalayer/handlers.py:49
unified scraper spider timeout	timeout	asyncio.wait_for(..., timeout=900.0)	sn-be-svc-unified-scraper	Built	songnodes/scrapers/unified_scraper_api.py:596
scrapy LOGSTATS interval feeding progress	poll	LOGSTATS_SECONDS = 10	sn-be-svc-unified-scraper	Built	songnodes/scrapers/unified_scraper_api.py:112
rest-api scrape progress fetch timeout	timeout	httpx.AsyncClient(timeout=2.0)	sn-be-svc-rest-api	Built	songnodes/services/rest_api/routers/jobs.py:216
worker heartbeat staleness	heartbeat	time.time()-mtime < 90	sn-be-svc-worker	Built	songnodes/services/worker/Dockerfile
worker Docker HEALTHCHECK interval	heartbeat	--interval=30s --timeout=10s --start-period=40s	sn-be-svc-worker	Built	songnodes/services/worker/Dockerfile
postgres healthcheck	heartbeat	interval 3s, retries 30	sn-be-store-postgres	Built	songnodes/compose.dev.yaml:21
redis healthcheck	heartbeat	interval 3s, retries 20	sn-be-store-redis	Built	songnodes/compose.dev.yaml:31
rest-api healthcheck	heartbeat	interval 5s, timeout 6s, retries 20	sn-be-svc-rest-api	Built	songnodes/compose.dev.yaml:80
rabbitmq healthcheck	heartbeat	interval 10s, retries 12	sn-be-store-rabbitmq	Built	songnodes/compose.dev.yaml:91
unified-scraper healthcheck	heartbeat	interval 10s, start_period 20s	sn-be-svc-unified-scraper	Built	songnodes/compose.dev.yaml:179
audio-analysis healthcheck	heartbeat	interval 10s, start_period 30s	sn-be-svc-audio-analysis	Built	songnodes/compose.dev.yaml:192

start-dev compose up wait timeout	timeout	--wait-timeout 120	sn-be-cli-start-dev	● Built	songnodes/scripts/start-dev.sh:56
ref_mb statement timeout	timeout	MB_LOCAL_STATEMENT_TIMEOUT_MS 2000	sn-be-store-refmb	● Built	songnodes/compose.dev.yaml:69
graph-api CACHE_TTL (unused decorator)	ttl	CACHE_TTL default 300	sn-be-svc-graph-api	● Built	songnodes/services/graph-visualization-api/main.py:83
graph-api batch status key TTL	ttl	setex(batch:{id}, 3600)	sn-be-store-redis	● Built	songnodes/services/graph-visualization-api/main.py:475
graph-api pool and query timeout	timeout	pool_timeout=30, command_timeout 30	sn-be-svc-graph-api	● Built	songnodes/services/graph-visualization-api/main.py:93 +1
graph-api pool recycle	ttl	pool_recycle=3600	sn-be-svc-graph-api	● Built	songnodes/services/graph-visualization-api/main.py:92
Spotify/TIDAL OAuth state TTL	ttl	setex(..., 600)	sn-be-store-redis	● Built	songnodes/services/rest_api/routers/music_auth.py:415 +1
music search proxy timeout	timeout	ClientTimeout(total=5)	sn-be-svc-rest-api	● Built	songnodes/services/rest_api/main.py:693
AI proxy upstream timeout	timeout	ClientTimeout(total=60)	sn-be-svc-rest-api	● Built	songnodes/services/rest_api/routers/ai_proxy.py:85
401 rate-limit window	timeout	BLOCK_DURATION_SEC = 60	sn-be-svc-rest-api	● Built	songnodes/services/rest_api/security/auth.py:15
spider daily quota key TTL	ttl	SCRAPER_RUN_TTL_HOURS '24'	sn-be-store-redis	● Built	songnodes/scrapers/spiders/mixesdb_spider.py:89 +1
setlist dedupe key TTL	ttl	SCRAPER_SOURCE_TTL_DAYS '30'	sn-be-store-redis	● Built	songnodes/scrapers/spiders/mixesdb_spider.py:87 +1
Scrapy default DOWNLOAD_DELAY (randomised 0.5-1.5x)	debounce	DOWNLOAD_DELAY = 2	sn-be-svc-unified-scraper	● Built	songnodes/scrapers/settings.py:31
1001tracklists DOWNLOAD_DELAY	debounce	'DOWNLOAD_DELAY': 90.0	sn-be-svc-unified-scraper	● Built	songnodes/scrapers/spiders/1001tracklists_spider.py:85
AUTOTHROTTLE_MAX_DELAY	retry	AUTOTHROTTLE_MAX_DELAY = 60	sn-be-svc-unified-scraper	● Built	songnodes/scrapers/settings.py:105
Memgraph connection timeout	timeout	connection_timeout=5.0	sn-be-store-memgraph	● Built	songnodes/services/common/datalayer/graph_projection.py:37
Duo session cookie expiry (queued)	expiry	SESSION_EXPIRY_SECONDS = 7 * 24 * 60 * 60	sn-be-q-user-auth-router	▲ Queued	PR #578 services/rest_api/routers/user_auth.py
Duo auth state TTL (queued)	ttl	STATE_EXPIRY_SECONDS = 600	sn-be-q-user-auth-router	▲ Queued	PR #578 services/rest_api/routers/user_auth.py
Duo router Redis connect timeout (queued)	timeout	socket_connect_timeout=0.2	sn-be-q-user-auth-router	▲ Queued	PR #578 services/rest_api/routers/user_auth.py
pipeline telemetry poll	poll	startPolling(interval = 10000)	sn-fe-pipeline	● Built	songnodes/frontend/src/features/pipeline/usePipelineTelemetry.ts:317
pipeline poll backoff on error (never engages, stale closure)	retry	error ? 30000 : 10000	sn-fe-pipeline	● Built	songnodes/frontend/src/features/pipeline/usePipelineTelemetry.ts:321
job stall threshold	timeout	STALL_MS = 10 * 60 * 1000	sn-fe-pipeline	● Built	songnodes/frontend/src/features/pipeline/usePipelineTelemetry.ts:109
scrape job stages poll (tab visible)	poll	STAGES_POLL_MS = 5000	sn-fe-pipeline	● Built	songnodes/frontend/src/features/pipeline/ScrapeJobStages.tsx:10

target tracks search-state poll	poll	SEARCH_POLL_MS = 10_000	sn-fe-sources-targets	Built	songnodes/frontend/src/components/TargetTracksManager.tsx:28,90
library ingest job poll	poll	pollMs = 2000	sn-fe-library	Built	songnodes/frontend/src/features/musicLibrary/components/LibraryImportDialog.tsx:33,172
tracklist import stall	timeout	STALL_MS = 60_000	sn-fe-graph	Built	songnodes/frontend/src/components/TracklistImporter.tsx:35
ApiClient default request timeout	timeout	timeout: 30000 (DEFAULT_CONFIG.api.timeout)	sn-fe-home	Built	songnodes/frontend/src/types/index.ts:379 +1
token refresh buffer before expiry	expiry	REFRESH_BUFFER_MS = 5 * 60 * 1000	sn-fe-store-zustand	Built	songnodes/frontend/src/hooks/useTokenRefresh.ts:35 +1
token expiry check interval	poll	CHECK_INTERVAL_MS = 60 * 1000	sn-fe-store-zustand	Built	songnodes/frontend/src/hooks/useTokenRefresh.ts:38,312
localStorage mutex lease	lease	lockTimeoutMs = 10000	sn-fe-store-browser-ls	Built	songnodes/frontend/src/hooks/useTokenRefresh.ts:53
mutex acquire give-up	timeout	Date.now() - start > 12000	sn-fe-store-browser-ls	Built	songnodes/frontend/src/hooks/useTokenRefresh.ts:74-76
mutex retry spin	retry	setTimeout(res, 100)	sn-fe-store-browser-ls	Built	songnodes/frontend/src/hooks/useTokenRefresh.ts:78
default token lifetime when expires_in missing	expiry	(tokenData.expires_in || 3600) * 1000	sn-fe-oauth-callback	Built	songnodes/frontend/src/components/OAuthCallback.tsx:212 +1
callback pop-up close / redirect delay	timeout	setTimeout(..., 1500)	sn-fe-oauth-callback	Built	songnodes/frontend/src/components/OAuthCallback.tsx:267-270
callback error redirect delay	timeout	setTimeout(..., 3000)	sn-fe-oauth-callback	Built	songnodes/frontend/src/components/OAuthCallback.tsx:109,287
toast visible time (8 s with action)	timeout	left = current.action ? 8000 : 4000	sn-fe-home	Built	songnodes/frontend/src/components/ds/index.tsx:1454
"slow" loading hint	timeout	setTimeout(() ⇒ setSlow(true), 1000)	sn-fe-library	Built	songnodes/frontend/src/features/musicLibrary/LibraryPage.tsx:99 +1
clear-draft double-press window	timeout	setTimeout(() ⇒ setArmedNew(false), 3000)	sn-fe-sets	Built	songnodes/frontend/src/features/musicLibrary/SetBuilderPage.tsx:228
minimap viewport debounce	debounce	VIEWPORT_UPDATE_DEBOUNCE_MS = 100	sn-fe-graph	Built	songnodes/frontend/src/components/GraphMiniMap.tsx:16
InfoTip hover delay	debounce	HOVER_DELAY = 500	sn-fe-home	Built	songnodes/design-system/components/bundle.js:657
PIXI init retry delay	retry	setTimeout(() ⇒ initializePixi(), 100)	sn-fe-graph	Built	songnodes/frontend/src/components/GraphVisualization.tsx:755
copied/saved status reset	timeout	setTimeout(..., 2000)	sn-fe-sources-import	Built	songnodes/frontend/src/features/aiSetlists/components/ui.tsx:20 +1
nginx DNS re-resolve TTL	ttl	resolver 127.0.0.11 valid=10s	sn-fe-store-nginx	Built	songnodes/frontend/Dockerfile:36
frontend container healthcheck	heartbeat	interval: 10s, timeout: 5s, retries: 6	sn-fe-store-nginx	Built	songnodes/compose.dev.yaml:204-209
hashed asset cache	ttl	expires 1y	sn-fe-store-nginx	Built	songnodes/frontend/Dockerfile:40-43

Table: **Size and count limits** 76 limits

76 rows

LIMIT	VALUE	CURRENT	WHERE	STATE	EVIDENCE
ABC plays page size	100 items	–	ml-cli-scrape	Built	musicLibrary/pipeline/djlib/scrapers.py:90
episodes per program default	2 episodes	–	ml-cli-scrape	Built	musicLibrary/pipeline/djlib/scrapers.py:54 +1
ARIA parse sanity floor	40 rows	50	ml-cli-scrape	Built	musicLibrary/pipeline/djlib/scrapers.py:122

LIMIT	VALUE	CURRENT	WHERE	STATE	EVIDENCE
TIDAL search results kept	20 tracks	–	ml-cli-resolve	Built	musicLibrary/pipeline/djlib/tidal.py:207
TIDAL /tracks filter[id] and ISRC batch	20 ids	–	ml-cli-resolve	Built	musicLibrary/pipeline/djlib/tidal.py:217 +1
TIDAL playlist add/remove chunk	50 tracks	–	ml-cli-sync-tidal	Built	musicLibrary/pipeline/djlib/tidal.py:258 +1
Deezer search limit per query	25 tracks	–	ml-cli-resolve	Built	musicLibrary/pipeline/djlib/deezer.py:20
resolve thread pool	4 workers	–	ml-cli-resolve	Built	musicLibrary/pipeline/run_weekly.py:456
resolve flush interval	20 songs	–	ml-cli-resolve	Built	musicLibrary/pipeline/run_weekly.py:459
attempts before UNRESOLVED	3 attempts	max 2 in pending	ml-cli-process	Built	musicLibrary/pipeline/run_weekly.py:236
rejected re-check threshold	2 appearances	–	ml-cli-ingest	Built	musicLibrary/pipeline/djlib/ledger.py:114
high_crate_cap	60 tracks per crate	7	ml-cli-process	Built	musicLibrary/config/rules.json +1
std_crate_cap (not enforced)	250 tracks per crate	131	ml-cli-process	Built	musicLibrary/config/rules.json
promote_min_sources	2 sources	–	ml-cli-process	Built	musicLibrary/config/rules.json
promote_min_saves	50 saves	–	ml-cli-process	Built	musicLibrary/config/rules.json
high_chart_saturation_weeks	6 weeks	–	ml-cli-process	Built	musicLibrary/config/rules.json
duration floors	{'MEL_House': 285, 'MEL_Techno': 285, 'UK_Garage': 225} seconds	–	ml-cli-process	Built	musicLibrary/config/rules.json +1
SET playlist name length	250 characters	–	ml-cli-sync-sets	Built	musicLibrary/pipeline/run_weekly.py:539
runs kept in library.json	26 runs	4	ml-cli-build-db	Built	musicLibrary/pipeline/build_db.py:224
ArtifactData list page	1000 documents	0	ml-agent-weekly-run	Built	musicLibrary/agent/weekly_run.md:5
Beatport actions before check-in	25 actions	–	ml-agent-beatport-sync	Built	musicLibrary/agent/beatport_sync.md:7
artifact page and file size limit (platform)	16 MB	{'app.js': 372921, 'library.json': 185443, 'style.css': 262326}	ml-store-artifact-files	Built	Artifact tool contract
songnodes library search limit used by the page	5000 tracks	413	sn-fe-library	Built	songnodes/frontend/src/features/musicLibrary/data.ts:114

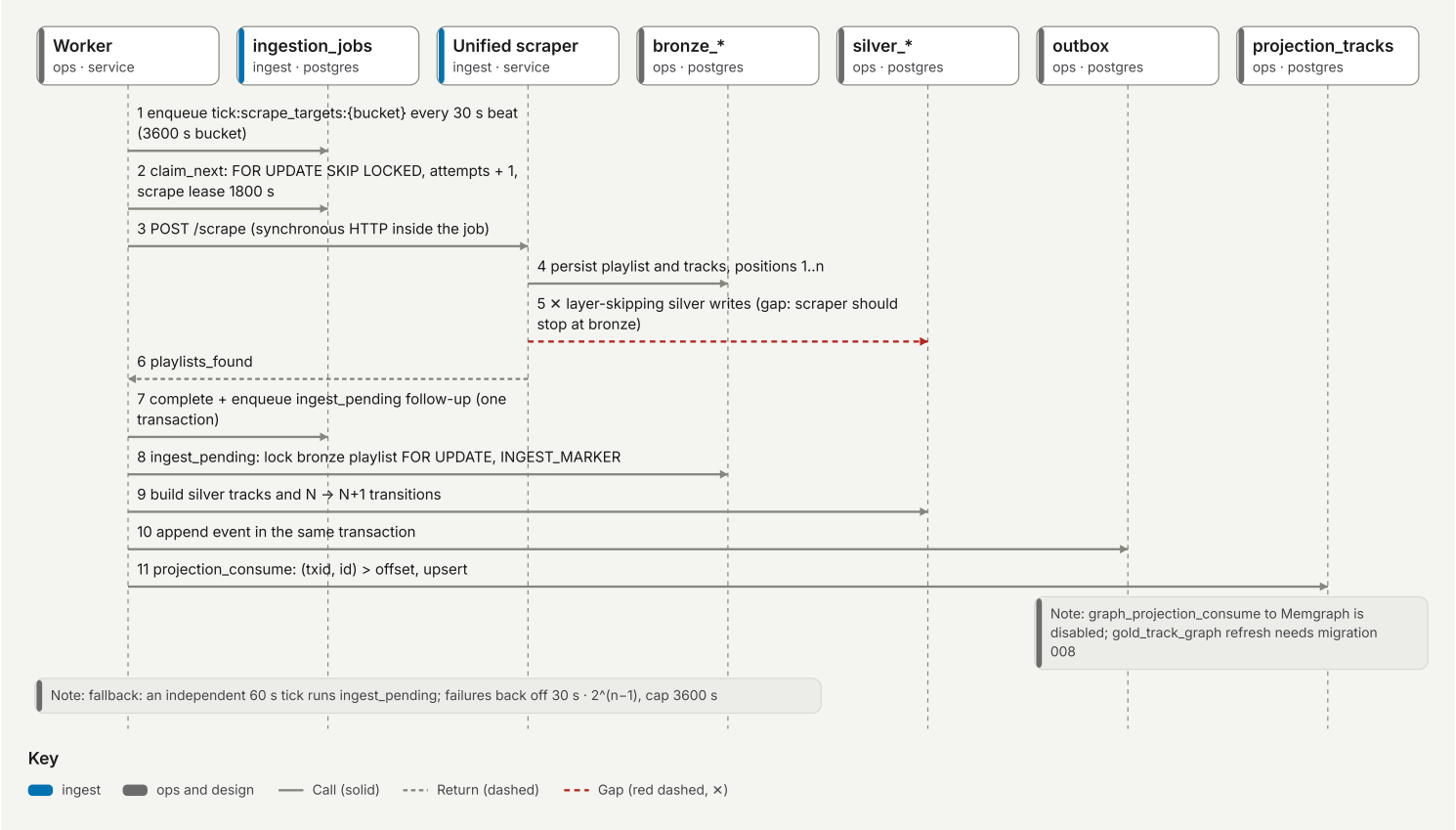
songnodes Idempotency-Key length	200 characters	39	sn-be-svc-rest-api	● Built	songnodes/services/rest_api/routers/library_ingest.py:12
ref-mb memory and buffers	{'mem_limit': '4g', 'shm_size': '1g', 'shared_buffers': '1GB', 'max_wal_size': '8GB', 'work_mem': '32MB'} bytes	–	ml-svc-ref-mb	● Built	musicLibrary/ref_mb/compose.yaml
MetaBrainz parallel downloads	1 connections	–	ml-cli-fetch-metabrainz	● Built	musicLibrary/pipeline/fetch_metabrainz.sh
data folder size	58 GB	–	ml-store-data	● Built	musicLibrary/_data_manifest.txt
MusicBrainz web API rate (brief rule)	1 request per second	–	ml-queued-wp7-pipeline-client	▲ Queued	musicLibrary/plan/agents/WP7_pipeline_client.md
MAX_JOBS_PER_DRAIN	25 jobs	–	sn-be-svc-worker	● Built	songnodes/services/worker/celery_app.py:31
Celery worker processes	2 processes	–	sn-be-svc-worker	● Built	songnodes/services/worker/Dockerfile
worker_prefetch_multiplier	1 tasks	–	sn-be-svc-worker	● Built	songnodes/services/worker/celery_app.py:45
job max_attempts default	3 attempts	–	sn-be-tbl-ingestion-jobs	● Built	songnodes/services/common/datalayer/jobs.py:27
outbox handler attempts before dead letter	3 attempts	–	sn-be-tbl-outbox	● Built	songnodes/services/common/datalayer/outbox.py:81
stored last_error length	2000 characters	–	sn-be-tbl-ingestion-jobs	● Built	songnodes/services/common/datalayer/jobs.py:78
ingest_pending playlists per job (standalone tracks x4)	25 playlists	–	sn-be-svc-worker	● Built	songnodes/services/common/datalayer/handlers.py:98 +1
playlist entries accepted	5000 tracks	–	sn-be-svc-worker	● Built	songnodes/services/common/datalayer/ingest.py:25
enrichment_backfill batch_size	25 tracks	–	sn-be-svc-worker	● Built	songnodes/services/common/datalayer/handlers.py:125
camelot_repair / transition_scoring batch	500 rows	–	sn-be-svc-worker	● Built	songnodes/services/common/datalayer/handlers.py:137 +1
projection_consume batch x max_batches	200 events (x50)	–	sn-be-svc-worker	● Built	songnodes/services/common/datalayer/handlers.py:160
graph_projection_consume batch x max_batches	100 events (x50)	–	sn-be-svc-worker	● Built	songnodes/services/common/datalayer/handlers.py:191
SCRAPE_TARGETS_PER_RUN	2 targets (x sources)	–	sn-be-svc-worker	● Built	songnodes/compose.dev.yaml:135
AUDIO_ANALYSIS_PER_RUN	10 tracks	–	sn-be-svc-worker	● Built	songnodes/compose.dev.yaml:129
scraper persistence batch_size	50 items	–	sn-be-svc-unified-scraper	● Built	songnodes/scrapers/pipelines/persistence_pipeline.py:188

CONCURRENT_REQUESTS	8 requests	–	sn-be-svc-unified-scraper	● Built	songnodes/scrapers/settings.py:28
GET /api/graph/nodes and /data limit max	15000 nodes	–	sn-be-svc-graph-api	● Built	songnodes/services/graph-visualization-api/main.py:557 +1
GET /api/graph/edges limit max	50000 edges	–	sn-be-svc-graph-api	● Built	songnodes/services/graph-visualization-api/main.py:565
neighbourhood view max tracks / steps	2000 tracks (steps ⩽ 6)	–	sn-be-svc-graph-api	● Built	songnodes/services/graph-visualization-api/main.py:579
WebSocket connections per room / total	100 connections (total 1000)	–	sn-be-svc-graph-api	● Built	songnodes/services/graph-visualization-api/main.py:110
rest-api asyncpg pool	15 connections (min 5)	–	sn-be-svc-rest-api	● Built	songnodes/services/rest_api/main.py:146
graph-api SQLAlchemy pool + overflow	30 connections (15+15)	–	sn-be-svc-graph-api	● Built	songnodes/services/graph-visualization-api/main.py:89
Idempotency-Key length	200 characters	–	sn-be-svc-rest-api	● Built	songnodes/services/rest_api/routers/jobs.py:69 +1
scrape search_query length	255 characters	–	sn-be-svc-rest-api	● Built	songnodes/services/rest_api/routers/jobs.py:68
AI proxy body size	102400 bytes	–	sn-be-svc-rest-api	● Built	songnodes/services/rest_api/routers/ai_proxy.py:14
AI proxy max_tokens cap	8000 tokens	–	sn-be-svc-rest-api	● Built	songnodes/services/rest_api/routers/ai_proxy.py:70
401 attempts before block	5 attempts/60s/IP	–	sn-be-svc-rest-api	● Built	songnodes/services/rest_api/security/auth.py:14
audio-analysis mem_limit	1 GiB	–	sn-be-svc-audio-analysis	● Built	songnodes/compose.dev.yaml:191
CLAUDE.md resource limits (DB 1-2GB, APIs 512MB, scrapers 1GB, frontend 256MB)	None not enforced	–	compose.dev.yaml	● Built	songnodes/CLAUDE.md:252 +1
Duo router PBKDF2 iterations (queued)	100000 iterations	–	sn-be-q-user-auth-router	▲ Queued	PR #578 services/rest_api/routers/user_auth.py
library search page size	5000 tracks	294 tracks in the pinned fixture (frontend/tests/e2e/data/library.fixture.json)	sn-fe-library	● Built	songnodes/frontend/src/features/musicLibrary/data.ts:114
library rows rendered first	150 rows	–	sn-fe-library	● Built	songnodes/frontend/src/features/musicLibrary/LibraryPage.tsx:27
library export upload max	50 MB	fixture export 132,486 bytes	sn-fe-library	● Built	songnodes/frontend/src/features/musicLibrary/components/LibraryImportDialog.tsx:24

frontend nginx request body (default, unset)	1 MB	–	sn-fe-store-nginx	Built	songnodes/frontend/Dockerfile:28-102 +1
tracklist upload max	5 MB	–	sn-fe-graph	Built	songnodes/frontend/src/components/TracklistImporter.tsx:306
graph nodes/edges per load	10000 rows	–	sn-fe-graph	Built	songnodes/frontend/src/hooks/useDataLoader.ts:28
neighbourhood view cap (graph-hood-max token)	200 nodes	–	sn-fe-graph	Built	songnodes/frontend/src/lib/graphView.ts:2-10 +1
jobs listed on pipeline page	200 jobs	–	sn-fe-pipeline	Built	songnodes/frontend/src/features/pipeline/usePipelineTelemetry.ts:275
scrape jobs listed	50 jobs	–	sn-fe-pipeline	Built	songnodes/frontend/src/features/pipeline/usePipelineTelemetry.ts:276
scrape stage cards kept per visit	5 cards	–	sn-fe-pipeline	Built	songnodes/frontend/src/pages/ScraperProgressPage.tsx:33-36
artist attribution page size	50 rows	–	sn-fe-graph	Built	songnodes/frontend/src/components/ArtistAttributionManager.tsx:79
recent searches kept per namespace	10 items	–	sn-fe-store-browser-ls	Built	songnodes/frontend/src/utils/recentSearches.ts:47
autocomplete recent / suggestions	8 items (5 recent, 8 suggestions)	–	sn-fe-paths	Built	songnodes/frontend/src/components/ds/AutocompleteSearchField.tsx:55-56
quick search results	12 items	–	sn-fe-graph	Built	songnodes/frontend/src/components/QuickSearch.tsx:174-191
AI prompt history kept	30 items	–	sn-fe-store-browser-ls	Built	songnodes/frontend/src/services/aiService.ts:386
edges fetched for details	2 edges	–	sn-fe-graph	Built	songnodes/frontend/src/components/EdgeDetailsModal.tsx:29
toasts queued	3 toasts	–	sn-fe-home	Built	songnodes/design-system/components/Toast/preview.html:1
unified-scraper memory (compose; frontend has none set, CLAUDE.md says 256MB)	1 GB	–	sn-fe-store-nginx	Built	songnodes/compose.dev.yaml:191 +1

D6a: **Sequence – Scrape to bronze, silver and projections**

11 messages across 7 participants, numbered in order; red marks where the protocol leaks.

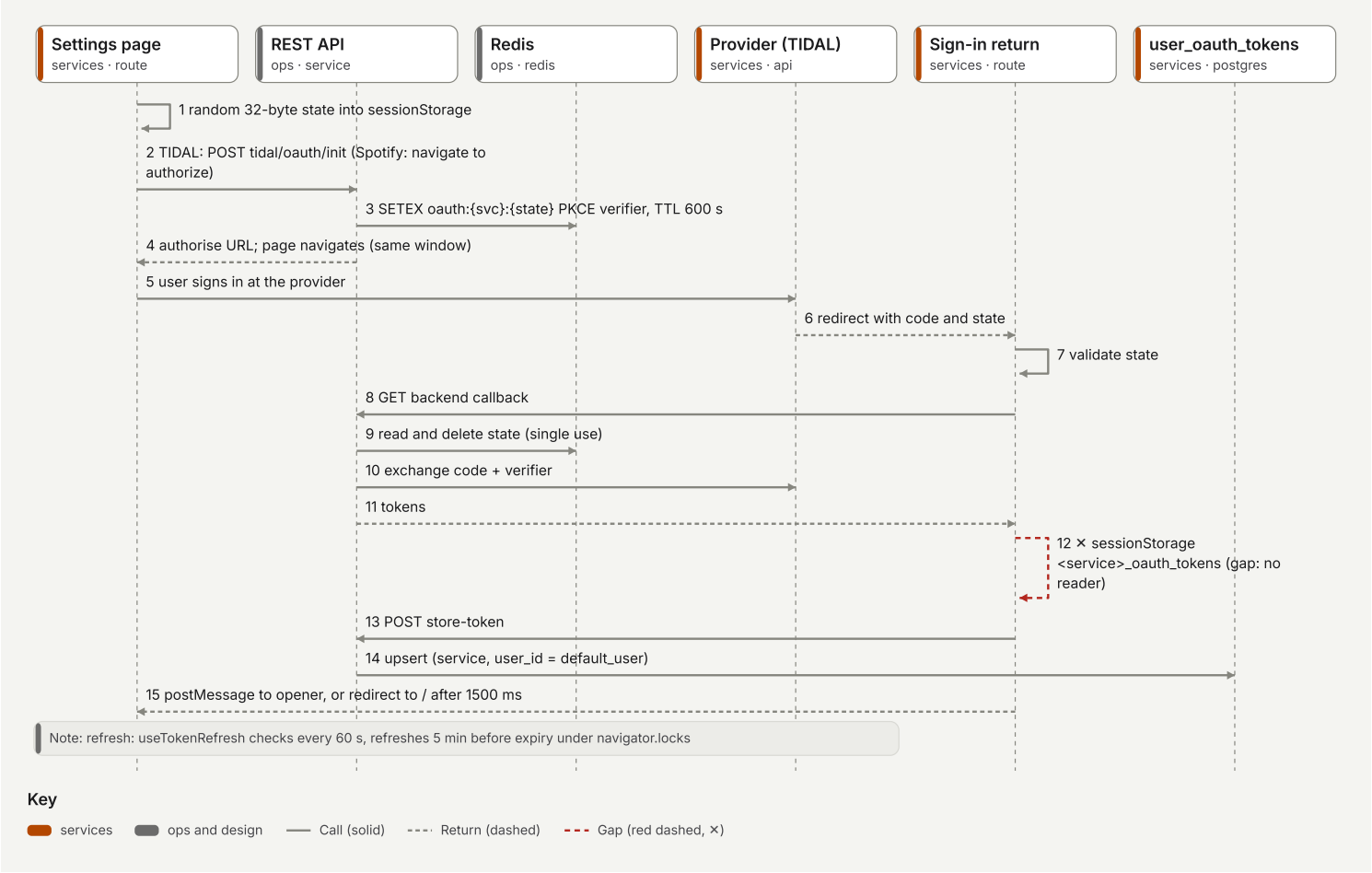


D6a · n = 11 · Unit: messages · ● Built

· Source: protocols[] sn-be-proto-recurring-schedule, sn-be-proto-job-queue, sn-be-proto-scrape-etl-order, sn-be-proto-outbox (algorithm, conflict_resolution, stale_fallback); timings[] · Method: hand-placed UML sequence (lifelines in first-use order, one row per message); protocols chosen per theme by flows touching their participants (sn-be-proto-recurring-schedule 6, sn-be-proto-job-queue 31, sn-be-proto-scrape-etl-order 10, sn-be-proto-outbox 0); 0 crossings

D6c: **Sequence – TIDAL and Spotify OAuth connect**

15 messages across 6 participants, numbered in order; red marks where the protocol leaks.

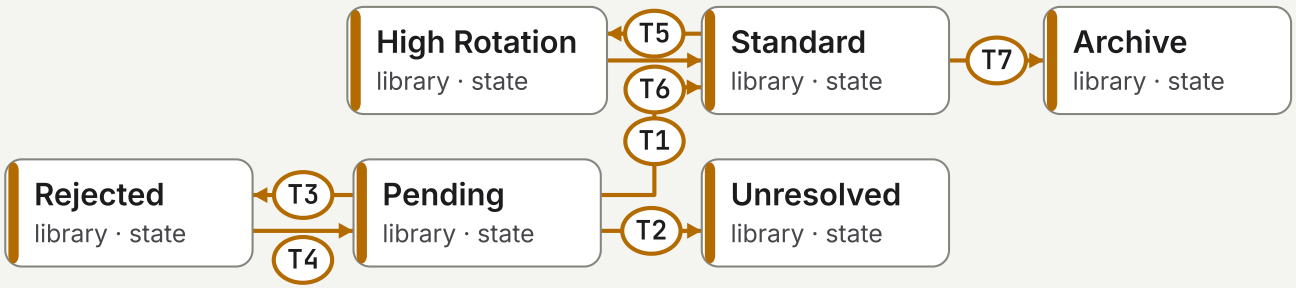


D6c · n = 15 · Unit: messages · ● Built

· Source: protocols[] sn-fe-proto-oauth, sn-be-proto-oauth-pkce (algorithm, conflict_resolution, stale_fallback); timings[] · Method: hand-placed UML sequence (lifelines in first-use order, one row per message); protocols chosen per theme by flows touching their participants (sn-fe-proto-oauth 4, sn-be-proto-oauth-pkce 25); 0 crossings

D7b: **State machine – Track rotation**

6 states, 7 transitions; every timing on an arrow comes from the timings table.



Transitions

T1 Pending → Standard: resolved and passes the gates

T2 Pending → Unresolved: 3 resolve attempts fail (resolve_attempts)

T3 Pending → Rejected: fails a gate (for example a duration floor)

T4 Rejected → Pending: re-checked after 2 sightings (recheck threshold)

T5 Standard → High Rotation: promote: min 2 sources, min 50 saves, window 30 days; chart entry window 21 days; cap 60

T6 High Rotation → Standard: tenure > 42 days, silence 45 days or chart saturation 6 weeks

T7 Standard → Archive: age ≥ 180 days and silence 60 days

Key

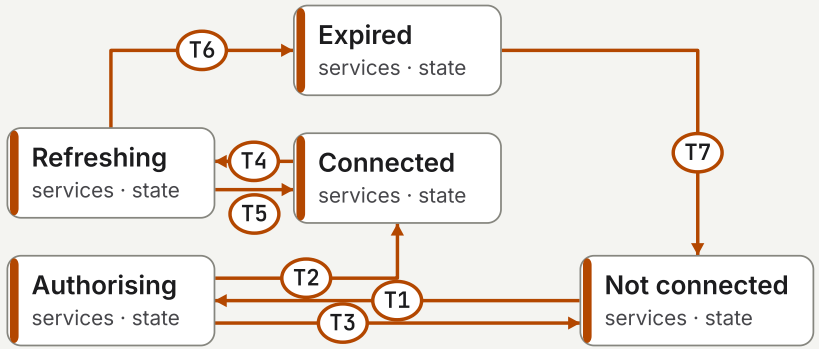
library — Transition (badge T = row in the transition list)

D7b · n = 7 · Unit: transitions · ● Built

· Source: timings ml-t-promote-window, ml-t-chart-entry, ml-t-high-tenure, ml-t-high-silence, ml-t-std-archive-age, ml-t-std-archive-silence; limits ml-l-promote-sources, ml-l-promote-saves, ml-l-high-cap, ml-l-chart-sat, ml-l-resolve-attempts, ml-l-recheck · Method: graphviz dot layered layout (LR, order reverse; 12 rank directions and input orders tried, scored by legibility then crossings), splines=ortho, own number badges, type floor 12 units; 0 crossings, 0 edges through nodes

D7c: **State machine – OAuth token**

5 states, 7 transitions; every timing on an arrow comes from the timings table.



Transitions

T1 Not connected → Authorising: connect: state in sessionStorage, PKCE verifier in Redis (TTL 600 s)

T2 Authorising → Connected: callback validates state, exchanges code, stores token (default_user)

T3 Authorising → Not connected: state expired or error: redirect after 3000 ms; musicLibrary PKCE link > 900 s

T4 Connected → Refreshing: checked every 60 s; 5 min before expiry (default lifetime 3600 s)

T5 Refreshing → Connected: refresh under navigator.locks (localStorage mutex fallback)

T6 Refreshing → Expired: refresh fails

T7 Expired → Not connected: user signs in again

Key

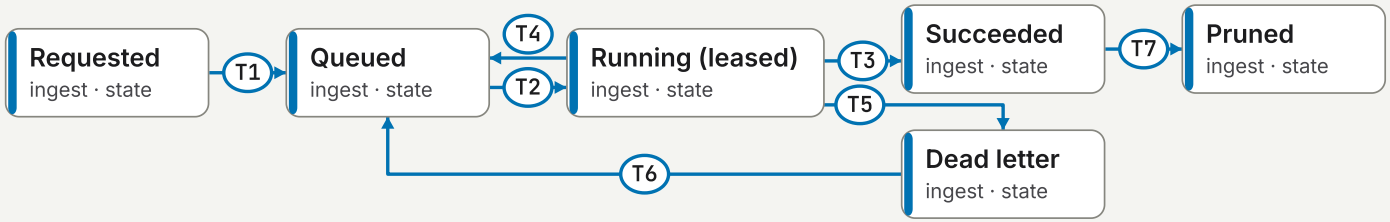
services — Transition (badge T = row in the transition list)

D7c · n = 7 · Unit: transitions · ● Built

· Source: protocols sn-fe-proto-oauth, sn-be-proto-oauth-pkce, ml-proto-tidal-login; timings sn-be-t-oauth-state, sn-fe-t-token-buffer, sn-fe-t-token-check, sn-fe-t-token-default-expiry, sn-fe-t-oauth-error-redirect, ml-t-pkce-age · Method: graphviz dot layered layout (LR, order reverse; 12 rank directions and input orders tried, scored by legibility then crossings), splines=ortho, own number badges, type floor 12 units; 0 crossings, 0 edges through nodes

D7a: **State machine – Scrape and ingestion job**

6 states, 7 transitions; every timing on an arrow comes from the timings table.



Transitions

T1 Requested → Queued: enqueue: INSERT ON CONFLICT (kind, idempotency_key) DO NOTHING returns the existing job

T2 Queued → Running (leased): claim_next: run_after ≤ now, FOR UPDATE SKIP LOCKED, attempts + 1

T3 Running (leased) → Succeeded: handler, follow-up enqueues and complete in one transaction

T4 Running (leased) → Queued: fail: requeue with backoff 30 s · 2^(n−1), cap 3600 s (×4 transient); or lease expired (300 s, scrape 1800 s) and reclaimed

T5 Running (leased) → Dead letter: attempts ≥ max_attempts, or permanent 4xx (not 408/429)

T6 Dead letter → Queued: POST /api/v1/jobs/{id}/retry

T7 Succeeded → Pruned: pruned after 14 days

Key

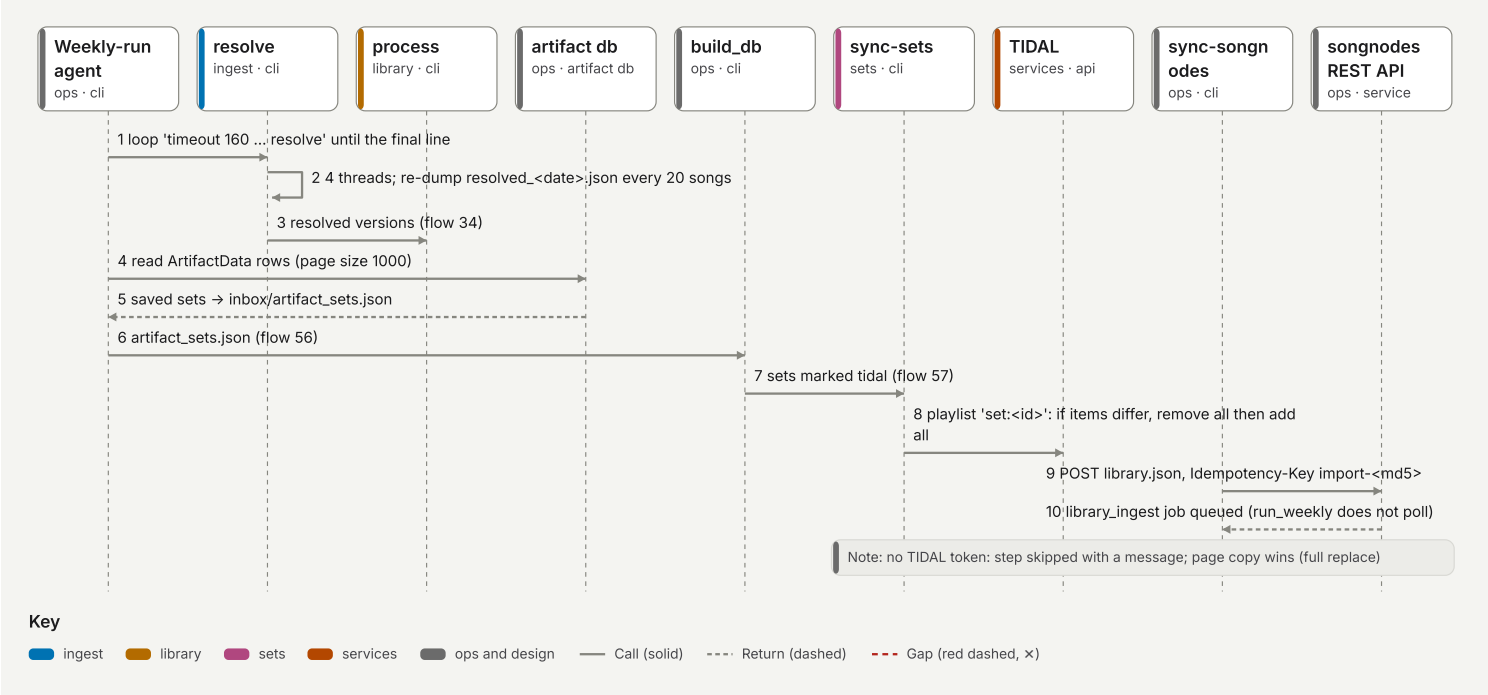
ingest — Transition (badge T = row in the transition list)

D7a · n = 7 · Unit: transitions · ● Built

· Source: protocols sn-be-proto-job-queue, sn-be-proto-scrape-etl-order; timings sn-be-t-lease-default, sn-be-t-lease-scrape, sn-be-t-backoff-base, sn-be-t-backoff-max · Method: graphviz dot layered layout (LR, order given; 12 rank directions and input orders tried, scored by legibility then crossings), splines=ortho, own number badges, type floor 12 units; 0 crossings, 0 edges through nodes

D6b: **Sequence – Weekly musicLibrary run to TIDAL and songnodes**

10 messages across 9 participants, numbered in order; red marks where the protocol leaks.

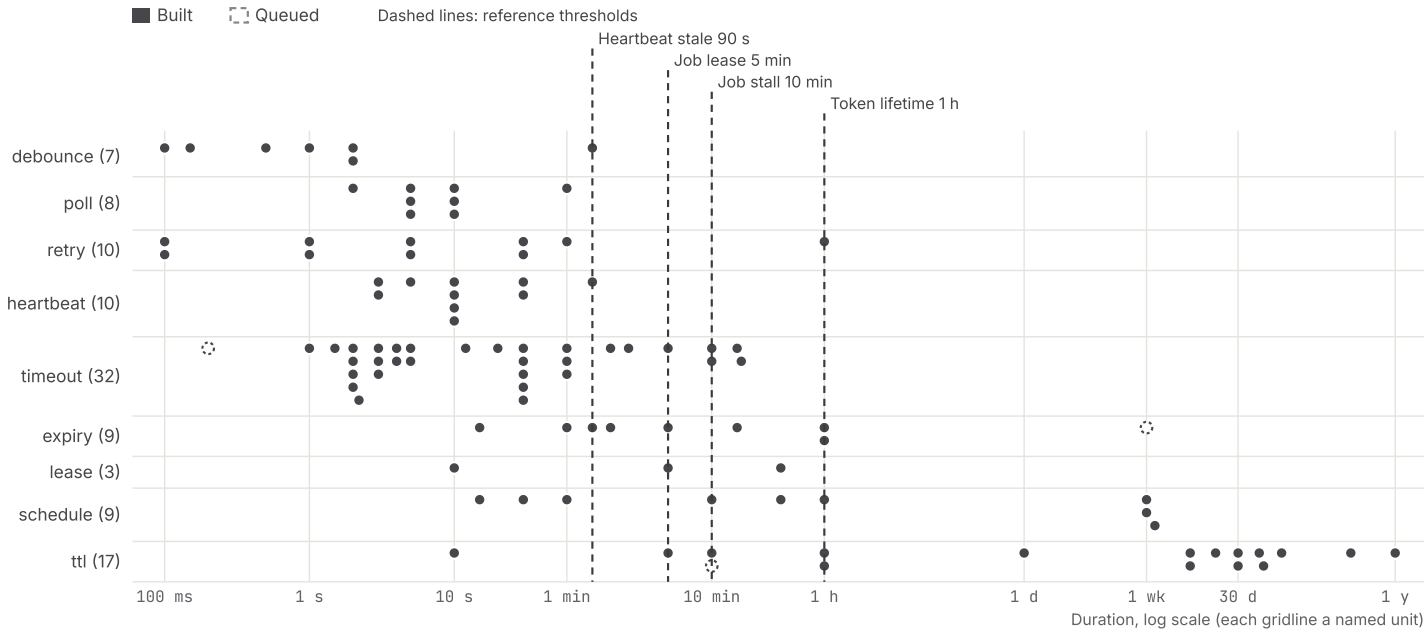


D6b · n = 10 · Unit: messages · ● Built

· Source: protocols[] ml-proto-resolve-resume, ml-proto-sync-sets, ml-proto-songnodes-ingest (algorithm, conflict_resolution, stale_fallback); timings[] · Method: hand-placed UML sequence (lifelines in first-use order, one row per message); protocols chosen per theme by flows touching their participants (ml-proto-resolve-resume 7, ml-proto-sync-sets 9, ml-proto-songnodes-ingest 2); 0 crossings

C5: **Timing constants on a log axis**

105 timing constants span 100 ms to 365 d; reference lines mark Heartbeat stale (sn-be-t-heartbeat-stale), Job lease (sn-be-t-lease-default), Job stall (sn-fe-t-stall), Token lifetime (ml-t-tidal-user-default).



C5 · n = 105 · Unit: milliseconds (log scale) · **i Projected**

· Source: timings[].value_ms, timings[].kind, timings[].state · Method: Each constant is one dot on a log10 axis, one lane per kind (kinds ordered by median value); coincident dots stack. Reference lines are the timings matched by name: heartbeat staleness, default job lease, job stall threshold and default token lifetime.

DJ performing

Objective: Pick the next track from proven transitions while a set is playing.

Device: laptop in the DJ booth (desktop browser, booth theme available)

Entry point: `sn-fe-graph`

Basis: Inferred from pages and docs

Routine

- open /graph?mode=play
- search a track (QuickSearch / Ctrl K)
- read neighbours and edge details
- check key on /harmony

3 rows

STEP	PAGE	ACTION	DATA WRITTEN	TRIGGER TO NEXT	HANDOFF	FALLBACK
1	sn-fe-graph	load graph and select current track	songnodes-store viewState, sn-graph-view, sn-recent-quick-search	open /graph	–	/api/graph fallback, then error state
2	sn-fe-graph	open edge details	–	pointer on edge → GET /api/graph/edges	–	none
3	sn-fe-harmony	check Camelot moves	–	NavDrawer Harmony	track id in route / set draft via /paths or Library add-to-set	none

Failure modes and timeouts

- graph API down: useDataLoader falls back to /api/graph then shows error
- TIDAL token expired: useTokenRefresh refresh under lock

Set planner

Objective: Shape a wave-shaped set from the crate, check every transition and send it to TIDAL.

Device: browser on claude.ai (desktop or phone); desktop browser at home (127.0.0.1:3006)

Entry point: `ml-page-set-builder`

Basis: Inferred from pages and docs

Routine

- open the Set Builder artifact, filter Library, add tracks
- switch to Set builder, choose shape and minutes, Build set
- Save set, tick Send to TIDAL on the weekly run
- wait for the next weekly run to create 'SET · <name>'
- filter the crate on /library
- Build set from crate
- shape and reorder on /sets
- save, export, send to TIDAL

8 rows

STEP	PAGE	ACTION	DATA WRITTEN	TRIGGER TO NEXT	HANDOFF	FALLBACK
1	`ml-page-set-builder`	filter and add to set	localStorage sn-filters-library, sn-set-draft	user	–	–
2	`ml-page-set-builder`	Save set with tidal flag	artifact db sets/<id>, localStorage sn-music-library-sets	user	SavedSet {name,shape,minutes,mix,pool,tidal,seed,tracks[{id,tgt,...,move}],updated,created} via artifact db collection sets	no db capability: set kept in localStorage only and never reaches the weekly run
3	`ml-agent-weekly-run`	export sets	inbox/artifact_sets.json	schedule	–	–
4	`ml-cli-sync-sets`	push to TIDAL	TIDAL playlist, secrets/tidal_playlists.json	schedule	–	TIDAL not logged in: skipped with a message
5	`sn-fe-library`	filter and add tracks	sn-filters-library, sn-set-draft, sn-favs-library-crate	filter rail	–	filters kept in memory if storage blocked
6	`sn-fe-sets`	build from crate and shape energy	sn-set-draft	/sets?pool=crate&...	–	draft kept in memory
7	`sn-fe-sets`	save set	PUT /api/v1/library/sets/{id}; sn-music-library-sets	Save	–	sets kept on device (source device)
8	`sn-fe-sets`	export or send	download .m3u8/.csv/.json/.xml or artifact_sets.json; POST /api/v1/tidal/sets/send	Export / Send to TIDAL	artifact_sets.json (SavedSet[]) via manual file copy to E:\musicLibrary\inbox	download works offline; TIDAL send shows HTTP error

Failure modes and timeouts

- up to a week of latency before a set reaches TIDAL
- Camelot moves and BPM gaps are inert: M3 render says none of the 413 tracks has a key and BPM
- a set edited while the db capability is unavailable is overwritten by the shared copy on next load (shared wins by id)
- songnodes /sets has its own Send to TIDAL that creates playlists directly, so the same set can end up as two TIDAL playlists
- server unreachable: sets kept on device and merged by content later
- set created twice if PUT /0 returns no id (content match fixes on next read)

Library curator

Objective — Keep the crates right each week: review, buy, override and import.

Device — laptop tuf-boi (E:\musicLibrary, Excel/DB Browser) and phone for the run message; desktop browser on the machine running the Compose stack

Entry point — `ml-agent-weekly-run`

Basis — Inferred from pages and docs

Routine

- read the weekly run message and outputs/<date>/weekly_report.md
- open review_queue.csv, fill energy_score_override, copy rows into overrides/energy_overrides.csv
- buy High Rotation tracks from buy_list.csv and put local_path into overrides
- add heard tracks to inbox/manual_adds.csv
- occasionally run calibrate.py and copy agreed values into config/rules.json
- browse db/dj_library.sqlite in DB Browser

- import the weekly library.json export
- review tracks needing an energy check
- mark missing tracks as targets
- manage targets on /sources/targets

8 rows

STEP	PAGE	ACTION	DATA WRITTEN	TRIGGER TO NEXT	HANDOFF	FALLBACK
1	ml-agent-weekly-run	read summary	–	scheduled task message	–	–
2	ml-cli-process	review_queue.csv to overrides CSV (manual copy)	overrides/energy_overrides.csv	weekly	energy_overrides.csv rows via file	none; rows ignored if not copied
3	ml-cli-scrape	manual adds	inbox/manual_adds.csv	heard a track	–	–
4	ml-cli-calibrate	fit rules	outputs/calibration_suggestions.json	ad hoc	–	–
5	sn-fe-library	Import library export	POST /api/v1/library/ingest; job polled	Import dialog FileDrop .json	–	Retry on dead job
6	sn-fe-library	review rev=true rows	sn-filters-library	Banner Review link	–	none
7	sn-fe-library	mark target	POST /api/v1/target-tracks; sn-target-tracks-v1	TargetActionButton	–	pendingSync=true kept locally
8	sn-fe-sources-targets	edit priorities and search	PUT/POST /api/v1/target-tracks	NavDrawer Sources	target_tracks rows via /api/v1/target-tracks → scrape jobs	toast error

Failure modes and timeouts

- review queue lists all 413 tracks because no track has a BPM, so it cannot be worked through
- 0 overrides, 0 local_path, no rekordbox_collection.xml: none of the three levers has been used, so calibrate has nothing to fit
- review_queue.csv override column is not read by any code; rows must be copied by hand into a CSV with different columns
- failed target sync stays pendingSync=true forever
- ingest job dead: Retry button

Pipeline operator

Objective: Keep the stack, scrapers and imports healthy.

Device: laptop tuf-boi (Windows, Git Bash + Docker Desktop); 127.0.0.1 ports; desktop browser on the Docker host

Entry point: `sn-be-cli-start-dev`

Basis: Inferred from pages and docs

Routine

- bash scripts/start-dev.sh after git pull on main
- bash scripts/import-music-library.sh after the musicLibrary weekly run
- Set the scrape window (PUT /api/v1/scrape/window) and target tracks
- Watch /api/v1/jobs and /api/v1/jobs/{id}/stages; retry dead jobs
- Back up Postgres with the CLAUDE.md pg_dump one-liner
- queue a scrape for a target
- watch medallion stages and jobs
- retry dead jobs
- set scrape window
- connect TIDAL/Spotify and API keys in Settings

- import AI-found setlists

10 rows

STEP	PAGE	ACTION	DATA WRITTEN	TRIGGER TO NEXT	HANDOFF	FALLBACK
1	sn-be-cli-start-dev	git switch main && git pull; bash scripts/start-dev.sh	images, schema migrations, ingestion_jobs transition_scoring seed	code change on main	–	re-run; never reset the Postgres volume
2	sn-be-cli-import-music-library	bash scripts/import-music-library.sh after the musicLibrary weekly run	ingestion_jobs library_ingest	weekly musicLibrary run	library.json (export shape) via sn-be-sig-import-music-library	script retries a dead job once, then prints the error
3	sn-be-svc-worker	worker runs library_ingest, projection and backfill follow-ups	library.*, target_tracks, outbox, projection_tracks	beat drain	outbox events via sn-be-sql-outbox	projection_reconcile job
4	sn-be-svc-rest-api	PUT /api/v1/scrape/window and manage /api/v1/target-tracks	scrape_window, target_tracks	planning	–	no window = scrape every hour
5	sn-be-svc-rest-api	watch /api/v1/jobs and /api/v1/jobs/{id}/stages; POST /retry for dead jobs	ingestion_jobs	dead letter or warning	–	docker compose logs worker
6	sn-be-store-postgres	pg_dump -Fc via compose exec	backup-YYYYMMDD.dump	before risky change	–	none: scripts/backup_database.sh targets the wrong database
7	sn-fe-pipeline	queue scrape	POST /api/v1/scrape/trigger	scrape form	–	toast with HTTP detail
8	sn-fe-pipeline	retry job	POST /api/v1/jobs/{id}/retry	ConfirmDialog	–	none
9	sn-fe-settings	connect TIDAL	sessionStorage state; backend token store	/settings/tidal-connection	–	error banner on Settings
10	sn-fe-sources-import	find and import AI setlists	POST /api/v1/tracks/import-tracklist; songnodes_ai_setlists_ledger	/sources/import	job rows via REST → RabbitMQ/Celery (backend scope)	ledger kept locally

Failure modes and timeouts

- import-music-library.sh sends no API token, so it gets 401 once SONGNODES_API_TOKEN is set
- backup scripts in scripts/ and root target musicdb/songnodes_* names and back up nothing
- music search returns 503 because metadata-enrichment is not in compose
- hourly 1001tracklists target scrapes after the first of the day hit the Redis daily quota and report found_nothing
- polling backoff never engages (stale error closure)
- same-window OAuth leaves browser store without tokens

Developer

Objective: Ship one-job packages through PRs without breaking contracts or the design system.

Device: laptop or cloud agent session (Jules/Claude) with Docker for E2E; developer workstation and claude.ai artifact viewer; laptop tuf-boi (Git Bash / Linux VM shell)

Entry point: `sn-be-cli-scope-guard-g1`

Basis: Inferred from pages and docs

Routine

- branch <code>-<slug> off main

- add a router module with register(app) (G3) or a migration in the owned range (contracts/ownership.yaml)
- run scope_guard, pytest, e2e-docker
- update own STATUS.md slot
- read docs/design-system/reference.html
- check component previews
- run the violations fixture through the linter
- track work packages on the work-queue artifact
- edit pipeline/djlib, run test_pipeline.py
- fix a scraper after a 0-row source and log in outputs/<date>/fixes.md
- rebuild app.js with build.mjs from songnodes/frontend and publish
- load and evaluate ref_mb

8 rows

STEP	PAGE	ACTION	DATA WRITTEN	TRIGGER TO NEXT	HANDOFF	FALLBACK
1	`sn-be-cli-scope-guard-g1`	branch <code>-<slug>, edit owned paths only, run scope_guard.py	git branch	work-package brief	–	two attempts then draft PR
2	`sn-be-svc-rest-api`	add a router module with register(app) or a migration in the owned number range	new file	feature need	–	list follow-ups in the PR
3	`sn-be-cli-start-dev`	add the migration to start-dev.sh and rebuild the stack	compose stack	new migration	–	migration silently missing if not added
4	`sn-be-cli-e2e-docker`	run E2E and pytest; update own STATUS.md slot	STATUS.md slot	before PR	PR with checks run via git	report out-of-scope failures
5	`sn-fe-ds-reference`	look up tokens and components	sn-density	open file	–	none
6	`sn-fe-ds-violations-fixture`	lint regression	–	validate.py	–	none
7	`sn-fe-work-queue-tracker`	copy job prompt	artifact db tasks.note	artifact page	prompt text via clipboard	none
8	`ml-page-artifact-app-entry`	node build.mjs <fe> <out> <tools>	artifact/build/app.js, assets/*	songnodes Library or Set builder pages change	app.js IIFE via Artifact publish	–

Failure modes and timeouts

- a new migration is not applied unless someone adds it to start-dev.sh (no migration runner/ledger in compose)
- router modules without register(app) are silently skipped (data_quality)
- CLAUDE.md schema and API examples disagree with code
- previews fail standalone without injected window.SongNodes
- build.mjs against current songnodes main will not compile: SetBuilderPage imports readSavedSetsAsync and writeSavedSetsAsync, which artifact-data.ts does not export
- copy patch strings no longer match songnodes wording
- build pinned to design system 1.1.0 while songnodes is at 1.10.0

Scraper (system actor)

Objective	Find complete playlists containing target tracks and persist them to bronze.
Device	worker and unified-scraper containers on the laptop compose stack
Entry point	`sn-be-svc-unified-scraper`
Basis	Inferred from pages and docs

Routine

- songnodes.schedule enqueues scrape_targets hourly
- scrape_targets picks 2 targets not searched for 7 days, one scrape job per source
- worker drain claims scrape job (lease 1800s) and POSTs /scrape
- spider writes bronze (and silver) via item pipelines
- handler records playlists_found and enqueues ingest_pending

6 rows

STEP	PAGE	ACTION	DATA WRITTEN	TRIGGER TO NEXT	HANDOFF	FALLBACK
1	sn-be-svc-worker	songnodes.schedule enqueues tick:scrape_targets:{hour}	ingestion_jobs scrape_targets	30s beat	–	bucket collision skips duplicates
2	sn-be-svc-worker	scrape_targets picks 2 active targets older than 7 days (SKIP LOCKED), one scrape job per source	target_tracks.last_searched, ingestion_jobs scrape	job run	–	outside_window returns without work
3	sn-be-svc-worker	claim scrape (lease 1800s) and POST /scrape	ingestion_jobs running	drain	ScrapeRequest json via sn-be-http-worker-to-scraper	transport error or 5xx/429 → retry with backoff
4	sn-be-svc-unified-scraper	scrapy crawl subprocess; item pipelines persist	bronze_*, silver_*, Redis quota and dedupe keys	POST /scrape	–	daily_quota_reached closes the spider with nothing found; 900s timeout → 504
5	sn-be-svc-worker	record playlists_found, complete job, enqueue ingest_pending	target_tracks.playlists_found, ingestion_jobs	scraper response	follow-up job ingest-after-scrape:{job_id} via sn-be-sql-ingestion-jobs	60s ingest tick
6	sn-be-svc-worker	ingest_pending builds silver and N → N+1 transitions	silver_*, silver_track_transitions, outbox	follow-up	silver_track_transitions via sn-be-sql-graph-read-silver	rejected playlist outbox event

Failure modes and timeouts

- scraper restart loses live progress
- daily quota closes spider
- scrape timeout 900s → 504 → retry with backoff
- long scrape occupies one of two worker processes

Weekly-run agent (system actor)

Objective	Run scrape to republish every week without a person present.
Device	cloud session driving laptop tuf-boi through device_bash
Entry point	`ml-agent-weekly-run`
Basis	Inferred from pages and docs

Routine

- test_pipeline
- scrape and ingest
- resolve in 160 s foreground chunks until done
- process, sync-tidal
- ArtifactData list sets to inbox/artifact_sets.json via device_commit_files with md5 check
- db (also posts to songnodes)
- sync-sets
- copy library.json to artifact/build, stage, read artifact, publish
- check scrape_log and ALERT files, SendUserMessage

3 rows

STEP	PAGE	ACTION	DATA WRITTEN	TRIGGER TO NEXT	HANDOFF	FALLBACK
1	ml-cli-resolve	chunked resume	inbox/resolved_<date>.json	loop until 'resolved N songs →'	–	repeat chunks until 'resolved N songs →'
2	ml-cli-build-db	rebuild	db, library.json	step 6	short report: headline numbers, top 15 High Rotation, review count, alerts, Beatport actions, sets pushed, artifact version via SendUserMessage	–
3	ml-page-set-builder	republish	artifact library.json	step 8	–	live shell differs: stop and report

Failure modes and timeouts

- playbook says Sunday 7:37am but runs are dated 24 Oct (Sat), 27 Oct (Tue), 31 Oct (Fri), 3 Nov (Tue)
- resolution yield fell from 614/777 TIDAL hits (24 Oct) to 6/81 (3 Nov) with no alert
- songnodes ingest failures are only printed as a 'note', never raised as an ALERT file

Beatport sync (assisted)

Objective: Check Beatport for tracks to buy, with Markus at the laptop.

Device: laptop tuf-boi Chrome with Markus present

Entry point: `ml-agent-beatport-sync`

Basis: Inferred from pages and docs

Routine

- read latest sync_plan.json, buy_list.csv, ledger beatport_check
- in Markus's Chrome add and remove tracks per action, pause every 25
- write resolved_<date>.json beatport_versions and sync_result_<date>.json
- run process and confirm

3 rows

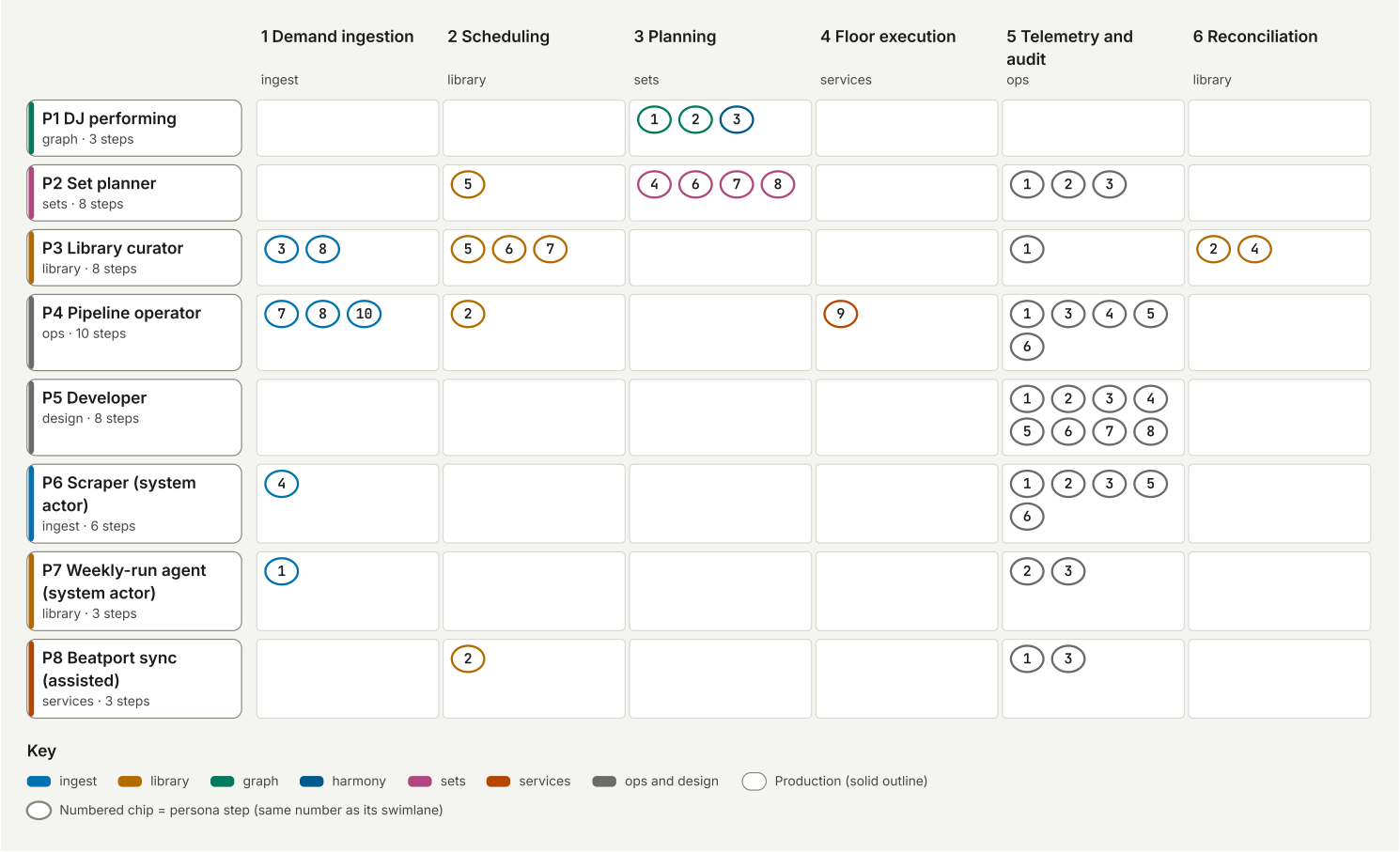
STEP	PAGE	ACTION	DATA WRITTEN	TRIGGER TO NEXT	HANDOFF	FALLBACK
1	ml-agent-beatport-sync	playlist edits	Beatport playlists	Markus asks	{norm_key: {beatport_versions[{id,url,mix_name,duration_seconds,bpm,key,genre,label}]}} via inbox/resolved_<date>.json	–
2	ml-cli-process	re-gate short-only songs	ledger	after beatport_versions written	–	–
3	ml-cli-confirm	mark beatport synced	ledger platform_sync_state	after sync_result	–	–

Failure modes and timeouts

- writing resolved_<date>.json and sync_result_<date>.json for a date the weekly run already used overwrites TIDAL resolve data and the TIDAL sync result
- sync_plan only holds this run's diff, so skipped weeks are never caught up on Beatport (unlike TIDAL's full reconcile)
- 243 beatport_check songs waiting; no Beatport sync has run (0 tracks with beatport ids, no SYNCED_BOTH)

D5: **Persona overview**

8 personas take 49 steps; only 2 of them touch Reconciliation.

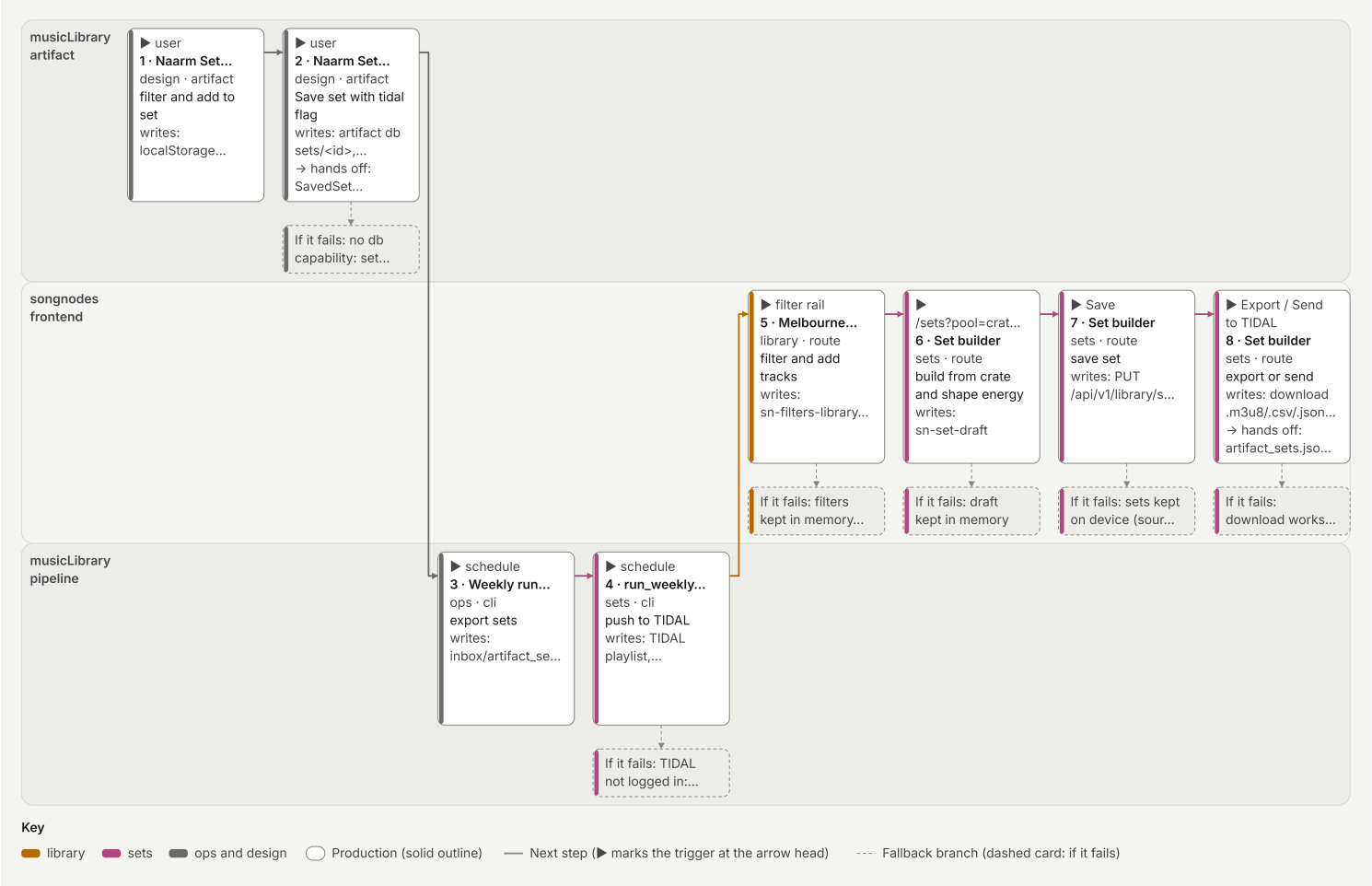


D5 · n = 49 · Unit: steps · ● Built

· Source: personas[] (steps[].page, action, writes); loop[]; pages[].domain · Method: hand-placed grid (lanes = personas, columns = loop stages); stage = page domain, override/calibrate/confirm steps = Reconciliation; 0 crossings

D5-P2: **Swimlane – P2 Set planner**

Set planner: 8 steps across 3 lanes; 6 steps name a fallback.

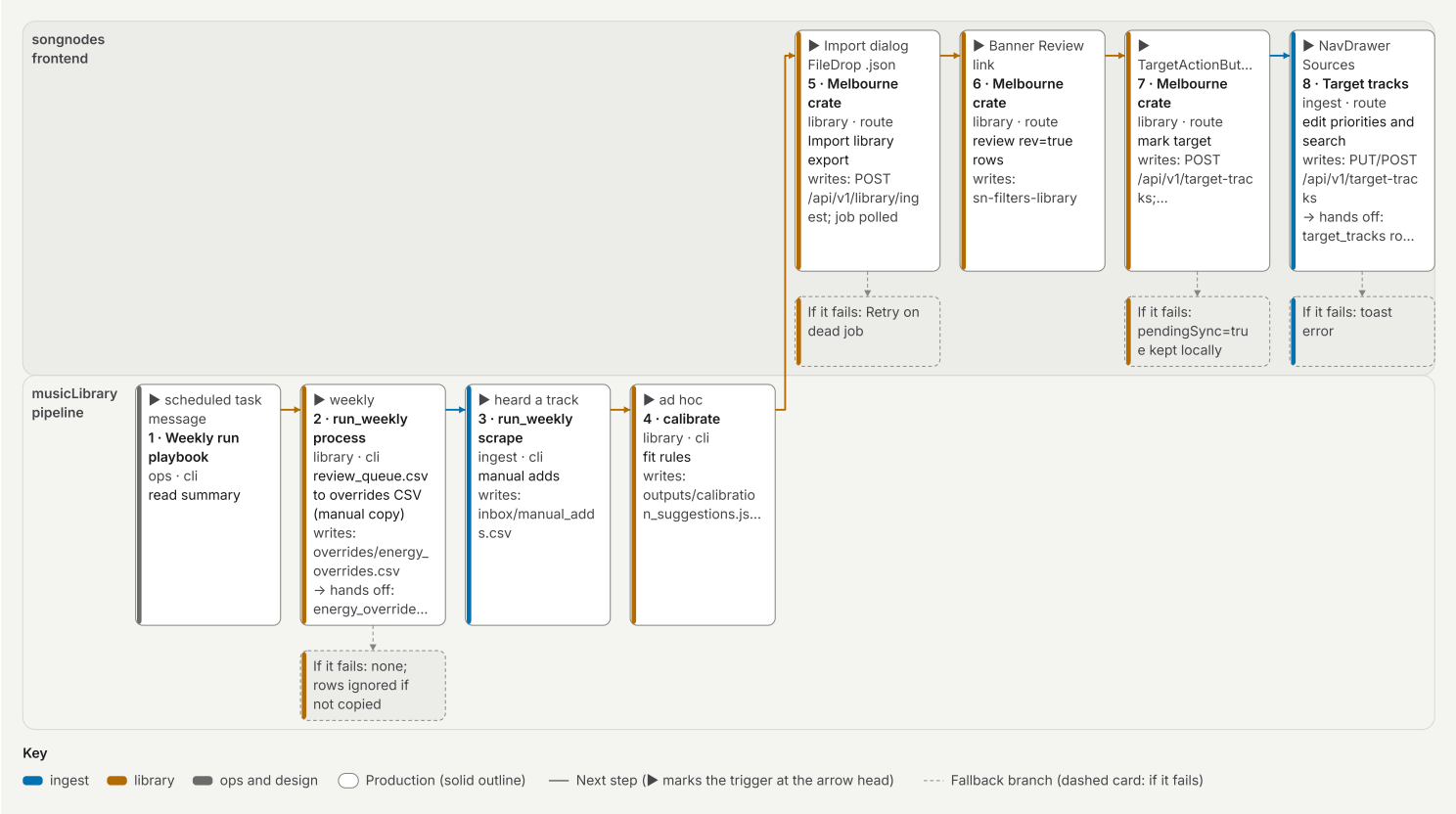


D5-P2 · n = 8 · Unit: steps · ● Built

· Source: personas[].steps[] (n, page, action, writes, trigger, handoff, fallback); pages[].domain · Method: hand-placed swimlane (lanes = systems, columns = steps, card width 160); orthogonal arrows in the column gaps; 0 crossings by construction

D5-P3: **Swimlane – P3 Library curator**

Library curator: 8 steps across 2 lanes; 4 steps name a fallback.

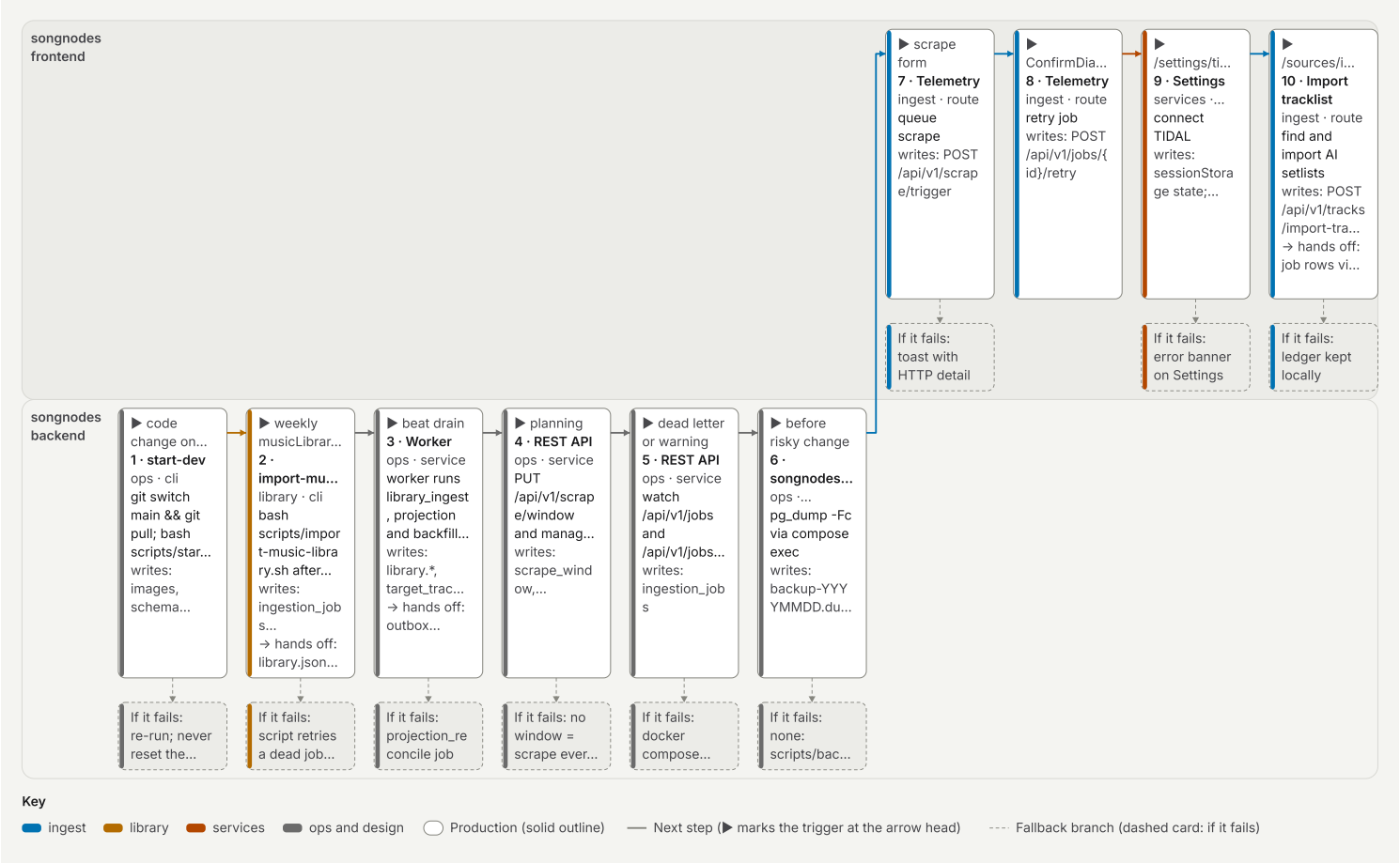


D5-P3 · n = 8 · Unit: steps · ● Built

· Source: personas[].steps[] (n, page, action, writes, trigger, handoff, fallback); pages[].domain · Method: hand-placed swimlane (lanes = systems, columns = steps, card width 160); orthogonal arrows in the column gaps; 0 crossings by construction

D5-P4: **Swimlane – P4 Pipeline operator**

Pipeline operator: 10 steps across 2 lanes; 9 steps name a fallback.

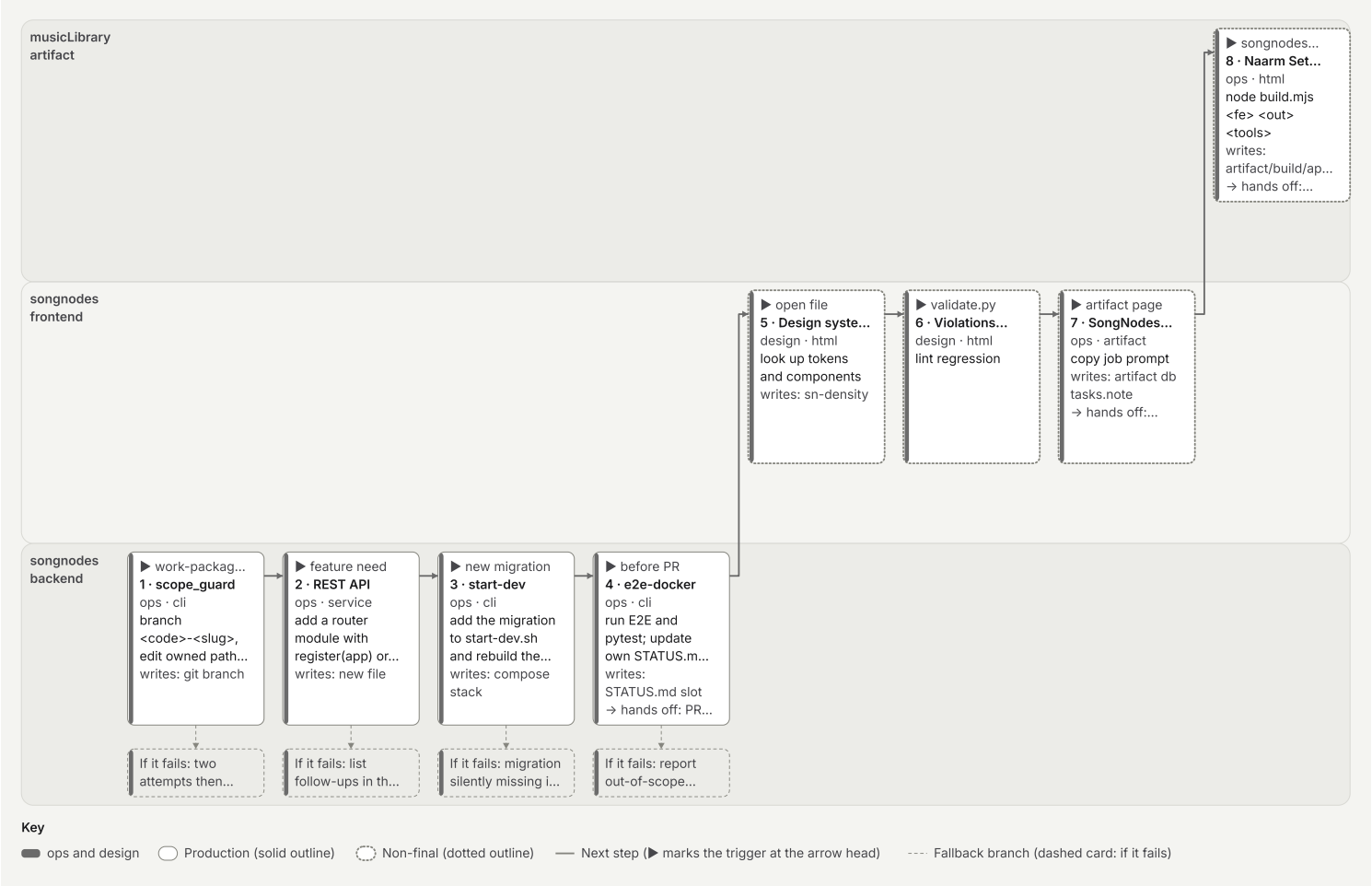


D5-P4 · n = 10 · Unit: steps · ● Built

· Source: personas[].steps[] (n, page, action, writes, trigger, handoff, fallback); pages[].domain · Method: hand-placed swimlane (lanes = systems, columns = steps, card width 124); orthogonal arrows in the column gaps; 0 crossings by construction

D5-P5: **Swimlane – P5 Developer**

Developer: 8 steps across 3 lanes; 4 steps name a fallback.

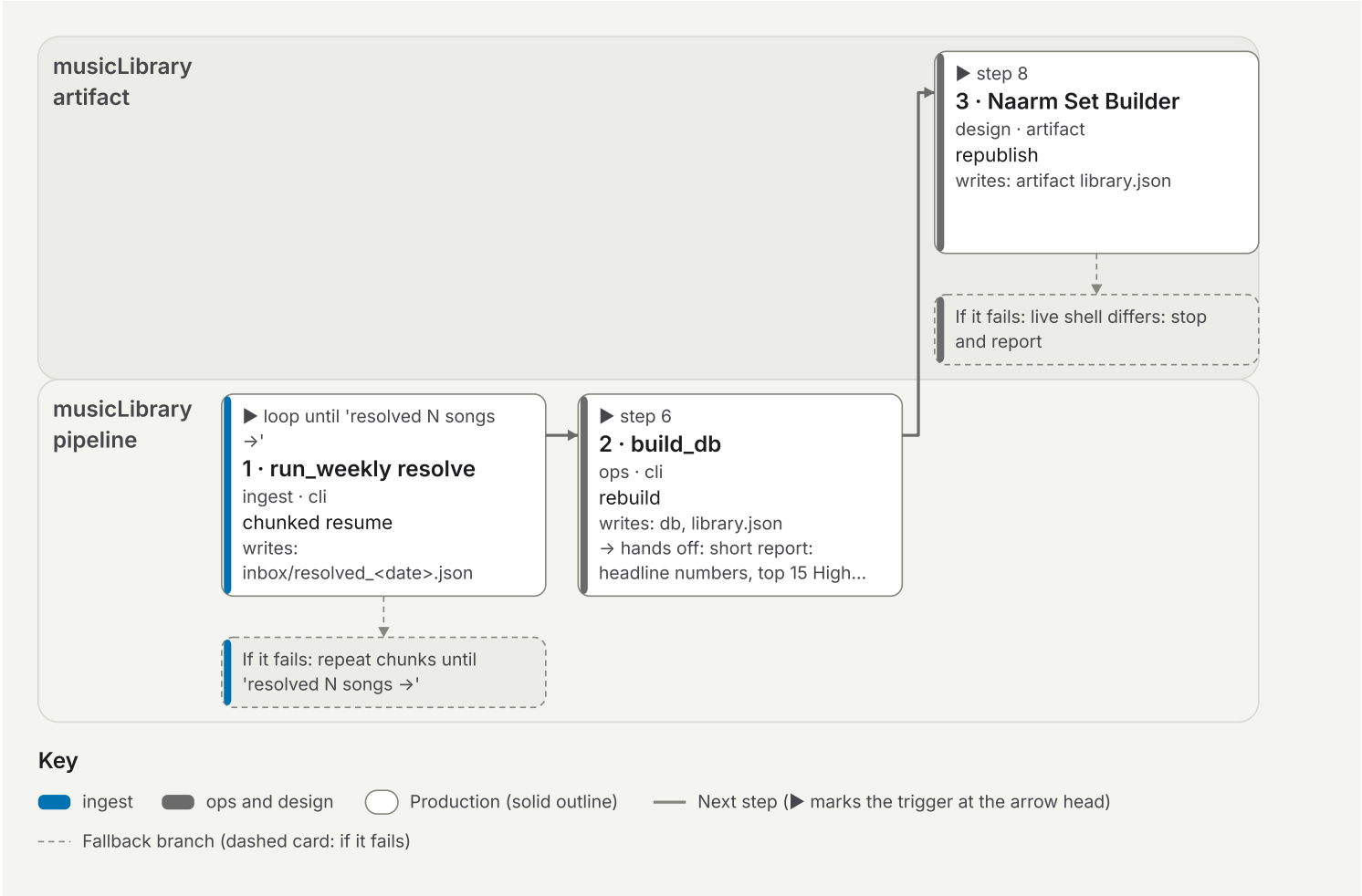


D5-P5 · n = 8 · Unit: steps · ● Built

· Source: personas[].steps[] (n, page, action, writes, trigger, handoff, fallback); pages[].domain · Method: hand-placed swimlane (lanes = systems, columns = steps, card width 160); orthogonal arrows in the column gaps; 0 crossings by construction

D5-P7: **Swimlane – P7 Weekly-run agent (system actor)**

Weekly-run agent (system actor): 3 steps across 2 lanes; 2 steps name a fallback.

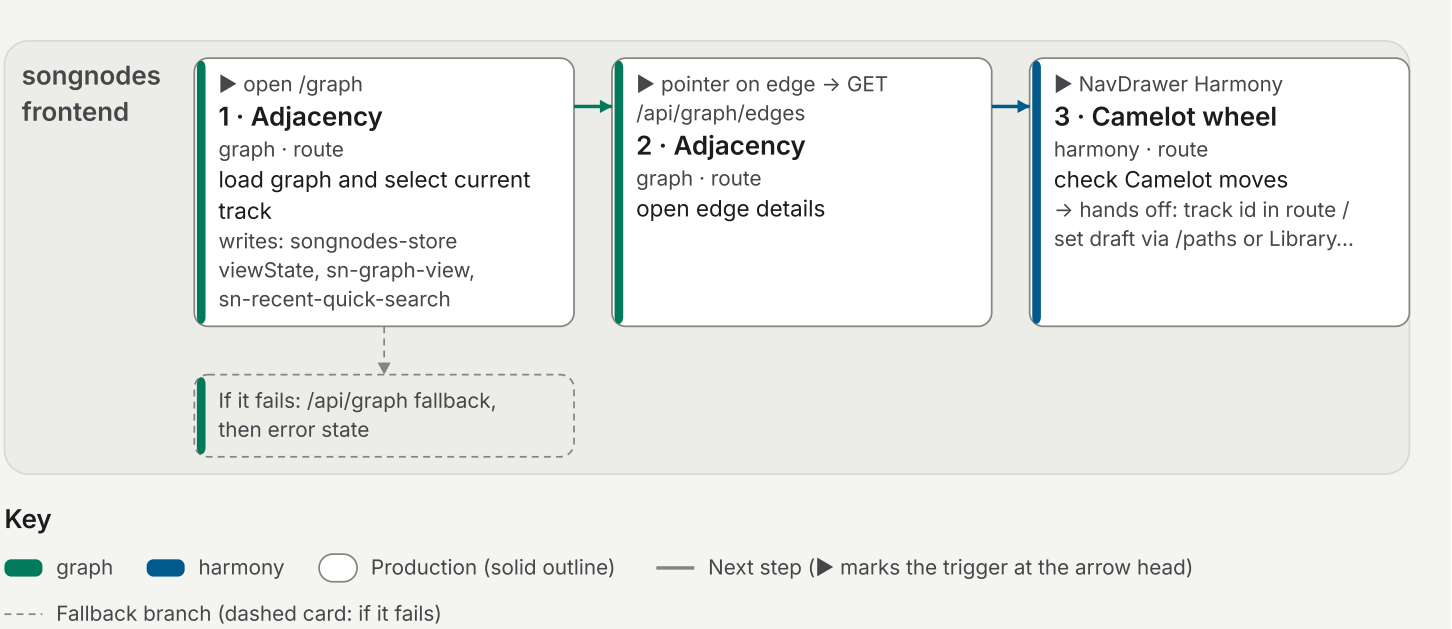


D5-P7 · n = 3 · Unit: steps · ● Built

· Source: personas[].steps[] (n, page, action, writes, trigger, handoff, fallback); pages[].domain · Method: hand-placed swimlane (lanes = systems, columns = steps, card width 220); orthogonal arrows in the column gaps; 0 crossings by construction

D5-P1: **Swimlane – P1 DJ performing**

DJ performing: 3 steps across 1 lane; 1 step name a fallback.



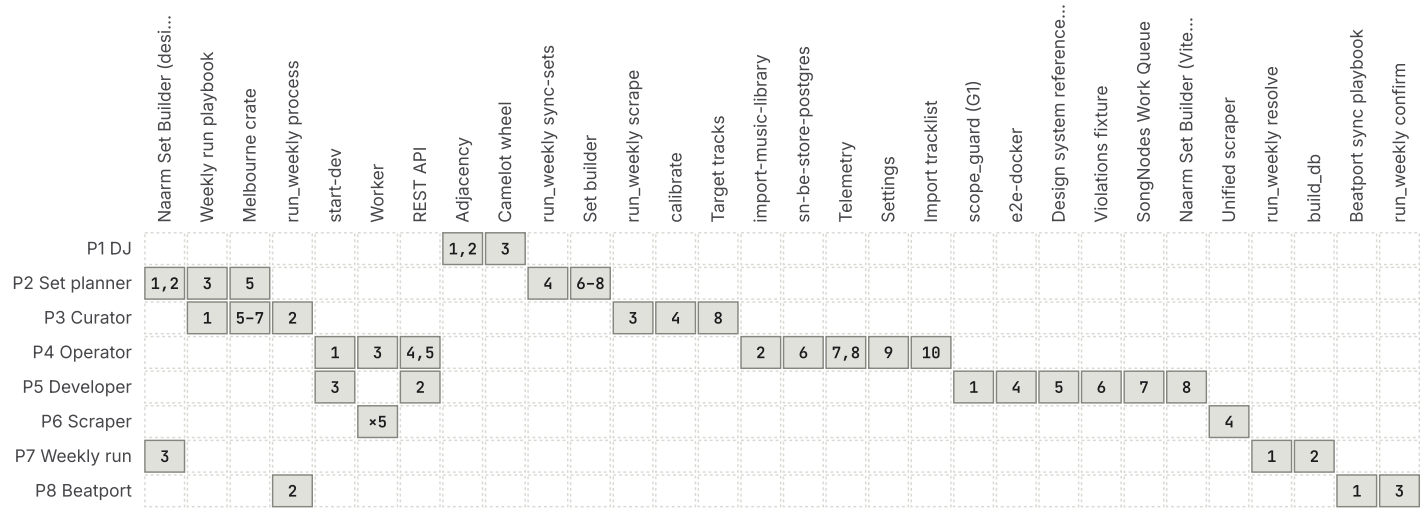
D5-P1 · n = 3 · Unit: steps · ● Built

· Source: personas[].steps[] (n, page, action, writes, trigger, handoff, fallback); pages[].domain · Method: hand-placed swimlane (lanes = systems, columns = steps, card width 220); orthogonal arrows in the column gaps; 0 crossings by construction

C9: **Personas by page, with step numbers**

8 personas take 49 steps over 30 pages; 7 pages serve more than one persona. No ease scores: findings.json holds no evidence for them.

Cell numbers are step numbers in that persona's journey; ×n = n steps (listed in the table)

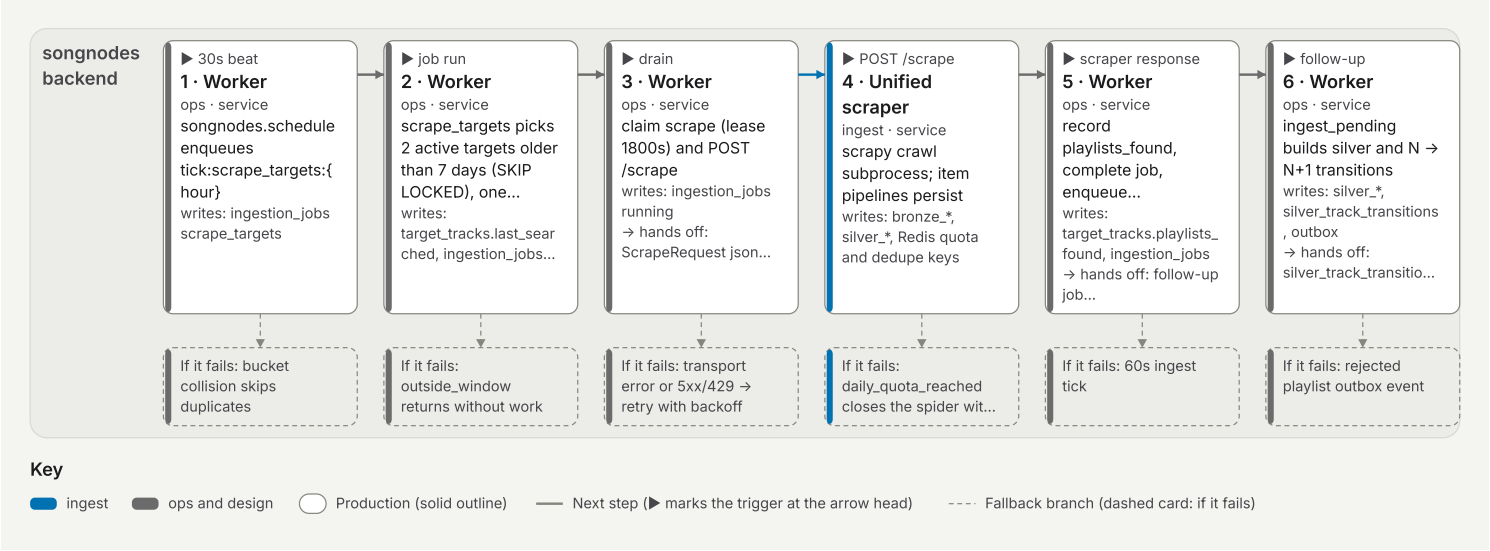


C9 · n = 49 · Unit: steps · ● Built

· Source: personas[].steps[] (n, page) · Method: Each cell lists the step numbers a persona takes on a page; columns ordered by number of personas, then by first appearance. Ease scores omitted (no evidence).

D5-P6: **Swimlane – P6 Scraper (system actor)**

Scraper (system actor): 6 steps across 1 lane; 6 steps name a fallback.

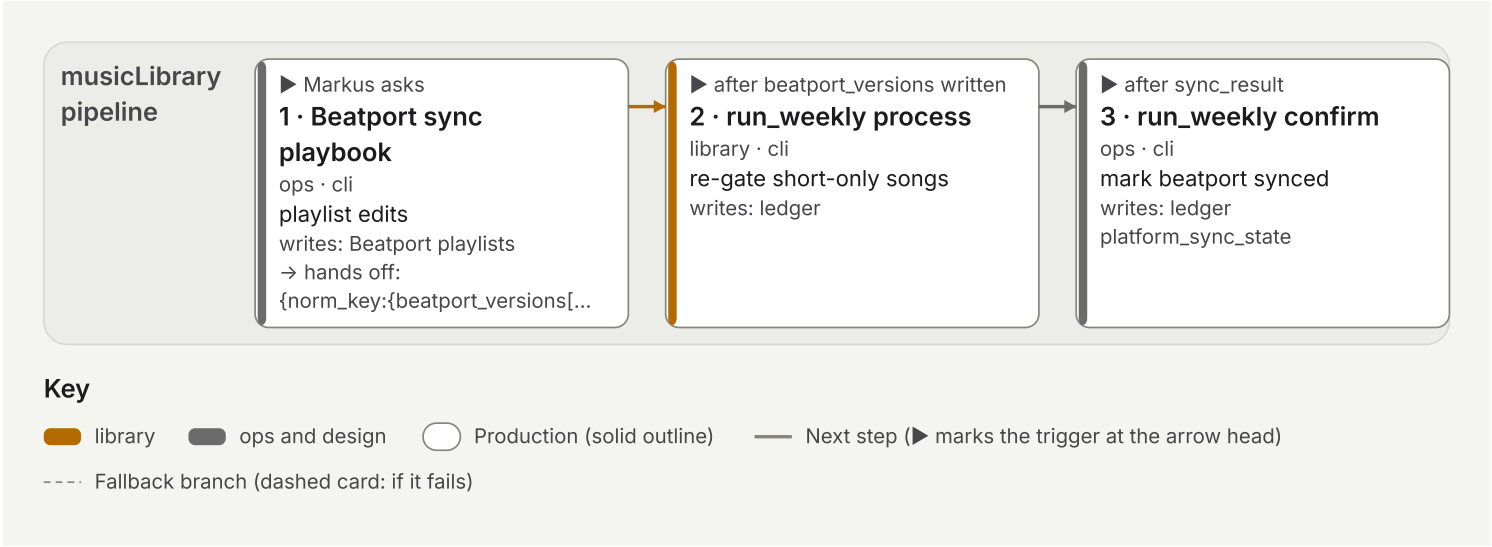


D5-P6 · n = 6 · Unit: steps · ● Built

· Source: personas[].steps[] (n, page, action, writes, trigger, handoff, fallback); pages[].domain · Method: hand-placed swimlane (lanes = systems, columns = steps, card width 160); orthogonal arrows in the column gaps; 0 crossings by construction

D5-P8: **Swimlane – P8 Beatport sync (assisted)**

Beatport sync (assisted): 3 steps across 1 lane; 0 steps name a fallback.



D5-P8 · n = 3 · Unit: steps · ● Built

· Source: personas[].steps[] (n, page, action, writes, trigger, handoff, fallback); pages[].domain · Method: hand-placed swimlane (lanes = systems, columns = steps, card width 220); orthogonal arrows in the column gaps; 0 crossings by construction

C10: **Risk matrix, severity by effort**

11 of 47 debt items are critical or high; 4 of those are small-effort fixes.

	Effort S (small)	Effort M (medium)	Effort L (large)
✕ Critical	0 items	1 item ml:no-bpm	1 item be:q-tunnel Q
▲ High	4 items ml:pending-rejected ml:resolve-yield be:open-admin fe:creds-localstorage	5 items ml:artifact-build be:backups-wrong be:daily-quota be:gold-operational be:scraper-silver	0 items
◖ Medium	7 items ml:idempotency-same ml:no-run ml:secrets-plain ml:songnodes-ingest be:auth-header be:migration-ledger be:worker-starvation	8 items be:default-encryption be:doc-drift be:library-events be:library-ingest be:long-transaction be:orphan-tables fe:api-ts fe:dead-graph	0 items
○ Low	17 items ml:build-charset ml:history-overwrite ml:legacy-page ml:schedule-drift ml:sqlite-copy ml:unused-std be:ingest-like be:memgraph-noop be:progress-polling be:route-shadowing be:set-delete be:two-setlist be:unmounted-router fe:poll-backoff fe:presets-only fe:test-key fe:tracker-hex	4 items be:q-sentinel Q fe:proxy-drift fe:source-label fe:unused-nginx	0 items

IDs: repo code (ml, be backend, fe frontend) and the first words of the debt id; Q = queued

C10 · n = 47 · Unit: debt items · i Projected

· Source: debt[] (severity, effort, state) · Method: Count of debt items per severity (ordinal rows) and effort S/M/L (ordinal columns), with each item's short id.

Table: **Technical debt, severity ranked** 47 items

47 rows

ID	SEVERITY	ITEM	KIND	EFFORT	DETAIL	STATE	EVIDENCE
ml-debt-no-bpm-key	× Critical	**No track has a BPM or key, so two of the three energy methods and all Camelot logic are dead**	drift	M	TIDAL v2 returned 0 bpm/key across 1007 versions; Deezer BPM (38 versions) is only used as the cross-check 'other' and never copied onto the chosen TIDAL version. Result: ledger bpm=0/413, camelot_key=0/413, every track needs_review (v_review_queue 413), Set Builder warns that key and BPM moves are off (M3).	● Built	musicLibrary/pipeline/run_weekly.py:219-230 +3
sn-be-debt-q-tunnel-before-auth	× Critical	**Queued tunnel would publish open admin routes and a default admin/admin login**	security	L	PR #577 exposes the frontend (which proxies all /api/ to rest-api and graph-api) publicly; rest-api has no auth middleware on main (SONGNODES_API_TOKEN unset) and PR #578 adds a login router without any middleware gating other routes. PR #578 falls back to AUTH_USERNAME/AUTH_PASSWORD defaults of admin/admin and accepts plain-text password matches. Merging #577 before an enforced auth layer (Cloudflare Access or middleware) opens jobs, merges, API keys and AI proxy to the internet. cloudflared:latest is also an unpinned tag.	▲ Queued	PR #577 compose.dev.yaml +2
ml-debt-artifact-build-drift	▲ High	**Artifact bundle cannot be rebuilt from songnodes main**	drift	M	SetBuilderPage now imports readSavedSetsAsync and writeSavedSetsAsync and loads from /api/v1/library/search; artifact-data.ts exports neither, and the copy-patch text changed. Published app.js is the 24 Oct build on design system 1.1.0 (songnodes now 1.10.0).	● Built	musicLibrary/artifact/app/artifact-data.ts:8-72 +4

ID	SEVERITY	ITEM	KIND	EFFORT	DETAIL	STATE	EVIDENCE
ml-debt-pending-rejected-dup	▲ High	**Rejected songs re-enter pending and stay there**	orphaned-data	S	ingest_raw setdefaults a pending entry for any replayed key even when it is in rejected, but queues it for resolve only at 2+ appearances. 297 of 442 pending keys are also in rejected with resolve_attempts 0; counts and the pending table are inflated.	● Built	musicLibrary/pipeline/djlib/ledger.py:108-116
ml-debt-resolve-yield-silent	▲ High	**Resolve failures are silent**	other	S	Exceptions are printed and an empty result is saved and never retried; TIDAL hits fell 614/777 → 6/81 between 24 Oct and 3 Nov with no ALERT. resolve_log.txt exists only for 24 Oct and is empty.	● Built	musicLibrary/pipeline/run_weekly.py:439-453 +1
sn-be-debt-backups-wrong-target	▲ High	**Backup scripts target names that do not exist**	drift	M	scripts/backup_database.sh dumps -U musicdb_user musicdb; backup_volumes.sh tars songnodes_postgres_data etc. The compose project uses user/db songnodes and volume songnodes-dev_dev-postgres-18. Only the CLAUDE.md one-liner works.	● Built	songnodes/scripts/backup_database.sh:52 +2
sn-be-debt-	▲ High	**1001tracklists daily Redis quota**	race	M	scrape_targets enqueues a 1001tracklists job per target each hour; the 1001tracklists spider does	● Built	songnodes/scrapers/spiders/1001tracklists_

daily-quota-vs-hourly-targets		**blocks hourly target scrapes**			not set force_run for search_query (MixesDB does), so after the first run in 24 h enforce_run_quota raises CloseSpider(daily_quota_reached), the job succeeds with found_nothing and the target is stamped last_searched for 7 days.		spider.py:309 +3
sn-be-debt-gold-operational-absent	▲ High	**Gold and operational layers do not exist in the running stack**	drift	M	CLAUDE.md describes Bronze → Silver → Gold → Operational; compose applies no gold migration (003) and runs no gold ETL; gold_track_graph is created but never read or refreshed; graph-api reads silver directly. Operational is effectively projection_tracks (library search) and silver reads.	● Built	songnodes/CLAUDE.md:11 +3
sn-be-debt-open-admin	▲ High	**Write and admin REST routes open in compose**	security	S	SONGNODES_API_TOKEN is not passed in compose.dev.yaml, so register_security returns without a middleware. Port bound to 127.0.0.1 limits exposure to the laptop.	● Built	songnodes/services/rest_api/security/auth.py:31 +1
sn-be-debt-scraper-silver-layer-skip	▲ High	**Unified scraper writes silver tables directly, bypassing datalayer ingest**	race	M	PersistencePipeline maps tracks/playlists/playlist_tracks batches to silver_enriched_* and inserts with no ON CONFLICT on tracks, defaults artist to Unknown Artist and appends no outbox event. datalayer ingest_pending then creates its own silver rows for the same bronze playlist (lookup-then-insert, no unique key), so duplicates and Unknown Artist rows are possible and projection_tracks never hears of the scraper rows. Violates CLAUDE.md "no layer skipping" and artist attribution.	● Built	songnodes/scrapers/pipelines/persistence_pipeline.py:910 +3
sn-fe-debt-creds-localstorage	▲ High	**OAuth tokens and client secrets persisted in localStorage and exposed on window.debugZustand**	security	S	songnodes-store partialize persists musicCredentials (access/refresh tokens, clientId/clientSecret); window.debugZustand.getMusicCredentials and injectTestCredentials ship in production builds. OAuthCallback deliberately moved tokens to sessionStorage, so the two disagree.	● Built	songnodes/frontend/src/store/useStore.ts:1507-1508,1566-1590 +1
ml-debt-idempotency-same-length	◐ Medium	**sync-sets add uses an idempotency key that ignores content**	race	S	add_tracks key is uuid5(add:pid:<date>:<set id>:<len>:i). Re-running on the same date after editing a set to the same length reuses the key, so TIDAL may replay the earlier add. Remove-all-then-add also loses duplicates because adds use onDuplicates SKIP, so a set with a repeated track is rewritten every run.	● Built	musicLibrary/pipeline/run_weekly.py:555-559 +1
ml-debt-no-run-lock	◐ Medium	**No run lock between weekly run, Beatport agent and hand runs**	race	S	Ledger, resolved_<date>.json and sync_result_<date>.json are whole-file rewrites; the Beatport playbook writes the same date-keyed files the weekly run owns.	● Built	musicLibrary/pipeline/djlib/ledger.py:62-70 +1

ml-debt-secrets-plain	◐ Medium	**TIDAL tokens and ref_mb passwords are plain files on a synced drive**	security	S	secrets/*.json and ref_mb/secrets/*.txt; ref_mb/.gitignore covers secrets but musicLibrary has no git; deletes are blocked on the synced drive so pending PKCE files are blanked, not removed.	● Built	musicLibrary/pipeline/djlib/tidal.py:112-113 +1
ml-debt-songnodes-ingest-swallowed	◐ Medium	**songnodes ingest and setlist posts swallow errors**	other	S	cmd_sync_songnodes prints a 'note' on ingest failure and 'except Exception: pass' on setlists; it never polls the job, unlike import-music-library.sh.	● Built	musicLibrary/pipeline/run_weekly.py:593-616
sn-be-debt-auth-header-collision	◐ Medium	**API token middleware reads Authorization Bearer before X-API-Token**	permission	S	TIDAL routes require Authorization: Bearer <tidal token>; when SONGNODES_API_TOKEN is set the middleware takes the TIDAL token as the API token and returns 401 even with a valid X-API-Token. SENSITIVE_GET_PREFIXES lists /api-keys but the route is /api/v1/api-keys, so GET of saved keys stays open.	● Built	songnodes/services/rest_api/security/auth.py:17 +2
sn-be-debt-default-encryption-key	◐ Medium	**Saved API keys encrypted with a hard-coded default key**	security	M	API_KEY_ENCRYPTION_SECRET is not set for rest-api or worker; store_api_key falls back to a hard-coded default key committed in the SQL file (value not reproduced here). OAuth tokens are plain text by design.	● Built	songnodes/sql/init/07-api-keys.sql:59 +2
sn-be-debt-doc-drift	◐ Medium	**CLAUDE.md contracts disagree with code**	drift	M	Unified scraper body is {source, search_query, ...} not {spider_name, search_params}; transitions columns are from_track_id/to_track_id not source/target; bronze uses raw_json/scraped_at not raw_metadata; target_tracks not target_track_searches; enrichment does not delegate to metadata-enrichment:8020 (8020 is audio-analysis); Discogs env is DISCOGS_TOKEN in worker and DISCOGS_API_TOKEN for the scraper, which reads DISCOGS_TOKEN.	● Built	songnodes/CLAUDE.md:44 +7
sn-be-debt-library-events-ignored	◐ Medium	**projection_tracks consumer ignores library_track events**	drift	M	import_library emits library.track.upserted.v1 / rotation_changed.v1 with aggregate_type library_track; projection.handle_event handles only track and artist, so library search reads stale projection rows until reconcile/rebuild. graph_projection does handle library_track.	● Built	songnodes/services/library/importer/import_library.py:162 +2

sn-be-debt-library-ingest-contract	◐ Medium	**Library ingest does not meet its contract**	drift	M	Contract: same key + different body → 409, and per-row results in the response. Code: jobs.enqueue returns the original job for any body and the endpoint returns only {job_id,status,created}. GET /api/v1/library/tracks/{id} is documented but only /preview exists.	● Built	songnodes/contracts/library_ingest.md:16 +2
sn-be-debt-long-transaction-scrape	◐ Medium	**Scrape handler holds a DB transaction open for up to 16 minutes**	race	M	runner.run_one wraps the handler in conn.transaction(); the scrape handler awaits a synchronous POST /scrape (timeout 960s) inside it, leaving an idle-in-transaction session and blocking pg_snapshot_xmin, which also stalls outbox consumers (events from newer transactions stay invisible while this xmin is held).	● Built	songnodes/services/common/datalayer/runner.py:37 +2
sn-be-debt-migration-ledger	◐ Medium	**No migration ledger in compose; start-dev.sh is the list**	drift	S	Migrations run by hand-maintained psql lines with ad-hoc sentinels; 000 schema_migrations, 003-005/007-008 medallion, 001-011 and 250 are not applied; worker migrate.py only knows 012-014.	● Built	songnodes/scripts/start-dev.sh:31 +1
sn-be-debt-orphan-tables	◐ Medium	**Tables written but never read, or read but never written**	orphaned-data	M	web_call_log (written, no reader); gold_track_graph (created, no reader); musicdb.user_setlists (written by graph-api, no reader, schema absent); song_adjacency (written by scraper, table absent); bronze_api_enrichments (neither); target_track_searches (CLAUDE.md says scraper reads it; only legacy SQL writes it); library.set.saved.v1 and setlist ingested/rejected outbox events have no consumer.	● Built	songnodes/services/common/providers/web_call_log.py:24 +4
sn-be-debt-worker-starvation	◐ Medium	**Two worker processes shared by 15-minute scrapes and 15-second projections**	other	S	Worker runs --concurrency=2 with beat in-process; two queued scrapes occupy both processes and projection/ingest ticks wait (their Celery tasks expire after 15s).	● Built	songnodes/services/worker/Dockerfile +1
sn-fe-debt-api-ts-dead	◐ Medium	**services/api.ts declares many clients for endpoints that do not exist**	drift	M	searchApi (/api/v1/search/*; backend has /api/search/tracks), pathfindingApi (/api/v1/pathfinding/*; backend is /api/v1/pathfinder/find-path), setlistApi POST/DELETE/export, /api/v1/tracks/batch, /api/graph/tracks, /api/graph/positions have no backend route and no caller.	● Built	songnodes/frontend/src/services/api.ts:242-556 +1

sn-fe-debt-dead-graph-modules	◐ Medium	**Unreachable modules still carrying endpoints**	orphaned-data	M	Import-graph walk from main.tsx: 50 of 184 non-test modules unreachable, including LiveDataLoader (/api/ws WebSocket), EditTrackModal (PATCH /api/tracks/{id}), NowPlayingDeck (trackApi preview), AIAssistantModal (songnodes_ai_prompt_history), StatsPanel (stats JSON/CSV download), PreferencesPanel, AITransitionHelper, useLiveDataLoader, useKeyboardShortcuts.	● Built	/home/claude/dfa/scratch/sn-frontend/unreachable.txt +1
ml-debt-build-charset	i Low	**Build shell has no charset**	other	S	artifact/build/index.html has no <meta charset>; rendered outside the claude.ai wrapper the middle dot and apostrophes become mojibake (M3: 'Â·'). Fine only while the platform wrapper adds a charset.	● Built	musicLibrary/artifact/build/index.html
ml-debt-history-overwrite	i Low	**Ledger snapshot is overwritten on a same-day re-run**	other	S	ledger_before_<date>.json is copied on every process save, so a second process on the same date replaces the real pre-run snapshot with the first run's output.	● Built	musicLibrary/pipeline/djlib/ledger.py:67-69
ml-debt-legacy-pageds	i Low	**Legacy artifact/index.html breaks the design system**	drift	S	48 hex colours, 181 px values, own Google fonts and tokens, 0 sn- classes; README still calls it the page source. The published build (app.js) has 0 hex and uses songnodes tokens.	● Built	musicLibrary/artifact/index.html +1
ml-debt-schedule-drift	i Low	**Run cadence does not match the documented schedule**	drift	S	Documented Sunday 7:37am; outputs exist for 24 Oct, 27 Oct, 31 Oct and 3 Nov (none on a Sunday), so the 8-day window overlaps and ARIA weeks are re-counted.	● Built	musicLibrary/agent/weekly_run.md +1
ml-debt-sqlite-copy	i Low	**SQLite swap is a copy, not an atomic rename**	other	S	shutil.copyfile over db/dj_library.sqlite; an open DB Browser session can see a half-copied file. Docstring claims an atomic swap.	● Built	musicLibrary/pipeline/build_db.py:6-8 +1
ml-debt-unused-std-cap	i Low	**std_crate_cap is configured but never enforced**	drift	S	rules.rotation.std_crate_cap=250 has no reader; only high_crate_cap is applied.	● Built	musicLibrary/pipeline/run_weekly.py:278-287 +1
sn-be-debt-ingest-like-scan	i Low	**already_linked check scans transitions with LIKE on jsonb text**	other	S	Re-ingest of an existing playlist runs playlist_occurrences::text LIKE %id% across silver_track_transitions (no index), cost grows with the graph.	● Built	songnodes/services/common/datalayer/ingest.py:266

sn-be-debt-memgraph-noop	i Low	**graph_projection consumer runs every 15 s against a store that does not exist**	orphaned-data	S	No memgraph service or MEMGRAPH_URI in compose; handle_event returns immediately but offsets advance, so enabling Memgraph later needs a rebuild.	● Built	songnodes/services/common/datalayer/graph_projection.py:27 +2
sn-be-debt-progress-polling	i Low	**Job stages poll in-memory scraper progress**	polling	S	Frontend polls GET /api/v1/jobs/{id}/stages, which polls GET /scrape/progress/{job_id} with a 2s timeout; progress lives in process memory and is lost on scraper restart.	● Built	songnodes/services/rest_api/routers/jobs.py:213 +1
sn-be-debt-q-sentinel-dead-route	i Low	**Queued PR #580 sanitises errors on routes no client can reach**	drift	M	PR #580 changes rest-api main.py /api/graph/nodes and /api/graph/edges error details; nginx sends /api/graph to graph-api, so those handlers are shadowed. The music-search change is effective.	▲ Queued	PR #580 services/rest_api/main.py +1
sn-be-debt-route-shadowing	i Low	**Duplicate /api/graph/nodes and /api/graph/edges in rest-api are unreachable**	drift	S	nginx sends ^/api/graph to graph-api, so rest-api main.py versions never receive frontend traffic; two implementations can diverge.	● Built	songnodes/services/rest_api/main.py:501 +2
sn-be-debt-set-delete-no-event	i Low	**Deleting a saved set emits no outbox event**	drift	S	save_set appends library.set.saved.v1; delete_set does not, so any projection of sets cannot remove it.	● Built	songnodes/services/library/model/library_db.py:26 +1
sn-be-debt-two-setlist-stores	i Low	**Two saved-setlist stores**	orphaned-data	S	graph-api POST /api/user-setlists (musicdb.user_setlists) and rest-api library.sets coexist; only library.sets has a reader.	● Built	songnodes/services/graph-visualization-api/main.py:808 +1
sn-be-debt-unmounted-router	i Low	**data_quality router never mounted**	orphaned-data	S	No register(app) and not included by main.py; router_registry skips it by AST check. Its routes are also commented out.	● Built	songnodes/services/rest_api/routers/data_quality.py:19 +1
sn-fe-debt-poll-backoff	i Low	**Pipeline poll backoff reads a stale error value**	polling	S	startPolling(error ? 30000 : 10000) captures error from the first render inside useEffect, so the 30 s backoff never engages.	● Built	songnodes/frontend/src/features/pipeline/usePipelineTelemetry.ts:317-322

sn-fe-debt-presets-only-cross-tab	i Low	**Only filter presets react to other tabs**	race	S	sn-set-draft and sn-music-library-sets are written by Library and Set builder without storage-event listeners; two tabs overwrite each other's draft (last write wins).	● Built	songnodes/frontend/src/features/musicLibrary/components/FilterPresets.tsx:38-45 +1
sn-fe-debt-proxy-drift	i Low	**Vite dev proxy and nginx route graph traffic differently**	drift	M	nginx sends /api/(graph|v1/visualization|user-setlists) to graph-api; Vite sends only /api/graph and /api/v1/graph (rewritten) to 8084, so /api/v1/visualization and /api/user-setlists hit rest-api in dev. rest-api also defines /api/graph/nodes and /api/graph/edges that nginx never reaches.	● Built	songnodes/frontend/Dockerfile:81-91 +2
sn-fe-debt-source-label	i Low	**Library rows always labelled 'library-fixture'**	drift	M	fromLibraryRow sets source 'library-fixture' on every row although loadLibrary only reads the backend (_source 'backend'); CamelotPage shows 'Music library'. DB1 brief planned the fixture fallback; current code has neither the fallback nor public/data/music-library.json.	● Built	songnodes/frontend/src/services/unifiedTrackService.ts:31-64 +2
sn-fe-debt-test-key-drift	i Low	**registry.test seeds 'sn-saved-sets' but code reads 'sn-music-library-sets'**	drift	S	Test mock storage key does not match SAVED_SETS_KEY, so the test never exercises saved sets on Home.	● Built	songnodes/frontend/src/routes/registry.test.tsx +1
sn-fe-debt-tracker-hex	i Low	**Work-queue tracker uses its own hex palette**	drift	S	plan/work-queue-2026-10-30/tracker.html defines --bg:#f6f5f9 etc. instead of design-system tokens, against CLAUDE.md §4.	● Built	songnodes/plan/work-queue-2026-10-30/tracker.html
sn-fe-debt-unused-nginx	i Low	**nginx/ and nginx-local-proxy.conf are not used by the Compose stack**	drift	M	compose.dev.yaml builds the frontend with an inline default.conf; nginx/conf.d/frontend.conf proxies to api-gateway:8080 and websocket-api:8083, nginx-local-proxy.conf to host.docker.internal:8088 — none exist in compose.dev.yaml. Design-system docs (13-page-patterns, 16-platform) still cite nginx/nginx.conf for error pages and frame headers.	● Built	songnodes/nginx/conf.d/frontend.conf:63-86 +3

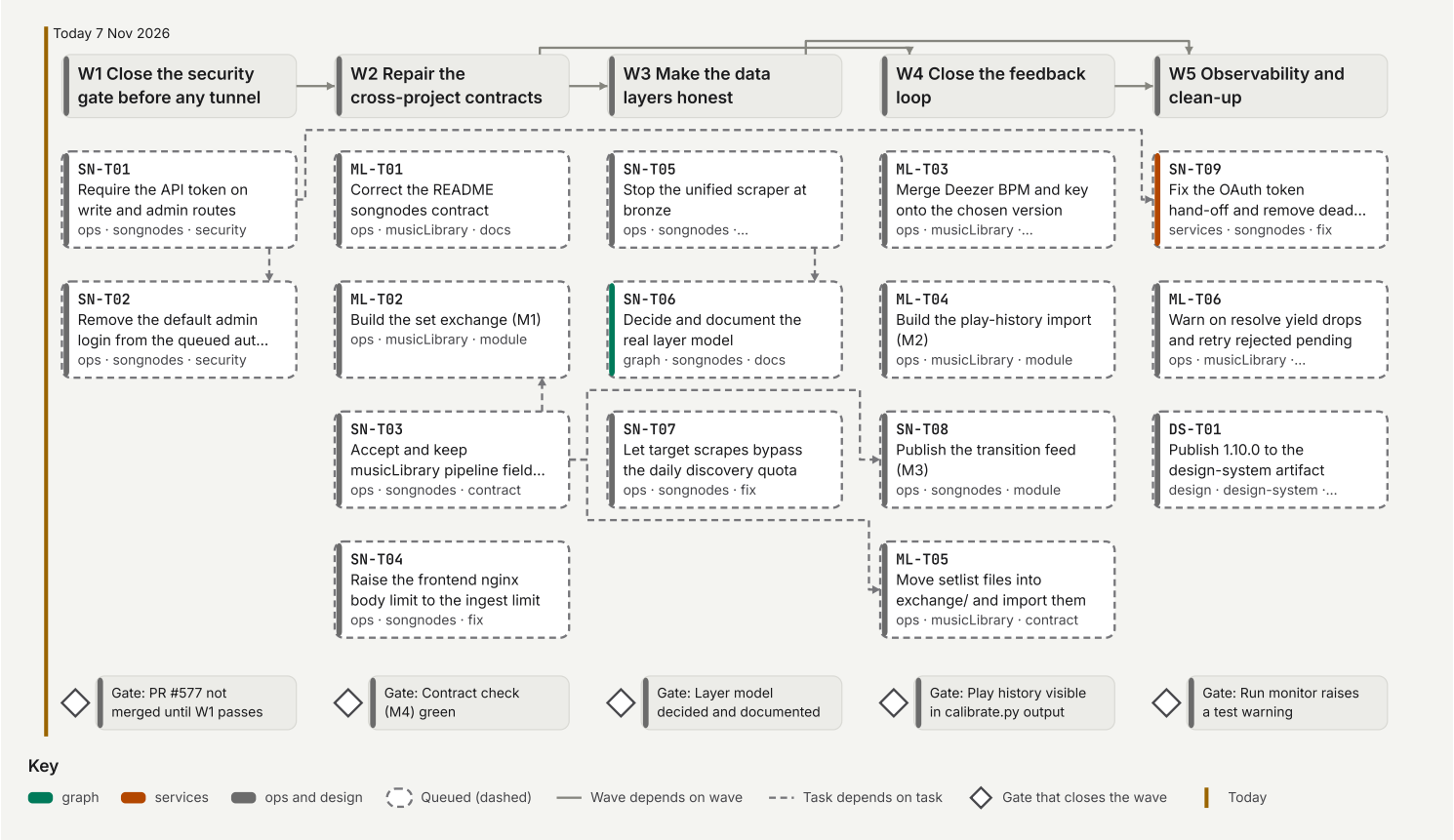
Table: **Roadmap waves** `5 waves`

5 rows

WAVE	GOAL	TASKS	GATE	DEPENDS ON
W1	**Close the security gate before any tunnel**	SN-T01, SN-T02	PR #577 not merged until W1 passes	–
W2	**Repair the cross-project contracts**	ML-T01, ML-T02, SN-T03, SN-T04	Contract check (M4) green	W1
W3	**Make the data layers honest**	SN-T05, SN-T06, SN-T07	Layer model decided and documented	W2
W4	**Close the feedback loop**	ML-T03, ML-T04, SN-T08, ML-T05	Play history visible in calibrate.py output	W2
W5	**Observability and clean-up**	SN-T09, ML-T06, DS-T01	Run monitor raises a test warning	W3, W4

D11: **Roadmap timeline**

5 waves and 16 tasks; the security gate in W1 comes before any tunnel.



D11 · n = 16 · Unit: tasks · ▲ Queued

· Source: roadmap[] (tasks, gate, depends); tasks[] (project, type, depends, gaps); meta.ml_snapshot (Today) · Method: hand-placed columns, dependency arrows routed through row corridors; 1 crossings

Section 8: **Gaps, proposed modules and storage backbone** 28 gaps · 5 modules

Gaps: **Critical data gaps** 28 gaps

Gap: **README contract names a songnodes script that no longer exists**

ml-gap-sync-script-missing · ◐ Medium · ● Built

README and .gitignore say scripts/sync-music-library.sh copies library.json to frontend/public/data/music-library.json. That script and folder are gone; songnodes now loads /api/v1/library/search fed by POST /api/v1/library/ingest (import-music-library.sh, and cmd_sync_songnodes inside cmd_db).

Fix — Update README 'In SongNodes' section to name scripts/import-music-library.sh and the run_weekly db step; drop the .gitignore lines for music-library.json in songnodes.

Data contract — `library.json v1 (build_db.export_artifact_json) → POST /api/v1/library/ingest, Idempotency-Key import-<md5>`

Affects — musicLibrary/README.md, songnodes/.gitignore

Evidence 3

```
musicLibrary/README.md
songnodes/.gitignore:251-253
songnodes/scripts/import-music-library.sh
```

Gap: **Saved-set shape is no longer shared between the two Set Builders**

ml-gap-saved-set-shape · × Critical · ● Built

musicLibrary sets carry tidal, shape, minutes, mix, pool, seed and ledger track ids (h:/bp:). songnodes stores only name, description and uuid track ids server-side and has no tidal flag. A songnodes 'Export sets for the weekly run' file imports into sets/set_tracks with uuid ids that match no ledger track, and sync-sets skips every set (no tidal). /api/v1/library/sets/export returns {sets:[...]} without ids, which load_sets turns into a list of lists and crashes on.

Fix — Version the document: add schema 'naarm.saved-set.v2' {id, name, shape, minutes, mix, pool, seed, tidal, tracks[{ml_id, sn_id, tgt, move}]}; songnodes stores the extra fields in library.sets (jsonb) and maps uuid→source_id via library.track_xref on export; load_sets accepts {sets:[...]} and resolves ids via xref.

Data contract — `naarm.saved-set.v2`

Affects — musicLibrary/pipeline/build_db.py, musicLibrary/pipeline/run_weekly.py, songnodes/frontend/src/features/musicLibrary/data.ts, songnodes/services/library/model/library_db.py

Evidence 4

```
musicLibrary/pipeline/build_db.py:93-106
musicLibrary/pipeline/run_weekly.py:529-547
songnodes/frontend/src/features/musicLibrary/data.ts:140-161
songnodes/services/library/model/library_db.py:50-62
```

Gap: **JohnCourse.json and kiki.json are written but never consumed**

`ml-gap-setlists-orphan` · × Critical · ● Built

Both are songnodes.setlists.v1 DJ tracklists (29 and 5 sets) at the musicLibrary root. cmd_sync_songnodes globs only the parent folder and inbox/, so they are never found; if found, the body {schema, setlists} does not match TracklistImportRequest {tracks, dj_artist, name} and the 422 is swallowed. JohnCourse.json.bak is invalid JSON.

Fix	Move them to inbox/setlists/; post one request per setlist as {name: event, dj_artist: dj, tracks: [{artist,title,position,...}]}; log failures to outputs/<date>/ALERT_songnodes.txt.
Data contract	`songnodes.setlists.v1 → TracklistImportRequest`
Affects	musicLibrary/pipeline/run_weekly.py, musicLibrary root JSON

Evidence 4

```
musicLibrary/pipeline/run_weekly.py:599-616
songnodes/services/rest_api/main.py:96-100
musicLibrary/JohnCourse.json
musicLibrary/kiki.json
```

Gap: **Rekordbox play history and overrides do not feed back automatically**

`ml-gap-rekordbox-feedback` · ◐ Medium · ● Built

Play history reaches only calibrate.py, and only from a hand-dropped overrides/rekordbox_collection.xml (absent); calibrate writes suggestions only and has never run. local_path must be typed into overrides, so the XML export holds 0 tracks and the buy list never shrinks.

Fix	Parse rekordbox_collection.xml in process: match by Location/Artist+Title to set local_path and last_played, add a play appearance (source 'Rekordbox') so rotation sees real play; keep calibrate as the suggestion step.
Data contract	`rekordbox DJ_PLAYLISTS TRACK{Location, Artist, Name, PlayCount, LastPlayed, DateAdded} → ledger local_path, appearances`
Affects	musicLibrary/pipeline/run_weekly.py, musicLibrary/pipeline/calibrate.py

Evidence 2

```
musicLibrary/pipeline/calibrate.py:55-83
musicLibrary/pipeline/djlib/export.py:356-358
```

Gap: **songnodes proven transitions never inform musicLibrary set building**

`ml-gap-transitions-not-used` · ◐ Medium · ● Built

Direction musicLibrary → songnodes exists (library.json ingest; load-library-targets.sql makes library tracks scrape targets). Nothing flows back: library.json and the artifact Set Builder have no transition data, and the songnodes /sets page does not call the graph either.

Fix	Add GET /api/v1/library/transitions?ids=... returning occurrence_count for library-track pairs; have build_db fetch it (optional) and add 'nx' (proven next track ids with counts) to library.json tracks; Set Builder ranks spares by it.
Data contract	`library.json tracks[].nx: [{id, n}]`
Affects	songnodes graph-api, musicLibrary/pipeline/build_db.py, Set Builder pages

Evidence 2

```
songnodes/scripts/load-library-targets.sql
musicLibrary/pipeline/build_db.py:190-229
```

Gap: **Second-source BPM and key are discarded**

`ml-gap-deezer-bpm-not-merged` · × Critical · ● Built

When the chosen version is TIDAL, Deezer's BPM is used only for the cross-check; ref_mb is not queried by the pipeline at all despite the 2026-10-25 decision to cut web calls with it.

Fix	In process, when v.bpm is null take the Deezer twin's bpm (fold to pillar range) and record bpm_source; add an optional ref_mb lookup by ISRC for key/label/release date.
Data contract	`ledger track bpm_source, key_source`
Affects	musicLibrary/pipeline/run_weekly.py

Evidence 2

```
musicLibrary/pipeline/run_weekly.py:226-230
musicLibrary/decisions/2026-10-25-metabrainz-integration-
baseline.md
```

Gap: **Outputs with no reader**

`ml-gap-written-never-read` · × Critical · ● Built

outputs/crates/*.csv, rejected.csv, ledger/history/*, db/dj_library.sqlite (only DB Browser), tests/reports/* (songnodes output) and the review_queue override column have no code reader; ledger beatport_check (243) is read only by the supervised Beatport playbook, which has never run.

Fix	Either delete the step or give it a consumer: have process read the newest review_queue.csv override column into energy_overrides; move tests/reports to songnodes.
Data contract	`review_queue.csv energy_score_override → overrides`
Affects	musicLibrary/outputs, musicLibrary/tests

Evidence 2

```
musicLibrary/pipeline/djlib/export.py:285-305
musicLibrary/pipeline/run_weekly.py:298-308
```

Gap: **TIDAL login collides with songnodes on 127.0.0.1:3006**

`ml-gap-tidal-port` · ◐ Medium · ● Built

Both use the same TIDAL app and redirect; songnodes' /oauth/callback route consumes the single-use code if the frontend is up. Two TIDAL send paths exist (sync-sets 'SET · name' and songnodes /api/v1/tidal/sets/send) with separate token stores and no shared playlist map.

Fix	Register a second redirect (http://127.0.0.1:3016/oauth/callback) on the TIDAL app for musicLibrary, or reuse songnodes' stored token through an endpoint; keep one playlist map keyed by set id.
Data contract	`TIDAL redirect per client; shared set→playlist map`
Affects	musicLibrary/pipeline/djlib/tidal.py, songnodes music_auth / tidal_playlists

Evidence 3

```
musicLibrary/pipeline/djlib/tidal.py:32
songnodes/frontend/src/App.tsx:33
songnodes/services/rest_api/routers/tidal_playlists.py:318
```

Gap: **songnodes drops pipeline fields on ingest**

`ml-gap-songnodes-field-loss` · ◐ Medium · ● Built

library_search returns bp NULL, generated null, empty sources and runs, ids as uuids; rev/ovr/local/sync come back only if mapped. The songnodes Library page therefore cannot show buy links or run history that the artifact shows.

Fix	Persist export extras (bp, sync, rev, ovr, local) from _from_export 'export' into evidence and select them in library_search; return generated and runs from library.runs.
Data contract	`LibraryTrack parity between library.json and /api/v1/library/search`
Affects	songnodes/services/rest_api/routers/library_search.py

Evidence 2

```
songnodes/services/rest_api/routers/library_search.py
songnodes/frontend/src/features/musicLibrary/data.ts:121-129
```

Gap: **Scraper must stop at bronze**

`sn-be-gap-scraper-bronze-only` · ▲ High · ● Built

The unified scraper writes silver directly (layer skip, Unknown Artist, no outbox).

Fix	Remove the tracks/playlists/playlist_tracks/song_adjacency branches from PersistencePipeline._flush_batch (keep bronze_* only) and rely on the ingest_pending follow-up; add a unique index on silver_enriched_tracks (bronze_id) or (lower(artist_name), lower(track_title)) and convert upsert_track to ON CONFLICT.
Data contract	`Scraper output = bronze_scraped_playlists + bronze_scraped_tracks with positions 1..n; silver is written only by common/datalayer/ingest.py inside a transaction that appends an outbox event.`
Affects	songnodes/scrapers/pipelines, songnodes/services/common/datalayer

Evidence 2

```
songnodes/scrapers/pipelines/persistence_pipeline.py:910
songnodes/services/common/datalayer/ingest.py:104
```

Gap: **Target scrapes must bypass the daily discovery quota**

`sn-be-gap-target-scrape-quota` · ▲ High · ● Built

1001tracklists targeted searches are blocked by scraped:setlists:1001tracklists:last_run.

Fix — Set self.force_run = True in OneThousandOneTracklistsSpider.__init__ when search_query is given (as MixesDB does at line 116), or have the handler pass force_run=1 via params; and only stamp last_searched / found_nothing when the spider did not close with daily_quota_reached (return a reason from /scrape).

Data contract — `POST /scrape response gains close_reason; handler treats daily_quota_reached as RetryableError.`

Affects — songnodes/scrapers/spiders, songnodes/services/common/datalayer/handlers.py

Evidence 2

```
songnodes/scrapers/spiders/1001tracklists_spider.py:189
songnodes/services/common/datalayer/handlers.py:85
```

Gap: **Run the scrape HTTP call outside the job transaction**

`sn-be-gap-scrape-outside-tx` · ◐ Medium · ● Built

A 16-minute open transaction holds xmin and stalls the outbox.

Fix — In runner.run_one, let handlers declare long_io=True; for scrape, perform the HTTP POST before opening conn.transaction(), then record_playlists_found + follow-ups + complete inside a short transaction; lease (1800s) still guards double-run.

Data contract — `Handler contract: (prepare outside tx) → (commit inside tx).`

Affects — songnodes/services/common/datalayer/runner.py, songnodes/services/common/datalayer/handlers.py

Evidence 1

```
songnodes/services/common/datalayer/runner.py:37
```

Gap: **Decide and document the real layer model**

`sn-be-gap-medallion-doc` · ▲ High · ● Built

Gold/operational exist only in docs and unused Helm cronjobs.

Fix — Either (a) update CLAUDE.md to Bronze → Silver → projection_tracks/graph reads, drop gold_track_graph, or (b) add a gold job kind (REFRESH MATERIALIZED VIEW CONCURRENTLY gold_track_graph after ingest, apply medallion 008) and point graph-api at it. Record the choice in docs/decisions.

Data contract — `Layer contract table: layer → table → sole writer → readers.`

Affects — songnodes/CLAUDE.md, songnodes/services/graph-visualization-api

Evidence 2

```
songnodes/CLAUDE.md:11
songnodes/sql/migrations/medallion/008_gold_track_graph_index_up.sql:3
```

Gap: **Make API token usable by scripts and TIDAL routes**

`sn-be-gap-auth-token-flow` · × Critical · ● Built

Token middleware collides with TIDAL Bearer; import script sends no token; prefix typo. Queued PRs #578/#579 add a separate user login (cookie) but no enforcement on existing routes.

Fix	In security/auth.py prefer X-API-Token over Authorization, fix the prefix to /api/v1/api-keys and add /api/v1/music-auth/spotify/token/status; pass SONGNODES_API_TOKEN and API_KEY_ENCRYPTION_SECRET through compose.dev.yaml; add -H "X-API-Token: $SONGNODES_API_TOKEN" to import-music-library.sh curl calls.
Data contract	`X-API-Token header is the only admin credential; Authorization is reserved for provider tokens.`
Affects	songnodes/services/rest_api/security, songnodes/compose.dev.yaml, songnodes/scripts/import-music-library.sh

Evidence 2

```
songnodes/services/rest_api/security/auth.py:58
songnodes/scripts/import-music-library.sh:45
```

Gap: **Project library_track events into projection_tracks**

`sn-be-gap-library-projection` · ◐ Medium · ● Built

Library search reads stale rows after a library ingest.

Fix	In projection.handle_event treat aggregate_type library_track like track (project_track on aggregate_id), and add a consumer or explicit ignore list for library_set and setlist events.
Data contract	`Outbox event registry: event_type → consumers (projection_tracks, graph_projection).`
Affects	songnodes/services/common/datalayer/projection.py

Evidence 1

```
songnodes/services/common/datalayer/projection.py:75
```

Gap: **Working backup for the compose volume**

`sn-be-gap-backups` · ▲ High · ● Built

No script backs up songnodes-dev_dev-postgres-18.

Fix	Rewrite scripts/backup_database.sh to docker compose -f compose.dev.yaml exec -T postgres pg_dump -U songnodes -Fc songnodes > backups/songnodes-$(date +%Y%m%d).dump with retention, and retire backup_volumes.sh.
Data contract	`Backup file: backups/songnodes-YYYYMMDD.dump (pg custom format).`
Affects	songnodes/scripts

Evidence 1

```
songnodes/scripts/backup_database.sh:52
```

Gap: **Single migration runner with a ledger**

`sn-be-gap-migration-ledger` · ◐ Medium · ● Built

Migrations are a hand list in start-dev.sh.

Fix	Apply 000_schema_migrations_up.sql and extend services/worker/migrate.py to apply every sql/migrations/*_up.sql in numeric order once, recording filename+checksum; start-dev.sh calls docker compose run --rm worker python migrate.py.
Data contract	`schema_migrations(version, checksum, applied_at).`
Affects	songnodes/services/worker/migrate.py, songnodes/scripts/start-dev.sh

Evidence 2

```
songnodes/services/worker/migrate.py:1
songnodes/scripts/start-dev.sh:31
```

Gap: **Enforce the library ingest body contract**

`sn-be-gap-library-ingest-409` · ◐ Medium · ● Built

Same key + different body silently returns the old job.

Fix	Store sha256 of the canonical JSON payload in ingestion_jobs.payload_hash (or compare payload) in jobs.enqueue for library_ingest and raise CatalogError(409) on mismatch.
Data contract	`Idempotency-Key + payload hash.`
Affects	songnodes/services/common/datalayer/jobs.py, songnodes/services/rest_api/routers/library_ingest.py

Evidence 1

```
songnodes/contracts/library_ingest.md:16
```

Gap: **Music search has no deployed backend**

`sn-be-gap-music-search` · ◐ Medium · ● Built

rest-api proxies to metadata-enrichment:8086, not in compose.

Fix	Move the S1 search router into rest-api using common/providers (tidal, deezer) or add metadata-enrichment to compose; until then return an explicit 503 with cause in the UI.
Data contract	`GET /api/v1/music-search/{service}?q= → [{id, artist, title, isrc}].`
Affects	songnodes/services/rest_api

Evidence 1

```
songnodes/services/rest_api/main.py:692
```

Gap: **Frontend calls routes that do not exist**

`sn-be-gap-frontend-dead-routes` · ◐ Medium · ● Built

DJInterface posts /api/v1/scrapers/target-tracks/search and /api/v1/scrapers/queue/clear.

Fix	Point DJInterface at POST /api/v1/target-tracks/{id}/search and drop queue/clear (no queue endpoint), or add the routes in a router module.
Data contract	`Use existing target-tracks router.`
Affects	songnodes/frontend/src/components/DJInterface.tsx

Evidence 2

```
songnodes/frontend/src/components/DJInterface.tsx:360
songnodes/frontend/src/components/DJInterface.tsx:432
```

Gap: **OAuth tokens written to sessionStorage have no reader**

`sn-fe-gap-oauth-session-tokens` · ▲ High · ● Built

Settings connects in the same window (window.location.href), so OAuthCallback has no opener: it writes sessionStorage <service>_oauth_tokens and redirects to /. GraphPage and SettingsPanel only read localStorage tidal_oauth_tokens/spotify_oauth_tokens (which OAuthCallback removes), so the browser store never receives the tokens; sendToTidal and useTokenRefresh then lack an access token.

Fix	In OAuthCallback, when window.opener is null, call useStore.getState().credentials.updateCredentials(service, {accessToken,refreshToken,expiresAt,isConnected:true}) before redirecting to /settings/<service>-connection, and delete the sessionStorage write; or have SettingsPage read and clear sessionStorage '<service>_oauth_tokens' on mount.
Data contract	`oauth-handoff/v1: {service, access_token, refresh_token, expires_at} consumed exactly once by the credentials slice`
Affects	songnodes/frontend/src/components/OAuthCallback.tsx, songnodes/frontend/src/pages/SettingsPage.tsx

Evidence 3

```
songnodes/frontend/src/components/OAuthCallback.tsx:226-270
songnodes/frontend/src/pages/SettingsPage.tsx:362-394
songnodes/frontend/src/pages/GraphPage.tsx:15-33
```

Gap: **Graph page calls scraper endpoints that do not exist**

`sn-fe-gap-scrapers-endpoints` · Medium · Built

DJInterface posts to /api/v1/scrapers/target-tracks/search and /api/v1/scrapers/queue/clear; no rest-api route matches (route scan of services/rest_api).

Fix	Point the actions at POST /api/v1/target-tracks/{id}/search and POST /api/v1/scrape/trigger, or remove the buttons; add a contract test listing every frontend path against the FastAPI OpenAPI document.
Data contract	`frontend-endpoint-manifest.json checked against rest-api /openapi.json in CI`
Affects	songnodes/frontend/src/components/DJInterface.tsx

Evidence 2

```
songnodes/frontend/src/components/DJInterface.tsx:360,395,432
/home/claude/dfa/scratch/sn-frontend/backend_routes.tsv
```

Gap: **No IndexedDB last-good-data cache despite design-system rule**

`sn-fe-gap-indexeddb-cache` · Medium · Built

design-system/16-platform.md:50 says SongNodes keeps the last graph, library and saved sets in IndexedDB and shows them with FreshnessStamp when the API is unreachable. No indexedDB usage exists in frontend/src; library load just errors and Home shows a dash.

Fix	Add frontend/src/lib/offlineCache.ts (idb-keyval style store "sn-cache" with keys graph, library, sets, each {data, fetchedAt}); write on successful loadLibrary/useDataLoader/readSavedSetsAsync, read on failure and pass fetchedAt to FreshnessStamp.
Data contract	`sn-cache/v1 {key, data, fetchedAt ISO, apiVersion}`
Affects	songnodes/frontend/src/features/musicLibrary/data.ts, songnodes/frontend/src/hooks/useDataLoader.ts

Evidence 2

```
songnodes/design-system/16-platform.md:50
songnodes/frontend/src/features/musicLibrary/data.ts:107-135
```

Gap: **artifact_sets.json lacks the tidal flag musicLibrary needs**

`sn-fe-gap-artifact-sets-tidal-flag` · × Critical · ● Built

Set builder exports SavedSet[] without a 'tidal' field; musicLibrary run_weekly.py sync-sets filters sets by s.get('tidal'), so sets exported from songnodes are never pushed to TIDAL by the weekly run (songnodes has its own Send to TIDAL instead). The file is also only a browser download that has to be copied by hand.

Fix — Either add 'tidal: boolean' (set when Send to TIDAL is requested) to SavedSet and the export, or document that songnodes sets go to TIDAL through /api/v1/tidal/sets/send and drop the inbox path; better, have musicLibrary read GET /api/v1/library/sets/export (exists, unused by the frontend).

Data contract — `artifact_sets/v2: SavedSet[] + {tidal:boolean}`

Affects — songnodes/frontend/src/features/musicLibrary/SetBuilderPage.tsx, musicLibrary/pipeline/run_weekly.py

Evidence 3

```
songnodes/frontend/src/features/musicLibrary/SetBuilderPage.tsx:223-
226
musicLibrary/pipeline/run_weekly.py:529
musicLibrary/pipeline/build_db.py:93-106
```

Gap: **Pending target syncs are never retried and override the server**

`sn-fe-gap-target-pending-sync` · ◐ Medium · ● Built

A failed POST/DELETE leaves pendingSync=true in sn-target-tracks-v1; nothing reads pendingSync, and useUnifiedTracks prefers the local entry over backend state, so the device can disagree with target_tracks forever.

Fix — On useUnifiedTracks.refresh(), replay entries with pendingSync=true through syncTargetTrack, and drop local entries whose backend match agrees; show a count of unsynced targets.

Data contract — `sn-target-tracks-v2 {isTarget, priority, pendingSync, lastAttempt}`

Affects — songnodes/frontend/src/services/trackSyncAdapter.ts, songnodes/frontend/src/hooks/useUnifiedTracks.ts

Evidence 2

```
songnodes/frontend/src/services/trackSyncAdapter.ts:31-100
songnodes/frontend/src/hooks/useUnifiedTracks.ts:93-103
```

Gap: **Library freshness stamp is always unknown**

`sn-fe-gap-library-generated` · ◐ Medium · ● Built

loadLibrary sets generated: null and runs: [] because /api/v1/library/search returns rows only; Home and Library show an en dash for 'Library data from' and the weekly-run launcher summary never appears.

Fix	Add GET /api/v1/library/meta returning {generated, ledger_updated, counts, sources, runs} from the ingest job result and merge it in loadLibrary.
Data contract	`library-meta/v1 = LibraryFile minus tracks`
Affects	songnodes/frontend/src/features/musicLibrary/data.ts, songnodes/services/rest_api/routers/library_search.py

Evidence 2

```
songnodes/frontend/src/features/musicLibrary/data.ts:120-130
songnodes/frontend/src/pages/HomePage.tsx:22-46
```

Gap: **Graph WebSocket has a server but no client**

`sn-fe-gap-graph-ws` · ◐ Medium · ● Built

graph-api exposes WEBSOCKET /api/graph/ws/{room_id} (nginx passes Upgrade headers for /api/graph). The only frontend WebSocket client (LiveDataLoader, /api/ws) is unreachable and targets a path rest-api does not serve.

Fix	Delete LiveDataLoader/useLiveDataLoader, or rewire it to /api/graph/ws/<room> and mount it from GraphPage.
Data contract	`graph-ws/v1 message {type: nodes|edges|delta, payload}`
Affects	songnodes/frontend/src/components/LiveDataLoader.tsx

Evidence 2

```
songnodes/frontend/src/components/LiveDataLoader.tsx:138-140
songnodes/frontend/Dockerfile:81-91
```

Gap: **Library ingest allows 50 MB but frontend nginx accepts 1 MB**

`sn-fe-gap-ingest-body-limit` · ◐ Medium · ● Built

LibraryImportDialog FileDrop limit is 50 MB and the body is the whole export; the frontend image nginx config sets no client_max_body_size, so nginx default 1 MB returns 413 for any larger export. STATUS.md already lists it as a follow-up.

Fix	Add client_max_body_size 64m; inside location /api/ (or a dedicated location = /api/v1/library/ingest) in the frontend Dockerfile default.conf, and map HTTP 413 to an explicit error in LibraryImportDialog.
Data contract	`ingest upload limit = FileDrop maxMb = nginx client_max_body_size = rest-api request limit`
Affects	songnodes/frontend/Dockerfile, songnodes/frontend/src/features/musicLibrary/components/LibraryImportDialog.tsx

Evidence 3

```
songnodes/frontend/src/features/musicLibrary/components/LibraryImportDialog.tsx:23-
24
songnodes/frontend/Dockerfile:91-99
songnodes/STATUS.md:1899
```

Modules: **Proposed single-purpose modules** `5 modules`

Module: **Set exchange**

Target users	Set planner, weekly-run agent
One verb	**Convert**
Data inputs	songnodes saved sets (/api/v1/library/sets/export), artifact sets (artifact db `sets`)
Core operation	Reads either saved-set shape and writes one versioned `songnodes.saved-set/2` document (name, shape, minutes, tracks[{isrc, tidal_id, ledger_id, tgt}], tidal) to musicLibrary inbox/artifact_sets.json, rejecting rows it cannot map.
Size	about 250 lines (Estimate)
Where it lives	`musicLibrary/pipeline/djlib/set_exchange.py (new)`
Design system	Built from the SongNodes design system in full (tokens, sn-* classes, page pattern).

Module: **Play-history import**

Target users	Library curator
One verb	**Import**
Data inputs	Rekordbox / Lexicon collection XML, play counts and dates
Core operation	Matches plays to ledger tracks by local_path then ISRC, appends `play` events to the ledger and writes rotation evidence that calibrate.py reads.
Size	about 300 lines (Estimate)
Where it lives	`musicLibrary/pipeline/djlib/play_history.py`
Design system	Built from the SongNodes design system in full (tokens, sn-* classes, page pattern).

Module: **Transition feed**

Target users	Set planner, DJ
One verb	**Export**
Data inputs	songnodes silver_track_transitions (occurrence_count) joined to library tracks
Core operation	Publishes `songnodes.transitions/1` (pairs of ISRCs with counts) beside library.json so both Set Builders can rank spares by proven pairings.
Size	about 200 lines (Estimate)
Where it lives	`songnodes/services/rest_api/routers/transitions_export.py`
Design system	Built from the SongNodes design system in full (tokens, sn-* classes, page pattern).

Module: **Contract check**

Target users	Developer, pipeline operator
One verb	**Verify**
Data inputs	contracts/*.schema.json, the newest musicLibrary outputs, the songnodes ingest fixture
Core operation	Validates every cross-project file and request body against its schema in CI and in the weekly run, and fails with the field that broke.
Size	about 180 lines (Estimate)
Where it lives	`songnodes/scripts/check_contracts.py`
Design system	Built from the SongNodes design system in full (tokens, sn-* classes, page pattern).

Module: **Run monitor**

Target users	Pipeline operator, library curator
One verb	**Alert**
Data inputs	musicLibrary runs table and scrape_log; songnodes ingestion_jobs and web_call_log
Core operation	Compares each run's resolve yield, source counts and job failures with the last four runs and raises one Banner-shaped warning with a source link when a figure drops by half.
Size	about 220 lines (Estimate)
Where it lives	`songnodes/frontend/src/features/pipeline/RunMonitor.tsx`
Design system	Built from the SongNodes design system in full (tokens, sn-* classes, page pattern).

Backbone: **Unified storage backbone**

Keep each project's own data folder and owner. Add one shared contracts folder and one exchange folder with versioned, schema-checked files, so every cross-project hop is a named file or endpoint with one writer.

```
songnodes/
  contracts/                      # single source of every cross-project schema (exists; extend)
    library.set.saved.v2.json      # new
    library_ingest.openapi.yaml    # exists
    transitions.v1.json            # new
    setlists.v1.json               # new
  Postgres (compose volume)       # owner: songnodes services only
    bronze_* → silver_* → (gold_* decided in DFA gap) → projection_*
musicLibrary/
  ledger/                         # owner: process step (append-only)
  db/dj_library.sqlite            # rebuilt each run from ledger
  inbox/                          # inputs written by people or other systems
  outputs/<date>/                 # run outputs, read by people
  exchange/                       # NEW: the only files the other project reads or writes
    library.json                  # musicLibrary → songnodes (also POST /api/v1/library/ingest)
```

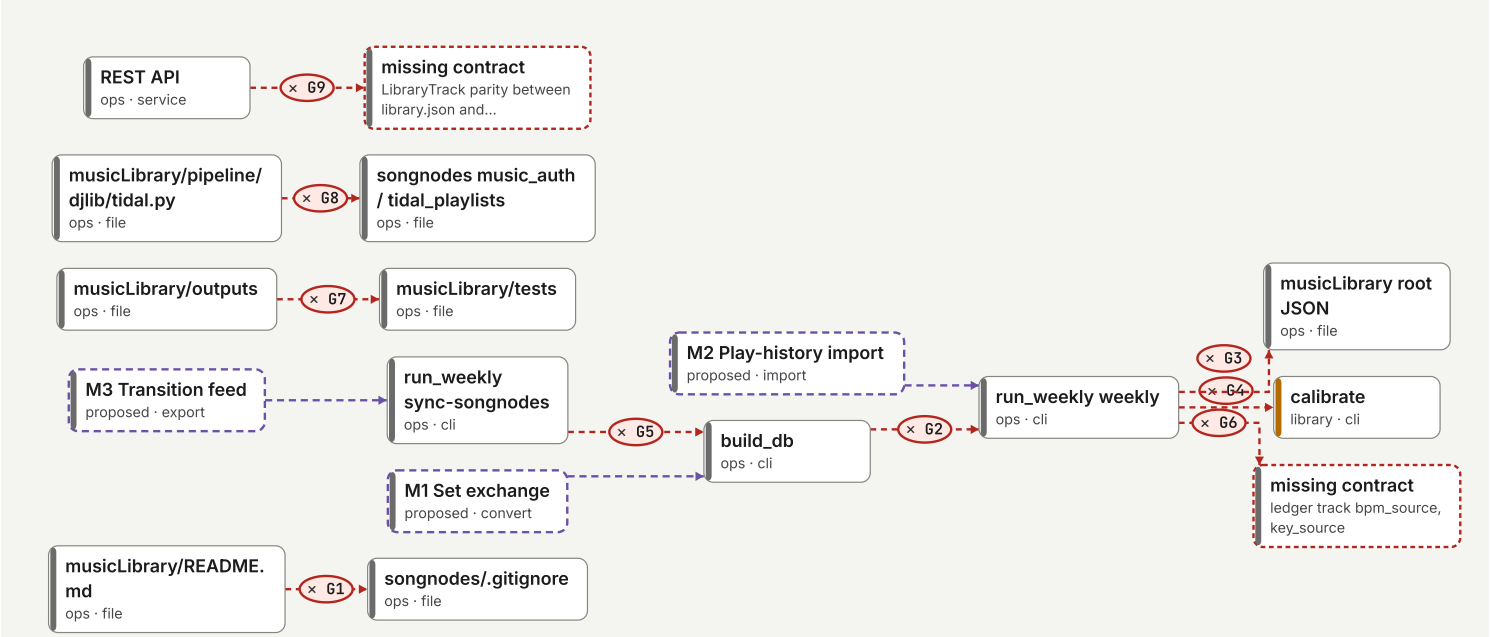
```
    saved-sets.json              # both ways, saved-set/2
    transitions.json             # songnodes → musicLibrary, transitions/1
    setlists/*.json              # tracklists for songnodes import (moved from the root)
```

4 rows

RISK	MITIGATION
Readers still point at old paths (README, import script, artifact)	Keep the old paths as copies for one release and log a warning when they are read.
Saved-set/1 documents in the artifact db	M1 reads both shapes; write only /2; keep /1 readable until no /1 set has been opened for four weeks.
Weekly run writes a file songnodes is reading	Write to a temporary name and rename; songnodes reads only complete files.
Postgres schema change for gold	One migration through the migration ledger gap first; never reset the volume (CLAUDE.md).

D8a: **Gap map – musicLibrary and cross-project**

9 musicLibrary and cross-project gaps, 4 of them critical; dashed brand cards are the modules that close them.



Gap contracts

- **G1** library.json v1 (build_db.export_artifact_json) → POST /api/v1/library/ingest, Idempotency-Key import-<md5> (medium)
- **G2** naarm.saved-set.v2 (critical)
- **G3** songnodes.setlists.v1 → TracklistImportRequest (critical)
- **G4** rekordbox DJ_PLAYLISTS TRACK{Location, Artist, Name, PlayCount, LastPlayed, DateAdded} → ledger local_path, appearances (medium)
- **G5** library.json tracks[].nx: [{id, n}] (medium)
- **G6** ledger track bpm_source, key_source (critical)
- **G7** review_queue.csv energy_score_override → overrides (critical)
- **G8** TIDAL redirect per client; shared set→playlist map (medium)
- **G9** LibraryTrack parity between library.json and /api/v1/library/search (medium)

Key

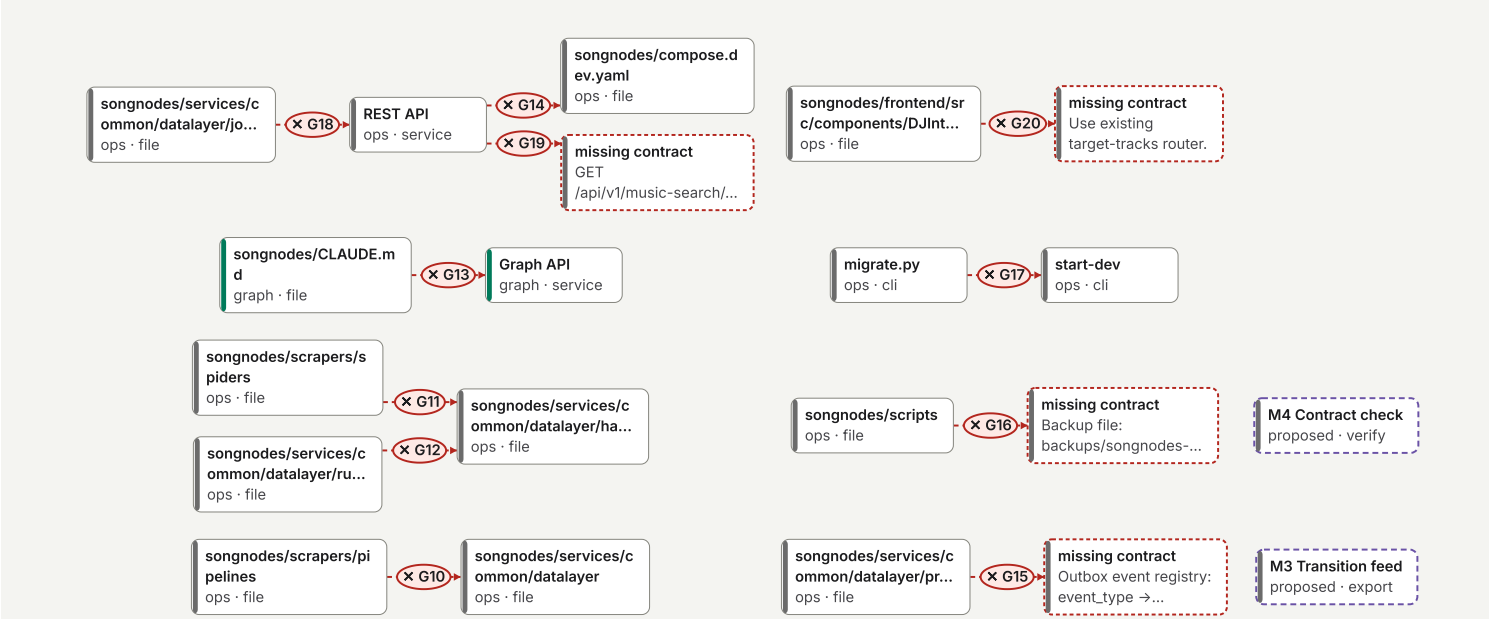
library · ops and design · Production (solid outline) · Proposed (brand, dashed) · Gap (red dashed, ×) · Proposed link (brand dashed)

D8a · n = 9 · Unit: gaps · ● Built

· Source: gaps[] (title, contract, affects, severity, effort); modules[]; tasks[] (gaps, title) · Method: graphviz dot layered layout (LR, order given; 12 rank directions and input orders tried, scored by legibility then crossings), splines=ortho, own number badges, type floor 12 units; 0 crossings, 0 edges through nodes; gap endpoints = first two affected paths matched to pages by path prefix

D8b: **Gap map – songnodes backend**

11 songnodes backend gaps, 1 of them critical; dashed brand cards are the modules that close them.



Gap contracts

G10 Scraper output = bronze_scraped_playlists + bronze_scraped_tracks with positions 1..n; silver is written only by common/datalayer/ingest.py inside a transaction that appends an...

G11 POST /scrape response gains close_reason; handler treats daily_quota_reached as RetryableError. (high)

G12 Handler contract: (prepare outside tx) → (commit inside tx). (medium)

G13 Layer contract table: layer → table → sole writer → readers. (high)

G14 X-API-Token header is the only admin credential; Authorization is reserved for provider tokens. (critical)

G15 Outbox event registry: event_type → consumers (projection_tracks, graph_projection). (medium)

G16 Backup file: backups/songnodes-YYYYMMDD.dump (pg custom format). (high)

G17 schema_migrations(version, checksum, applied_at). (medium)

G18 Idempotency-Key + payload hash. (medium)

G19 GET /api/v1/music-search/{service}?q= → [{id, artist, title, isrc}]. (medium)

G20 Use existing target-tracks router. (medium)

Key

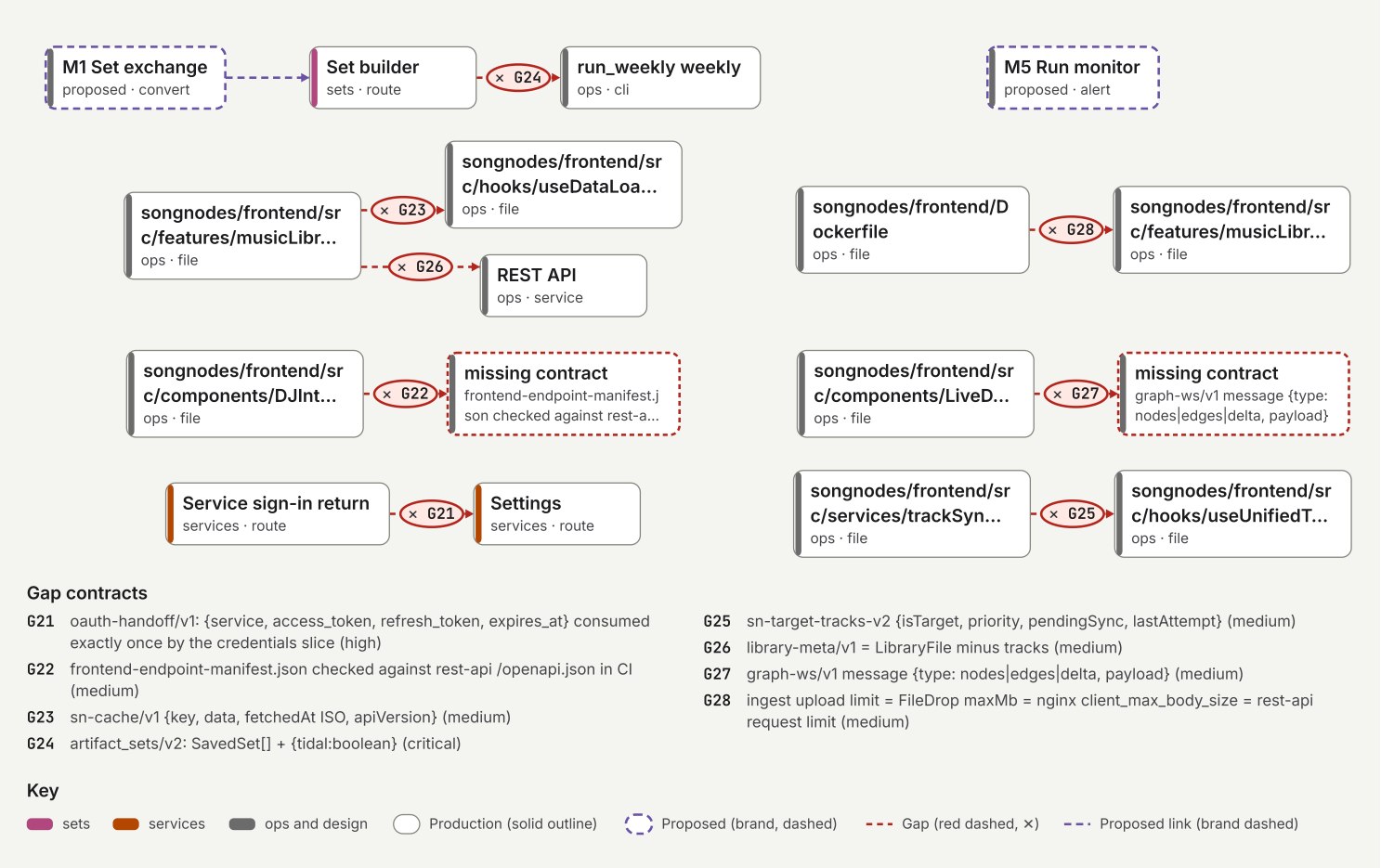
graph · ops and design · Production (solid outline) · Proposed (brand, dashed) · Gap (red dashed, ×) · Proposed link (brand dashed)

D8b · n = 11 · Unit: gaps · ● Built

· Source: gaps[] (title, contract, affects, severity, effort); modules[]; tasks[] (gaps, title) · Method: graphviz dot layered layout (LR, order given, components packed 4 across; 12 rank directions and input orders tried, scored by legibility then crossings), splines=ortho, own number badges, type floor 15 units; 0 crossings, 0 edges through nodes; gap endpoints = first two affected paths matched to pages by path prefix

D8c: **Gap map – songnodes frontend**

8 songnodes frontend gaps, 1 of them critical; dashed brand cards are the modules that close them.



Gap contracts

- **G21** oauth-handoff/v1: {service, access_token, refresh_token, expires_at} consumed exactly once by the credentials slice (high)
- **G22** frontend-endpoint-manifest.json checked against rest-api /openapi.json in CI (medium)
- **G23** sn-cache/v1 {key, data, fetchedAt ISO, apiVersion} (medium)
- **G24** artifact_sets/v2: SavedSet[] + {tidal:boolean} (critical)
- **G25** sn-target-tracks-v2 {isTarget, priority, pendingSync, lastAttempt} (medium)
- **G26** library-meta/v1 = LibraryFile minus tracks (medium)
- **G27** graph-ws/v1 message {type: nodes|edges|delta, payload} (medium)
- **G28** ingest upload limit = FileDrop maxMb = nginx client_max_body_size = rest-api request limit (medium)

Key

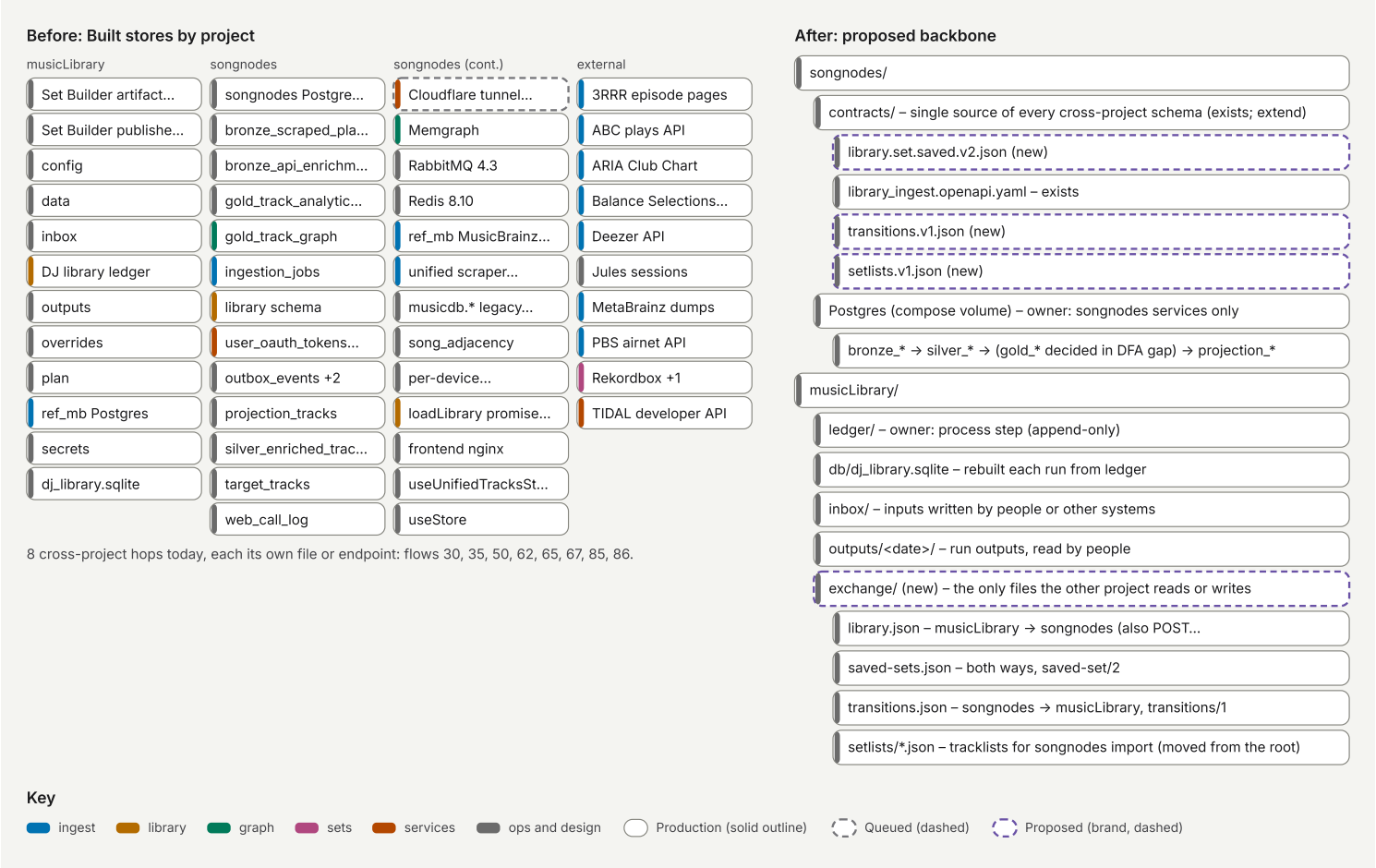
sets · services · ops and design · Production (solid outline) · Proposed (brand, dashed) · Gap (red dashed, ×) · Proposed link (brand dashed)

D8c · n = 8 · Unit: gaps · ● Built

· Source: gaps[] (title, contract, affects, severity, effort); modules[]; tasks[] (gaps, title) · Method: graphviz dot layered layout (LR, order given, components packed 4 across; 12 rank directions and input orders tried, scored by legibility then crossings), splines=ortho, own number badges, type floor 12 units; 0 crossings, 0 edges through nodes; gap endpoints = first two affected paths matched to pages by path prefix

D10: **Storage backbone before and after**

Each project keeps its own data folder; the change is 4 new contract and exchange entries that carry the 8 cross-project hops.



D10 · n = 66 · Unit: stores and backbone entries · **i built+projected**

· Source: stores[] (repo, kind, domain, state); backbone.tree; flows[] crossing repos · Method: hand-placed two-panel tree; 0 edges

Section 9: **Built and Projected** 37 queued items

Built = songnodes main @7429821 and the musicLibrary snapshot of 7 Nov 2026. Projected = Built with every queued item applied.

Table: **Queued work** 37 items

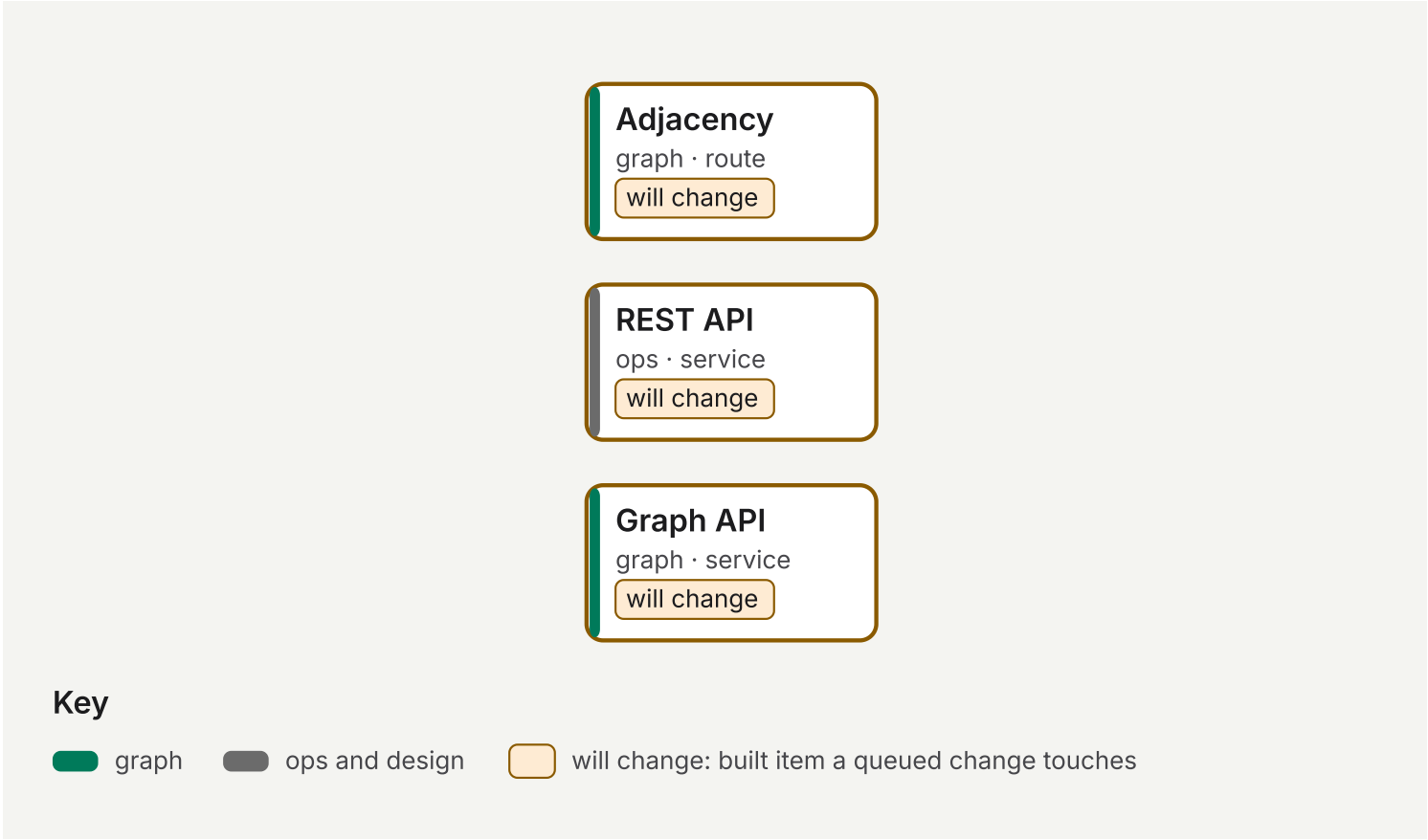
37 rows

ID	WHAT	SOURCE	EFFECT WHEN APPLIED	TOUCHES
ml-queued-be1	Real TIDAL and Spotify track search	musicLibrary/plan/jules_payloads/BE1.json	–	–
ml-queued-be2	Python 3.12 upgrade audit	musicLibrary/plan/jules_payloads/BE2.json	–	–
ml-queued-fe1	Unified track data bridge	musicLibrary/plan/jules_payloads/FE1.json	–	–
ml-queued-fe10	AI setlist discovery and import page	musicLibrary/plan/jules_payloads/FE10.json	–	–
ml-queued-fe11	Navigation audit	musicLibrary/plan/jules_payloads/FE11.json	–	–
ml-queued-fe12	Pipeline page on real job and outbox data	musicLibrary/plan/jules_payloads/FE12.json	–	–
ml-queued-fe2	Target Tracks module	musicLibrary/plan/jules_payloads/FE2.json	–	–
ml-queued-fe3	Route finder page	musicLibrary/plan/jules_payloads/FE3.json	–	–
ml-queued-fe4	Set Builder: crate export formats and reordering	musicLibrary/plan/jules_payloads/FE4.json	–	–
ml-queued-fe5	Key and mood Camelot module	musicLibrary/plan/jules_payloads/FE5.json	–	–
ml-queued-fe6	Pipeline health page	musicLibrary/plan/jules_payloads/FE6.json	–	–
ml-queued-fe7	Service settings	musicLibrary/plan/jules_payloads/FE7.json	–	–
ml-queued-fe8	Graph explorer workspace	musicLibrary/plan/jules_payloads/FE8.json	–	–
ml-queued-fe9	Add library tracks to Target Tracks	musicLibrary/plan/jules_payloads/FE9.json	–	–
ml-queued-few	Route wiring and cleanup	musicLibrary/plan/jules_payloads/FEW.json	–	–
ml-queued-ref0	ref_mb as a separate database	musicLibrary/plan/jules_payloads/REF0.json	–	–
ml-queued-sec1	Remove eval and audit subprocess use	musicLibrary/plan/jules_payloads/SEC1.json	–	–
ml-queued-sec2	CORS allowlist	musicLibrary/plan/jules_payloads/SEC2.json	–	–
ml-queued-sec3	Dependency security updates	musicLibrary/plan/jules_payloads/SEC3.json	–	–
ml-queued-sec4	Secrets and dependency scanning in CI	musicLibrary/plan/jules_payloads/SEC4.json	–	–
ml-queued-sec5	Opt-in token auth for write and admin routes	musicLibrary/plan/jules_payloads/SEC5.json	–	–
ml-queued-sec6	Server-side AI provider proxy	musicLibrary/plan/jules_payloads/SEC6.json	–	–
ml-queued-wp0	Contracts, DDL, fixtures	musicLibrary/plan/jules_payloads/WP0.json	–	–

ml-queued-wp2	MusicBrainz reference ETL into ref_mb	musicLibrary/plan/jules_payloads/WP2.json	–	–
ml-queued-wp3	Local MusicBrainz resolver	musicLibrary/plan/jules_payloads/WP3.json	–	–
ml-queued-wp4	Local MusicBrainz provider in enrichment	musicLibrary/plan/jules_payloads/WP4.json	–	–
ml-queued-wp5	Library overlay model, importer, ingest job and saved-sets endpoints	musicLibrary/plan/jules_payloads/WP5.json	–	–
ml-queued-wp6	Ingest API and transactional outbox	musicLibrary/plan/jules_payloads/WP6.json	–	–
ml-queued-wp8	Graph projection as outbox consumer	musicLibrary/plan/jules_payloads/WP8.json	–	–
ml-queued-wp9	Server-side search and Library / Set builder pages on the API	musicLibrary/plan/jules_payloads/WP9.json	–	–
ml-queued-wp7-pipeline-client	MetaBrainz reference data + musicLibrary → songnodes pipeline client	musicLibrary/plan/agents/WP7_pipeline_client.md	–	–
ml-queued-open-prs	Open songnodes PRs #576-#580 (Duo auth, tunnel, Sentinel)	gh api repos/zvirb/songnodes/pulls/576..580/files	–	–
PR-576	DUOAUTH: Cisco Duo 2FA and Cloudflare Tunnel architecture guide	github zvirb/songnodes#576	Docs only	docs
PR-577	DUO3: Cloudflare Tunnel connector integration	github zvirb/songnodes#577	Publishes the stack through a public tunnel	compose.dev.yaml, sn-be-q-cloudflared
PR-578	DUO1: Cisco Duo Web SDK v4 auth router	github zvirb/songnodes#578	Adds /api/v1/auth routes with a default admin login fallback	sn-be-q-user-auth-router
PR-579	DUO2: Frontend login page and auth context	github zvirb/songnodes#579	Adds /login	sn-fe-login
PR-580	Sentinel: sanitise REST API main exception details	github zvirb/songnodes#580	Sanitises errors, partly on unreachable routes	sn-be-svc-rest-api

D9: **Built to Projected change map**

Queued work adds 10 cards and 6 edges and changes 3 built items; 32 briefs wait behind them.



D9 · n = 51 · Unit: items · i built+queued

· Source: queued[] (PR touches and effect, brief notes); flows[] and pages[]/stores[] with state queued · Method: graphviz dot layered layout (LR, order given; 12 rank directions and input orders tried, scored by legibility then crossings), splines=ortho, own number badges, type floor 12 units; 0 crossings, 0 edges through nodes

Section 10: **Sister-project tasks** 16 tasks in 3 projects

C12: **Sister tasks by project, type and status**

16 queued tasks over 3 projects; 10 are ready now and 6 wait on another task.

	songnodes (9)	musicLibrary (6)	design-system (1)
fix (3)	**SN-T04** Ready **SN-T07** Ready **SN-T09** Waits on SN-T01		
module (3)	**SN-T08** Waits on SN-T03	**ML-T02** Waits on SN-T03 **ML-T04** Ready	
contract (2)	**SN-T03** Ready	**ML-T05** Waits on SN-T03	
data-integrity (2)	**SN-T05** Ready	**ML-T03** Ready	
docs (2)	**SN-T06** Waits on SN-T05	**ML-T01** Ready	
security (2)	**SN-T01** Ready **SN-T02** Waits on SN-T01		
design-system (1)			**DS-T01** Ready
observability (1)		**ML-T06** Ready	

Every task is Queued (dashed cell). Ready = no dependency; Waits on = blocked by the named task

C12 · n = 16 · Unit: tasks · ▲ Queued

· Source: tasks[] (project, type, depends) · Method: Tasks placed by type (rows, by count) and project (columns, by count). tasks[] has no status field: status is derived from depends (Ready when empty).

Table: **Task matrix** `16 tasks`

16 rows

TASK	PROJECT	PACKAGE	TYPE	OWNED PATHS	DEPENDS ON	VERIFICATION	STATUS
SN-T01	songnodes	**Require the API token on write and admin routes**	security	services/rest_api/security/auth.py, compose.dev.yaml	–	curl without token gets 401 on every POST/PUT/DELETE; scripts pass with the token	To do
SN-T02	songnodes	**Remove the default admin login from the queued auth router**	security	services/rest_api/routers/user_auth.py (in PR #578)	SN-T01	no login succeeds without a configured user	To do
SN-T03	songnodes	**Accept and keep musicLibrary pipeline fields on library ingest**	contract	services/rest_api/routers/library_ingest.py, contracts/library_ingest.openapi.yaml	–	fixture round-trip keeps energy, rotation, pillar and ISRC	To do
SN-T04	songnodes	**Raise the frontend nginx body limit to the ingest limit**	fix	frontend/Dockerfile	–	a 5 MB library import returns 202, not 413	To do
SN-T05	songnodes	**Stop the unified scraper at bronze**	data-integrity	scrapers/pipelines/persistence_pipeline.py	–	a scrape writes bronze rows only; ingest creates silver once	To do
SN-T06	songnodes	**Decide and document the real layer model**	docs	docs/DATA_LAYER.md, CLAUDE.md	SN-T05	docs and compose name the same jobs	To do
SN-T07	songnodes	**Let target scrapes bypass the daily discovery quota**	fix	scrapers/spiders/1001tracklists_spider.py	–	two target scrapes in an hour both run	To do
SN-T08	songnodes	**Publish the transition feed (M3)**	module	services/rest_api/routers/transitions_export.py (new), contracts/transitions.v1.json (new)	SN-T03	GET returns transitions/1 for library ISRCs	To do
SN-T09	songnodes	**Fix the OAuth token hand-off and remove dead scraper calls**	fix	frontend/src/components/OAuthCallback.tsx, frontend/src/pages/SettingsPage.tsx, frontend/src/pages/GraphPage.tsx	SN-T01	Send to TIDAL finds a token after a fresh login; no 404 in the graph page network log	To do
ML-T01	musicLibrary	**Correct the README songnodes contract**	docs	README.md	–	every path in the README exists	To do
ML-T02	musicLibrary	**Build the set exchange (M1)**	module	pipeline/djlib/set_exchange.py (new), pipeline/run_weekly.py	SN-T03	a songnodes set reaches TIDAL through sync-sets	To do
ML-T03	musicLibrary	**Merge Deezer BPM and key onto the chosen version**	data-integrity	pipeline/run_weekly.py, pipeline/djlib/deezer.py	–	more than 0 tracks have BPM and key after one run	To do

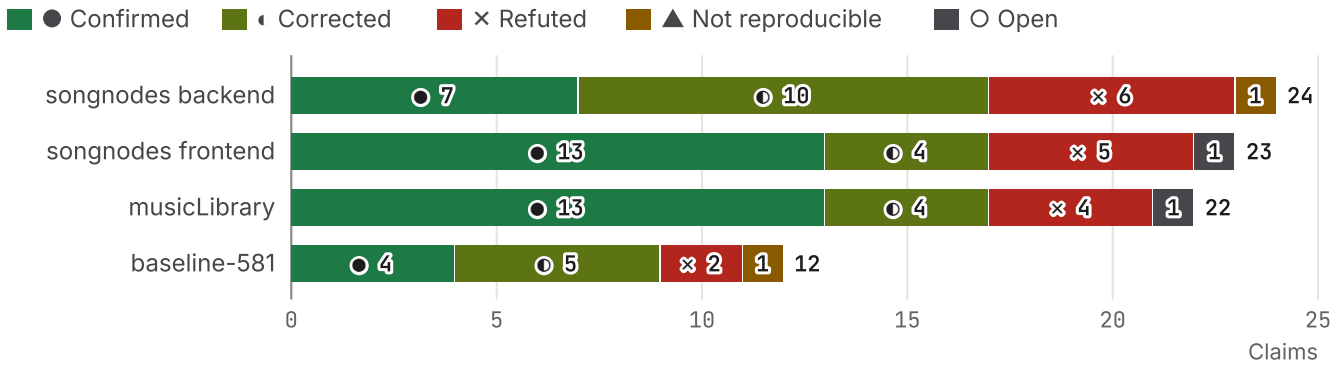
ML-T04	musicLibrary	**Build the play-history import (M2)**	module	pipeline/djlib/play_history.py (new)	–	calibrate.py reports rotation timings from plays	To do
ML-T05	musicLibrary	**Move setlist files into exchange/ and import them**	contract	exchange/setlists/ (new), pipeline/run_weekly.py	SN-T03	JohnCourse and kiki appear as songnodes setlists	To do
ML-T06	musicLibrary	**Warn on resolve yield drops and retry rejected pending**	observability	pipeline/run_weekly.py, pipeline/djlib/export.py	–	a run with half the usual yield writes a warning	To do
DS-T01	design-system	**Publish 1.10.0 to the design-system artifact**	design-system	design-system artifact 7f37707d	–	artifact tokens.json version is 1.10.0	To do

Section 11: **Verification register** 81 claims

Baseline: PR #581 (area baseline-581), re-verified against this run; the register also re-checks the repos' own documentation.

C11: **Verification outcome by area**

37 of 81 repo claims hold as written; 23 needed correcting and 17 are refuted.



C11 · n = 81 · Unit: claims · Built · Source: claims[] (outcome, _agent) · Method: Count of claims per scanning area split by outcome; sorted by total.

81 rows

ID	CLAIM	SOURCE	OUTCOME	FINDING	EVIDENCE
ml-claim-stdlib	Standard-library Python 3.10+. Nothing to install.	musicLibrary/README.md:9	● Confirmed	pipeline imports only stdlib modules; djlib has no third-party imports	musicLibrary/pipeline/run_weekly.py:11-27 +1
ml-claim-schedule	Weekly flow runs as a Sunday 7:37am scheduled task	musicLibrary/README.md:13	× Refuted	runs dated 24 Oct (Sat), 27 Oct (Tue), 31 Oct (Fri), 3 Nov (Tue); none on a Sunday	musicLibrary/outputs/ +2
ml-claim-resolve-fields	resolve writes TIDAL and Deezer versions with duration, version, BPM, key, ISRC, genre	musicLibrary/README.md (step table)	◐ Corrected	0 of 1007 TIDAL versions carry BPM or key; Deezer BPM on 38 of 732 and never copied to the chosen version; ledger BPM 0/413, key 0/413	musicLibrary/inbox/resolved_2026-10-24.json +5

ID	CLAIM	SOURCE	OUTCOME	FINDING	EVIDENCE
ml-claim-resumable	resolve is resumable	musicLibrary/README.md (step table)	● Confirmed	skips keys already in resolved_<date>.json, flushes every 20	musicLibrary/pipeline/run_weekly.py:418-461 +3
ml-claim-db-tables	DB tables tracks … runs; views v_crates, v_source_yield, v_review_queue, v_buy_list	musicLibrary/README.md (Database)	● Confirmed	sqlite_master matches (plus a meta table not listed)	musicLibrary/db/dj_library.sqlite +1
ml-claim-page-source	Page source: artifact/index.html; the weekly run republishes it	musicLibrary/README.md (Set Builder)	× Refuted	the published page is artifact/build/index.html with app.js; artifact/index.html has no library.json beside it	musicLibrary/agent/weekly_run.md:8 +3
ml-claim-sync-script	scripts/sync-music-library.sh copies library.json to frontend/public/data/music-library.json	musicLibrary/README.md (In SongNodes)	× Refuted	neither exists in songnodes 7429821; contract is POST /api/v1/library/ingest (import-music-library.sh, cmd_sync_songnodes)	songnodes/scripts/import-music-library.sh +4
ml-claim-same-set-shape	Sets saved in SongNodes use the same document shape as this artifact's saved sets	musicLibrary/README.md (In SongNodes)	× Refuted	songnodes server stores name, description and uuid track ids only; no tidal, shape or minutes on the server	songnodes/frontend/src/features/musicLibrary/data.ts:271-283 +4
ml-claim-review-queue	Tracks go to review_queue.csv when BPM and source methods disagree by 3+ points	musicLibrary/README.md (Energy model)	◐ Corrected	needs_review is also set when fewer than two methods exist, so all 413 tracks are in the queue	musicLibrary/pipeline/djlib/classify.py:77 +3
ml-claim-weights	Energy blend BPM 45%, source 40%, version cues 15%	musicLibrary/README.md (Energy model)	● Confirmed	rules.energy.weights = 0.45/0.40/0.15; in practice only the source weight applies	musicLibrary/config/rules.json +2
ml-claim-overrides	Overrides are re-applied every run and the auto score is kept	musicLibrary/README.md (levers)	● Confirmed	apply_overrides stores energy_auto once; 0 overrides so far	musicLibrary/pipeline/run_weekly.py:152-182 +3

ml-claim-manual-stamp	manual_adds rows get stamped processed	musicLibrary/README.md (levers)	● Confirmed	stamp written before raw_<date>.json	musicLibrary/pipeline/run_weekly.py:386-395 +1
ml-claim-calibrate	calibrate writes suggestions only	musicLibrary/README.md (levers)	● Confirmed	writes outputs/calibration_suggestions.json; file absent, never run	musicLibrary/pipeline/calibrate.py:85 +2
ml-claim-xml-local	Rekordbox XML only includes tracks with a local_path	musicLibrary/README.md (one-off setup)	● Confirmed	0 local paths, XML has Entries=0	musicLibrary/outputs/rekordbox_musicLibrary.xml +3
ml-claim-tidal-fallback	Until TIDAL setup is done resolve uses Deezer only and TIDAL sync is skipped	musicLibrary/README.md (one-off setup)	● Confirmed	guards on secrets/tidal_app.json and tidal_user_token.json; sync_result files show 403 tracks on TIDAL, so setup was done	musicLibrary/pipeline/run_weekly.py:420-424 +4
ml-claim-db-loud	db fails loudly if the row count or integrity check is wrong	musicLibrary/agent/weekly_run.md:6	● Confirmed	sys.exit on mismatch	musicLibrary/pipeline/build_db.py:185-186 +2
ml-claim-atomic	database is written to a temp file and swapped in atomically	musicLibrary/pipeline/build_db.py:6-7	◐ Corrected	shutil.copyfile over the target, not an atomic rename	musicLibrary/pipeline/build_db.py:179 +2
ml-claim-shared-wins	Shared sets win by id, and sets only in this browser are uploaded	musicLibrary/artifact/app/README.md	● Confirmed	initSavedSets merges remote first then local-only uploads	musicLibrary/artifact/app/artifact-data.ts:55-72 +1
ml-claim-ds-version	Published Set Builder is built from SongNodes design system 1.1.0	musicLibrary/artifact/app/README.md	i Open	no version string in style.css to confirm; songnodes is now 1.10.0 and the build cannot be redone from main	songnodes/design-system/tokens.json +3
ml-claim-isrc-319	Resolver thresholds to be fitted to the 319 ISRC-bearing library tracks	musicLibrary/decisions/2026-10-25-metabrainz-integration-baseline.md	◐ Corrected	all 413 ledger tracks now carry an ISRC	musicLibrary/ledger/dj_library_ledger.json +2
ml-claim-goodstatic	Good Static is always 0 because it publishes no playlists	musicLibrary/agent/weekly_run.md:9	● Confirmed	0 rows in all four scrape logs; Elektrosophy and Passing Notes also fell to 0 on 3 Nov after identical 46-row counts three runs running	musicLibrary/outputs/2026-11-03/scrape_log.txt +3
ml-claim-only-library-json	The weekly run only replaces library.json	musicLibrary/agent/weekly_run.md:8	● Confirmed	publish step passes only index.html and library.json; build/library.json equals outputs/artifact/library.json	musicLibrary/agent/weekly_run.md:8 +3

sn-be-c-unified-only	unified-scraper is the ONLY scraper container	songnodes/CLAUDE.md:58	● Confirmed	Only scraper service in compose.dev.yaml.	songnodes/compose.dev.yaml:145 +1
sn-be-c-medallion	Scraper → Bronze → Silver → Gold → Operational → Frontend	songnodes/CLAUDE.md:11	× Refuted	No gold migration or job runs in compose; graph-api reads silver directly; gold_track_graph has no reader.	songnodes/compose.dev.yaml:13 +4
sn-be-c-no-layer-skip	Bronze → Silver → Gold flow, no layer skipping	songnodes/CLAUDE.md:28	× Refuted	Scraper persistence pipeline writes silver tables directly.	songnodes/scrapers/pipelines/persistence_pipeline.py:910 +2
sn-be-c-etl-sql	Transition ETL inserts (source_track_id, target_track_id, playlist_id)	songnodes/CLAUDE.md:215	◐ Corrected	Columns are from_track_id/to_track_id with playlist_occurrences jsonb and occurrence_count upsert, done in Python.	songnodes/services/common/data layer/ingest.py:28 +2
sn-be-c-target-searches	Scraper queries target_track_searches	songnodes/CLAUDE.md:44	◐ Corrected	Scheduler reads target_tracks (migration 012); target_track_searches is only in legacy SQL.	songnodes/services/common/data layer/target_scrape.py:72 +3
sn-be-c-bronze-schema	Bronze tables carry raw_metadata JSONB	songnodes/CLAUDE.md:199	◐ Corrected	Code writes raw_json (and scraped_at, scraper_version).	songnodes/scrapers/pipelines/persistence_pipeline.py:1117 +1
sn-be-c-scraper-api	Unified Scraper API body {spider_name, search_params}	songnodes/CLAUDE.md:226	◐ Corrected	ScrapeRequest is {source, url, search_query, artist_name, limit, max_pages, job_id, params}.	songnodes/scrapers/unified_scraper_api.py:94 +2
sn-be-c-enrich-8020	Enrichment delegates to metadata-enrichment:8020	songnodes/CLAUDE.md:174	× Refuted	metadata-enrichment is not in compose; 8020 is audio-analysis; enrichment runs in the worker via common/providers.	songnodes/compose.dev.yaml:186 +4
sn-be-c-ollama	Resilience includes Ollama CAPTCHA solving	songnodes/CLAUDE.md:176	▲ Not reproducible	No Ollama service in compose and the Ollama modules are not in ITEM_PIPELINES.	songnodes/scrapers/settings.py:90 +3
sn-be-c-artist-enforce	Artist validity enforced Database view → API filter → Frontend → ETL	songnodes/CLAUDE.md:37	◐ Corrected	API filter (graph-api VISIBLE_TRACK) and ETL (is_placeholder_artist) hold; no enforcing database view exists in applied migrations, and the scraper writes Unknown Artist into silver.	songnodes/services/graph-visualization-api/catalog.py:17 +6
sn-be-c-trigger-scrape	POST /api/v1/scrape/trigger with Idempotency-Key queues a scrape	songnodes/CLAUDE.md:354	● Confirmed	jobs.trigger_scrape enqueues kind scrape keyed by the header.	songnodes/services/rest_api/routers/jobs.py:66 +1

sn-be-c-pools	pool_size=5, max_overflow=10	songnodes/CLAUDE.md:248	◐ Corrected	graph-api uses pool_size 15 + max_overflow 15; rest-api asyncpg min 5 max 15.	songnodes/services/graph-visualization-api/main.py:89 +4
sn-be-c-resource-limits	Resource limits DB 1-2GB, APIs 512MB, scrapers 1GB, frontend 256MB	songnodes/CLAUDE.md:252	× Refuted	Only audio-analysis has mem_limit (1g) in compose.	songnodes/compose.dev.yaml:191 +1
sn-be-c-pg-unpublished	postgres, redis, rabbitmq not published	songnodes/CLAUDE.md:120	● Confirmed	No ports on those services.	songnodes/compose.dev.yaml:4 +3
sn-be-c-backup	pg_dump -U songnodes -Fc songnodes backup one-liner	songnodes/CLAUDE.md:300	● Confirmed	Matches compose user and db; the scripts/backup_database.sh alternative targets musicdb.	songnodes/compose.dev.yaml:7 +3
sn-be-c-scrape-writes-bronze	POST /scrape is synchronous (up to 15 minutes) and writes bronze itself	songnodes/compose.dev.yaml:146	◐ Corrected	It also writes silver tables and song_adjacency.	songnodes/scrapers/pipelines/persistence_pipeline.py:910 +1
sn-be-c-audio-no-db	audio-analysis needs no database or RabbitMQ for /analyze-url	songnodes/compose.dev.yaml:187	● Confirmed	Startup tolerates missing DATABASE_URL and RabbitMQ; queue consumer stays off.	songnodes/services/audio-analysis/main.py:112 +4
sn-be-c-jobs-idem	A retried request returns the original job	songnodes/services/common/datalayer/jobs.py:7	● Confirmed	ON CONFLICT (kind,idempotency_key) DO NOTHING then SELECT.	songnodes/services/common/datalayer/jobs.py:26 +1
sn-be-c-outbox-skip	A slow writer that commits late can never be skipped	songnodes/services/common/datalayer/outbox.py:4	● Confirmed	Reads only txid < pg_snapshot_xmin; side effect: long transactions stall consumers.	songnodes/services/common/datalayer/outbox.py:30 +2
sn-be-c-ingest-409	Same Idempotency-Key with a different body yields 409	songnodes/contracts/library_ingest.md:16	× Refuted	enqueue returns the existing job without comparing the body.	songnodes/services/common/datalayer/jobs.py:26 +2
sn-be-c-ingest-rows	Ingest returns per-row results	songnodes/contracts/library_ingest.md:14	× Refuted	Endpoint returns {job_id,status,created}; per-row stats land in the job result.	songnodes/services/rest_api/routers/library_ingest.py:17 +1
sn-be-c-orchestrator	services/scraper-orchestrator/main_enhanced.py imports automated_scheduler	songnodes/STATUS.md:1567	◐ Corrected	No services/scraper-orchestrator directory on main; root scraper-orchestrator holds only enrich_existing_tracks.py.	songnodes/scraper-orchestrator/enrich_existing_tracks.py:1 +2

sn-be-c-mock-search	REST API still returns Mock Artist until S2	songnodes/plan/RESTRUCTURE.md:14	◐ Corrected	No Mock strings in rest-api main.py; it proxies to metadata-enrichment and returns 503 when unreachable (always, in compose).	songnodes/services/rest_api/main.py:692 +4
sn-be-c-migrate-py	Migrations 012-014 are applied by python migrate.py	songnodes/services/worker/Dockerfile	◐ Corrected	start-dev.sh applies them with psql; migrate.py is used by the Helm hook or a manual compose run.	songnodes/scripts/start-dev.sh:31 +4
sn-fe-c-port	frontend at http://127.0.0.1:3006, nginx proxies /api to the APIs	songnodes/CLAUDE.md §2 Services table	● Confirmed	compose publishes 127.0.0.1:3006:8080; inline default.conf sends /api/(graph|v1/visualization|user-setlists) to graph-api and /api/ to rest-api	songnodes/compose.dev.yaml:202 +3
sn-fe-c-import-order	Import order: legacy CSS → tokens.css → bundle.css → legacy-aliases.css → app.css	songnodes/CLAUDE.md:66	◐ Corrected	main.tsx imports no legacy CSS before tokens.css; order is tokens, bundle, legacy-aliases, app	songnodes/frontend/src/main.tsx:4-10 +1
sn-fe-c-g2	feature routes self-register via G2	songnodes/CLAUDE.md §2 Stack	◐ Corrected	true for library, sets, paths, harmony, sources/import; /, /graph, /sources/targets, /pipeline, /settings and * are still static entries in routes/registry.tsx	songnodes/frontend/src/routes/registry.tsx:13-45 +2
sn-fe-c-nav	The NavDrawer is the only site navigation	songnodes/CLAUDE.md §4 Interaction	● Confirmed	AppShell renders only AppHeader (menu + search) and NavDrawer; App.tsx comment says no visible nav bar	songnodes/frontend/src/shell/AppShell.tsx +3
sn-fe-c-stack	React 19, TypeScript 7, PIXI.js v8, Zustand; Node 26 frontend image	songnodes/CLAUDE.md §2 Stack, §1 Prerequisites	● Confirmed	package.json react ^19.3.1, typescript ^7.0.2, pixi.js ^8.22.0, zustand ^5.0.16; Dockerfile FROM node:26-alpine	songnodes/frontend/package.json:62-100 +3
sn-fe-c-ds-sync	frontend design-system copy is synced and --check fails on drift	songnodes/CLAUDE.md §4	● Confirmed	sync-design-system.sh --check: "design-system 1.10.0: frontend copy in sync"; VERSION 1.10.0 = tokens.json 1.10.0	songnodes/frontend/src/styles/design-system/VERSION +4
sn-fe-c-no-alert	Never alert/confirm/prompt	songnodes/CLAUDE.md §4 Interaction	● Confirmed	0 calls in frontend/src outside tests	M1 grep +1
sn-fe-c-one-primary	one primary button per view	songnodes/CLAUDE.md §4 Interaction	× Refuted	Set builder and Pathfinder show two primaries at once; the repo checker does not test this	songnodes/frontend/src/features/musicLibrary/SetBuilderPage.tsx:450,590 +3
sn-fe-c-en-dash	Missing values are an en dash; no "Unknown"	songnodes/CLAUDE.md §4 Writing, §8	× Refuted	AI setlists show 'Unknown Date'; OAuth page 'Unknown error'	songnodes/frontend/src/features/aiSetlists/AISetlistsPage.tsx:338 +2
sn-fe-c-pending-list	Files not yet on the design system are listed in design-system-pending.txt; the conformance job scans every other file	songnodes/frontend/design-system-pending.txt:1-3	● Confirmed	all 37 lines are commented out as migrated; check_design_system.py --check exits 0 with no findings (but misses R2 and R5 violations found by grep)	songnodes/frontend/design-system-pending.txt +3

sn-fe-c-pointer-events	Graph uses pointerdown/pointerup, not click	songnodes/CLAUDE.md §5.4	● Confirmed	GraphVisualization: 8 pointerdown/pointerup uses, 0 .on('click')	songnodes/frontend/src/components/GraphVisualization.tsx +1
sn-fe-c-shortcuts	Shortcuts D (debug), Space (pause), Escape (clear), Ctrl+A (select all)	songnodes/CLAUDE.md §5.4	i Open	only case 'd' found in GraphVisualization; useKeyboardShortcuts is unreachable; Space/Escape/Ctrl+A handlers not located	songnodes/frontend/src/components/GraphVisualization.tsx:3123 +4
sn-fe-c-artist-filter	Frontend validation drops NULL/Unknown artists from the graph	songnodes/CLAUDE.md Critical requirement 1	● Confirmed	useDataLoader filters nodes with hasValidArtist on both primary and fallback loads (fallback edge filtering gap noted in R8)	songnodes/frontend/src/hooks/useDataLoader.ts:40,88 +3
sn-fe-c-frontend-mem	Resource limits: Frontend 256MB	songnodes/CLAUDE.md §5.3	× Refuted	compose.dev.yaml frontend service sets no mem_limit	songnodes/compose.dev.yaml:200-211 +1
sn-fe-c-indexeddb	SongNodes keeps the last graph, library and saved sets in IndexedDB	songnodes/design-system/16-platform.md:50	× Refuted	no indexedDB use anywhere in frontend/src	M1 grep indexedDB = 0 +2
sn-fe-c-frame-deny	nginx/nginx.conf sends X-Frame-Options: DENY and frame-ancestors none	songnodes/design-system/16-platform.md:72	◐ Corrected	nginx/nginx.conf is not used by compose; the live frontend default.conf sets no frame headers	songnodes/frontend/Dockerfile:28-102 +2
sn-fe-c-error-pages	404/50x static files do not exist yet	songnodes/design-system/13-page-patterns.md:136	● Confirmed	no error_page in the frontend default.conf; SPA 404 route handles unknown paths	songnodes/frontend/Dockerfile:28-102 +3
sn-fe-c-db1-fallback	loadLibrary falls back to /data/music-library.json only when the API fails	songnodes/plan/agents/DB1_one_track_source.md:23	× Refuted	no fallback in data.ts and frontend/public has no data/ folder; failure surfaces as an error	songnodes/frontend/src/features/musicLibrary/data.ts:107-135 +3
sn-fe-c-stages-poll	Scrape stage cards poll every 5 s while the tab is visible, at most 5 cards	songnodes/STATUS.md:2161	● Confirmed	STAGES_POLL_MS = 5000; slice(0, 5) on watched runs	songnodes/frontend/src/features/pipeline/ScrapeJobStages.tsx:10,45 +3
sn-fe-c-gv2-next	GV2 next: load the neighbourhood from the API (GV3) instead of the whole graph	songnodes/STATUS.md:2505	● Confirmed	useDataLoader still fetches /api/graph/data?limit=10000 with no view param (queued)	songnodes/frontend/src/hooks/useDataLoader.ts:28 +1
sn-fe-c-body-limit	frontend nginx 1 MB default body limit is a follow-up	songnodes/STATUS.md:1899	● Confirmed	no client_max_body_size in frontend default.conf while FileDrop allows 50 MB	songnodes/frontend/Dockerfile:28-102 +3

sn-fe-c-ui9-debug	debug console.log calls in GraphVisualization and useTokenRefresh stay out of production	songnodes/STATUS.md (UI9 slot)	◐ Corrected	true for those two files, but OAuthCallback still console.logs code/state prefixes and storage keys in production	songnodes/frontend/src/lib/debugLog.ts +4
sn-fe-c-banner-day	Banner dismissal should also end at the next Melbourne day (DR-051)	songnodes/STATUS.md:2236	● Confirmed	bannerKey stores only the count; follow-up still queued	songnodes/frontend/src/features/musicLibrary/storage.ts:29-30 +1
BL-B01	12 pages	documentation/data-flow-analysis/findings.json pages[] @1f296c3	◐ Corrected	12 is the songnodes frontend route count (13 production routes found here, including the shadowed /paths placeholder); the estate holds 115 pages and entry points across both repos.	PR #581 (branch docs/data-flow-investigation @1f296c3) +1
BL-B02	8 stores	findings.json stores[] @1f296c3	◐ Corrected	39 internal stores plus 9 external systems (48) found by three scanners; #581 grouped Postgres by layer and omitted RabbitMQ, the artifact database, ref_mb and external APIs.	PR #581 (branch docs/data-flow-investigation @1f296c3) +1
BL-B03	26 edges	findings.json edges[] @1f296c3	◐ Corrected	87 numbered flows, every one drawn in D2a–D2g.	PR #581 (branch docs/data-flow-investigation @1f296c3) +1
BL-B04	5 personas	findings.json personas[] @1f296c3	◐ Corrected	8 personas (5 human, 2 system actors, 1 assisted); #581's selector, librarian, data engineer, resolution specialist and devops map onto P1/P2, P3, P4/P5, P3/P7 and P4.	PR #581 (branch docs/data-flow-investigation @1f296c3) +1
BL-B05	5 gaps	findings.json gaps[] @1f296c3	◐ Corrected	28 gaps; all five #581 gaps are kept: GAP-01 Rekordbox feedback = ml-gap-rekordbox-feedback; GAP-02 dual storage = ml-gap-saved-set-shape and ml-gap-songnodes-field-loss; GAP-03 audio fingerprint segmenter is not evidenced in either repo (kept as Unverified); GAP-04 unwired target queue feedback = sn-be-gap-target-scrape-quota and sn-fe-gap-target-pending-sync; GAP-05 outbox projection lag = sn-be-gap-library-projection.	PR #581 (branch docs/data-flow-investigation @1f296c3) +1
BL-B06	Gate 1: 0 hard-coded colours or fonts outside token definitions	PR #581 description	× Refuted	243 literal hex colours in SVG fill/stroke attributes of report/index.html outside style blocks (this run's gate 2 regex over tags).	PR #581 (branch docs/data-flow-investigation @1f296c3) +1
BL-B07	All 14 internal navigation anchors resolve	PR #581 description	● Confirmed	14 href=#... anchors, all with a matching id.	PR #581 (branch docs/data-flow-investigation @1f296c3) +1
BL-B08	Standalone SVGs at viewBox 1800×1100	PR #581 description	● Confirmed	system-architecture-data-flow.svg viewBox 0 0 1800 1100.	PR #581 (branch docs/data-flow-investigation @1f296c3) +1

BL-B09	enhanced-architecture-connections-and-gaps.svg and ARCHITECTURE_ENHANCEMENTS_AND_GAPS_REPORT.md produced	STATUS.md SLOT:ARCH1 @1f296c3	× Refuted	Neither file is in the branch (git diff --stat origin/main...1f296c3 lists 17 files without them); both exist in this run.	PR #581 (branch docs/data-flow-investigation @1f296c3) +1
BL-B10	Sister TODO written to E:\musicLibrary\TODO-ARCHITECTURAL-IMPROVEMENTS.md	STATUS.md SLOT:ARCH1 @1f296c3	● Confirmed	File present on tuf-boi (7,332 bytes, 6 Nov 2026 22:18 UTC); kept and extended, not replaced.	PR #581 (branch docs/data-flow-investigation @1f296c3) +1
BL-B11	PDF 1,775 KB, vector, searchable	PR #581 description	● Confirmed	1,817,715 bytes in the branch.	PR #581 (branch docs/data-flow-investigation @1f296c3) +1
BL-B12	Rebuild pipeline passes in 3.2 s	STATUS.md SLOT:ARCH1 @1f296c3	▲ Not reproducible	Not re-run: its build scripts are kept in the branch but superseded by build.py; timing is the author's figure.	PR #581 (branch docs/data-flow-investigation @1f296c3) +1

Section 12: **Methods, assumptions and open decisions**

Profile: **Project profile as applied**

Scope	songnodes (E:\songnodes = github zvirb/songnodes main) and musicLibrary (E:\musicLibrary). Sub-projects scanned: songnodes frontend, services, scrapers, design-system; musicLibrary pipeline, artifact, ref_mb, plan.
Page source of truth	songnodes source tree on main @7429821 (no built bundle is kept in the repo); musicLibrary artifact/build/index.html is the live page, artifact/index.html is the legacy page (agent ml, M3 render).
Queued work	Open PRs #576–#580 (Duo 2FA, Cloudflare tunnel, login page, Sentinel sanitisation); musicLibrary plan/jules_payloads (30) and the unmerged WP7 brief.
Baseline report	PR #581 (ARCH1, opened 7 Nov 2026 09:23 by a parallel run on branch docs/data-flow-investigation): its numbers and gate claims are re-verified in the register (area baseline-581) and every topic it covers is kept; this report is merged into that branch. The register also re-checks the repos' own documentation.
Standing rules	R1–R8 from songnodes CLAUDE.md §2–§4 and AGENTS.md rule 12 (listed in section 3).
Retired or excluded	deploy/ and k8s/ (kept but unused, CLAUDE.md §1); musicLibrary _to_delete/ and data/ (data/ listed only).
Locale	en-AU, A4, Australia/Melbourne 24-hour time (design system 01-writing).
Personas	Inferred from pages and docs; merged across the three scans into one set (section 6).
Design system	SongNodes design system, artifact 7f37707d and repo mirror songnodes/design-system, 1.9.1 raised to 1.10.0 by this investigation.

Pre-flight: **Tools and access**

songnodes git	main @7429821, clean except untracked docs/DEPLOYMENT_E2E_STATUS_2026_11_06.md on the laptop; push via the session's GitHub app (REST only, GraphQL blocked)
musicLibrary git	Not a git repository: files only, no commits or PRs possible
Tools	node 22.23, python 3.13, graphviz dot 2.43, Chromium 153 (Playwright), poppler 24.02 (pdftoppm, pdffonts, pdftotext), pypdf 6.17; missing: PyMuPDF (not needed)
Laptop shell	tuf-boi Linux VM over the bridge: deletes blocked, no gh; used for the musicLibrary snapshot only
Existing reports	None of this kind. Related: songnodes docs/DATA_LAYER.md, STATUS.md, plan/RESTRUCTURE.md; musicLibrary README.md, docs/mb/*

Methods: **Where the methods agreed and disagreed** 37 notes

37 rows

ITEM	METHODS	RESULT	NOTE
artifact localStorage keys	M1, M3	● Agree	M1 found sn-music-library-sets, sn-set-draft, sn-filters-library, sn-banner-keyed-*; M3 also saw sn-density (written by SN.prefs, not in feature code). Legacy nsb:* keys found by M1 only, since the legacy page never reaches the draft write without library.json.
window.claude calls	M1, M3	● Agree	Build: use('db') then collection('sets').get; legacy: use('db'), use('user'), user.can('data.write'), onSnapshot('sets'). Without a stub no calls and no errors.
library.json beside the legacy page	M1, M3	▲ Disagree	Code fetches library.json relative to artifact/index.html (M1); file does not exist there, M3 got 404/CORS and 'Library file missing'. Confirms superseded status.

ITEM	METHODS	RESULT	NOTE
file:// render	M3	● Agree	Both pages fail to load library.json from file:// (CORS, origin null); console errors only from that. Over http the build page renders 413 tracks with 0 console errors (stubbed claude) or one favicon 404.
songnodes sync contract	M1, M2	▲ Disagree	README names scripts/sync-music-library.sh and frontend/public/data/music-library.json; neither exists. Code (M2: run_weekly.cmd_db → cmd_sync_songnodes; songnodes import-music-library.sh) uses POST /api/v1/library/ingest.
artifact page source	M1, M2	▲ Disagree	README says artifact/index.html is the page source; weekly_run.md and app/README say the published page is artifact/build/index.html built from songnodes pages.
BPM and key coverage	M1, M3	▲ Disagree	Code and README describe a BPM-band method and Camelot moves; data (ledger, resolved files, SQLite) and the rendered page show 0 of 413 tracks with BPM or key.
sync-sets tidal_id fallback	M1	▲ Disagree	sync-sets falls back to tr.tidal_id, but both page versions save tr.tidal (a URL), so the fallback never fires.
SQLite schema	M1, M3	● Agree	SCHEMA string matches sqlite_master of db/dj_library.sqlite (13 tables, 4 views); counts recorded from the file.
ref_mb consumers	M1, M2	● Agree	Only songnodes (refmb/settings.py, MB_LOCAL_* in compose.dev.yaml) reads ref_mb; no musicLibrary pipeline module does.
HTTP endpoints	M1, M2, M3	● Agree	AST parse found 215 decorated routes (rest-api 98 incl. main.py, graph-api 18, unified-scraper 5, audio-analysis 4, rest are non-compose services). Registry check: every rest-api router with routes is mounted explicitly or via register(); data_quality is neither (its decorators are commented out, AST found none).
rest-api /api/graph/nodes and /edges	M2, M3	▲ Disagree	AST lists them on rest-api; nginx config routes ^/api/graph to graph-api, so they are unreachable from the frontend.
Silver writers	M1, M2	▲ Disagree	Docs say silver comes from bronze ETL only; grep finds the scraper persistence pipeline, rest-api catalog writes, and one-off scripts also writing silver_enriched_tracks.
Gold layer	M1, M2	▲ Disagree	grep finds gold ETL code and Helm cronjobs; compose/start-dev never run or migrate them; gold_track_graph has zero readers.
RabbitMQ usage	M1, M2	▲ Disagree	grep finds aio_pika in audio-analysis, metadata-enrichment, dlq-manager, websocket-api and a scraper publisher; compose gives RabbitMQ only to worker (Celery) and scraper env; audio-analysis has no RabbitMQ env so its consumer stays off.
Redis graph cache	M1	▲ Disagree	cache_result decorator and CACHE_TTL exist, but no route uses the decorator; only batch:{id} keys are written.
Frontend-to-backend routes	M1, M3	▲ Disagree	Frontend references /api/v1/scrapers/target-tracks/search and /api/v1/scrapers/queue/clear; AST route list has neither.

Memgraph	M1, M2	▲ Disagree	graph_projection code and job kinds exist; compose has no memgraph service or MEMGRAPH_* env.
Migrations applied	M2	● Agree	compose initdb mounts (001,002,006,012,013,014) and start-dev.sh psql lines (101,170,180-185,220,240,260,270,280, init/07) cross-checked with worker/migrate.py REQUIRED list.
Open PRs #576-#580	M1, M2	● Agree	Read via gh REST pulls/{n}/files and patches; items marked state queued. #578 adds no route-gating middleware; #577 adds public ingress.
R6 router registration	M2, M3	▲ Disagree	AST check for top-level register() (router_registry._defines_register) vs main.py include_router list: 11 routers lack register() and rely on explicit wiring.
routes	M1, M2	▲ Disagree	Grep finds PlannedModulePage for /paths in STATIC_ROUTES; registry replacement logic (M2) shows features/pathfinder/route.tsx overrides it, so the planned page is dead.
LiveDataLoader /api/ws	M1, M2	▲ Disagree	Grep finds a WebSocket client; import graph from main.tsx shows LiveDataLoader and useLiveDataLoader are never imported.
services/api.ts clients	M1, M2	▲ Disagree	Grep lists ~20 endpoints; only api.get/post/put/delete (target-tracks client) and trackApi (from unreachable NowPlayingDeck) are imported; backend route scan shows most paths do not exist.
songnodes_ai_prompt_history	M1, M2	▲ Disagree	Key found by grep; its only consumer AIAssistantModal is unreachable.
OAuth token storage	M1, M2	▲ Disagree	Writers use sessionStorage <service>_oauth_tokens; readers use localStorage tidal_oauth_tokens; same-window redirect means no postMessage path.
display preference keys	M1, M3	● Agree	bundle.js and index.html both use sn-theme/sn-text-size/sn-density/sn-motion; Playwright render of reference.html wrote sn-density only.
design-system previews	M1, M3	▲ Disagree	Grep shows previews write sn-disclosure:* and sn-skip-moved-targets; Playwright standalone render throws before writing because window.SongNodes is injected by the card host.
nginx proxy target	M2	● Agree	compose.dev.yaml builds frontend/Dockerfile whose inline default.conf is the only live proxy; nginx/ and nginx-local-proxy.conf are not referenced by compose (only Helm values mount a default.conf).
frontend endpoints vs backend	M1, M2	▲ Disagree	Static route scan of services/rest_api and graph-visualization-api (120 decorators) found no match for /api/v1/scrapers/*, /api/v1/search/*, /api/v1/pathfinding/*, /api/graph/tracks|positions, /api/ws; prefixes defined via include_router in main.py were not resolved, so a few may exist (see unverified).
public/data/music-library.json	M1, M2	▲ Disagree	Plan briefs (DB1, L8, QA7) describe the static file and its fallback; the file is absent from frontend/public and no code fetches /data/*.json. Only tests/e2e/data/library.fixture.json remains (test-only).
R2/R5 conformance	M1, repo checker	▲ Disagree	check_design_system.py --check and design-system/linters/check.py report 0 findings on all route page files, but grep finds two simultaneous primaries (Set builder, Pathfinder), Title Case labels and "Unknown" fallbacks.
static HTML conformance	design-system linter, M3 Playwright	● Agree	linter flags tracker.html hex (12) and reference.html emoji/caps; Playwright shows tracker loads only Google Fonts and needs the artifact runtime.

claim ml-claim-ds-version	scanner: ml, verifier: GitHub REST history of design-system/tokens.json vs build file mtimes; token-name set diff of build style.css against tokens.css at 29619fe5b	▲ Disagree	Disagree with open: the build was made 11 minutes after 1.1.0 landed and before 1.1.1, and its token set matches 1.1.x (1.1.0-1.2.0 tokens.css are identical, so tokens alone cannot pin the patch). Also: main @7429821 is 1.9.1, not 1.10.0; 1.10.0 is the unpushed local commit 78d1d4f.
claim sn-be-c-enrich-8020	scanner: sn-backend, verifier: Followed the registered scraper pipeline to its default URL	▲ Disagree	Disagree with refuted: the scraper does still delegate to metadata-enrichment:8020 as documented; the host just does not exist in compose, so every call lands in service_unavailable and real enrichment happens in the worker.
claim sn-be-c-ollama	scanner: sn-backend, verifier: Checked the downloader-middleware layer (where CAPTCHA solving lives), spider custom_settings and compose	▲ Disagree	Disagree with not-reproducible: it is checkable and false. Code exists (middlewares/captcha_middleware.py, ollama_captcha_solver.py) but is never registered. The scanner looked at ITEM_PIPELINES, the wrong layer.
claim sn-fe-c-shortcuts	scanner: sn-frontend, verifier: Read the window keydown handler in full	▲ Disagree	Disagree with open: all four handlers exist in the same switch the scanner cited. Caveat: Ctrl+A calls toggleNodeSelection, so already-selected nodes get deselected.

Secrets: **Secrets scan (location and type only)** `32 findings`

32 rows

LOCATION	TYPE	RISK
musicLibrary/secrets/tidal_app.json (not in snapshot)	TIDAL client id and client secret copied from songnodes .env TIDAL_CLIENT_ID/TIDAL_CLIENT_SECRET	▲ Real-looking
musicLibrary/secrets/tidal_user_token.json (not in snapshot)	TIDAL OAuth access and refresh token	▲ Real-looking
musicLibrary/secrets/tidal_pkce_pending.json (not in snapshot)	PKCE verifier and state (blanked to {} after use)	▲ Real-looking

LOCATION	TYPE	RISK
musicLibrary/secrets/tidal_playlists.json (not in snapshot)	crate and set to TIDAL playlist id map (not a credential)	● Placeholder
musicLibrary/ref_mb/secrets/ref_mb_admin_password.txt	Postgres admin password file (32 bytes, Docker secret)	▲ Real-looking
musicLibrary/ref_mb/secrets/ref_mb_reader_password.txt	Postgres read-only role password file (32 bytes)	▲ Real-looking
musicLibrary/ref_mb/.env	REF_MB_PORT only, no secret	● Placeholder
env JULES_API_KEY (shell only)	Google Jules API key	● Placeholder
songnodes/.env.example:40	TIDAL client id/secret placeholders	● Placeholder
songnodes/.env.example:34	Spotify client id/secret placeholders	● Placeholder
songnodes/.env.example:59	API_KEY_ENCRYPTION_SECRET placeholder (not wired into compose)	● Placeholder
songnodes/compose.dev.yaml:60	provider credentials passed from host env with empty defaults	● Placeholder
songnodes/compose.dev.yaml:101	RabbitMQ guest/guest default broker credentials in CELERY_BROKER_URL	× Committed
songnodes/compose.dev.yaml:11	Postgres POSTGRES_HOST_AUTH_METHOD trust (no password) inside compose network	× Committed
songnodes/sql/init/07-api-keys.sql:61	hard-coded fallback pgp_sym encryption key used when no secret is supplied	× Committed
songnodes/sql/init/07-api-keys.sql:308	POSTGRES_ENCRYPTION_SECRET example in comment	● Placeholder
songnodes/sql/migrations/006_user_oauth_tokens_up.sql:126	OAuth access/refresh tokens stored plain text in user_oauth_tokens (runtime data, not committed)	▲ Real-looking
songnodes/k8s/base/secret.yaml:20	Kubernetes Secret with CHANGE_ME placeholders (DB, Redis, Spotify, TIDAL, Last.fm, YouTube, Discogs, Anthropic)	● Placeholder
songnodes/deploy/helm/songnodes/values.yaml:777	Helm secretKeyRef names only	● Placeholder

songnodes/services/api-gateway/.env.example:6	JWT secret and Redis password examples	● Placeholder
songnodes/services/db-connection-pool/start.sh:25	default DB password fallback in shell parameter expansion	▲ Real-looking
songnodes/scripts/run_integration_tests.sh:138	default DB password fallback	▲ Real-looking
songnodes/scrapers/middlewares/captcha_middleware.py:58	CAPTCHA API key placeholder literal	● Placeholder
PR #578 user_auth.py	DUO_CLIENT_ID/SECRET, JWT_SECRET_KEY, AUTH_USERS from env; AUTH_USERNAME/AUTH_PASSWORD default admin/admin in code	× Committed
PR #577 compose.dev.yaml	CLOUDFLARE_TUNNEL_TOKEN from host env	● Placeholder
browser localStorage songnodes-store.state.musicCredentials (runtime, not in repo)	TIDAL/Spotify access and refresh tokens, client id and client secret	▲ Real-looking
browser sessionStorage <service>_oauth_tokens (runtime)	OAuth access/refresh tokens	▲ Real-looking
songnodes/frontend/src/store/useStore.ts:1577-1581	debugZustand.injectTestCredentials placeholder client id/secret	● Placeholder
songnodes/frontend/src/components/APIKeyManager.tsx:529,569	AI provider API key input placeholders (prefix hints only); keys POSTed to /api/v1/api-keys, legacy raw keys under songnodes_key_*/ai_key_*/apiKey_* removed on migration	● Placeholder
songnodes/frontend/src/features/setBuilder/sendToTidal.ts:38-44	browser-held TIDAL access token sent as Authorization Bearer to rest-api	▲ Real-looking
songnodes/nginx/ssl/ (ca.crt, musicdb.crt, musicdb-chain.crt, ca.srl)	TLS certificates only; musicdb.key referenced by nginx/conf.d/blue-production.conf but not committed	● Placeholder
PR #579 frontend/src/features/auth/AuthContext.tsx	session cookie (credentials: include) for Duo login	● Placeholder

Unverified: **Unverified items** 20 items

- musicLibrary/secrets/ is not in this copy: tidal_app.json, tidal_user_token.json, tidal_playlists.json contents and whether TIDAL login is live are inferred from sync_result files (403 tracks TIDAL_ONLY) and code only
- Artifact db capability config and the 'only you write' rule are from app/README.md, not read from the live artifact
- Whether TIDAL honours a reused Idempotency-Key by replaying the earlier response
- Why TIDAL v2 returns no bpm/key (scope, country or catalogue coverage)
- data/ contents beyond _data_manifest.txt (58 GB not copied); ref-mb running state inferred from postmaster.pid date
- Scheduled task configuration itself (cron, device, folders) was not inspected, only agent/weekly_run.md
- songnodes Send to TIDAL backend behaviour beyond the route signature
- Runtime behaviour: nothing was executed against a running stack (no Docker here); daily-quota and duplicate-silver effects are inferred from code.
- scripts/sync-music-library.sh named in the brief does not exist; the music library import script is scripts/import-music-library.sh.
- External API host list is from URL literals; per-provider rate limits and auth were not traced line by line.
- Legacy service endpoints are listed from decorators only; their table use was not traced.
- common/datalayer/audio.py exact call line to /analyze-url not pinned.
- Backend route existence was checked by decorator scan only; routers mounted with include_router(prefix=...) in services/rest_api/main.py were not resolved, so

/api/v1/scrapers/* could exist under a different mount.

- Whether sendToTidal reads the token from songnodes-store musicCredentials.tidal.accessToken was inferred from the Authorization header code path, not traced to its caller.
- Behaviour of POST /api/v1/library/ingest (writes library.tracks) belongs to the backend agent.
- design-system/source/previews files were grouped by folder count; individual titles not extracted.
- Pathfinder export JSON branch content (pathfinder.machine.ts:80-90) not read in full.
- PR #579 login page reviewed by grep of its files at the PR head only; not rendered.
- sn-fe-sources-import R2 third-primary visibility inferred from markup, not rendered.
- #581 GAP-03 (audio fingerprint and AcoustID transition segmenter): no code, schema or doc in either repo supports it; kept as a topic, not as a gap.

Decisions: **Assumptions and open decisions**

- The laptop copy of songnodes equals origin/main @7429821 (git status showed only one untracked doc).
- musicLibrary data/ (58 GB) holds raw dumps and caches; it was listed, not read.
- Effort sizes (S, M, L) are Estimates from evidence spread against the AGENTS.md package budget.
- Gap severity is the highest severity of debt sharing evidence with the gap, otherwise medium (Estimate).
- Open decision: keep a gold layer or document silver-plus-projection as the model (SN-T06).
- Open decision: which Set Builder is primary once both read the same saved-set shape.

Section 13: **Handover**

Handover: **Card**

Pull request (master)	https://github.com/zvirb/songnodes/pull/581 (github.com)
Branch	docs/data-flow-investigation (merged with the parallel run's PR #581 branch)
Deliverables	documentation/data-flow-analysis/: report/index.html, report/print-sheet.html, DATA_FLOW_AND_PAGE_INTERACTION_REPORT.pdf and .md, PAGE_CLASSIFICATION_AUDIT.md, findings.json, figures.json, 51 SVGs, build/
Design system	1.10.0 (source changes in design-system/, frontend copy re-synced)
Tool gaps	None blocking. PyMuPDF absent (not needed); musicLibrary has no git, so its files are written to E:\musicLibrary directly

Gates: **Verification gates** `10 gates`

✓ **1 linter**
0 findings from design-system/linters/check.py

✓ **2 tokens**
0 literal colours, sizes, spacing or radii outside design-system definitions

✓ **3 parity**
0 items missing from a figure or table in web, sheet or PDF; 0 flows undrawn

✓ **4 diagrams**
51 SVGs checked; 0 with overflow, collision or text under 9.5 px; 0 edges through nodes; crossings {'D1': 0, 'D2a': 1, 'D2b': 7, 'D2c': 3, 'D2d': 0, 'D2e': 1, 'D2f': 0, 'D2g': 0, 'D3a': 2, 'D3b': 0, 'D3c': 10, 'D3d': 12, 'D3e': 0, 'D3f': 0, 'D4a': 4, 'D4b': 0, 'D4c': 0, 'D5': 0, 'D5-P1': 0, 'D5-P2': 0, 'D5-P3': 0, 'D5-P4': 0, 'D5-P5': 0, 'D5-P6': 0, 'D5-P7': 0, 'D5-P8': 0, 'D6a': 0, 'D6b': 0, 'D6c': 0, 'D7a': 0, 'D7b': 0, 'D7c': 0, 'D8a': 0, 'D8b': 0, 'D8c': 0, 'D9': 0, 'D10': 0, 'D11': 1, 'D12': 0}

✗ 5 pdf
196 pages; pages over 40% empty (pages before an orientation change exempt): [(47, 0.43), (145, 0.41), (146, 0.44), (180, 0.44)]

✓ 6 colour
0 module-colour uses in figures; design-system contrast report passes; greyscale shots for QA in build/evidence

✓ 7 conformance
18 screenshots (report and reference, 3 widths × themes) for QA

✓ 8 offline
0 console errors, 0 unresolved anchors, 0 network requests with the network blocked

✓ 9 secrets
0 secret-shaped strings in outputs

✓ 10 claims
1 PR URLs in the handover, 0 not confirmed by the GitHub API

Next: **Recommended next steps**

- Publish design system 1.10.0 to the design-system artifact so it stays mirrored (task DS-T01).
- Close the security gate before PR #577's tunnel merges (wave W1: SN-T01, SN-T02).
- Repair the cross-project contracts (wave W2), then run build.py again: a refresh is a findings.json swap.
- Decide the two open questions: gold layer or silver plus projection; which Set Builder is primary.